

Topological Energy Condensate Theory (TECT): 33

Jusang Lee*
Independent Researcher, Incheon, Korea
 (Dated: March, 2026)

I. ENLARGED RESIDUAL SYSTEM ON THE EXACT FOUR-CLASS BASIS

II. GOAL

At the corrected residual stage, the correct reduced object is the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}, \quad \delta = \begin{pmatrix} a \\ e \\ o \end{pmatrix},$$

not a naive reduced free-energy stationary equation.

The frozen operational point is

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta_f := \delta^{\text{final}} = (0.0096, -0.0288, 0.0096)^T.$$

The exact four-class quartic shell data are

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813.$$

The purpose of this section is:

1. to write the enlarged residual system directly on the four-class basis;
2. to prove that a coarse A/E -symmetric completion cannot explain the corrected branch;
3. to identify the representation type of the first admissible microscopic completion.

III. 1. EXACT COARSE (A, E, O) QUARTIC RESIDUAL BASELINE

Define

$$r(\delta) := \begin{pmatrix} a^2 \\ e^2 \\ o^2 \end{pmatrix}, \quad D(\delta) := \text{diag}(a, e, o).$$

The exact coarse four-class quartic matrix is

$$M^{(4c)} = \begin{pmatrix} \frac{U_d + U_{\text{opp}}}{2} & U_{\perp} & U_{\text{obl}} \\ U_{\perp} & \frac{U_d + U_{\text{opp}}}{2} & U_{\text{obl}} \\ U_{\text{obl}} & U_{\text{obl}} & \frac{U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}}}{8} \end{pmatrix}.$$

Numerically,

$$M^{(4c)} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}.$$

* jusanglee@kakao.com

The exact quartic residual baseline is then

$$\boxed{G^{(4)}(\delta) = D(\delta) M^{(4c)} r(\delta).} \quad (1)$$

Equivalently, in components,

$$G_A^{(4)}(\delta) = a \left[\frac{U_d + U_{\text{opp}}}{2} a^2 + U_{\perp} e^2 + U_{\text{obl}} o^2 \right], \quad (2)$$

$$G_E^{(4)}(\delta) = e \left[U_{\perp} a^2 + \frac{U_d + U_{\text{opp}}}{2} e^2 + U_{\text{obl}} o^2 \right], \quad (3)$$

$$G_O^{(4)}(\delta) = o \left[U_{\text{obl}}(a^2 + e^2) + \frac{U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}}}{8} o^2 \right]. \quad (4)$$

At the corrected seed

$$\delta_f = (0.0096, -0.0288, 0.0096)^T, \quad r_f = (9.216 \times 10^{-5}, 8.2944 \times 10^{-4}, 9.216 \times 10^{-5})^T,$$

one obtains

$$Q_f := M^{(4c)} r_f = \begin{pmatrix} 8.494671573456258 \\ 8.241391656489691 \\ 2.580391735022780 \end{pmatrix},$$

and therefore

$$\boxed{G^{(4)}(\delta_f) = D(\delta_f) Q_f = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}.} \quad (5)$$

IV. 2. WHY THE FREE-ENERGY ROUTE STAYS DEAD

The corresponding coarse stationary equations are obtained by dividing by $a, e, o \neq 0$:

$$0 = \frac{U_d + U_{\text{opp}}}{2} a^2 + U_{\perp} e^2 + U_{\text{obl}} o^2 + \dots, \quad (6)$$

$$0 = U_{\perp} a^2 + \frac{U_d + U_{\text{opp}}}{2} e^2 + U_{\text{obl}} o^2 + \dots, \quad (7)$$

$$0 = U_{\text{obl}}(a^2 + e^2) + \frac{U_d + U_{\text{opp}} + 2U_{\perp} + 4U_{\text{obl}}}{8} o^2 + \dots. \quad (8)$$

But the already-extracted exact four-class algebraic stationary solution has

$$r_A^{\text{crit}} > 0, \quad r_E^{\text{crit}} < 0, \quad r_O^{\text{crit}} < 0,$$

so the free-energy stationary reconstruction is physically inadmissible. Hence the correct next object remains the residual map itself:

$$\boxed{G(\delta) = 0 \quad \text{must be handled directly, not through } r\text{-stationarity.}}$$

V. 3. SHARP COARSE IMPOSSIBILITY OF AN A/E -SYMMETRIC LOCAL COMPLETION

The four-class quartic matrix is exactly invariant under

$$A \leftrightarrow E.$$

Assume, for contradiction, that the local completion at coarse level also preserves A/E -symmetry and contributes only equally to the A - and E -channels at linear order. Then the corrected-seed stationarity conditions for the A - and E -channels have identical nonquartic parts, so after division by $a \neq 0$ and $e \neq 0$ and subtraction one gets

$$\left(\frac{U_d + U_{\text{opp}}}{2} - U_{\perp} \right) (a^2 - e^2) = 0. \quad (9)$$

Now

$$\frac{U_d + U_{\text{opp}}}{2} = 8683.640035688459, \quad U_{\perp} = 9027.17290917827,$$

so

$$\frac{U_d + U_{\text{opp}}}{2} - U_{\perp} = -343.532873489811 \neq 0.$$

Therefore (9) forces

$$a^2 = e^2.$$

But at the corrected seed,

$$a^2 = (0.0096)^2 = 9.216 \times 10^{-5}, \quad e^2 = (0.0288)^2 = 8.2944 \times 10^{-4} = 9a^2.$$

Hence

$$a^2 \neq e^2,$$

a contradiction.

Therefore:

a coarse A/E -symmetric local completion cannot explain the corrected branch.

(10)

So a nonzero A/E splitting is already necessary at the local reduced level.

VI. 4. EXACT QUARTIC JACOBIAN AND WHY THE RESIDUAL MAP IS LOCALLY WELL-CONDITIONED

Differentiating (1) gives

$J^{(4)}(\delta) = \text{diag}(M^{(4c)}r(\delta)) + 2D(\delta) M^{(4c)} D(\delta).$

(11)

At the corrected seed this becomes

$$J^{(4)}(\delta_f) = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix},$$

with eigenvalues

$$3.52129589, \quad 8.35219818, \quad 24.47164948.$$

So although the quartic coefficient matrix itself is indefinite, the quartic *residual Jacobian* at the corrected seed is already positive definite. This proves that the local residual map is well-conditioned even though the naive free-energy reconstruction is not.

VII. 5. DIRECT ENLARGED RESIDUAL SYSTEM

The correct enlarged residual system is not

$$G(\delta) = G^{(4)}(\delta)$$

but rather

$G(\delta) = G^{(4)}(\delta) + \mathcal{N}(\delta),$

(12)

where

$$\mathcal{N}(\delta)$$

collects everything not captured by the exact four-class quartic baseline:

$$\boxed{\mathcal{N}(\delta) = G(\delta) - D(\delta)M^{(4c)}r(\delta).} \quad (13)$$

At the frozen corrected seed,

$$\boxed{\mathcal{N}(\delta_f) = G(\delta_f) - G^{(4)}(\delta_f).} \quad (14)$$

Since

$$\|G(\delta_f)\| \approx 0.0379185, \quad \|G^{(4)}(\delta_f)\| \approx 0.2521901,$$

one necessarily has

$$\boxed{\|\mathcal{N}(\delta_f)\| \geq 0.2142716098179067.} \quad (15)$$

Thus the nonquartic remainder is not perturbatively tiny at the corrected branch.

VIII. 6. FIRST ADMISSIBLE LOCAL COMPLETION: SYMMETRIC NON-DIAGONAL LINEAR REMAINDER

The simplest local residual completion is to approximate

$$\mathcal{N}(\delta) \approx S \delta$$

near the corrected seed, with $S = S^T$ a real symmetric 3×3 matrix. Then the local completed residual model is

$$\boxed{G_{\text{loc}}(\delta) = D(\delta)M^{(4c)}r(\delta) + S\delta.} \quad (16)$$

If one asks only for exact stationarity at the frozen corrected seed, then the condition is

$$\boxed{S\delta_f = -G^{(4)}(\delta_f).} \quad (17)$$

Among all symmetric matrices satisfying (17), the minimum-Frobenius solution is

$$\boxed{S_{\min} = \frac{yx^T + xy^T}{\|x\|^2} - \frac{x^T y}{\|x\|^4} xx^T, \quad x := \delta_f, \quad y := -G^{(4)}(\delta_f).} \quad (18)$$

Evaluating this gives

$$S_{\min} = \begin{pmatrix} -0.839960218174805 & 2.450804313533534 & -0.302298414680853 \\ 2.450804313533534 & -7.145183917627953 & 0.837818903051675 \\ -0.302298414680853 & 0.837818903051675 & 0.235363388813100 \end{pmatrix},$$

which indeed satisfies

$$S_{\min}\delta_f = -G^{(4)}(\delta_f).$$

This is the first viable corrected residual completion already found numerically.

IX. 7. WHY DIAGONAL-ONLY COMPLETION IS IMPOSSIBLE

If one restricts to diagonal completion only,

$$S = \text{diag}(s_A, s_E, s_O),$$

then (17) forces

$$s_A = -Q_{f,A}, \quad s_E = -Q_{f,E}, \quad s_O = -Q_{f,O},$$

that is,

$$s_A = -8.494671573456258, \quad s_E = -8.241391656489691, \quad s_O = -2.580391735022780.$$

But then

$$J^{(4)}(\delta_f) + S$$

has a negative eigenvalue. Therefore diagonal-only completion cannot produce an admissible local corrected residual model.

So one obtains the sharp statement:

$$\boxed{\text{the first admissible local completion must be genuinely non-diagonal.}} \quad (19)$$

X. 8. HONEST MICROSCOPIC INTERPRETATION

What has been identified at theorem level is not yet the unique microscopic operator, but the *representation type* of the missing completion.

The missing microscopic source must satisfy all of the following:

1. it must contribute directly to the residual map, not merely to a reconstructed free energy;
2. it must break the coarse A/E symmetry locally;
3. it cannot be purely diagonal in (A, E, O) ;
4. its dominant mixing channel must involve the A/E sector.

Thus the first microscopic completion beyond the exact four-class quartic baseline must be an orbit-mixing residual operator of schematic type

$$\boxed{\mathcal{N}_{\text{mic}} : (A, E, O) \longrightarrow (A, E, O), \quad \mathcal{N}_{\text{mic}} \not\propto \text{diag}.} \quad (20)$$

At the present stage, the exact microscopic parent of this operator is still open. What is already fixed is that any candidate microscopic source must reduce, after projection, to a local 3×3 residual operator with nonzero A/E splitting and genuine non-diagonal mixing.

XI. 9. FINAL LOCAL MAINLINE STATEMENT

Therefore the corrected residual quartic closure program has now reached the following sharp form:

$$\boxed{G(\delta) = D(\delta)M^{(4c)}r(\delta) + \mathcal{N}(\delta),} \quad (21)$$

with

$$\boxed{\mathcal{N}(\delta_f) \neq 0, \quad \|\mathcal{N}(\delta_f)\| \geq 0.2142716098179067,} \quad (22)$$

and the first admissible local completion requires

$$\boxed{\text{non-diagonal } A/E\text{-breaking orbit mixing.}} \quad (23)$$

So the quartic basis problem is solved, but the true remaining frontier is now exactly:

$$\boxed{\text{identify the microscopic origin of the projected non-diagonal residual operator } \mathcal{N}_{\text{mic}}.}$$

XII. FINAL SUMMARY

The corrected branch cannot be explained by the failed (u, v, κ) kernel, and even the exact four-class quartic enlargement

$$U_{\alpha\beta}^{(4c)}$$

does not close the branch by itself.

The correct reduced object is the projected residual map

$$G(\delta),$$

and its exact quartic baseline is

$$G^{(4)}(\delta) = D(\delta)M^{(4c)}r(\delta).$$

At the frozen corrected seed

$$\delta_f = (0.0096, -0.0288, 0.0096)^T,$$

the quartic residual Jacobian

$$J^{(4)}(\delta_f) = \text{diag}(M^{(4c)}r_f) + 2D_fM^{(4c)}D_f$$

is positive definite, but the quartic baseline itself overshoots strongly:

$$\|G(\delta_f) - G^{(4)}(\delta_f)\| \geq 0.2142716098179067.$$

Therefore the missing completion is not a tiny correction. It must be a genuinely residual, non-diagonal, A/E -breaking orbit-mixing operator.

A coarse A/E -symmetric completion is impossible, because it would force

$$a^2 = e^2,$$

whereas the corrected seed has

$$e^2 = 9a^2.$$

Diagonal-only completion is also impossible, because it leaves a negative eigenvalue in the completed Jacobian.

Hence the first admissible local corrected completion must already be non-diagonal, and the remaining mainline problem is to identify its microscopic projected origin.

XIII. JACOBIAN-LEVEL MAINLINE BEYOND THE EXACT FOUR-CLASS QUARTIC BASELINE

XIV. GOAL

The corrected residual quartic mainline has already fixed the following data:

$$\delta_f := \delta^{\text{final}} = (0.0096, -0.0288, 0.0096)^T,$$

$$G_f^{(4)} = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix},$$

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix},$$

with

$$\lambda_{\min}(J_f^{(4)}) \approx 3.5212959 > 0.$$

The exact remaining local object is not the shell means

$$(\sigma_A, \sigma_E, \sigma_O),$$

but the full corrected residual Jacobian

$$J_f := DG(\delta_f), \quad R_f := J_f - J_f^{(4)}.$$

The exact executable next extraction is the centered-difference audit

$$J_{\cdot Y} \approx \frac{G(\delta_f + \varepsilon e_Y) - G(\delta_f - \varepsilon e_Y)}{2\varepsilon}, \quad Y \in \{A, E, O\},$$

and, if desired, the Richardson upgrade

$$\tilde{J}_f^{\text{Rich}} = \frac{4\tilde{J}_f^{[\varepsilon/2]} - \tilde{J}_f^{[\varepsilon]}}{3} = J_f + O(\varepsilon^4).$$

Thus the next exact problem is:

$$\text{extract } J_f, \text{ compute } R_f = J_f - J_f^{(4)}, \text{ and classify } R_f.$$

XV. 1. GEOMETRY-ADAPTED BASIS AT THE CORRECTED SEED

The corrected seed has the exact shape

$$\delta_f = 0.0096(1, -3, 1)^T.$$

Hence its norm is

$$\rho := \|\delta_f\| = 0.0096\sqrt{11} \approx 0.03183959798741184.$$

Define the normalized tangent direction

$$t := \frac{\delta_f}{\|\delta_f\|} = \frac{1}{\sqrt{11}}(1, -3, 1)^T.$$

Choose two orthonormal transverse directions

$$n_1 := \frac{1}{\sqrt{2}}(1, 0, -1)^T, \quad n_2 := \frac{1}{\sqrt{22}}(3, 2, 3)^T,$$

so that

$$B := (t \mid n_1 \mid n_2) \in O(3).$$

Then

$$B^T \delta_f = \begin{pmatrix} \rho \\ 0 \\ 0 \end{pmatrix}.$$

This is the correct local basis for the corrected-branch Jacobian audit, because it separates:

1. the branch-tangent direction t ,
2. one A/O -antisymmetric transverse direction n_1 ,
3. one complementary coarse transverse direction n_2 .

XVI. 2. EXACT QUARTIC BASELINE IN THE ADAPTED BASIS

Transform the known quartic Jacobian:

$$J_{f,B}^{(4)} := B^T J_f^{(4)} B.$$

Direct evaluation gives

$$J_{f,B}^{(4)} = \begin{pmatrix} 23.24934224 & 3.78278610 & -3.12859268 \\ 3.78278610 & 6.43553954 & 1.80527992 \\ -3.12859268 & 1.80527992 & 6.66026178 \end{pmatrix}.$$

So even before extracting the full Jacobian, the quartic baseline already shows:

$$\boxed{\text{the tangent direction } t \text{ is the stiffest local direction of the quartic residual geometry.}} \quad (24)$$

Moreover, the quartic baseline already couples t to both transverse directions:

$$(J_{f,B}^{(4)})_{tn_1} \neq 0, \quad (J_{f,B}^{(4)})_{tn_2} \neq 0.$$

XVII. 3. EXACT MINIMUM-FROBENIUS COMPLETION AND ITS ADAPTED-BASIS STRUCTURE

The first viable local corrected completion already known in the files is the minimum-Frobenius symmetric matrix

$$S_{\min} = \begin{pmatrix} -0.839960218174805 & 2.450804313533534 & -0.302298414680853 \\ 2.450804313533534 & -7.145183917627953 & 0.837818903051675 \\ -0.302298414680853 & 0.837818903051675 & 0.235363388813100 \end{pmatrix},$$

satisfying

$$S_{\min} \delta_f = -G_f^{(4)}.$$

Let

$$y := -G_f^{(4)}.$$

Expand y in the adapted basis:

$$y = y_t t + y_1 n_1 + y_2 n_2, \quad \begin{pmatrix} y_t \\ y_1 \\ y_2 \end{pmatrix} = B^T y = \begin{pmatrix} -0.24674990 \\ -0.04014746 \\ 0.03320438 \end{pmatrix}.$$

Because $x := \delta_f = \rho t$, the exact minimum-Frobenius formula

$$S_{\min} = \frac{yx^T + xy^T}{\|x\|^2} - \frac{x^T y}{\|x\|^4} xx^T$$

simplifies drastically. In the adapted basis one gets

$$\boxed{(S_{\min})_B = \begin{pmatrix} y_t/\rho & y_1/\rho & y_2/\rho \\ y_1/\rho & 0 & 0 \\ y_2/\rho & 0 & 0 \end{pmatrix}.} \quad (25)$$

Numerically,

$$(S_{\min})_B = \begin{pmatrix} -7.74978075 & -1.26092870 & 1.04286422 \\ -1.26092870 & 0 & 0 \\ 1.04286422 & 0 & 0 \end{pmatrix}.$$

This is the key local geometric result.

A. Interpretation

Equation (25) proves that the first admissible local completion has the following structure:

$$\boxed{\text{no pure transverse-transverse correction at all;}} \quad (26)$$

$$(S_{\min})_{n_1 n_1} = (S_{\min})_{n_1 n_2} = (S_{\min})_{n_2 n_2} = 0.$$

Instead, the entire completion is of the form

$$\boxed{S_{\min} = \frac{y_t}{\rho} tt^T + \frac{y_1}{\rho} (tn_1^T + n_1 t^T) + \frac{y_2}{\rho} (tn_2^T + n_2 t^T).} \quad (27)$$

Therefore the first viable non-diagonal completion is not a renormalization of the transverse 2×2 block itself. It is a *tangent-coupling completion*.

XVIII. 4. COMPLETED LOCAL PRIOR JACOBIAN

If one uses the minimum-Frobenius completion as the best local prior, the completed Jacobian is

$$J_f^{\text{prior}} := J_f^{(4)} + S_{\min}.$$

In the adapted basis:

$$J_{f,B}^{\text{prior}} = J_{f,B}^{(4)} + (S_{\min})_B = \begin{pmatrix} 15.49956149 & 2.52185740 & -2.08572845 \\ 2.52185740 & 6.43553954 & 1.80527992 \\ -2.08572845 & 1.80527992 & 6.66026178 \end{pmatrix}.$$

Its eigenvalues are

$$3.82469902, \quad 8.35243128, \quad 16.41823251,$$

so it remains positive definite.

Thus the first admissible completion does two things and only two things:

1. it reduces the tangent stiffness

$$23.24934224 \longrightarrow 15.49956149,$$

2. it weakens the tangent-to-transverse couplings, while leaving the pure transverse block unchanged.

XIX. 5. CONSEQUENCE FOR THE MICROSCOPIC ORIGIN PROBLEM

This already sharpens the microscopic target considerably.

A diagonal-only correction would correspond, in the adapted basis, to adding diagonal terms to the transverse block as well. But the first viable completion does not do that.

So the missing microscopic source is not naturally interpreted as a simple orbitwise self-energy

$$A \mapsto q_A A, \quad E \mapsto q_E E, \quad O \mapsto q_O O.$$

Instead, the observed local pattern points to a projected operator whose dominant coarse effect is

$$\boxed{\text{branch-tangent coupling } t \leftrightarrow n_1, \ t \leftrightarrow n_2, \text{ with negligible intrinsic transverse renormalization at first order.}} \quad (28)$$

Equivalently, the microscopic origin should be sought among projected residual operators of the schematic form

$$\mathcal{N}_{\text{mic}} \sim \Pi_t \mathcal{N} \Pi_t + \Pi_t \mathcal{N} \Pi_{\perp} + \Pi_{\perp} \mathcal{N} \Pi_t, \quad \Pi_{\perp} \mathcal{N} \Pi_{\perp} \approx 0$$

at the first admissible local level.

XX. 6. EXACT EXTRACTION PROTOCOL FOR THE REAL JACOBIAN REMAINDER

The prior analysis above is only a local geometric inference from the already-known viable completion. The actual next theorem-level audit still requires the real full Jacobian:

$$J_f = DG(\delta_f).$$

Once J_f is extracted, define

$$R_f := J_f - J_f^{(4)}.$$

Then split

$$R_f^{(s)} := \frac{1}{2}(R_f + R_f^T), \quad R_f^{(a)} := \frac{1}{2}(R_f - R_f^T).$$

In the adapted basis,

$$R_{f,B} := B^T R_f B = \begin{pmatrix} r_{tt} & r_{t1} & r_{t2} \\ r_{1t} & r_{11} & r_{12} \\ r_{2t} & r_{21} & r_{22} \end{pmatrix}.$$

This gives the exact classification tree:

Case I: small remainder

If

$$\|R_f\|_2 \ll \lambda_{\min}(J_f^{(4)}) \approx 3.5212959,$$

then the quartic baseline controls the local Jacobian.

Case II: tangent-coupling remainder

If

$$r_{11}, r_{12}, r_{22} \approx 0$$

but

$$(r_{tt}, r_{t1}, r_{t2}) \neq 0,$$

then the real microscopic source matches the $S_{\min}$ -type tangent-coupling pattern.

Case III: coarse full symmetric remainder

If

$$R_f^{(a)} \approx 0$$

but the transverse block

$$\begin{pmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{pmatrix}$$

is also large, then the next correct reduced model is a full symmetric coarse 3×3 normalization layer.

Case IV: genuinely non-gradient local remainder

If

$$\|R_f^{(a)}\| \not\ll \|R_f^{(s)}\|,$$

then the corrected reduced residual map is non-gradient already at Jacobian level.

Case V: coarse (A, E, O) basis failure

If the adapted-basis decomposition shows large unstable mixing that cannot be compressed into the coarse (A, E, O) basis, then sign-resolved suborbit refinement becomes unavoidable.

XXI. 7. SHARP CONCLUSION OF THE PRESENT STEP

Even before the actual six-point centered-difference extraction is rerun, one theorem-level local statement is already available:

the first viable corrected residual completion is a tangent-coupling completion, not a pure diagonal or pure transverse correction (29)

So the microscopic origin problem has already narrowed to:

find the projected microscopic residual source whose first coarse effect is $(t-t, t-\perp)$ mixing with negligible $\perp-\perp$ term. (30)

XXII. FINAL SUMMARY

The corrected residual quartic mainline has now been sharpened from vector level to local geometry level. The exact next theorem-level object remains

$$J_f = DG(\delta_f), \quad R_f := J_f - J_f^{(4)},$$

to be extracted by centered differences around

$$\delta_f = (0.0096, -0.0288, 0.0096)^T.$$

However, the already-known minimum-Frobenius admissible completion $S_{\min}$ can be analyzed exactly in the branch-adapted basis

$$t = \frac{1}{\sqrt{11}}(1, -3, 1), \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1), \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3).$$

Since

$$\delta_f = \|\delta_f\| t,$$

the completion takes the exact adapted-basis form

$$(S_{\min})_B = \begin{pmatrix} y_t/\rho & y_1/\rho & y_2/\rho \\ y_1/\rho & 0 & 0 \\ y_2/\rho & 0 & 0 \end{pmatrix}, \quad y := -G_f^{(4)}, \quad \rho = \|\delta_f\|.$$

Therefore the first admissible local corrected completion has no intrinsic transverse-transverse block:

$$(S_{\min})_{\perp\perp} = 0.$$

Its only effect is:

$$\text{tangent self-correction} + \text{tangent-transverse mixing}.$$

So the missing microscopic source is already constrained to be a projected tangent-coupling residual operator, not a diagonal orbitwise self-energy and not a pure transverse renormalization.

The next exact audit is therefore uniquely fixed:

extract J_f , compute $R_f = J_f - J_f^{(4)}$, and test whether R_f really has the same tangent-coupling structure.

XXIII. EXACT LOCAL JACOBIAN EXTRACTION SHEET AT THE CORRECTED RESIDUAL STAGE

XXIV. SHORT KOREAN EXPLANATION

The corrected residual stage is frozen at

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta_f = (0.0096, -0.0288, 0.0096),$$

and the primary reduced object is

$$G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T.$$

The exact four-class quartic baseline is already fixed, and its Jacobian at the corrected seed is

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix},$$

with

$$\lambda_* := \lambda_{\min}(J_f^{(4)}) \approx 3.52129589.$$

The next exact task is:

extract $J_f = DG(\delta_f)$, $R_f := J_f - J_f^{(4)}$,

then compute the norm classifiers

$$\alpha_f, \quad \varepsilon_{\text{diag}}, \quad \varepsilon_{\text{sym}}, \quad \varepsilon_{\text{mix}},$$

and branch accordingly. This is now a finite 3×3 matrix audit, not a conceptual theorem-building problem.

XXV. 1. EXACT SIX-POINT CENTERED-DIFFERENCE EXTRACTION

Let

$$e_A = (1, 0, 0)^T, \quad e_E = (0, 1, 0)^T, \quad e_O = (0, 0, 1)^T.$$

Fix a small step $\varepsilon > 0$. For each $Y \in \{A, E, O\}$, define

$$G_Y^{(+)} := G(\delta_f + \varepsilon e_Y), \quad G_Y^{(-)} := G(\delta_f - \varepsilon e_Y).$$

Then the Y -th column of the Jacobian is extracted by

$(J_f)_{\cdot Y} \approx \frac{G_Y^{(+)} - G_Y^{(-)}}{2\varepsilon}.$

(31)

Equivalently,

$$J_f^{[\varepsilon]} = \begin{pmatrix} \frac{G_A(\delta_f + \varepsilon e_A) - G_A(\delta_f - \varepsilon e_A)}{2\varepsilon} & \frac{G_A(\delta_f + \varepsilon e_E) - G_A(\delta_f - \varepsilon e_E)}{2\varepsilon} & \frac{G_A(\delta_f + \varepsilon e_O) - G_A(\delta_f - \varepsilon e_O)}{2\varepsilon} \\ \frac{G_E(\delta_f + \varepsilon e_A) - G_E(\delta_f - \varepsilon e_A)}{2\varepsilon} & \frac{G_E(\delta_f + \varepsilon e_E) - G_E(\delta_f - \varepsilon e_E)}{2\varepsilon} & \frac{G_E(\delta_f + \varepsilon e_O) - G_E(\delta_f - \varepsilon e_O)}{2\varepsilon} \\ \frac{G_O(\delta_f + \varepsilon e_A) - G_O(\delta_f - \varepsilon e_A)}{2\varepsilon} & \frac{G_O(\delta_f + \varepsilon e_E) - G_O(\delta_f - \varepsilon e_E)}{2\varepsilon} & \frac{G_O(\delta_f + \varepsilon e_O) - G_O(\delta_f - \varepsilon e_O)}{2\varepsilon} \end{pmatrix}.$$

So the raw finite audit requires exactly six residual evaluations:

$$\delta_f \pm \varepsilon e_A, \quad \delta_f \pm \varepsilon e_E, \quad \delta_f \pm \varepsilon e_O.$$

XXVI. 2. RICHARDSON UPGRADE

To reduce truncation error, extract both

$$J_f^{[\varepsilon]}, \quad J_f^{[\varepsilon/2]}.$$

Then define

$$J_f^{\text{Rich}} := \frac{4J_f^{[\varepsilon/2]} - J_f^{[\varepsilon]}}{3}. \quad (32)$$

This cancels the leading $O(\varepsilon^2)$ centered-difference error and yields

$$J_f^{\text{Rich}} = J_f + O(\varepsilon^4).$$

Thus the practical extraction object should be

$$J_f := J_f^{\text{Rich}}.$$

a. Practical note. Since

$$\|\delta_f\| \approx 0.0318396,$$

a numerically sensible first pair is

$$\varepsilon = 10^{-4}, \quad \varepsilon/2 = 5 \times 10^{-5}.$$

This is a practical numerical choice, not a theorem-level constant.

XXVII. 3. EXACT QUARTIC BASELINE AND REMAINDER

The quartic residual is cubic-homogeneous, so

$$J_f^{(4)} \delta_f = 3G_f^{(4)}.$$

The already fixed quartic baseline quantities are

$$G_f^{(4)} = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix},$$

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix},$$

with eigenvalues

$$3.52129589, \quad 8.35219818, \quad 24.47164948.$$

Now define the exact local remainders

$$\boxed{R_f := J_f - J_f^{(4)}, \quad E_f := G_f - G_f^{(4)}}. \quad (33)$$

The vector-level theorem already shows

$$0.8496431059127669 \leq \frac{\|E_f\|}{\|G_f^{(4)}\|} \leq 1.1503568940872333,$$

so strong quartic-local sufficiency is already excluded. Therefore the only remaining question is whether the Jacobian remainder R_f is still small enough for weak quartic dominance.

XXVIII. 4. FIRST CLASSIFIER: INVERTIBILITY CONTROL

Define

$$\boxed{\alpha_f := \frac{\|R_f\|_2}{\lambda_*}, \quad \lambda_* := \lambda_{\min}(J_f^{(4)}) \approx 3.52129589}. \quad (34)$$

Interpretation:

$$\alpha_f < 1 \implies J_f = J_f^{(4)} + R_f \text{ remains invertible by a Neumann-type bound,}$$

whereas

$$\alpha_f \geq 1$$

means the quartic baseline no longer controls local invertibility.

Thus α_f is the first hard branch discriminator.

XXIX. 5. BRANCH-ADAPTED BASIS

Because

$$\delta_f = 0.0096(1, -3, 1)^T,$$

define the normalized tangent direction

$$t := \frac{1}{\sqrt{11}}(1, -3, 1)^T.$$

Choose two orthonormal transverse directions

$$n_1 := \frac{1}{\sqrt{2}}(1, 0, -1)^T, \quad n_2 := \frac{1}{\sqrt{22}}(3, 2, 3)^T.$$

Let

$$B := (t | n_1 | n_2) \in O(3).$$

Now rotate the Jacobian remainder into the branch-adapted basis:

$$\boxed{R_{f,B} := B^T R_f B}. \quad (35)$$

Write

$$R_{f,B} = \begin{pmatrix} r_{tt} & r_{t1} & r_{t2} \\ r_{1t} & r_{11} & r_{12} \\ r_{2t} & r_{21} & r_{22} \end{pmatrix}.$$

This basis is the correct one for deciding whether the missing completion is:

1. diagonal coarse renormalization,
2. full symmetric coarse renormalization,
3. tangent-coupling renormalization,
4. genuinely non-gradient local structure,
5. or coarse-basis failure.

XXX. 6. EXACT NORM DECOMPOSITIONS

A. 6.1 Diagonal coarse approximation

Define the diagonal projection

$$\Pi_{\text{diag}}(X) := \text{diag}(X_{11}, X_{22}, X_{33}).$$

Then define the diagonal defect ratio

$$\varepsilon_{\text{diag}} := \frac{\|R_f - \Pi_{\text{diag}}(R_f)\|_F}{\max(\|R_f\|_F, \epsilon_0)}. \quad (36)$$

Here $\epsilon_0 > 0$ is a tiny fixed regularizer, e.g. 10^{-15} .

If $\varepsilon_{\text{diag}}$ is small, a diagonal coarse quadratic layer is sufficient.

B. 6.2 Symmetric-gradient approximation

Define

$$R_f^{(s)} := \frac{1}{2}(R_f + R_f^T), \quad R_f^{(a)} := \frac{1}{2}(R_f - R_f^T).$$

Then define the antisymmetry defect ratio

$$\varepsilon_{\text{sym}} := \frac{\|R_f^{(a)}\|_F}{\max(\|R_f\|_F, \epsilon_0)}. \quad (37)$$

If ε_{sym} is small, the local residual map is effectively gradient-type at Jacobian level. If not, the map is already locally non-gradient.

C. 6.3 Tangent-coupling coarse model

The first viable local completion found earlier had exactly the pattern

$$\text{tangent self-correction} + \text{tangent-transverse mixing}, \quad \perp - \perp \text{ block negligible.}$$

So define the tangent-coupling subspace

$$\mathcal{T} := \left\{ \begin{pmatrix} x_{tt} & x_{t1} & x_{t2} \\ x_{1t} & 0 & 0 \\ x_{2t} & 0 & 0 \end{pmatrix} \right\}.$$

Its orthogonal projection is

$$\Pi_{\mathcal{T}}(R_{f,B}) = \begin{pmatrix} r_{tt} & r_{t1} & r_{t2} \\ r_{1t} & 0 & 0 \\ r_{2t} & 0 & 0 \end{pmatrix}.$$

Now define the mixing defect ratio

$$\varepsilon_{\text{mix}} := \frac{\|R_{f,B} - \Pi_{\mathcal{T}}(R_{f,B})\|_F}{\max(\|R_f\|_F, \epsilon_0)}. \quad (38)$$

Interpretation:

$$\varepsilon_{\text{mix}} \ll 1$$

means the missing local completion is essentially of tangent-coupling type;

$$\varepsilon_{\text{mix}} \not\ll 1$$

means the coarse (A, E, O) basis is already insufficient locally, and sign-resolved suborbit refinement is indicated.

XXXI. 7. EXACT CLASSIFICATION RULE

Let $\tau_{\text{cert}} \in (0, 1)$ be the chosen certification threshold. Then the clean local branching rule is:

$\alpha_f \geq 1$	$\implies$ quartic baseline no longer controls local invertibility,
$\alpha_f < 1, \varepsilon_{\text{diag}} \leq \tau_{\text{cert}}$	$\implies$ diagonal coarse quadratic layer,
$\alpha_f < 1, \varepsilon_{\text{diag}} > \tau_{\text{cert}}, \varepsilon_{\text{sym}} \leq \tau_{\text{cert}}, \varepsilon_{\text{mix}} \leq \tau_{\text{cert}}$	$\implies$ full symmetric tangent-coupling coarse layer,
$\alpha_f < 1, \varepsilon_{\text{sym}} > \tau_{\text{cert}}$	$\implies$ locally non-gradient residual map,
$\alpha_f < 1, \varepsilon_{\text{mix}} > \tau_{\text{cert}}$	$\implies$ coarse branch basis fails locally; move to sign-resolved suborbits.

(39)

This matches the already established logical structure: the next branch is completely determined once J_f is extracted and R_f is formed.

XXXII. 8. AUDIT TABLE TO POPULATE

The exact computation sheet to fill is:

$$\delta_f + \varepsilon e_A, \quad \delta_f - \varepsilon e_A,$$

$$\delta_f + \varepsilon e_E, \quad \delta_f - \varepsilon e_E,$$

$$\delta_f + \varepsilon e_O, \quad \delta_f - \varepsilon e_O,$$

and the same six points again for $\varepsilon/2$.

From these 12 residual evaluations compute:

$$J_f^{[\varepsilon]}, \quad J_f^{[\varepsilon/2]}, \quad J_f^{\text{Rich}},$$

then

$$R_f, \quad \alpha_f, \quad \varepsilon_{\text{diag}}, \quad \varepsilon_{\text{sym}}, \quad \varepsilon_{\text{mix}}.$$

After that, the mathematically correct next move is fixed by (39).

XXXIII. 9. FINAL CONSEQUENCE

At this stage, the correct frontier is no longer another quartic basis enlargement. The four-class quartic kernel and the coarse orbit combinatorics are already closed, and the coarse quartic sector is exactly $A \leftrightarrow E$ symmetric. So the missing structure must come from the local remainder:

$$R_f = J_f - J_f^{(4)}.$$

The honest next theorem-level object is therefore not

$$(u, v, \kappa), \quad (u_d, u_{\text{opp}}, u_{\perp}, u_{\text{obl}}),$$

but

$$J_f, \quad R_f, \quad (\alpha_f, \varepsilon_{\text{diag}}, \varepsilon_{\text{sym}}, \varepsilon_{\text{mix}}).$$

XXXIV. FINAL SUMMARY

The corrected residual stage is frozen at

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta_f = (0.0096, -0.0288, 0.0096),$$

and the primary reduced object is

$$G(\delta) = (G_A(\delta), G_E(\delta), G_O(\delta))^T.$$

The exact four-class quartic baseline is already fixed, with

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix},$$

and

$$\lambda_{\min}(J_f^{(4)}) \approx 3.52129589.$$

The strong quartic-sufficiency route is already excluded at vector level. Hence the only honest remaining local question is the Jacobian remainder

$$R_f := J_f - J_f^{(4)}.$$

The exact next finite audit is:

$$\delta_f \pm \varepsilon e_A, \quad \delta_f \pm \varepsilon e_E, \quad \delta_f \pm \varepsilon e_O,$$

plus the same six evaluations at $\varepsilon/2$, followed by

$$J_f^{\text{Rich}} = \frac{4J_f^{[\varepsilon/2]} - J_f^{[\varepsilon]}}{3}.$$

Then one computes

$$\alpha_f = \frac{\|R_f\|_2}{\lambda_*}, \quad \varepsilon_{\text{diag}}, \quad \varepsilon_{\text{sym}}, \quad \varepsilon_{\text{mix}},$$

where ε_{mix} is measured in the branch-adapted basis

$$t = \frac{1}{\sqrt{11}}(1, -3, 1)^T, \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1)^T, \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3)^T.$$

These four numbers determine the next mathematically correct move:

$$\alpha_f \geq 1 \implies \text{quartic Jacobian no longer controls invertibility,}$$

$$\alpha_f < 1, \varepsilon_{\text{diag}} \text{ small} \implies \text{diagonal coarse layer,}$$

$$\alpha_f < 1, \varepsilon_{\text{sym}} \text{ small, } \varepsilon_{\text{mix}} \text{ small} \implies \text{tangent-coupling symmetric coarse layer,}$$

$$\varepsilon_{\text{sym}} \text{ large} \implies \text{locally non-gradient residual map,}$$

$$\varepsilon_{\text{mix}} \text{ large} \implies \text{coarse basis failure and sign-resolved suborbit refinement.}$$

Therefore the next exact frontier is no longer another quartic ansatz. It is the direct finite extraction and classification of the corrected local Jacobian.

XXXV. EXPLICIT NUMERICAL CERTIFICATE FROM THE CURRENTLY AVAILABLE ADMISSIBLE LOCAL COMPLETION

XXXVI. SHORT EXPLANATION

At the corrected residual stage, the uploaded files do *not* yet contain the actual centered-difference Jacobian

$$J_f := DG(\delta_f)$$

at the corrected seed

$$\delta_f = (0.0096, -0.0288, 0.0096).$$

So the exact solver-level certificate

$$(\alpha_f, \varepsilon_{\text{diag}}, \varepsilon_{\text{sym}}, \varepsilon_{\text{mix}})$$

cannot yet be claimed for the true J_f .

However, the files do contain the exact quartic baseline

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix},$$

and the first admissible symmetric local completion

$$R_f^{\text{prior}} := S_{\min} = \begin{pmatrix} -0.839960218174805 & 2.450804313533534 & -0.302298414680853 \\ 2.450804313533534 & -7.145183917627953 & 0.837818903051675 \\ -0.302298414680853 & 0.837818903051675 & 0.235363388813100 \end{pmatrix}.$$

Therefore the strongest honest numerical step that can be completed *now* is:

compute the certificate for $R_f^{\text{prior}} = S_{\min}$.

XXXVII. 1. QUARTIC BASELINE SPECTRAL DATA

The eigenvalues of $J_f^{(4)}$ are

$$\lambda(J_f^{(4)}) = (3.52129589, 8.35219818, 24.47164949),$$

so

$$\lambda_* := \lambda_{\min}(J_f^{(4)}) = 3.52129589.$$

This confirms that the quartic baseline Jacobian is positive definite.

XXXVIII. 2. NORMS OF THE ADMISSIBLE PRIOR REMAINDER

For

$$R_f^{\text{prior}} := S_{\min},$$

the spectral and Frobenius norms are

$$\|R_f^{\text{prior}}\|_2 = 8.081109861025224, \quad \|R_f^{\text{prior}}\|_F = 8.087899329725035.$$

Therefore the invertibility-control ratio is

$$\alpha_f^{\text{prior}} := \frac{\|R_f^{\text{prior}}\|_2}{\lambda_*} = \frac{8.081109861025224}{3.52129589} = 2.294924967594988. \quad (40)$$

Hence

$$\alpha_f^{\text{prior}} > 1.$$

This means that the quartic baseline does *not* control the completed local Jacobian by the small-remainder norm criterion.

XXXIX. 3. DIAGONAL-DEFECT FRACTION

Let

$$D_f^{\text{prior}} := \text{diag}(R_f^{\text{prior}}).$$

Then

$$D_f^{\text{prior}} = \text{diag}(-0.839960218174805, -7.145183917627953, 0.235363388813100).$$

The diagonal-defect fraction is

$$\varepsilon_{\text{diag}}^{\text{prior}} := \frac{\|R_f^{\text{prior}} - D_f^{\text{prior}}\|_F}{\|R_f^{\text{prior}}\|_F} = 0.455959495840998. \quad (41)$$

So about 45.6% of the prior remainder is intrinsically non-diagonal in the coarse (A, E, O) basis. Therefore a diagonal quadratic correction is not an accurate description of the admissible local completion.

XL. 4. SYMMETRY-DEFECT FRACTION

Since $S_{\min}$ is exactly symmetric,

$$(R_f^{\text{prior}})^T = R_f^{\text{prior}},$$

so

$$(R_f^{\text{prior}})^{(a)} = \frac{1}{2} \left(R_f^{\text{prior}} - (R_f^{\text{prior}})^T \right) = 0.$$

Hence

$$\varepsilon_{\text{sym}}^{\text{prior}} = \frac{\|(R_f^{\text{prior}})^{(a)}\|_F}{\|R_f^{\text{prior}}\|_F} = 0. \quad (42)$$

Thus the currently admissible local completion is perfectly gradient-type at Jacobian level.

XLI. 5. ADAPTED-BASIS DECOMPOSITION

Define the corrected-seed adapted orthonormal basis

$$t = \frac{1}{\sqrt{11}}(1, -3, 1)^T, \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1)^T, \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3)^T.$$

Let

$$B = (t | n_1 | n_2) \in O(3).$$

Then

$$\tilde{R}_f^{\text{prior}} := B^T R_f^{\text{prior}} B$$

is

$$\tilde{R}_f^{\text{prior}} = \begin{pmatrix} -7.74978075 & -1.26092870 & 1.04286422 \\ -1.26092870 & 0 & 0 \\ 1.04286422 & 0 & 0 \end{pmatrix}, \quad (43)$$

up to roundoff at the 10^{-16} level.

Therefore the transverse 2×2 block vanishes identically:

$$P_{\perp\perp}^{\text{prior}} = 0.$$

This means that the admissible prior remainder is *exactly* of tangent-coupling type:

$$\tilde{R}_f^{\text{prior}} = \begin{pmatrix} r_{tt} & * & * \\ * & 0 & 0 \\ * & 0 & 0 \end{pmatrix}.$$

XLII. 6. MIXING-DEFECT FRACTION

Let $\Pi_{\mathcal{T}}$ denote the tangent-coupling projection in the adapted basis:

$$\Pi_{\mathcal{T}} \left(\begin{pmatrix} x_{tt} & x_{t1} & x_{t2} \\ x_{1t} & x_{11} & x_{12} \\ x_{2t} & x_{21} & x_{22} \end{pmatrix} \right) = \begin{pmatrix} x_{tt} & x_{t1} & x_{t2} \\ x_{1t} & 0 & 0 \\ x_{2t} & 0 & 0 \end{pmatrix}.$$

Then

$$\Pi_{\mathcal{T}}(\tilde{R}_f^{\text{prior}}) = \tilde{R}_f^{\text{prior}},$$

up to machine roundoff. Hence

$$\varepsilon_{\text{mix}}^{\text{prior}} := \frac{\|\tilde{R}_f^{\text{prior}} - \Pi_{\mathcal{T}}(\tilde{R}_f^{\text{prior}})\|_F}{\|\tilde{R}_f^{\text{prior}}\|_F} = 9.547253466277641 \times 10^{-17}. \quad (44)$$

So the admissible local completion is, numerically, an *exact tangent-coupling remainder*.

XLIII. 7. COMPARISON WITH THE DIAGONAL TARGET

The files also give the minimal diagonal target

$$q_A = -8.494671573456258, \quad q_E = -8.241391656489691, \quad q_O = -2.580391735022780.$$

Define

$$Q := \text{diag}(q_A^{,q}, q_E^{,q}, q_O^{,q}).$$

Then

$$\lambda(J_f^{(4)} + Q) = (-0.11554365, 0.91059735, 16.23363489).$$

So the diagonal-only completion is not admissible, because it destroys positive definiteness.

By contrast, the symmetric non-diagonal completion gives

$$\lambda(J_f^{(4)} + S_{\min}) = (3.82469901, 8.35243128, 16.41823252),$$

which is strictly positive.

XLIV. 8. FINAL NUMERICAL CLASSIFICATION AVAILABLE NOW

Hence the strongest certificate that can honestly be completed *now* is:

$$\boxed{\alpha_f^{\text{prior}} = 2.294924967594988,} \quad (45)$$

$$\boxed{\varepsilon_{\text{diag}}^{\text{prior}} = 0.455959495840998,} \quad (46)$$

$$\boxed{\varepsilon_{\text{sym}}^{\text{prior}} = 0,} \quad (47)$$

$$\boxed{\varepsilon_{\text{mix}}^{\text{prior}} = 9.547253466277641 \times 10^{-17}.} \quad (48)$$

Therefore:

1. the quartic baseline is not norm-dominant at local Jacobian level;
2. diagonal correction is insufficient;
3. the admissible local completion is symmetric;
4. the admissible local completion is essentially a pure tangent-coupling remainder.

So the currently computable numerical conclusion is

$$\boxed{\text{the next reduced model is a symmetric non-diagonal tangent-coupling layer, not a diagonal correction.}} \quad (49)$$

XLV. FINAL SUMMARY

At the corrected residual stage, the actual centered-difference Jacobian

$$J_f = DG(\delta_f)$$

has not yet been supplied in the uploaded files, so the exact solver-level certificate

$$(\alpha_f, \varepsilon_{\text{diag}}, \varepsilon_{\text{sym}}, \varepsilon_{\text{mix}})$$

for the true $R_f = J_f - J_f^{(4)}$ cannot yet be claimed.

However, the uploaded files do contain the exact quartic baseline $J_f^{(4)}$ and the first admissible local symmetric completion $S_{\min}$. Using

$$R_f^{\text{prior}} := S_{\min},$$

one obtains the explicit numerical prior certificate

$$\alpha_f^{\text{prior}} = 2.294924967594988, \quad \varepsilon_{\text{diag}}^{\text{prior}} = 0.455959495840998, \quad \varepsilon_{\text{sym}}^{\text{prior}} = 0, \quad \varepsilon_{\text{mix}}^{\text{prior}} = 9.547253466277641 \times 10^{-17}.$$

Thus the currently computable local classification is sharp:

not diagonal, not skew, but essentially exact tangent-coupling symmetric remainder.

Moreover, the diagonal-only target

$$Q = \text{diag}(q_A^{,q_E}, q_O)$$

fails, because

$$\lambda_{\min}(J_f^{(4)} + Q) = -0.11554365 < 0,$$

whereas

$$\lambda(J_f^{(4)} + S_{\min}) = (3.82469901, 8.35243128, 16.41823252) > 0.$$

Therefore the strongest honest numerical conclusion available now is:

the corrected local completion is forced into a symmetric non-diagonal tangent-coupling form.

XLVI. SHORT KOREAN EXPLANATION

At the corrected residual stage, the uploaded files do not yet provide the actual centered-difference Jacobian

$$J_f := DG(\delta_f) \quad \text{at} \quad \delta_f = (0.0096, -0.0288, 0.0096).$$

What is already available is:

1. the corrected seed δ_f ,
2. the corrected residual norm $\|G(\delta_f)\| \approx 0.0379185$,
3. the exact four-class quartic baseline Jacobian $J_f^{(4)}$,
4. the minimal diagonal target $Q = \text{diag}(q_A^{,q_E}, q_O)$,
5. the minimum-Frobenius symmetric completion $S_{\min}$.

Therefore the strongest honest computation that can be completed now is the explicit certificate for

$$R_f^{\text{prior}} := S_{\min}.$$

The result is:

$$\alpha_f^{\text{prior}} = 2.294924967594988, \quad \varepsilon_{\text{diag}}^{\text{prior}} = 0.455959495840998,$$

$$\varepsilon_{\text{sym}}^{\text{prior}} = 0, \quad \varepsilon_{\text{mix}}^{\text{prior}} = 9.547253466277641 \times 10^{-17}.$$

So the quartic baseline is not norm-dominant, diagonal correction is insufficient, and the first admissible local completion is an essentially exact symmetric tangent-coupling remainder.

XLVII. EXPLICIT NUMERICAL CERTIFICATE FROM THE AVAILABLE CORRECTED RESIDUAL DATA

XLVIII. 1. FROZEN CORRECTED SEED AND EXACT FOUR-CLASS QUARTIC BASELINE

The corrected local seed is frozen as

$$\delta_f = (0.0096, -0.0288, 0.0096)^T.$$

The exact four-class quartic baseline Jacobian at the corrected seed is

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix}.$$

Its eigenvalues are

$$\lambda(J_f^{(4)}) = (3.52129589, 8.35219818, 24.47164949),$$

so

$$\lambda_* := \lambda_{\min}(J_f^{(4)}) = 3.52129589.$$

Thus the quartic baseline is positive definite.

XLIX. 2. AVAILABLE ADMISSIBLE LOCAL COMPLETION

The minimum-Frobenius symmetric completion already extracted in the files is

$$S_{\min} = \begin{pmatrix} -0.839960218174805 & 2.450804313533534 & -0.302298414680853 \\ 2.450804313533534 & -7.145183917627953 & 0.837818903051675 \\ -0.302298414680853 & 0.837818903051675 & 0.235363388813100 \end{pmatrix}.$$

We therefore define the currently computable local remainder by

$$R_f^{\text{prior}} := S_{\min}.$$

L. 3. INVERTIBILITY-CONTROL RATIO

The operator and Frobenius norms of R_f^{prior} are

$$\|R_f^{\text{prior}}\|_2 = 8.081109861025224, \quad \|R_f^{\text{prior}}\|_F = 8.087899329725035.$$

Hence

$$\alpha_f^{\text{prior}} := \frac{\|R_f^{\text{prior}}\|_2}{\lambda_*} = \frac{8.081109861025224}{3.52129589} = 2.294924967594988. \quad (50)$$

Therefore

$$\alpha_f^{\text{prior}} > 1.$$

So the quartic baseline is not small-remainder dominant at local Jacobian level.

LI. 4. DIAGONAL-DEFECT RATIO

Let

$$D_f^{\text{prior}} := \text{diag}(R_f^{\text{prior}}).$$

Then

$$D_f^{\text{prior}} = \text{diag}(-0.839960218174805, -7.145183917627953, 0.235363388813100).$$

The diagonal-defect ratio is

$$\varepsilon_{\text{diag}}^{\text{prior}} := \frac{\|R_f^{\text{prior}} - D_f^{\text{prior}}\|_F}{\|R_f^{\text{prior}}\|_F} = 0.455959495840998. \quad (51)$$

Thus about 45.6% of the admissible prior remainder is intrinsically non-diagonal in the coarse (A, E, O) basis. So a diagonal quadratic correction is not a faithful description.

LII. 5. SYMMETRY-DEFECT RATIO

Since $S_{\min}$ is exactly symmetric,

$$(R_f^{\text{prior}})^T = R_f^{\text{prior}},$$

hence its antisymmetric part vanishes:

$$(R_f^{\text{prior}})^{(a)} = \frac{1}{2} (R_f^{\text{prior}} - (R_f^{\text{prior}})^T) = 0.$$

Therefore

$$\varepsilon_{\text{sym}}^{\text{prior}} := \frac{\|(R_f^{\text{prior}})^{(a)}\|_F}{\|R_f^{\text{prior}}\|_F} = 0. \quad (52)$$

So the admissible local completion is perfectly gradient-type at Jacobian level.

LIII. 6. ADAPTED-BASIS DECOMPOSITION

Define the corrected-seed adapted orthonormal basis

$$t = \frac{1}{\sqrt{11}}(1, -3, 1)^T, \quad n_1 = \frac{1}{\sqrt{2}}(1, 0, -1)^T, \quad n_2 = \frac{1}{\sqrt{22}}(3, 2, 3)^T.$$

Let

$$B = (t \mid n_1 \mid n_2) \in O(3).$$

Then

$$\tilde{R}_f^{\text{prior}} := B^T R_f^{\text{prior}} B$$

is

$$\tilde{R}_f^{\text{prior}} = \begin{pmatrix} -7.74978075 & -1.26092870 & 1.04286422 \\ -1.26092870 & 0 & 0 \\ 1.04286422 & 0 & 0 \end{pmatrix}$$

up to machine roundoff.

Therefore the transverse 2×2 block vanishes:

$$(\tilde{R}_f^{\text{prior}})_{\perp\perp} = 0.$$

So the admissible prior remainder is exactly of tangent-coupling type:

$$\tilde{R}_f^{\text{prior}} = \begin{pmatrix} r_{tt} & * & * \\ * & 0 & 0 \\ * & 0 & 0 \end{pmatrix}.$$

LIV. 7. MIXING-DEFECT RATIO

Let $\Pi_{\mathcal{T}}$ denote the tangent-coupling projection in the adapted basis:

$$\Pi_{\mathcal{T}} \left(\begin{pmatrix} x_{tt} & x_{t1} & x_{t2} \\ x_{1t} & x_{11} & x_{12} \\ x_{2t} & x_{21} & x_{22} \end{pmatrix} \right) = \begin{pmatrix} x_{tt} & x_{t1} & x_{t2} \\ x_{1t} & 0 & 0 \\ x_{2t} & 0 & 0 \end{pmatrix}.$$

Then

$$\Pi_{\mathcal{T}}(\tilde{R}_f^{\text{prior}}) = \tilde{R}_f^{\text{prior}}$$

up to machine precision. Hence

$$\varepsilon_{\text{mix}}^{\text{prior}} := \frac{\|\tilde{R}_f^{\text{prior}} - \Pi_{\mathcal{T}}(\tilde{R}_f^{\text{prior}})\|_F}{\|\tilde{R}_f^{\text{prior}}\|_F} = 9.547253466277641 \times 10^{-17}. \quad (53)$$

This is numerically zero. So the available admissible local completion is an essentially exact tangent-coupling remainder.

LV. 8. DIAGONAL-ONLY COMPLETION FAILS

The diagonal-only stationarity target is

$$Q = \text{diag}(q_A^{,q_E}, q_O^{\text{diag}}) = \text{diag}(-8.494671573456258, -8.241391656489691, -2.580391735022780).$$

Then

$$\lambda(J_f^{(4)} + Q) = (-0.11554365, 0.91059735, 16.23363489).$$

So the diagonal-only completion is not admissible, because it destroys positive definiteness. By contrast,

$$\lambda(J_f^{(4)} + S_{\min}) = (3.82469901, 8.35243128, 16.41823252),$$

which is strictly positive.

Therefore the first viable completion is not diagonal but genuinely non-diagonal.

LVI. 9. FINAL COMPUTABLE CERTIFICATE

The strongest numerical certificate that can honestly be completed from the currently uploaded corrected-residual data is:

$$\alpha_f^{\text{prior}} = 2.294924967594988,$$

$$\varepsilon_{\text{diag}}^{\text{prior}} = 0.455959495840998,$$

$$\varepsilon_{\text{sym}}^{\text{prior}} = 0,$$

$$\varepsilon_{\text{mix}}^{\text{prior}} = 9.547253466277641 \times 10^{-17}.$$

Thus the admissible local corrected completion is:

1. not small in norm relative to the quartic baseline,
2. not diagonal,
3. perfectly symmetric,
4. and essentially exact tangent-coupling in the adapted branch basis.

So the correct next reduced model is

symmetric non-diagonal tangent-coupling completion,

not a diagonal correction.

LVII. FINAL SUMMARY

At the corrected residual quartic stage, the uploaded files already fix:

$$\delta_f = (0.0096, -0.0288, 0.0096), \quad \|G(\delta_f)\| \approx 0.03791852545979427,$$

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665531859216 & 0.4137780199567714 \\ -4.991665531859216 & 22.646508448870943 & -1.2413340598703142 \\ 0.4137780199567714 & -1.2413340598703142 & 3.603395005929136 \end{pmatrix},$$

and the admissible symmetric completion

$$S_{\min} = \begin{pmatrix} -0.839960218174805 & 2.450804313533534 & -0.302298414680853 \\ 2.450804313533534 & -7.145183917627953 & 0.837818903051675 \\ -0.302298414680853 & 0.837818903051675 & 0.235363388813100 \end{pmatrix}.$$

Using

$$R_f^{\text{prior}} := S_{\min},$$

the explicit numerical prior certificate is

$$\alpha_f^{\text{prior}} = 2.294924967594988, \quad \varepsilon_{\text{diag}}^{\text{prior}} = 0.455959495840998,$$

$$\varepsilon_{\text{sym}}^{\text{prior}} = 0, \quad \varepsilon_{\text{mix}}^{\text{prior}} = 9.547253466277641 \times 10^{-17}.$$

Therefore the current corrected local completion is decisively:

not diagonal, not skew, but essentially exact symmetric tangent-coupling.

Moreover, the diagonal-only target fails because

$$\lambda_{\min}(J_f^{(4)} + Q) = -0.11554365 < 0,$$

while

$$\lambda(J_f^{(4)} + S_{\min}) = (3.82469901, 8.35243128, 16.41823252) > 0.$$

Hence the strongest honest numerical conclusion already available is:

the corrected residual closure requires a symmetric non-diagonal tangent-coupling completion.

LVIII. QUANTITATIVE NEWTON-MISMATCH BARRIER AT THE CORRECTED RESIDUAL STAGE

LIX. SHORT KOREAN EXPLANATION

Even though the actual centered-difference Jacobian

$$J_f := DG(\delta_f)$$

has not yet been explicitly supplied in the uploaded files, the current notes already contain a stronger result than a mere qualitative statement.

They fix:

$$\delta_f = (0.0096, -0.0288, 0.0096), \quad \|\delta_f\| \approx 0.03183959798741184,$$

$$\|G(\delta_f)\| \approx 0.03791852545979427,$$

the exact quartic baseline Jacobian

$$J_f^{(4)}, \quad \lambda_* := \lambda_{\min}(J_f^{(4)}) \approx 3.5212959, \quad \|J_f^{(4)}\|_2 \approx 24.4716494789482,$$

and the vector-level lower bound

$$\|E_f\| = \|G_f - G_f^{(4)}\| \geq 0.21427160981790663.$$

Moreover, the quartic Newton predictor is exactly

$$\Delta_f^{(4)} = -\frac{1}{3}\delta_f.$$

Thus the file already gives a quantitative lower bound on the actual Newton mismatch

$$\mathfrak{D}_{\text{Newt}} := \frac{\|\Delta_f + \delta_f/3\|}{\|\delta_f\|}, \quad \Delta_f := -J_f^{-1}G_f,$$

in terms of the unknown Jacobian remainder norm $\|R_f\|_2$, where

$$R_f := J_f - J_f^{(4)}.$$

So the next sharp result is not another ansatz. It is a numerical exclusion theorem: if the actual Jacobian remainder lies in a strong or moderate quartic-dominance regime, then the full Newton step must stay a definite nonzero distance away from the quartic predictor $-\delta_f/3$.

LX. FULL DETAILED DERIVATION

LXI. 1. FROZEN CORRECTED-BRANCH DATA

The corrected branch is frozen at

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta_f = (0.0096, -0.0288, 0.0096)^T.$$

Its corrected residual norm is

$$\|G(\delta_f)\| \approx 0.03791852545979427.$$

The exact quartic baseline Jacobian at the corrected seed is

$$J_f^{(4)} = \begin{pmatrix} 10.095240102467715 & -4.991665534361308 & 0.41377801570768584 \\ -4.991665534361308 & 22.64650843997611 & -1.2413340471230576 \\ 0.41377801570768584 & -1.2413340471230576 & 3.6033950146440785 \end{pmatrix},$$

with eigenvalues

$$\lambda_1^{(4)} \approx 3.5212959, \quad \lambda_2^{(4)} \approx 8.35219818, \quad \lambda_3^{(4)} \approx 24.47164948.$$

Hence

$$\lambda_* := \lambda_{\min}(J_f^{(4)}) \approx 3.5212959, \quad \|J_f^{(4)}\|_2 \approx 24.4716494789482.$$

The vector remainder satisfies

$$0.21427160981790663 \leq \|E_f\| \leq 0.2901086607374952, \quad E_f := G_f - G_f^{(4)}.$$

In particular,

$$\frac{\|E_f\|}{\|G_f^{(4)}\|} \geq 0.8496431059127669,$$

so strong quartic sufficiency is already excluded at vector level.

LXII. 2. EXACT QUARTIC NEWTON PREDICTOR

Because $G^{(4)}$ is homogeneous of degree 3,

$$J_f^{(4)} \delta_f = 3G_f^{(4)}.$$

Therefore the quartic-only Newton correction is exactly

$$\Delta_f^{(4)} := -(J_f^{(4)})^{-1} G_f^{(4)} = -\frac{1}{3} \delta_f.$$

Hence

$$\Delta_f^{(4)} = (-0.0032, 0.0096, -0.0032).$$

So the correct comparison quantity for the full Newton step

$$\Delta_f := -J_f^{-1} G_f$$

is

$$\mathfrak{D}_{\text{Newt}} := \left\| \Delta_f + \frac{1}{3} \delta_f \right\|$$

or, in normalized form,

$$\widehat{\mathfrak{D}}_{\text{Newt}} := \frac{\mathfrak{D}_{\text{Newt}}}{\|\delta_f\|}.$$

LXIII. 3. EXACT LOWER BOUND FROM THE NOTES

The uploaded notes already derive

$$\widehat{\mathfrak{D}}_{\text{Newt}} \geq \frac{\|E_f\|/\|\delta_f\| - \|R_f\|_2/3}{\|J_f^{(4)}\|_2 + \|R_f\|_2}, \quad R_f := J_f - J_f^{(4)}.$$

Using

$$\|\delta_f\| \approx 0.03183959798741184, \quad \|E_f\| \geq 0.21427160981790663,$$

one gets

$$\frac{\|E_f\|}{\|\delta_f\|} \geq 6.729308620179073.$$

Hence

$$\boxed{\widehat{\mathfrak{D}}_{\text{Newt}} \geq \frac{6.729308620179073 - \|R_f\|_2/3}{24.4716494789482 + \|R_f\|_2}.} \quad (54)$$

This is already a sharp quantitative exclusion formula.

LXIV. 4. STRONG, VERY STRONG, AND MODERATE QUARTIC-DOMINANCE REGIMES

The notes further evaluate (54) in three regimes.

A. Very strong quartic Jacobian dominance

If

$$\|R_f\|_2 \leq 0.1 \lambda_* = 0.35212959,$$

then

$$\widehat{\mathfrak{D}}_{\text{Newt}} \geq 0.2663713855973499.$$

Multiplying by

$$\|\delta_f\| \approx 0.03183959798741184,$$

gives the absolute barrier

$$\boxed{\mathfrak{D}_{\text{Newt}} \geq 0.008481157832769485.} \quad (55)$$

B. Strong quartic Jacobian dominance

If

$$\|R_f\|_2 \leq 0.2 \lambda_* = 0.70425918,$$

then

$$\widehat{\mathfrak{D}}_{\text{Newt}} \geq 0.2579834549109606,$$

hence

$$\boxed{\mathfrak{D}_{\text{Newt}} \geq 0.008214089491768573.} \quad (56)$$

C. Moderate quartic Jacobian dominance

If

$$\|R_f\|_2 \leq 0.5 \lambda_* = 1.76064795,$$

then

$$\widehat{\mathfrak{D}}_{\text{Newt}} \geq 0.23417080882818148,$$

hence

$$\boxed{\mathfrak{D}_{\text{Newt}} \geq 0.007455904413476369.} \quad (57)$$

LXV. 5. EXACT MEANING OF THESE BARRIERS

The three bounds (55)–(57) are not heuristic. They mean:

$$\boxed{\|R_f\|_2 \leq 1.76064795 \implies \left\| \Delta_f + \frac{1}{3} \delta_f \right\| \geq 0.007455904413476369.} \quad (58)$$

Equivalently, if the actual extracted Newton step satisfies

$$\left\| \Delta_f + \frac{1}{3}\delta_f \right\| < 0.007455904413476369,$$

then the moderate quartic-dominance regime is impossible.

Likewise,

$$\left\| \Delta_f + \frac{1}{3}\delta_f \right\| < 0.008214089491768573$$

rules out strong quartic Jacobian dominance, and

$$\left\| \Delta_f + \frac{1}{3}\delta_f \right\| < 0.008481157832769485$$

rules out very strong quartic Jacobian dominance.

Thus the actual six-point Jacobian extraction does not merely decide whether R_f is small. It also yields a direct dynamical exclusion test through the observed Newton step.

LXVI. 6. CONNECTION WITH THE CURRENTLY AVAILABLE ADMISSIBLE PRIOR REMAINDER

The already available admissible local prior completion satisfies

$$\|R_f^{\text{prior}}\|_2 = 8.081109861025224 > \lambda_* \approx 3.5212959.$$

So the prior admissible completion lies completely outside the small-remainder invertibility-control regime.

Therefore the existing prior evidence and the Newton-barrier theorem point in the same direction:

$$\boxed{\text{the corrected residual branch is not naturally consistent with quartic-dominant local control.}} \quad (59)$$

The remaining local structure is instead controlled by a genuinely nontrivial completion beyond the exact four-class quartic baseline.

LXVII. FINAL SUMMARY

The uploaded files still do not contain the actual centered-difference Jacobian

$$J_f = DG(\delta_f),$$

but they already contain a stronger quantitative result than a qualitative no-go.

At the corrected seed

$$\delta_f = (0.0096, -0.0288, 0.0096), \quad \|\delta_f\| \approx 0.03183959798741184,$$

the quartic Newton predictor is exactly

$$\Delta_f^{(4)} = -\frac{1}{3}\delta_f.$$

Moreover, the vector-level mismatch satisfies

$$\|E_f\| \geq 0.21427160981790663,$$

and the exact quartic baseline Jacobian has

$$\lambda_* = \lambda_{\min}(J_f^{(4)}) \approx 3.5212959, \quad \|J_f^{(4)}\|_2 \approx 24.4716494789482.$$

Therefore the notes already imply the Newton-mismatch barrier

$$\widehat{\mathfrak{D}}_{\text{Newt}} \geq \frac{6.729308620179073 - \|R_f\|_2/3}{24.4716494789482 + \|R_f\|_2}.$$

In particular:

$$\|R_f\|_2 \leq 0.1\lambda_* \implies \left\| \Delta_f + \frac{1}{3}\delta_f \right\| \geq 0.008481157832769485,$$

$$\|R_f\|_2 \leq 0.2\lambda_* \implies \left\| \Delta_f + \frac{1}{3}\delta_f \right\| \geq 0.008214089491768573,$$

$$\|R_f\|_2 \leq 0.5\lambda_* \implies \left\| \Delta_f + \frac{1}{3}\delta_f \right\| \geq 0.007455904413476369.$$

So the next six-point extraction has an immediate sharp interpretation:

if the actual Newton step comes out closer than 7.456×10^{-3} to $-\delta_f/3$, then even moderate quartic Jacobian dominance is already impossible.

Thus the next frontier is no longer another quartic ansatz. It is the actual finite extraction of

$$J_f, \quad R_f = J_f - J_f^{(4)}, \quad \Delta_f = -J_f^{-1}G_f,$$

followed by the Newton-barrier test above.

LXVIII. EXACT COARSE-ORBIT CLOSURE OF THE CORRECTED FOUR-CLASS QUARTIC KERNEL

LXIX. SHORT EXPLANATION

At the corrected residual quartic stage, the extracted four-class quartic data are already fixed:

$$u_d = U_d = 2493.1087739398513, \quad u_{\text{opp}} = U_{\text{opp}} = 14874.171297437068,$$

$$u_{\perp} = U_{\perp} = 9027.17290917827, \quad u_{\text{obl}} = U_{\text{obl}} = 2244.8894311890813.$$

Moreover, the coarse orbit split of the selected BCC shell is already fixed as

$$|A| = 2, \quad |E| = 2, \quad |O| = 8,$$

and the ordered-pair class-count matrices are known exactly:

$$N^{(d)}, \quad N^{(\text{opp})}, \quad N^{(\perp)}, \quad N^{(\text{obl})}.$$

Therefore the orbit-reduced quartic matrix can now be computed explicitly. The resulting matrix is exactly A/E -symmetric. Hence the corrected-branch A/E asymmetry cannot come from four-class pair-class quartics alone. It must arise either from the quadratic/local residual sheet or from a refinement beyond coarse pair classes.

LXX. 1. EXACT FOUR-CLASS DATA

The corrected projected quartic class means are

$$u_d = 2493.1087739398513, \quad u_{\text{opp}} = 14874.171297437068, \quad u_{\perp} = 9027.17290917827, \quad u_{\text{obl}} = 2244.8894311890813. \quad (60)$$

The old three-parameter kernel is ruled out because

$$4U_{\text{obl}} - 3U_{\perp} - U_{\text{opp}} \neq 0,$$

so the minimal exact quartic kernel is the four-class one:

$$U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})}. \quad (61)$$

LXXI. 2. COARSE ORBIT SPLIT AND EXACT ORDERED-PAIR COUNTS

The selected BCC shell is split into coarse orbits

$$A, \quad E, \quad O,$$

with

$$\boxed{n_A = 2, \quad n_E = 2, \quad n_O = 8.} \quad (62)$$

The ordered-pair class-count matrices are exactly

$$\boxed{N^{(d)} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix}, \quad N^{(\text{opp})} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix},} \quad (63)$$

$$\boxed{N^{(\perp)} = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 16 \end{pmatrix}, \quad N^{(\text{obl})} = \begin{pmatrix} 0 & 0 & 16 \\ 0 & 0 & 16 \\ 16 & 16 & 32 \end{pmatrix}.} \quad (64)$$

The consistency check is

$$N^{(d)} + N^{(\text{opp})} + N^{(\perp)} + N^{(\text{obl})} = \begin{pmatrix} 4 & 4 & 16 \\ 4 & 4 & 16 \\ 16 & 16 & 64 \end{pmatrix} = (n_X n_Y)_{X,Y \in \{A,E,O\}}.$$

LXXII. 3. RAW ORDERED-PAIR ORBIT MATRIX

Under the raw ordered-pair convention,

$$\boxed{M_{XY}^{\text{raw}} = N_{XY}^{(d)} u_d + N_{XY}^{(\text{opp})} u_{\text{opp}} + N_{XY}^{(\perp)} u_{\perp} + N_{XY}^{(\text{obl})} u_{\text{obl}}.} \quad (65)$$

Substituting (60), (63), and (64), one gets:

$$M_{AA}^{\text{raw}} = 2u_d + 2u_{\text{opp}} = 34734.560142753835,$$

$$M_{EE}^{\text{raw}} = 2u_d + 2u_{\text{opp}} = 34734.560142753835,$$

$$M_{AE}^{\text{raw}} = 4u_{\perp} = 36108.69163671308,$$

$$M_{AO}^{\text{raw}} = M_{EO}^{\text{raw}} = 16u_{\text{obl}} = 35918.23089902530,$$

$$M_{OO}^{\text{raw}} = 8u_d + 8u_{\text{opp}} + 16u_{\perp} + 32u_{\text{obl}} = 355209.46891591826.$$

Hence

$$\boxed{M^{\text{raw}} = \begin{pmatrix} 34734.560142753835 & 36108.69163671308 & 35918.23089902530 \\ 36108.69163671308 & 34734.560142753835 & 35918.23089902530 \\ 35918.23089902530 & 35918.23089902530 & 355209.46891591826 \end{pmatrix}.} \quad (66)$$

LXXIII. 4. ORBIT-AVERAGED QUARTIC MATRIX

For the reduced amplitude equations, the natural coarse matrix is the orbit-averaged one:

$$M_{XY}^{\text{avg}} := \frac{M_{XY}^{\text{raw}}}{n_X n_Y}. \quad (67)$$

Using (62) and (66), this gives

$$M^{\text{avg}} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}. \quad (68)$$

This is exactly the coarse four-class quartic matrix previously appearing in the residual-level formulas.

LXXIV. 5. EXACT QUARTIC RESIDUAL CONTRIBUTION

Let

$$a_A, \quad a_E, \quad a_O$$

be the real coarse amplitudes, and define

$$r_A = a_A^2, \quad r_E = a_E^2, \quad r_O = a_O^2.$$

Then the quartic reduced free form is

$$F_4^{\text{red}} = \frac{1}{4} \sum_{X,Y \in \{A,E,O\}} M_{XY}^{\text{avg}} r_X r_Y. \quad (69)$$

Therefore the quartic contribution to the residual map is

$$G_X^{(4)} = a_X \sum_{Y \in \{A,E,O\}} M_{XY}^{\text{avg}} r_Y. \quad (70)$$

Explicitly:

$$G_A^{(4)} = a_A \left[8683.640035688459 r_A + 9027.17290917827 r_E + 2244.8894311890813 r_O \right], \quad (71)$$

$$G_E^{(4)} = a_E \left[9027.17290917827 r_A + 8683.640035688459 r_E + 2244.8894311890813 r_O \right], \quad (72)$$

$$G_O^{(4)} = a_O \left[2244.8894311890813 (r_A + r_E) + 5550.147951811223 r_O \right]. \quad (73)$$

LXXV. 6. EXACT STRUCTURAL CONSEQUENCE: AUTOMATIC A/E SYMMETRY

From (66) or (68),

$$M_{AA} = M_{EE}, \quad M_{AO} = M_{EO}, \quad M_{AE} = M_{EA}. \quad (74)$$

Thus the four-class pair-class closure is exactly invariant under

$$A \leftrightarrow E.$$

So at coarse-orbit level,

$$\boxed{\text{four-class quartics alone cannot generate an intrinsic } A/E \text{ asymmetry.}} \quad (75)$$

This implies that any corrected-branch asymmetry between the A - and E -channels must come from one of the following:

1. the quadratic/local residual sheet, schematically

$$G_X^{(2)} \sim a_X \sigma_X, \quad \sigma_A \neq \sigma_E;$$

2. a refinement beyond coarse pair-class closure, e.g. sign-resolved suborbit quartics;

3. higher residual structure not compressible to the coarse (A, E, O) level.

LXXVI. 7. FINAL THEOREM-LEVEL STATEMENT

Theorem LXXVI.1. *For the corrected residual quartic closure based on the exact four-class BCC shell kernel and the coarse orbit split*

$$|A| = 2, \quad |E| = 2, \quad |O| = 8,$$

the orbit-reduced quartic matrix is explicitly

$$M^{\text{avg}} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix},$$

and the quartic residual contribution is

$$G_X^{(4)} = a_X \sum_Y M_{XY}^{\text{avg}} r_Y.$$

Moreover this quartic closure is exactly A/E-symmetric at coarse-orbit level. Therefore any corrected-branch A/E asymmetry must come from the quadratic sector or from a refinement beyond coarse pair classes.

Proof. The first statement follows by direct insertion of the exact class-count matrices into the four-class kernel. The symmetry statement follows because the matrices

$$N^{(d)}, \quad N^{(\text{opp})}, \quad N^{(\perp)}, \quad N^{(\text{obl})}$$

are all invariant under $A \leftrightarrow E$, so their linear combination M is invariant as well. $\square$

LXXVII. FINAL SUMMARY

The corrected four-class quartic data are

$$u_d = 2493.1087739398513, \quad u_{\text{opp}} = 14874.171297437068, \quad u_{\perp} = 9027.17290917827, \quad u_{\text{obl}} = 2244.8894311890813.$$

Using the exact coarse-orbit split

$$n_A = 2, \quad n_E = 2, \quad n_O = 8,$$

and the exact ordered-pair class counts

$$N^{(d)} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix}, \quad N^{(\text{opp})} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix},$$

$$N^{(\perp)} = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 16 \end{pmatrix}, \quad N^{(\text{obl})} = \begin{pmatrix} 0 & 0 & 16 \\ 0 & 0 & 16 \\ 16 & 16 & 32 \end{pmatrix},$$

one obtains the raw ordered-pair orbit matrix

$$M^{\text{raw}} = \begin{pmatrix} 34734.560142753835 & 36108.69163671308 & 35918.23089902530 \\ 36108.69163671308 & 34734.560142753835 & 35918.23089902530 \\ 35918.23089902530 & 35918.23089902530 & 355209.46891591826 \end{pmatrix},$$

and the orbit-averaged coarse quartic matrix

$$M^{\text{avg}} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}.$$

Therefore the quartic residual contribution is explicitly

$$G_X^{(4)} = a_X \sum_Y M_{XY}^{\text{avg}} r_Y.$$

The decisive structural fact is:

$$M_{AA} = M_{EE}, \quad M_{AO} = M_{EO}, \quad M_{AE} = M_{EA}.$$

So the exact four-class pair-class closure is automatically A/E -symmetric.

Hence the corrected-branch A/E asymmetry cannot come from coarse four-class quartics alone. It must come from:

$$\boxed{\sigma_A \neq \sigma_E \text{ in the quadratic/local residual sheet, or from sign-resolved suborbit quartics.}}$$

Thus the next nonredundant frontier is no longer the coarse four-class closure itself. That part is now closed. The next frontier is the first A/E -breaking microscopic completion beyond coarse pair classes.

LXXVIII. COARSE RESIDUAL INSERTION AT THE CORRECTED SEED

LXXIX. GOAL

We now stop at the first nonredundant residual-level insertion problem.

The frozen corrected data are

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad (a_A, a_E, a_O) = (0.0096, -0.0288, 0.0096).$$

The exact coarse four-class quartic matrix is

$$\bar{M} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix}.$$

The corrected extractor outputs

$$\sigma_0 = -109.41971030446824, \quad \eta = -6.502583100163014,$$

and

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933.$$

The exact question is:

$$\boxed{\text{What coarse quadratic layer is required by the corrected seed,}} \tag{76}$$

and can it be generated by the natural two-parameter shell data (σ_0, η) ?

LXXX. 1. MINIMAL COARSE RESIDUAL ANSATZ

At coarse (A, E, O) level, write the minimal residual ansatz

$$\boxed{G_X(a_A, a_E, a_O) = a_X \left(q_X + \sum_{Y \in \{A, E, O\}} \bar{M}_{XY} r_Y \right), \quad r_Y := a_Y^2, \quad X \in \{A, E, O\}.} \tag{77}$$

This is the minimal diagonal quadratic completion of the exact coarse four-class quartic sector.
Define

$$r_A = a_A^2, \quad r_E = a_E^2, \quad r_O = a_O^2.$$

At the corrected seed one has

$$r_A = (0.0096)^2 = 9.216 \times 10^{-5}, \quad r_E = (0.0288)^2 = 8.2944 \times 10^{-4}, \quad r_O = (0.0096)^2 = 9.216 \times 10^{-5}.$$

LXXXI. 2. EXACT QUARTIC LOADS AT THE CORRECTED SEED

Define

$$Q_X := \sum_Y \bar{M}_{XY} r_Y.$$

Then

$$Q = \bar{M} \begin{pmatrix} r_A \\ r_E \\ r_O \end{pmatrix} = \begin{pmatrix} 8.494671573456258 \\ 8.241391656489691 \\ 2.580391735022780 \end{pmatrix}.$$

Hence

$$\boxed{Q_A = 8.494671573456258, \quad Q_E = 8.241391656489691, \quad Q_O = 2.580391735022780.} \quad (78)$$

The quartic residual contribution is therefore

$$G^{(4)}(\delta_f) = \begin{pmatrix} a_A Q_A \\ a_E Q_E \\ a_O Q_O \end{pmatrix} = \begin{pmatrix} 0.08154884710518008 \\ -0.23735207970690309 \\ 0.024771760656218685 \end{pmatrix}.$$

LXXXII. 3. NECESSITY OF COARSE A/E QUADRATIC SPLITTING

Assume first that the coarse quadratic sector is A/E -symmetric:

$$q_A = q_E =: q.$$

If the corrected seed were an exact stationary point of (77), then since

$$a_A \neq 0, \quad a_E \neq 0,$$

we would need

$$q + Q_A = 0, \quad q + Q_E = 0.$$

Subtracting gives

$$Q_A - Q_E = 0.$$

But

$$Q_A - Q_E = 8.494671573456258 - 8.241391656489691 = 0.25327991696656785 \neq 0.$$

Equivalently,

$$Q_A - Q_E = (\bar{M}_{AA} - \bar{M}_{AE})(r_A - r_E),$$

and

$$\bar{M}_{AA} - \bar{M}_{AE} = 8683.640035688459 - 9027.17290917827 = -343.5328734898113 \neq 0,$$

while

$$r_A - r_E = 9.216 \times 10^{-5} - 8.2944 \times 10^{-4} = -7.3728 \times 10^{-4} \neq 0.$$

Therefore the corrected seed cannot satisfy the coarse residual stationarity equations under

$$q_A = q_E.$$

So one obtains:

$$\boxed{\text{any exact coarse residual realization of the corrected seed requires } q_A \neq q_E.} \quad (79)$$

LXXXIII. 4. EXACT REQUIRED COARSE QUADRATIC COEFFICIENTS

If one insists that the corrected seed be an exact stationary point of the coarse ansatz, then

$$q_A^* = -Q_A, \quad q_E^* = -Q_E, \quad q_O^* = -Q_O.$$

Hence

$$\boxed{q_A^* = -8.494671573456258, \quad q_E^* = -8.241391656489691, \quad q_O^* = -2.580391735022780.} \quad (80)$$

In particular, the required coarse A/E splitting is

$$\boxed{q_A^* - q_E^* = -(Q_A - Q_E) = -0.25327991696656785.} \quad (81)$$

So the corrected seed requires

$$q_A^* < q_E^*.$$

LXXXIV. 5. DIRECTIONAL AGREEMENT WITH THE EXTRACTED ORBIT MEANS

The residual-based projection already gives

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113.$$

Therefore

$$\sigma_A - \sigma_E = -9.97814553821504 < 0.$$

Thus the numerical extractor already shows the same directional asymmetry:

$$\boxed{\sigma_A < \sigma_E.}$$

This does *not* yet prove that the normalization map is trivial, i.e.

$$q_A^* - q_E^* = \sigma_A - \sigma_E,$$

but it shows that the corrected branch is aligned with the same A/E -breaking direction required by the coarse residual insertion.

LXXXV. 6. FAILURE OF THE NATURAL TWO-PARAMETER (σ_0, η) MAP

The natural shell-level two-parameter pattern is

$$\sigma_A = \sigma_0 + \eta, \quad \sigma_E = \sigma_0 - \frac{\eta}{2}, \quad \sigma_O = \sigma_0 - \frac{\eta}{8}.$$

So the most natural linear map from shell data to coarse quadratic coefficients is

$$\boxed{q_A = c_0\sigma_0 + c_2\eta, \quad q_E = c_0\sigma_0 - \frac{1}{2}c_2\eta, \quad q_O = c_0\sigma_0 - \frac{1}{8}c_2\eta.} \quad (82)$$

Fit q_A^* and q_E^* from (80). This gives

$$\boxed{c_0 = 0.07609066293733878, \quad c_2 = 0.025967108051795404.} \quad (83)$$

Then the predicted O -coefficient is

$$q_O^{\text{pred}} = c_0\sigma_0 - \frac{1}{8}c_2\eta = -8.304711635731334.$$

But the corrected seed actually requires

$$q_O^* = -2.580391735022780.$$

So the defect is

$$\boxed{q_O^{\text{pred}} - q_O^* = -5.724319900708554.} \quad (84)$$

This is far too large to be attributed to the small corrected residual norm

$$\|G(\delta_f)\| \approx 0.03791852545979427.$$

Therefore no linear quadratic layer depending only on (σ_0, η) in the canonical

$$(\sigma_0 + \eta, \sigma_0 - \eta/2, \sigma_0 - \eta/8)$$

pattern can reproduce the corrected coarse stationary data.

LXXXVI. 7. EXACT CONSEQUENCE FOR THE NEXT BRANCH

We have now proved three sharp facts:

1. coarse four-class quartics are exactly A/E -symmetric;
2. the corrected seed requires a nonzero coarse A/E quadratic splitting

$$q_A^* - q_E^* = -0.25327991696656785;$$

3. the natural two-parameter shell map $(\sigma_0, \eta) \mapsto (q_A, q_E, q_O)$ fails badly in the O -channel.

Therefore the remaining microscopic source cannot be captured by the coarse two-parameter shell pattern alone. The next irreducible possibilities are:

$$\boxed{\text{(i) a genuinely three-parameter coarse quadratic layer, \quad or \quad (ii) sign-resolved suborbit refinement.}} \quad (85)$$

Because the project has already identified the minimal sign-resolved first-shell basis

$$(A_+, A_-, E_+, E_-, O_{++}, O_{+-}, O_{-+}, O_{--}),$$

the correct next mainline move is now immediate:

$$\boxed{\text{upgrade from coarse } (A, E, O) \text{ to the sign-resolved suborbit basis.}}$$

LXXXVII. CONCLUSION

The corrected seed can be supported at coarse level only if the quadratic layer already breaks A/E -symmetry, but that splitting cannot be explained by the naive two-parameter shell map (σ_0, η) .

So the coarse residual insertion is now closed, and it points uniquely to the next frontier:

the microscopic origin of the corrected branch must be sought beyond the coarse (A, E, O) shell pattern.

LXXXVIII. FINAL SUMMARY

At the corrected residual stage, the coarse four-class quartic matrix is

$$\bar{M} = \begin{pmatrix} 8683.640035688459 & 9027.17290917827 & 2244.8894311890813 \\ 9027.17290917827 & 8683.640035688459 & 2244.8894311890813 \\ 2244.8894311890813 & 2244.8894311890813 & 5550.147951811223 \end{pmatrix},$$

and the corrected seed is

$$(a_A, a_E, a_O) = (0.0096, -0.0288, 0.0096).$$

This gives the quartic loads

$$Q_A = 8.494671573456258, \quad Q_E = 8.241391656489691, \quad Q_O = 2.580391735022780.$$

Hence any exact coarse stationary realization of the corrected seed requires

$$q_A^* = -8.494671573456258, \quad q_E^* = -8.241391656489691, \quad q_O^* = -2.580391735022780,$$

with necessary coarse A/E -splitting

$$q_A^* - q_E^* = -0.25327991696656785.$$

So

$$q_A = q_E$$

is impossible.

Moreover, fitting the natural two-parameter shell map

$$q_A = c_0\sigma_0 + c_2\eta, \quad q_E = c_0\sigma_0 - \frac{1}{2}c_2\eta, \quad q_O = c_0\sigma_0 - \frac{1}{8}c_2\eta$$

to q_A^*, q_E^* gives

$$c_0 = 0.07609066293733878, \quad c_2 = 0.025967108051795404,$$

but then predicts

$$q_O^{\text{pred}} = -8.304711635731334,$$

whereas the corrected seed requires

$$q_O^* = -2.580391735022780.$$

The mismatch

$$q_O^{\text{pred}} - q_O^* = -5.724319900708554$$

is far too large.

Therefore the coarse residual insertion is now closed:

the corrected seed requires A/E -breaking quadratic structure,

but

that structure cannot be generated by the naive two-parameter (σ_0, η) shell map.

So the next nonredundant frontier is no longer coarse pair-class closure. It is the microscopic completion beyond the coarse (A, E, O) shell pattern, most naturally the sign-resolved suborbit basis.

LXXXIX. MINIMAL SIGN-RESOLVED SUBORBIT QUARTIC COMPLETION AND THE FIRST A/E -BREAKING ENTRY PATTERN

XC. SHORT KOREAN EXPLANATION

The sign-resolved first-shell basis is already fixed as

$$(A_+, A_-, E_+, E_-, O_{++}, O_{+-}, O_{-+}, O_{--}),$$

and the frozen corrected seed lifts to

$$\delta_{\text{sub}}^{\text{corr}} = (0.0096, 0.0096, -0.0288, -0.0288, 0.0096, 0.0096, 0.0096, 0.0096)^T.$$

The exact transverse rows are

$$r_{\text{sub}}^{(ct_1)} = \left(0, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right),$$

$$r_{\text{sub}}^{(ct_2)} = \left(0, 0, 0, 0, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}\right),$$

and the lifted corrected seed satisfies

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.$$

So the first role of the suborbit refinement is not yet to generate a linear transverse shift. It is first to provide the minimal quartic slot that can break the coarse A/E symmetry while preserving the sign symmetry inside the $\mathcal{O}$ -suborbits.

The minimal such completion is a rank-two symmetric matrix

$$\Delta M_{AE} = \xi(u_{AE}u_O^T + u_Ou_{AE}^T),$$

with

$$u_{AE} = (1, 1, -1, -1, 0, 0, 0, 0)^T, \quad u_O = (0, 0, 0, 0, 1, 1, 1, 1)^T.$$

Its coefficient is fixed by the required corrected-seed coarse splitting and comes out exactly as

$$\xi = -343.5328734898113.$$

Thus the first explicit suborbit quartic completion is now closed.

XCI. 1. FIXED SIGN-RESOLVED BASIS AND CORRECTED SEED LIFT

The minimal sign-resolved first-shell basis is

$$(A_+, A_-, E_+, E_-, O_{++}, O_{+-}, O_{-+}, O_{--}),$$

with the explicit shell representatives

$$A_{\pm} : \pm \frac{(1, 1, 0)}{\sqrt{2}}, \quad E_{\pm} : \pm \frac{(1, -1, 0)}{\sqrt{2}},$$

and

$$O_{++}, O_{+-}, O_{-+}, O_{--}$$

defined by the sign pattern of

$$c_{\alpha}t_{\alpha 1}, \quad c_{\alpha}t_{\alpha 2}.$$

The exact transverse rows are

$$r_{\text{sub}}^{(ct_1)} = \left(0, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right),$$

$$r_{\text{sub}}^{(ct_2)} = \left(0, 0, 0, 0, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}\right).$$

The frozen corrected seed

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096)$$

lifts naturally to

$$\boxed{\delta_{\text{sub}}^{\text{corr}} = (a, a, e, e, o, o, o, o)^T = (0.0096, 0.0096, -0.0288, -0.0288, 0.0096, 0.0096, 0.0096, 0.0096)^T.} \quad (86)$$

Applying the transverse rows gives

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.$$

Therefore the orbit-symmetric corrected seed produces no linear first-shell transverse shift.

XCII. 2. WHY THE FIRST SUBORBIT QUARTIC COMPLETION SHOULD BE A/E -BREAKING BUT O -UNIFORM

The coarse four-class quartic closure is exactly $A \leftrightarrow E$ symmetric. Hence it cannot account for the corrected-branch asymmetry by itself.

At the coarse residual insertion stage, the corrected seed requires

$$q_A^* - q_E^* = -0.25327991696656785.$$

Thus the first suborbit quartic correction should satisfy three requirements:

1. it must distinguish the A -sector from the E -sector;
2. it should preserve the internal sign symmetry inside each (A_+, A_-) and (E_+, E_-) pair;
3. it should keep the four O -suborbits uniform at the first step, because the present corrected seed itself is O -uniform and the first-shell transverse rows still vanish.

This uniquely points to a coupling between the coarse A/E -difference direction and the uniform O -direction.

XCIII. 3. CANONICAL SUBORBIT VECTORS

Define the two canonical 8-vectors

$$\boxed{u_{AE} := (1, 1, -1, -1, 0, 0, 0, 0)^T, \quad u_O := (0, 0, 0, 0, 1, 1, 1, 1)^T.} \quad (87)$$

The vector u_{AE} is the minimal coarse A/E -breaking direction. The vector u_O is the uniform O -sheet direction. The minimal symmetric quartic completion that couples them is therefore

$$\boxed{\Delta M_{AE} = \xi(u_{AE}u_O^T + u_Ou_{AE}^T).} \quad (88)$$

In the ordered basis

$$(A_+, A_-, E_+, E_-, O_{++}, O_{+-}, O_{-+}, O_{--}),$$

this matrix is explicitly

$$\Delta M_{AE} = \xi \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (89)$$

This is the first nontrivial sign-resolved quartic completion.

XCIV. 4. LOAD SHIFTS INDUCED BY ΔM_{AE}

Let

$$r_{\text{sub}} := (a^2, a^2, e^2, e^2, o^2, o^2, o^2, o^2)^T.$$

Then

$$u_O \cdot r_{\text{sub}} = 4o^2, \quad u_{AE} \cdot r_{\text{sub}} = 2a^2 - 2e^2.$$

Therefore the load shift generated by ΔM_{AE} is

$$\Delta Q_{\text{sub}} = \Delta M_{AE} r_{\text{sub}} = \xi \left[u_{AE}(4o^2) + u_O(2a^2 - 2e^2) \right].$$

So, componentwise,

$$\Delta Q_{A+} = \Delta Q_{A-} = 4\xi o^2, \quad \Delta Q_{E+} = \Delta Q_{E-} = -4\xi o^2, \quad (90)$$

and

$$\Delta Q_{O++} = \Delta Q_{O+-} = \Delta Q_{O-+} = \Delta Q_{O--} = 2\xi(a^2 - e^2). \quad (91)$$

Hence this completion breaks A/E at coarse level but keeps the four O -suborbits identical.

XCV. 5. FIXING ξ FROM THE REQUIRED CORRECTED COARSE SPLITTING

At the coarse level the baseline quartic loads were

$$Q_A = 8.494671573456258, \quad Q_E = 8.241391656489691, \quad Q_O = 2.580391735022780.$$

So

$$Q_A - Q_E = 0.25327991696656785.$$

Under the correction ΔM_{AE} ,

$$Q_A \mapsto Q_A + 4\xi o^2, \quad Q_E \mapsto Q_E - 4\xi o^2.$$

Therefore the corrected coarse difference becomes

$$(Q_A - Q_E)_{\text{new}} = (Q_A - Q_E) + 8\xi o^2.$$

If one wants the quartic suborbit correction to cancel the baseline coarse A/E mismatch, then one must impose

$$(Q_A - Q_E) + 8\xi o^2 = 0.$$

Since

$$o = 0.0096, \quad o^2 = 9.216 \times 10^{-5},$$

this gives

$$\xi = -\frac{Q_A - Q_E}{8o^2} = -\frac{0.25327991696656785}{8 \times 9.216 \times 10^{-5}} = -343.5328734898113. \quad (92)$$

So the first explicit A/E -breaking quartic suborbit entry scale is now completely fixed.

XCVI. 6. CORRECTED LOADS AFTER THE MINIMAL SUBORBIT COMPLETION

Using (92),

$$4\xi o^2 = -0.12663995848328392.$$

Hence

$$Q_A^{\text{new}} = Q_E^{\text{new}} = 8.368031614972974. \quad (93)$$

For the O -suborbits,

$$a^2 = (0.0096)^2 = 9.216 \times 10^{-5}, \quad e^2 = (0.0288)^2 = 8.2944 \times 10^{-4},$$

so

$$2\xi(a^2 - e^2) = 0.5065598339331357.$$

Thus each O -suborbit load becomes

$$Q_{O\bullet}^{\text{new}} = Q_O + 0.5065598339331357 = 3.0869515689559157. \quad (94)$$

Therefore the minimal suborbit quartic completion removes the coarse A/E load asymmetry while leaving the O -sheet sign-symmetric.

XCVII. 7. WHY THIS STILL DOES NOT GENERATE A LINEAR TRANSVERSE WITNESS

Because the four O -suborbit loads remain equal,

$$Q_{O++}^{\text{new}} = Q_{O+-}^{\text{new}} = Q_{O-+}^{\text{new}} = Q_{O--}^{\text{new}},$$

the transverse row contractions still vanish:

$$r_{\text{sub}}^{(ct_1)} \cdot Q_O^{\text{new}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot Q_O^{\text{new}} = 0.$$

So ΔM_{AE} is indeed the first A/E -breaking quartic completion, but it is *not yet* the first transverse-generating one.

This is exactly consistent with the earlier theorem-level conclusion that the present orbit-symmetric corrected seed produces no linear first-shell transverse correction.

XCVIII. 8. THE NEXT GENUINELY TRANSVERSE-SENSITIVE PATTERNS

Once ΔM_{AE} is fixed, the next irreducible sign-sensitive directions are

$$s_1 := (0, 0, 0, 0, 1, 1, -1, -1)^T, \quad s_2 := (0, 0, 0, 0, 1, -1, 1, -1)^T. \quad (95)$$

These are dual to the exact transverse rows

$$r_{\text{sub}}^{(ct_1)}, \quad r_{\text{sub}}^{(ct_2)}.$$

Therefore the first genuinely transverse-sensitive quartic completions beyond ΔM_{AE} are of the schematic form

$$\boxed{\Delta M_{t_1} = \kappa_1(u_{AE}s_1^T + s_1u_{AE}^T), \quad \Delta M_{t_2} = \kappa_2(u_{AE}s_2^T + s_2u_{AE}^T),} \quad (96)$$

or, alternatively, direct O -sheet splittings built from s_1, s_2 themselves.

These do not contribute on the present lifted seed unless the O -sheet is actually sign-split, which is precisely why the current corrected seed yields zero linear first-shell transverse shift.

XCIX. CONCLUSION

The first minimal quartic completion beyond the coarse (A, E, O) closure is the symmetric rank-two matrix

$$\Delta M_{AE} = \xi(u_{AE}u_O^T + u_Ou_{AE}^T), \quad \xi = -343.5328734898113.$$

This is the first explicit A/E -breaking sign-resolved quartic entry pattern. It is minimal, symmetric, and O -uniform. It closes the first suborbit-level quartic refinement without yet generating a linear first-shell transverse witness.

So the hierarchy is now sharp:

$$\boxed{\text{first suborbit quartic correction} = \Delta M_{AE},}$$

$$\boxed{\text{first transverse-sensitive corrections} = \Delta M_{t_1}, \Delta M_{t_2} \text{ or direct } O\text{-sheet sign splitting.}}$$

C. FINAL SUMMARY

The minimal sign-resolved first-shell basis is

$$(A_+, A_-, E_+, E_-, O_{++}, O_{+-}, O_{-+}, O_{--}),$$

and the frozen corrected seed lifts to

$$\delta_{\text{sub}}^{\text{corr}} = (0.0096, 0.0096, -0.0288, -0.0288, 0.0096, 0.0096, 0.0096, 0.0096)^T.$$

The exact transverse rows are

$$r_{\text{sub}}^{(ct_1)} = \left(0, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right), \quad r_{\text{sub}}^{(ct_2)} = \left(0, 0, 0, 0, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}\right),$$

and they annihilate the lifted corrected seed:

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.$$

Therefore the first sign-resolved quartic refinement should not yet be a transverse-sensitive one. It should first be the minimal A/E -breaking, O -uniform completion.

That completion is

$$\Delta M_{AE} = \xi(u_{AE}u_O^T + u_Ou_{AE}^T),$$

with

$$u_{AE} = (1, 1, -1, -1, 0, 0, 0, 0)^T, \quad u_O = (0, 0, 0, 0, 1, 1, 1, 1)^T,$$

and exact coefficient

$$\boxed{\xi = -343.5328734898113.}$$

In matrix form,

$$\Delta M_{AE} = \xi \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

This correction cancels the coarse A/E quartic load mismatch,

$$Q_A - Q_E = 0.25327991696656785,$$

but still leaves all four O -suborbit loads equal. Hence it does not yet generate a linear first-shell transverse witness.

So the suborbit refinement hierarchy is now exact:

first suborbit quartic correction = ΔM_{AE} ,

first genuinely transverse-sensitive completions come only at the next step,

through the sign-sensitive O -sheet directions

$$s_1 = (0, 0, 0, 0, 1, 1, -1, -1)^T, \quad s_2 = (0, 0, 0, 0, 1, -1, 1, -1)^T.$$

CI. FIRST EXPLICIT TRANSVERSE-SENSITIVE QUARTIC ENTRY PATTERNS ON THE SIGN-RESOLVED SUBORBIT BASIS

CII. SHORT KOREAN EXPLANATION

The sign-resolved first-shell basis is already fixed as

$$(A_+, A_-, E_+, E_-, O_{++}, O_{+-}, O_{-+}, O_{--}),$$

and the corrected lifted seed is

$$\delta_{\text{sub}}^{\text{corr}} = (a, a, e, e, o, o, o, o)^T = (0.0096, 0.0096, -0.0288, -0.0288, 0.0096, 0.0096, 0.0096, 0.0096)^T.$$

The exact transverse rows are

$$r_{\text{sub}}^{(ct_1)} = \left(0, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right),$$

$$r_{\text{sub}}^{(ct_2)} = \left(0, 0, 0, 0, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}\right).$$

The minimal A/E -breaking but O -uniform quartic completion

$$\Delta M_{AE} = \xi(u_{AE}u_O^T + u_Ou_{AE}^T)$$

was already fixed. The next irreducible step is therefore to add the first *sign-sensitive* O -sheet directions

$$s_1 = (0, 0, 0, 0, 1, 1, -1, -1)^T, \quad s_2 = (0, 0, 0, 0, 1, -1, 1, -1)^T,$$

which are dual to the two transverse rows.

The resulting entry patterns

$$\Delta M_{t_1} = \kappa_1(u_{AE}s_1^T + s_1u_{AE}^T), \quad \Delta M_{t_2} = \kappa_2(u_{AE}s_2^T + s_2u_{AE}^T)$$

turn out to be exactly channel-selective:

$$\Delta M_{t_1} \Rightarrow \text{nonzero } ct_1 \text{ and zero } ct_2,$$

$$\Delta M_{t_2} \Rightarrow \text{nonzero } ct_2 \text{ and zero } ct_1.$$

So the first transverse-sensitive quartic trigger theorem is now closed at the shell-sign level.

CIII. 1. FIXED SIGN-RESOLVED DATA

The minimal sign-resolved first-shell basis is

$$(A_+, A_-, E_+, E_-, O_{++}, O_{+-}, O_{-+}, O_{--}),$$

with corrected lifted seed

$$\delta_{\text{sub}}^{\text{corr}} = (a, a, e, e, o, o, o, o)^T = (0.0096, 0.0096, -0.0288, -0.0288, 0.0096, 0.0096, 0.0096, 0.0096)^T. \quad (97)$$

The exact transverse response rows are

$$r_{\text{sub}}^{(ct_1)} = \left(0, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right), \quad (98)$$

$$r_{\text{sub}}^{(ct_2)} = \left(0, 0, 0, 0, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}\right). \quad (99)$$

The orbit-symmetric corrected seed satisfies

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0,$$

so there is no linear transverse shift before sign-sensitive refinement.

CIV. 2. CANONICAL SIGN-SENSITIVE *O*-SHEET VECTORS

Define the two sign-sensitive *O*-sheet directions

$$s_1 := (0, 0, 0, 0, 1, 1, -1, -1)^T, \quad s_2 := (0, 0, 0, 0, 1, -1, 1, -1)^T. \quad (100)$$

Also retain the minimal *A/E*-difference vector

$$u_{AE} := (1, 1, -1, -1, 0, 0, 0, 0)^T. \quad (101)$$

These vectors satisfy the exact duality relations:

$$r_{\text{sub}}^{(ct_1)} \cdot s_1 = 1, \quad r_{\text{sub}}^{(ct_1)} \cdot s_2 = 0, \quad (102)$$

$$r_{\text{sub}}^{(ct_2)} \cdot s_1 = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot s_2 = \sqrt{2}. \quad (103)$$

So s_1 and s_2 are exactly the two independent transverse trigger directions.

CV. 3. FIRST TRANSVERSE-SENSITIVE QUARTIC ENTRY PATTERNS

The minimal symmetric quartic completions coupling the coarse *A/E*-difference sector to these sign-sensitive directions are

$$\Delta M_{t_1} = \kappa_1 (u_{AE} s_1^T + s_1 u_{AE}^T), \quad (104)$$

$$\Delta M_{t_2} = \kappa_2 (u_{AE} s_2^T + s_2 u_{AE}^T). \quad (105)$$

Their explicit matrix forms are:

$$\Delta M_{t_1} = \kappa_1 \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ -1 & -1 & 1 & 1 & 0 & 0 & 0 & 0 \\ -1 & -1 & 1 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (106)$$

$$\Delta M_{t_2} = \kappa_2 \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ -1 & -1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ -1 & -1 & 1 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (107)$$

CVI. 4. ACTION ON THE CORRECTED LIFTED SEED

Let

$$r_{\text{sub}} = (a^2, a^2, e^2, e^2, o^2, o^2, o^2, o^2)^T.$$

Then

$$s_1 \cdot r_{\text{sub}} = 0, \quad s_2 \cdot r_{\text{sub}} = 0,$$

because the corrected lifted seed is O -uniform. Also

$$u_{AE} \cdot r_{\text{sub}} = 2a^2 - 2e^2.$$

With

$$a = 0.0096, \quad e = -0.0288,$$

one has

$$2a^2 - 2e^2 = 2(0.0096^2) - 2(0.0288^2) = -0.00147456. \quad (108)$$

Therefore the suborbit load shifts are

$$\Delta Q_{t_1} = \Delta M_{t_1} r_{\text{sub}} = \kappa_1 (2a^2 - 2e^2) s_1 = -0.00147456 \kappa_1 s_1, \quad (109)$$

$$\Delta Q_{t_2} = \Delta M_{t_2} r_{\text{sub}} = \kappa_2 (2a^2 - 2e^2) s_2 = -0.00147456 \kappa_2 s_2. \quad (110)$$

So these completions do not feed back into the coarse A/E mismatch at first order. They inject directly into the sign-sensitive O -sheet sector.

CVII. 5. EXACT WITNESS SELECTION RULES

Now contract ΔQ_{t_1} and ΔQ_{t_2} with the transverse rows.

For ΔM_{t_1} ,

$$r_{\text{sub}}^{(ct_1)} \cdot \Delta Q_{t_1} = -0.00147456 \kappa_1 (r_{\text{sub}}^{(ct_1)} \cdot s_1) = -0.00147456 \kappa_1,$$

while

$$r_{\text{sub}}^{(ct_2)} \cdot \Delta Q_{t_1} = -0.00147456 \kappa_1 (r_{\text{sub}}^{(ct_2)} \cdot s_1) = 0.$$

Thus

$$\boxed{\Delta M_{t_1} \text{ activates only the } ct_1 \text{ channel.}} \quad (111)$$

For ΔM_{t_2} ,

$$r_{\text{sub}}^{(ct_1)} \cdot \Delta Q_{t_2} = -0.00147456 \kappa_2 (r_{\text{sub}}^{(ct_1)} \cdot s_2) = 0,$$

while

$$r_{\text{sub}}^{(ct_2)} \cdot \Delta Q_{t_2} = -0.00147456 \kappa_2 (r_{\text{sub}}^{(ct_2)} \cdot s_2) = -\sqrt{2} 0.00147456 \kappa_2.$$

Numerically,

$$\boxed{r_{\text{sub}}^{(ct_2)} \cdot \Delta Q_{t_2} = -0.002085342750532871 \kappa_2.} \quad (112)$$

Thus

$$\boxed{\Delta M_{t_2} \text{ activates only the } ct_2 \text{ channel.}} \quad (113)$$

So the two channels are exactly separated.

CVIII. 6. UNIT-NORMALIZED TRIGGER STRENGTHS

If one normalizes to unit shell-level witness shift, then

$$-0.00147456 \kappa_1 = 1 \implies \kappa_1^{\text{unit}} = -678.168402777778,$$

and

$$-0.002085342750532871 \kappa_2 = 1 \implies \kappa_2^{\text{unit}} = -479.5374763906165.$$

Hence the unit-normalized transverse-sensitive quartic insertions are

$$\boxed{\kappa_1^{\text{unit}} = -678.168402777778, \quad \kappa_2^{\text{unit}} = -479.5374763906165.} \quad (114)$$

CIX. 7. WHAT IS AND IS NOT CLOSED BY THIS COMPUTATION

What is now closed is the shell-sign selection theorem:

$$\boxed{\Delta M_{t_1} \Rightarrow (ct_1 \neq 0, ct_2 = 0), \quad \Delta M_{t_2} \Rightarrow (ct_1 = 0, ct_2 \neq 0).} \quad (115)$$

So the first sign-sensitive quartic trigger patterns are completely explicit.

However, one must not overclaim. The files also state that on the explicit canonical pure-even working branch

$$\alpha \neq 0, \quad \beta = 0,$$

and, in the minimal rigid-internal first-shell truncation,

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.$$

Therefore the present quartic suborbit trigger computation must be interpreted as:

$$\boxed{\text{a shell-level sign-sensitive quartic activation pattern,}} \quad (116)$$

not yet as a theorem-level derivation of the actual odd microscopic coefficient

$$\beta^{\text{corr}} \neq 0$$

of the full working branch.

That odd channel still requires the explicit nonquartic / transverse microscopic insertion identified in the notes.

CX. CONCLUSION

The first genuinely transverse-sensitive quartic completions beyond the A/E -breaking but O -uniform correction are

$$\Delta M_{t_1} = \kappa_1(u_{AE}s_1^T + s_1u_{AE}^T), \quad \Delta M_{t_2} = \kappa_2(u_{AE}s_2^T + s_2u_{AE}^T).$$

They satisfy the exact channel-selection rules:

$$\Delta M_{t_1} \text{ activates only } ct_1, \quad \Delta M_{t_2} \text{ activates only } ct_2.$$

Numerically, at the corrected lifted seed,

$$\delta(ct_1) = -0.00147456 \kappa_1, \quad \delta(ct_2) = -0.002085342750532871 \kappa_2.$$

So the first transverse-sensitive sign-resolved quartic entry patterns are now completely explicit and fully separated.

CXI. FINAL SUMMARY

The sign-resolved first-shell basis and corrected lifted seed are fixed as

$$(A_+, A_-, E_+, E_-, O_{++}, O_{+-}, O_{-+}, O_{--}),$$

$$\delta_{\text{sub}}^{\text{corr}} = (0.0096, 0.0096, -0.0288, -0.0288, 0.0096, 0.0096, 0.0096, 0.0096)^T.$$

The exact transverse rows are

$$r_{\text{sub}}^{(ct_1)} = \left(0, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right), \quad r_{\text{sub}}^{(ct_2)} = \left(0, 0, 0, 0, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}\right).$$

Define

$$u_{AE} = (1, 1, -1, -1, 0, 0, 0, 0)^T, \quad s_1 = (0, 0, 0, 0, 1, 1, -1, -1)^T, \quad s_2 = (0, 0, 0, 0, 1, -1, 1, -1)^T.$$

Then the first two transverse-sensitive quartic completions are

$$\Delta M_{t_1} = \kappa_1(u_{AE}s_1^T + s_1u_{AE}^T), \quad \Delta M_{t_2} = \kappa_2(u_{AE}s_2^T + s_2u_{AE}^T).$$

At the corrected lifted seed,

$$2a^2 - 2e^2 = -0.00147456,$$

so they produce

$$\Delta Q_{t_1} = -0.00147456 \kappa_1 s_1, \quad \Delta Q_{t_2} = -0.00147456 \kappa_2 s_2.$$

Therefore the exact channel-selection rules are

$$\Delta M_{t_1} \Rightarrow \delta(ct_1) = -0.00147456 \kappa_1, \quad \delta(ct_2) = 0,$$

$$\Delta M_{t_2} \Rightarrow \delta(ct_1) = 0, \quad \delta(ct_2) = -0.002085342750532871 \kappa_2.$$

So the first transverse-sensitive sign-resolved quartic entry patterns are completely explicit and fully separated. However, this still does *not* by itself prove

$$\beta^{\text{corr}} \neq 0$$

for the actual explicit working branch, because the notes also state that on the minimal rigid-internal first-shell truncation

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.$$

Thus what is now closed is the shell-level sign-sensitive quartic trigger structure. The actual odd transverse microscopic coefficient still requires the explicit nonquartic / transverse insertion identified as the remaining frontier.

CXII. CORRECTED-SEED INSERTION OF THE ODD TRANSVERSE HESSIAN RESPONSE

CXIII. SHORT KOREAN EXPLANATION

At the present sharpest explicit working-branch level, the odd transverse channel does not turn on linearly on the frozen corrected seed. The exact first active odd source is the quadratic Hessian response of the mixed $JK + KJ$ channel:

$$X_2(\delta) = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)], \quad H_{rs}^{(2)} = \partial_{\delta_r} \partial_{\delta_s} X_2(\delta)|_{\delta=0}.$$

The corrected odd coefficient is

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3),$$

with

$$Q_2(\delta) = \frac{1}{2} \sum_{r,s \in \{A,E,O\}} H_{rs}^{(2)} \delta_r \delta_s.$$

For the frozen corrected seed

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096),$$

this quadratic form can be written explicitly as a fixed numerical linear combination of the six symmetric coefficients

$$H_{AA}^{(2)}, H_{EE}^{(2)}, H_{OO}^{(2)}, H_{AE}^{(2)}, H_{AO}^{(2)}, H_{EO}^{(2)}.$$

This reduces the odd-channel frontier to a single weighted six-coefficient sum.

CXIV. 1. EXACT STARTING POINT

The exact transport-level odd source is

$$X_2(\delta) = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)]. \quad (117)$$

Its Hessian coefficients are defined by

$$H_{rs}^{(2)} := \partial_{\delta_r} \partial_{\delta_s} X_2(\delta)|_{\delta=0}, \quad r, s \in \{A, E, O\}. \quad (118)$$

The leading one-mediator working-branch realization is

$$H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{rs} K_0^\dagger + J_r K_s^\dagger + J_s K_r^\dagger + J_0 K_{rs}^\dagger + K_{rs} J_0^\dagger + K_r J_s^\dagger + K_s J_r^\dagger + K_0 J_{rs}^\dagger \right] + \Delta_{rs}^{\text{comp}}. \quad (119)$$

So no structural ambiguity remains: the odd corrected channel is controlled by six symmetric second-response coefficients.

CXV. 2. EXACT CORRECTED ODD COEFFICIENT

On the explicit canonical pure-even branch one has

$$X_2^{(0)} = 0,$$

and therefore the corrected odd coefficient starts only at quadratic order:

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3), \quad (120)$$

where

$$Q_2(\delta) = \frac{1}{2} \sum_{r,s \in \{A,E,O\}} H_{rs}^{(2)} \delta_r \delta_s. \quad (121)$$

Because $H_{rs}^{(2)} = H_{sr}^{(2)}$, this can be written as

$$Q_2(\delta) = \frac{1}{2} H_{AA}^{(2)} \delta_A^2 + \frac{1}{2} H_{EE}^{(2)} \delta_E^2 + \frac{1}{2} H_{OO}^{(2)} \delta_O^2 + H_{AE}^{(2)} \delta_A \delta_E + H_{AO}^{(2)} \delta_A \delta_O + H_{EO}^{(2)} \delta_E \delta_O. \quad (122)$$

CXVI. 3. FROZEN CORRECTED SEED INSERTION

Insert the frozen corrected seed

$$\delta_{\text{corr}} = (\delta_A, \delta_E, \delta_O) = (0.0096, -0.0288, 0.0096). \quad (123)$$

The exact monomials are

$$\delta_A^2 = (0.0096)^2 = 9.216 \times 10^{-5}, \quad \delta_E^2 = (0.0288)^2 = 8.2944 \times 10^{-4}, \quad \delta_O^2 = (0.0096)^2 = 9.216 \times 10^{-5},$$

$$\delta_A \delta_E = (0.0096)(-0.0288) = -2.7648 \times 10^{-4},$$

$$\delta_A \delta_O = (0.0096)(0.0096) = 9.216 \times 10^{-5},$$

$$\delta_E \delta_O = (-0.0288)(0.0096) = -2.7648 \times 10^{-4}.$$

Therefore

$$\begin{aligned} Q_2(\delta_{\text{corr}}) = & \frac{1}{2} H_{AA}^{(2)} (9.216 \times 10^{-5}) + \frac{1}{2} H_{EE}^{(2)} (8.2944 \times 10^{-4}) + \frac{1}{2} H_{OO}^{(2)} (9.216 \times 10^{-5}) \\ & + H_{AE}^{(2)} (-2.7648 \times 10^{-4}) + H_{AO}^{(2)} (9.216 \times 10^{-5}) + H_{EO}^{(2)} (-2.7648 \times 10^{-4}). \end{aligned} \quad (124)$$

So the exact corrected-seed quadratic form is

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)}. \quad (125)$$

Factoring out 9.216×10^{-5} , one gets the cleaner exact form

$$Q_2(\delta_{\text{corr}}) = 9.216 \times 10^{-5} \left[\frac{1}{2} H_{AA}^{(2)} + \frac{9}{2} H_{EE}^{(2)} + \frac{1}{2} H_{OO}^{(2)} - 3 H_{AE}^{(2)} + H_{AO}^{(2)} - 3 H_{EO}^{(2)} \right]. \quad (126)$$

This is the exact weighted six-coefficient combination controlling the odd corrected coefficient.

CXVII. 4. EXPLICIT CORRECTED ODD COEFFICIENT IN TERMS OF THE SIX HESSIAN ENTRIES

Substituting (126) into (120) yields

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 9.216 \times 10^{-5} \left[\frac{1}{2} H_{AA}^{(2)} + \frac{9}{2} H_{EE}^{(2)} + \frac{1}{2} H_{OO}^{(2)} - 3 H_{AE}^{(2)} + H_{AO}^{(2)} - 3 H_{EO}^{(2)} \right] + O(\delta^3). \quad (127)$$

Thus the full odd corrected response has been reduced to a single scalar combination:

$$\mathcal{H}_O^{\text{eff}} := \frac{1}{2} H_{AA}^{(2)} + \frac{9}{2} H_{EE}^{(2)} + \frac{1}{2} H_{OO}^{(2)} - 3 H_{AE}^{(2)} + H_{AO}^{(2)} - 3 H_{EO}^{(2)}. \quad (128)$$

Then

$$Q_2(\delta_{\text{corr}}) = 9.216 \times 10^{-5} \mathcal{H}_O^{\text{eff}}, \quad \beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 9.216 \times 10^{-5} \mathcal{H}_O^{\text{eff}} + O(\delta^3). \quad (129)$$

CXVIII. 5. IMMEDIATE SIGN CRITERION

Assuming

$$\theta_0 \kappa A_0 > 0,$$

the sign of the leading corrected odd coefficient is opposite to the sign of

$$\mathcal{H}_O^{\text{eff}}.$$

That is,

$$\text{sgn } \beta^{\text{corr}} = -\text{sgn } \mathcal{H}_O^{\text{eff}} \quad (\theta_0 \kappa A_0 > 0). \quad (130)$$

If instead $\theta_0 \kappa A_0 < 0$, the sign is reversed. So the corrected odd channel is fully controlled by:

1. the sign of the overlap factor $\theta_0 \kappa A_0$,
2. the sign of the weighted Hessian combination $\mathcal{H}_O^{\text{eff}}$.

CXIX. 6. MINIMAL RIGID-INTERNAL FIRST-SHELL TRUNCATION

The notes already state that in the minimal rigid-internal first-shell truncation,

$$K_A = K_E = K_O = 0,$$

and consequently

$$H_{AA}^{(2)} = H_{EE}^{(2)} = H_{OO}^{(2)} = H_{AE}^{(2)} = H_{AO}^{(2)} = H_{EO}^{(2)} = 0.$$

Hence

$$\boxed{\mathcal{H}_O^{\text{eff}} = 0, \quad Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0 \quad \text{in the minimal rigid-internal first-shell truncation.}} \quad (131)$$

Therefore orbit-amplitude quartic corrections alone cannot generate the odd transverse witness.

CXX. 7. PRECISE REMAINING FRONTIER

The remaining frontier is now completely compressed:

$$\boxed{\text{find the first completion-level microscopic slot for which } \mathcal{H}_O^{\text{eff}} \neq 0.} \quad (132)$$

Since the rigid first-shell orbit-amplitude sector gives

$$\mathcal{H}_O^{\text{eff}} = 0,$$

the first nonzero contribution must come from

$$\Delta_{rs}^{\text{comp}},$$

namely from completion-level sign-sensitive nonquartic / transverse insertions beyond the minimal rigid-internal orbit ansatz.

Equivalently:

$$\boxed{\beta^{\text{corr}} \neq 0 \iff \mathcal{H}_O^{\text{eff}} \neq 0 \iff \Delta_{rs}^{\text{comp}} \text{ supplies a nonzero odd weighted sum.}} \quad (133)$$

CXXI. 8. LOAD-BEARING ORDER OF IMPORTANCE AMONG THE SIX ENTRIES

Because the corrected seed has

$$|\delta_E| = 3|\delta_A| = 3|\delta_O|,$$

the coefficient weights in (128) show the exact priority ordering:

$$\boxed{H_{EE}^{(2)}, H_{AE}^{(2)}, H_{EO}^{(2)} \text{ carry the largest leverage on } \beta^{\text{corr}},} \quad (134)$$

while

$$H_{AA}^{(2)}, H_{OO}^{(2)}, H_{AO}^{(2)}$$

enter with smaller weights.

So the first practical search order for a nonzero odd corrected response is:

$$\boxed{H_{EE}^{(2)}, \quad H_{AE}^{(2)}, \quad H_{EO}^{(2)} \quad \text{first.}} \quad (135)$$

This is a genuine quantitative conclusion from the frozen corrected seed.

CXXII. CONCLUSION

The odd corrected coefficient is now reduced exactly to

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa A_0 9.216 \times 10^{-5} \mathcal{H}_O^{\text{eff}} + O(\delta^3),$$

with

$$\mathcal{H}_O^{\text{eff}} = \frac{1}{2}H_{AA}^{(2)} + \frac{9}{2}H_{EE}^{(2)} + \frac{1}{2}H_{OO}^{(2)} - 3H_{AE}^{(2)} + H_{AO}^{(2)} - 3H_{EO}^{(2)}.$$

Thus the remaining odd-channel problem is no longer structural. It is the finite evaluation of one weighted six-coefficient combination.

CXXIII. FINAL SUMMARY

At the frozen corrected seed

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096),$$

the odd transverse coefficient starts only at quadratic order:

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3).$$

The exact quadratic form is

$$Q_2(\delta) = \frac{1}{2} \sum_{r,s \in \{A,E,O\}} H_{rs}^{(2)} \delta_r \delta_s,$$

with

$$H_{rs}^{(2)} = \partial_{\delta_r} \partial_{\delta_s} \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)]|_{\delta=0}.$$

After inserting the corrected seed, one gets

$$Q_2(\delta_{\text{corr}}) = 9.216 \times 10^{-5} \left[\frac{1}{2}H_{AA}^{(2)} + \frac{9}{2}H_{EE}^{(2)} + \frac{1}{2}H_{OO}^{(2)} - 3H_{AE}^{(2)} + H_{AO}^{(2)} - 3H_{EO}^{(2)} \right].$$

Therefore

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa A_0 9.216 \times 10^{-5} \left[\frac{1}{2}H_{AA}^{(2)} + \frac{9}{2}H_{EE}^{(2)} + \frac{1}{2}H_{OO}^{(2)} - 3H_{AE}^{(2)} + H_{AO}^{(2)} - 3H_{EO}^{(2)} \right] + O(\delta^3).$$

In the minimal rigid-internal first-shell truncation all six coefficients vanish, so

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.$$

Hence orbit-amplitude quartic corrections alone cannot generate the odd transverse witness. The remaining frontier is exactly to identify the first completion-level microscopic slot producing a nonzero value of the weighted combination

$$\frac{1}{2}H_{AA}^{(2)} + \frac{9}{2}H_{EE}^{(2)} + \frac{1}{2}H_{OO}^{(2)} - 3H_{AE}^{(2)} + H_{AO}^{(2)} - 3H_{EO}^{(2)}.$$

Because $|\delta_E| = 3|\delta_A| = 3|\delta_O|$, the most load-bearing entries are

$$H_{EE}^{(2)}, \quad H_{AE}^{(2)}, \quad H_{EO}^{(2)}.$$

So the next exact microscopic search should target those three coefficients first.

CXXIV. FIRST NONZERO COMPLETION SLOT FOR THE CORRECTED ODD TRANSVERSE RESPONSE

CXXV. SHORT KOREAN EXPLANATION

The corrected odd transverse response is already fixed structurally as

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3),$$

with

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096).$$

At this point the remaining frontier is not structural. It is to determine which of the six symmetric second-response coefficients

$$H_{AA}^{(2)}, H_{EE}^{(2)}, H_{OO}^{(2)}, H_{AE}^{(2)}, H_{AO}^{(2)}, H_{EO}^{(2)}$$

turns on first once the rigid first-shell truncation is lifted.

The exact weight hierarchy at the frozen corrected seed is

$$|EE| : |AE| : |EO| : |AO| : |AA| : |OO| = 9 : 6 : 6 : 2 : 1 : 1.$$

Therefore the first nonzero completion slot should be searched in $H_{EE}^{(2)}$ first, then $H_{AE}^{(2)}, H_{EO}^{(2)}$, and only afterwards in the remaining three entries.

CXXVI. 1. EXACT CORRECTED ODD COEFFICIENT AND THE WEIGHTED SIX-ROW REDUCTION

The corrected odd coefficient is

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3), \quad (136)$$

with

$$Q_2(\delta) = \frac{1}{2} \sum_{r,s \in \{A,E,O\}} H_{rs}^{(2)} \delta_r \delta_s. \quad (137)$$

For

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096),$$

the exact numeric insertion is

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)}. \quad (138)$$

Hence the absolute weight hierarchy is

$$|EE| : |AE| : |EO| : |AO| : |AA| : |OO| = 9 : 6 : 6 : 2 : 1 : 1. \quad (139)$$

So if the six coefficients are of comparable microscopic size after lifting the rigid truncation, the dominant odd correction is forced to start in

$$H_{EE}^{(2)},$$

then

$$H_{AE}^{(2)}, \quad H_{EO}^{(2)},$$

and only later in

$$H_{AO}^{(2)}, \quad H_{AA}^{(2)}, \quad H_{OO}^{(2)}.$$

CXXVII. 2. EXACT SIX-ROW FORMULAS AND THE FAST WORKING APPROXIMATION

The common exact working-branch formula is

$$H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{rs} K_0^\dagger + J_r K_s^\dagger + J_s K_r^\dagger + J_0 K_{rs}^\dagger + K_{rs} J_0^\dagger + K_r J_s^\dagger + K_s J_r^\dagger + K_0 J_{rs}^\dagger \right] + \Delta_{rs}^{\text{comp}}. \quad (140)$$

Under the fast practical approximation

$$J_{rs} \approx 0, \quad K_{rs} \approx 0, \quad \Delta_{rs}^{\text{comp}} \approx 0,$$

this collapses to

$$H_{rs}^{(2)} \approx -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger \right]. \quad (141)$$

Therefore the diagonal E -row becomes

$$H_{EE}^{(2)} \approx -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_E K_E^\dagger + K_E J_E^\dagger \right]. \quad (142)$$

Likewise the first off-diagonal candidates are

$$H_{AE}^{(2)} \approx -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_A K_E^\dagger + J_E K_A^\dagger + K_A J_E^\dagger + K_E J_A^\dagger \right], \quad (143)$$

$$H_{EO}^{(2)} \approx -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_E K_O^\dagger + J_O K_E^\dagger + K_E J_O^\dagger + K_O J_E^\dagger \right]. \quad (144)$$

CXXVIII. 3. MINIMAL NON-RIGID COMPLETION HYPOTHESIS

The rigid first-shell truncation gives

$$K_A = K_E = K_O = 0 \quad \implies \quad H_{rs}^{(2)} = 0 \quad \forall r, s,$$

hence

$$\beta^{\text{corr}} = 0.$$

So the first nonzero odd source must come from the first orbit direction in which the K -sector ceases to vanish. Because of the exact hierarchy (139), the *minimal* non-rigid completion hypothesis is:

$$K_E \neq 0 \quad \text{first}, \quad K_A \approx 0, \quad K_O \approx 0. \quad (145)$$

This is not yet a theorem. It is the sharpest minimal working hypothesis compatible with the exact coefficient hierarchy.

Under (145), the first nonzero slot is

$$H_{EE}^{(2)},$$

while

$$H_{AE}^{(2)}, H_{EO}^{(2)}$$

remain subleading or vanish if K_A, K_O are still absent.

CXXIX. 4. FIRST EXPLICIT ODD COEFFICIENT UNDER THE MINIMAL E -SLOT HYPOTHESIS

Keeping only the leading EE -row in (138),

$$Q_2(\delta_{\text{corr}}) \approx 4.1472 \times 10^{-4} H_{EE}^{(2)}.$$

Substituting (142),

$$Q_2(\delta_{\text{corr}}) \approx -4.1472 \times 10^{-4} \frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger] \quad (146)$$

$$= -8.2944 \times 10^{-4} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger]. \quad (147)$$

Therefore

$$\beta_{\text{first}}^{\text{corr}} \approx \frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 8.2944 \times 10^{-4} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger]. \quad (148)$$

Numerically,

$$\frac{\sqrt{3}\pi}{16} (8.2944 \times 10^{-4}) = 2.820820771257055 \times 10^{-4},$$

so

$$\beta_{\text{first}}^{\text{corr}} \approx 2.820820771257055 \times 10^{-4} \theta_0 \kappa A_0 \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger]. \quad (149)$$

This is the first explicit nonzero-slot formula for the odd corrected response.

CXXX. 5. EXACT SIGN CRITERION FOR THE FIRST SLOT

If

$$\theta_0 \kappa A_0 \frac{\alpha_X \beta_X}{M_X^2} > 0,$$

then the sign of the first odd coefficient is exactly the sign of

$$\mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger].$$

Thus the first microscopic decision problem is reduced to the sign and nonvanishing of one finite odd projector overlap:

$$\boxed{\mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger]}. \quad (150)$$

If this quantity is nonzero, the odd channel is already activated at the EE -slot.

If it vanishes by an additional parity or projector symmetry, then one drops to the next two candidates

$$H_{AE}^{(2)}, \quad H_{EO}^{(2)},$$

which carry the next-largest exact weights.

CXXXI. 6. THE EXACT FALLBACK LADDER

Therefore the precise odd-channel search ladder is now:

$$\boxed{\text{Tier 1: } H_{EE}^{(2)}}, \quad (151)$$

tested by

$$\mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger].$$

If Tier 1 vanishes, move to

$$\boxed{\text{Tier 2: } H_{AE}^{(2)}, H_{EO}^{(2)}}, \quad (152)$$

tested by

$$\mathfrak{P}_O[J_A K_E^\dagger + J_E K_A^\dagger + K_A J_E^\dagger + K_E J_A^\dagger],$$

$$\mathfrak{P}_O[J_E K_O^\dagger + J_O K_E^\dagger + K_E J_O^\dagger + K_O J_E^\dagger].$$

If those also vanish, the next tier is

$$\boxed{\text{Tier 3: } H_{AO}^{(2)}}, \quad (153)$$

and only after that

$$H_{AA}^{(2)}, \quad H_{OO}^{(2)}.$$

This ordering is not heuristic; it is forced by the exact corrected-seed weights.

CXXXII. CONCLUSION

The corrected odd transverse frontier is now compressed to a single leading slot:

$$\boxed{H_{EE}^{(2)} \text{ is the first nonzero completion candidate.}}$$

Under the minimal non-rigid completion hypothesis $K_E \neq 0$ first, the leading odd corrected coefficient is

$$\beta_{\text{first}}^{\text{corr}} \approx 2.820820771257055 \times 10^{-4} \theta_0 \kappa A_0 \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger].$$

So the next microscopic task is no longer structural. It is the direct finite evaluation of the one overlap test

$$\mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger].$$

CXXXIII. FINAL SUMMARY

At the frozen corrected seed

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096),$$

the corrected odd transverse coefficient is

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3),$$

with exact weight hierarchy

$$|EE| : |AE| : |EO| : |AO| : |AA| : |OO| = 9 : 6 : 6 : 2 : 1 : 1.$$

Under the fast working approximation,

$$H_{EE}^{(2)} \approx -\frac{2\alpha_X\beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger].$$

Therefore the first nonzero completion slot is

$$H_{EE}^{(2)}.$$

Under the minimal non-rigid completion hypothesis $K_E \neq 0$ first, one gets

$$\beta_{\text{first}}^{\text{corr}} \approx 2.820820771257055 \times 10^{-4} \theta_0 \kappa A_0 \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger].$$

So the odd-channel problem has now been reduced to one finite matrix-element test:

$$\mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger] \neq 0 ?$$

If that test vanishes, the exact fallback order is

$$(H_{AE}^{(2)}, H_{EO}^{(2)}) \rightarrow H_{AO}^{(2)} \rightarrow (H_{AA}^{(2)}, H_{OO}^{(2)}).$$

CXXXIV. FIRST EXPLICIT FINITE OVERLAP TEST FOR THE LEADING ODD E -SECTOR SLOT

CXXXV. SHORT KOREAN EXPLANATION

The first nonzero odd completion candidate has already been reduced to

$$H_{EE}^{(2)}.$$

The remaining task is to open this coefficient into concrete branch fields.

This is now possible because the notes already provide the explicit frozen-branch orbit-current pair:

$$J_{E,i}^a = \partial_i M_E^a, \quad K_{E,i}^a = \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_E.$$

Therefore the leading odd source splits exactly into two parts:

$$H_{EE}^{(2),\text{fast}} = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)},$$

namely projector/internal texture and background charge-gradient contributions.

This reduces the odd frontier to a literal two-term finite overlap test.

CXXXVI. 1. STARTING POINT

The first nonzero odd completion slot is the EE -row:

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X\beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger]. \quad (154)$$

The corrected odd coefficient then starts as

$$\beta_{\text{first}}^{\text{corr}} \approx \frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 (4.1472 \times 10^{-4}) \frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger]. \quad (155)$$

Numerically,

$$\beta_{\text{first}}^{\text{corr}} \approx 2.820820771257055 \times 10^{-4} \theta_0 \kappa A_0 \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger]. \quad (156)$$

So the whole odd-channel question is reduced to the single projected Hermitian overlap

$$\mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger].$$

CXXXVII. 2. EXPLICIT FROZEN-BRANCH E -SECTOR CURRENTS

The notes already provide the exact frozen-branch orbit-current pair:

$$J_{E,i}^a = \partial_i M_E^a, \quad (157)$$

$$K_{E,i}^a = \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_E. \quad (158)$$

Here:

- M_E^a is the E -sector orbit-current amplitude,
- N_E is the associated scalar density fluctuation in the E -sector,
- $\bar{q}^a$ is the frozen background internal charge profile,
- $\bar{\rho}$ is the frozen background density,
- $\bar{\chi}$ is the frozen background defect/tangent profile,
- χ_E is the E -sector tangent fluctuation.

Thus the first odd slot is already computable once these background fields are known.

CXXXVIII. 3. EXACT SPLIT INTO PROJECTOR/INTERNAL AND CHARGE-GRADIENT PIECES

Substitute (158) into (154). Then

$$J_E K_E^\dagger + K_E J_E^\dagger$$

splits exactly into two Hermitian pieces:

$$\begin{aligned} J_E K_E^\dagger + K_E J_E^\dagger &= \left[(\partial_i M_E^a) \left(\partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} \right)^\dagger + \left(\partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} \right) (\partial_i M_E^a)^\dagger \right] \\ &\quad + \left[(\partial_i M_E^a) ((\partial_i \bar{q}^a) \chi_E)^\dagger + (\partial_i \bar{q}^a) \chi_E (\partial_i M_E^a)^\dagger \right]. \end{aligned} \quad (159)$$

Therefore define

$$\mathcal{O}_{E,\text{proj}} := (\partial_i M_E^a) \left(\partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} \right)^\dagger + \left(\partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} \right) (\partial_i M_E^a)^\dagger, \quad (160)$$

$$\mathcal{O}_{E,\text{qgrad}} := (\partial_i M_E^a) ((\partial_i \bar{q}^a) \chi_E)^\dagger + (\partial_i \bar{q}^a) \chi_E (\partial_i M_E^a)^\dagger. \quad (161)$$

Then the first odd slot splits exactly as

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \left[\mathfrak{P}_O(\mathcal{O}_{E,\text{proj}}) + \mathfrak{P}_O(\mathcal{O}_{E,\text{qgrad}}) \right]. \quad (162)$$

So the two literal finite tests are:

$$\mathfrak{P}_O(\mathcal{O}_{E,\text{proj}}) \neq 0 \text{ ?}, \quad \mathfrak{P}_O(\mathcal{O}_{E,\text{qgrad}}) \neq 0 \text{ ?}. \quad (163)$$

CXXXIX. 4. CORRESPONDING SPLIT OF THE CORRECTED ODD COEFFICIENT

Substituting (162) into (155) gives

$$\beta_{\text{first}}^{\text{corr}} = \beta_{\text{proj}}^{\text{corr}} + \beta_{\text{qgrad}}^{\text{corr}}, \quad (164)$$

with

$$\beta_{\text{proj}}^{\text{corr}} = 2.820820771257055 \times 10^{-4} \theta_0 \kappa A_0 \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O(\mathcal{O}_{E,\text{proj}}), \quad (165)$$

$$\beta_{\text{qgrad}}^{\text{corr}} = 2.820820771257055 \times 10^{-4} \theta_0 \kappa A_0 \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O(\mathcal{O}_{E,\text{qgrad}}). \quad (166)$$

Hence the first odd activation is exactly the sum of a projector/internal texture part and a background charge-gradient part.

CXL. 5. MINIMAL RIGID TRUNCATION AND WHY IT KILLS THE ODD SOURCE

The notes explicitly state that in the minimal rigid-internal first-shell truncation,

$$F_E^a = 0, \quad \partial_i F_E^a = 0, \quad \partial_i \bar{q}^a = 0.$$

Translated into the present notation, this means there is no sign-sensitive internal E -sector projector texture and no background charge gradient. Hence

$$\mathcal{O}_{E,\text{proj}} = 0, \quad \mathcal{O}_{E,\text{qgrad}} = 0,$$

and therefore

$$H_{EE}^{(2),\text{fast}} = 0, \quad Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0 \quad \text{in the minimal rigid first-shell truncation.} \quad (167)$$

So orbit-amplitude quartic corrections alone cannot open the odd transverse channel.

CXLI. 6. FIRST NONZERO COMPLETION CRITERIA

Because (167) holds in the rigid truncation, the first actual odd activation must come from one of the following two completion mechanisms:

(A) projector/internal-texture completion

$$\mathfrak{P}_O(\mathcal{O}_{E,\text{proj}}) \neq 0 \implies \beta_{\text{proj}}^{\text{corr}} \neq 0. \quad (168)$$

This means that the E -sector internal projector texture becomes sign-sensitive after lifting the rigid truncation.

(B) background charge-gradient completion

$$\boxed{\mathfrak{P}_O(\mathcal{O}_{E,\text{qgrad}}) \neq 0 \implies \beta_{\text{qgrad}}^{\text{corr}} \neq 0.} \quad (169)$$

This means that the frozen branch acquires a nontrivial internal charge-gradient profile

$$\partial_i \bar{q}^a \neq 0$$

which couples to the E -sector fluctuation.

Therefore the exact first nonzero completion criterion is simply

$$\boxed{\mathfrak{P}_O(\mathcal{O}_{E,\text{proj}}) + \mathfrak{P}_O(\mathcal{O}_{E,\text{qgrad}}) \neq 0.} \quad (170)$$

CXLII. 7. SHARP STATUS OF THE ODD-CHANNEL FRONTIER

The odd-channel frontier is now no longer structural.

It has been reduced to the explicit two-term finite overlap audit:

$$\boxed{\mathcal{O}_{E,\text{proj}}, \quad \mathcal{O}_{E,\text{qgrad}}.} \quad (171)$$

So the next exact microscopic task is:

1. compute the frozen-branch E -sector current profile M_E^a ;
2. compute the frozen-branch scalar density profile N_E ;
3. compute the background charge profile $\bar{q}^a$ and its gradient $\partial_i \bar{q}^a$;
4. insert them into (160) and (161);
5. evaluate the odd projector traces.

Once that is done,

$$\beta^{\text{corr}} \neq 0$$

is decided without further structural ambiguity.

CXLIII. CONCLUSION

The first odd completion slot has now been opened explicitly all the way down to branch fields.

The leading odd coefficient is

$$\beta_{\text{first}}^{\text{corr}} = 2.820820771257055 \times 10^{-4} \theta_0 \kappa A_0 \frac{\alpha_X \beta_X}{M_X^2} \left[\mathfrak{P}_O(\mathcal{O}_{E,\text{proj}}) + \mathfrak{P}_O(\mathcal{O}_{E,\text{qgrad}}) \right].$$

Therefore the first actual odd source is nonzero if and only if the frozen corrected branch supports either:

1. a sign-sensitive E -sector projector/internal texture, or
2. a nontrivial background charge-gradient coupling.

This is the sharpest currently available explicit closure of the odd transverse frontier.

CXLIV. FINAL SUMMARY

The first nonzero odd completion candidate is the EE -slot:

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X\beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger].$$

Using the explicit frozen-branch currents

$$J_{E,i}^a = \partial_i M_E^a, \quad K_{E,i}^a = \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_E,$$

this splits exactly into

$$H_{EE}^{(2),\text{fast}} = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)},$$

where

$$\mathcal{O}_{E,\text{proj}} = (\partial_i M_E^a) \left(\partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} \right)^\dagger + \text{h.c.},$$

$$\mathcal{O}_{E,\text{qgrad}} = (\partial_i M_E^a) ((\partial_i \bar{q}^a) \chi_E)^\dagger + \text{h.c.}$$

Thus

$$\beta_{\text{first}}^{\text{corr}} = 2.820820771257055 \times 10^{-4} \theta_0 \kappa A_0 \frac{\alpha_X \beta_X}{M_X^2} \left[\mathfrak{P}_O(\mathcal{O}_{E,\text{proj}}) + \mathfrak{P}_O(\mathcal{O}_{E,\text{qgrad}}) \right].$$

In the minimal rigid first-shell truncation,

$$\mathcal{O}_{E,\text{proj}} = 0, \quad \mathcal{O}_{E,\text{qgrad}} = 0,$$

so

$$\beta^{\text{corr}} = 0.$$

Therefore the first actual odd transverse activation requires one of two explicit completion mechanisms:

sign-sensitive E -sector projector/internal texture

or

nontrivial background charge-gradient coupling.

So the odd frontier is now reduced to a literal two-term finite overlap computation.

CXLV. HONEST CLOSURE OF THE JK -ROUTE AND THE FORCED MOVE TO THE EXPLICIT ODD EXTRA-CARRIER ROUTE

CXLVI. SHORT KOREAN EXPLANATION

At the present stage, the JK -route has already been reduced as far as possible with the currently instantiated data. The leading odd corrected coefficient is controlled first by the EE -slot:

$$H_{EE}^{(2),\text{fast}} = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)}.$$

Among these two pieces, the notes identify the q -gradient source as the first generic candidate beyond the rigid truncation:

$$H_{EE,\text{qgrad}}^{(2)} \propto \mathfrak{P}_O \left[(\partial_\xi M_E^a) (\theta'(\xi) \partial_\theta m_a(\theta) \chi_E)^\dagger + \text{h.c.} \right].$$

Equivalently, the first decisive microscopic sign test is

$$\boxed{\mathfrak{P}_O\left[(\partial_\xi M_E^a)\chi_E^\dagger + \chi_E(\partial_\xi M_E^a)^\dagger\right] \stackrel{?}{\neq} 0.}$$

However, the same notes also state that the literal JK -route sign test still requires the actual frozen harmonics

$$\bar{A}_G, \quad B_G^{(E)},$$

which are not instantiated in the present conversation context.

Therefore the JK -route is not structurally open anymore, but it is still data-blocked.

The honest forward move is then the explicit odd extra-carrier route, which already gives a concrete odd source:

$$q_O(\theta) = \cos \theta,$$

and therefore

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right].$$

So the odd transverse channel is already explicitly realizable on the complete lowest-order carrier route.

CXLVII. 1. THE JK -ROUTE HAS ALREADY BEEN REDUCED TO A SINGLE LITERAL OVERLAP TEST

The current leading odd slot is

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger].$$

The exact frozen-branch orbit-current pair is

$$J_{E,i}^a = \partial_i M_E^a, \quad K_{E,i}^a = \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_E.$$

Hence the EE -slot splits exactly into

$$H_{EE}^{(2),\text{fast}} = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)}.$$

The projector/internal-texture piece is

$$H_{EE,\text{proj}}^{(2)} \propto \mathfrak{P}_O\left[(\partial_i M_E^a)((\partial_i Q_E^a)\bar{\chi})^\dagger + \text{h.c.}\right],$$

while the q -gradient piece is

$$H_{EE,\text{qgrad}}^{(2)} \propto \mathfrak{P}_O\left[(\partial_\xi M_E^a)(\theta'(\xi)\partial_\theta m_a(\theta)\chi_E)^\dagger + \text{h.c.}\right].$$

The notes then identify the q -gradient source as the first generic candidate beyond the rigid truncation. Therefore the first decisive microscopic test is exactly

$$\boxed{\mathfrak{P}_O\left[(\partial_\xi M_E^a)\chi_E^\dagger + \chi_E(\partial_\xi M_E^a)^\dagger\right] \stackrel{?}{\neq} 0.} \tag{172}$$

If (172) is nonzero and no sign cancellation occurs in the prefactor, then the odd channel opens already at the EE -slot.

CXLVIII. 2. WHY THE JK -ROUTE STILL STALLS WITH THE PRESENT DATA

Although (172) is now the exact first sign bottleneck, the notes also state that the literal JK -route overlap evaluation still requires the actual frozen harmonics

$$\bar{A}_G, \quad B_G^{(E)}.$$

Equivalently, the current q-gradient audit has been reduced to the single shell-overlap test

$$\Im\left(\widehat{M}_E^a(G_*)\overline{\widehat{\chi}_E(G_*)}\right) \stackrel{?}{\neq} 0,$$

but that test cannot be numerically decided unless those actual frozen harmonics are instantiated.

Therefore the honest status is:

the JK -route is no longer structurally open, but it is still blocked by unavailable frozen-harmonic data.

 (173)

This is a data-availability obstruction, not a theorem obstruction.

CXLIX. 3. FORCED FORWARD MOVE: THE EXPLICIT ODD EXTRA-CARRIER ROUTE

Because the JK -route sign test is blocked by unavailable frozen harmonics, the correct next move is the explicit odd extra-carrier route.

The notes state that this route already produces a direct odd transverse carrier without relying on the unavailable JK -harmonic overlap. Its exact odd carrier is

$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi.$

The induced transport block is

$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\widetilde{\Lambda}_E E^\mu{}_a \Sigma_a + \widetilde{\Lambda}_O J^\mu{}_a \Sigma_a \right].$

 (174)

Thus the odd transverse density is explicitly nonzero whenever

$$\widetilde{\Lambda}_O \neq 0, \quad \theta'(\xi) \neq 0, \quad \cos \theta(\xi) \neq 0.$$

So the honest forward-computable odd channel on the present data is not the frozen-harmonic JK -route, but the explicit extra-carrier route.

In reduced coefficient language this means

$$\beta \sim I_\theta \widetilde{\Lambda}_O,$$

with local density proportional to

$$\theta'(\xi) \cos \theta(\xi).$$

CL. 4. EXACT PROJECT STATUS AFTER THIS STEP

The status is now sharp:

1. The JK -route odd source has been reduced to a single decisive overlap test:

$$\mathfrak{P}_O\left[(\partial_\xi M_E^a)\chi_E^\dagger + \chi_E(\partial_\xi M_E^a)^\dagger\right] \stackrel{?}{\neq} 0.$$

2. That test cannot yet be evaluated numerically in the present conversation, because the actual frozen harmonics

$$\bar{A}_G, \quad B_G^{(E)}$$

are not instantiated here.

3. Therefore, to continue now rather than stall, the correct route is the explicit odd extra-carrier branch:

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right].$$

4. This route already yields an explicit nonzero odd channel whenever

$$\tilde{\Lambda}_O \neq 0, \quad \theta'(\xi) \neq 0, \quad \cos \theta(\xi) \neq 0.$$

Therefore the honest next-stage statement is:

stop stalling on the unavailable frozen-harmonic JK -overlap and move to the explicit odd extra-carrier route.
--

(175)

CLI. FINAL SUMMARY

The JK -route has now been reduced to its sharpest literal test:

$$\mathfrak{P}_O \left[(\partial_\xi M_E^a) \chi_E^\dagger + \chi_E (\partial_\xi M_E^a)^\dagger \right] \stackrel{?}{\neq} 0.$$

This is the first decisive microscopic sign test for the E -sector q -gradient odd source. However, its explicit shell-overlap evaluation still requires the actual frozen harmonics

$$\bar{A}_G, \quad B_G^{(E)},$$

which are not available in the present conversation context.

Therefore the JK -route is structurally reduced but data-blocked.

The correct forward-computable route is instead the explicit odd extra-carrier one, whose transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right].$$

Hence the odd transverse channel is already explicitly realizable whenever

$$\tilde{\Lambda}_O \neq 0, \quad \theta'(\xi) \neq 0, \quad \cos \theta(\xi) \neq 0.$$

So the honest next move is no longer to keep polishing the blocked JK -route, but to continue on the explicit odd extra-carrier branch.

CLII. MINIMAL EXPLICIT MICROSCOPIC COUPLING THAT OPENS THE ODD EXTRA-CARRIER ROUTE

CLIII. SHORT KOREAN EXPLANATION

The JK -route has been reduced to a single frozen-harmonic overlap test, but that test is still data-blocked in the present conversation. Therefore the honest forward-computable route is the explicit odd extra-carrier branch.

The minimal microscopic completion that opens this route is the addition of the odd Class-II carrier slot

$$\gamma_O X_i^a \mathcal{O}_i^a[P, \psi].$$

Once this slot is present, the odd reduced coefficient is no longer schematic:

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1,$$

and the branch-level odd coefficient is

$$\beta = \tilde{\Lambda}_O I_c.$$

So the first explicit odd-channel opening mechanism is already fixed at operator level.

CLIV. 1. MINIMAL ONE-MEDIATOR MICROSCOPIC COMPLETION

The minimal one-mediator working truncation that is capable of carrying both even and odd transverse sectors is

$$\mathcal{L}_{\text{trunc}} = \mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} + \frac{1}{2} M_X^2 X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \psi] + \gamma_E X_i^a \mathcal{E}_i^a[P, \psi] + \gamma_O X_i^a \mathcal{O}_i^a[P, \psi]. \quad (176)$$

The new ingredient relative to the previously stalled routes is exactly

$$\gamma_O X_i^a \mathcal{O}_i^a[P, \psi]. \quad (177)$$

This is the minimal explicit microscopic odd extra-carrier coupling.

CLV. 2. EXPLICIT ODD CARRIER GENERATED BY THE EXTRA SLOT

The explicit odd extra-carrier route gives

$$q_O(\theta) = \text{Tr}(S_O M_O) = \cos \theta. \quad (178)$$

Therefore the odd explicit carrier is

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi. \quad (179)$$

Its induced tangent transport block is

$$V_{EO}^{\mu} = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^{\mu}_a \Sigma_a + \tilde{\Lambda}_O J^{\mu}_a \Sigma_a \right]. \quad (180)$$

Hence the local odd transverse density is

$$b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi). \quad (181)$$

So the odd channel is locally present whenever

$$\tilde{\Lambda}_O \neq 0, \quad \theta'(\xi) \neq 0, \quad \cos \theta(\xi) \neq 0.$$

CLVI. 3. MICROSCOPIC COEFFICIENT READING

At operator level, the explicit odd route gives

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1. \quad (182)$$

If the shell basis is canonically normalized,

$$C_{\parallel} = C_E = C_O = 1,$$

then

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} J_1. \quad (183)$$

In the canonical transport-side explicit branch notation, the normalization layer is absorbed into the coefficient point, so one writes simply

$$\tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}. \quad (184)$$

Thus the minimal odd microscopic coupling produces a nonzero odd coefficient already at one-mediator level whenever the mediator coupling pair (β_X, γ_O) is nonzero and the overlap factor does not vanish.

CLVII. 4. REDUCED BRANCH-LEVEL ODD COEFFICIENT

The explicit extra odd carrier has already been fully inserted into the reduced coefficient sheet:

$$\boxed{\beta = \tilde{\Lambda}_O I_c, \quad I_c = \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi).} \quad (185)$$

Substituting (182) gives

$$\boxed{\beta = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_c.} \quad (186)$$

Under canonical shell normalization,

$$\boxed{\beta = -\frac{\beta_X \gamma_O}{M_X^2} J_1 I_c.} \quad (187)$$

So the odd coefficient is now reduced to a direct product of:

1. the mediator-sector couplings β_X, γ_O ,
2. the heavy scale M_X^2 ,
3. the microscopic odd overlap J_1 ,
4. the geometric defect overlap I_c .

CLVIII. 5. EXACT NONZERO CONDITION

From (186), the odd transverse channel is nonzero if and only if

$$\boxed{\beta_X \neq 0, \quad \gamma_O \neq 0, \quad J_1 \neq 0, \quad I_c \neq 0.} \quad (188)$$

Equivalently, in reduced reading form,

$$\boxed{\tilde{\Lambda}_O \neq 0 \quad \text{and} \quad I_c \neq 0.} \quad (189)$$

This is the first fully explicit branch-level odd-channel criterion available on the present data.

CLIX. 6. WHY THIS IS THE CORRECT FORWARD ROUTE

The JK -route still requires unavailable frozen harmonics

$$\bar{A}_G, \quad B_G^{(E)}$$

to evaluate its final sign test explicitly. By contrast, the explicit odd extra-carrier route already gives the coefficient reading

$$\beta = \tilde{\Lambda}_O I_c$$

without needing those unavailable harmonics.

Therefore the microscopic odd frontier is no longer “whether an odd route exists.” It is only the actual value/sign extraction of

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.$$

CLX. CONCLUSION

The minimal explicit microscopic coupling that opens the odd channel is the odd extra-carrier slot

$$\boxed{\gamma_O X_i^a \mathcal{O}_i^a[P, \psi].}$$

Once this is added to the one-mediator branch, one gets

$$\boxed{\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1, \quad \beta = \tilde{\Lambda}_O I_c.}$$

So the odd transverse channel is explicitly realizable at lowest order on the complete carrier route, and the remaining microscopic problem is only the actual sign/value of J_1 (equivalently of $\tilde{\Lambda}_O$).

CLXI. FINAL SUMMARY

The minimal explicit microscopic coupling that opens the odd extra-carrier route is

$$\boxed{\gamma_O X_i^a \mathcal{O}_i^a[P, \psi].}$$

With this term included, the explicit odd carrier is

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi,$$

and the induced tangent transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right].$$

The microscopic odd coefficient is read off as

$$\boxed{\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.}$$

Therefore the reduced odd transverse coefficient is

$$\boxed{\beta = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_c, \quad I_c = \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi).}$$

In canonical normalization this becomes

$$\boxed{\beta = -\frac{\beta_X \gamma_O}{M_X^2} J_1 I_c.}$$

Hence the exact minimal nonzero condition is

$$\boxed{\beta_X \neq 0, \quad \gamma_O \neq 0, \quad J_1 \neq 0, \quad I_c \neq 0.}$$

So the odd route is no longer conceptually open. It is explicitly available at one-mediator level, and the remaining microscopic frontier is only the actual value/sign of J_1 (equivalently of $\tilde{\Lambda}_O$).

CLXII. EXACT SIGN CONVENTION FOR THE EXPLICIT ODD COEFFICIENT

CLXIII. SHORT KOREAN EXPLANATION

The odd extra-carrier route is already fully inserted into the reduced coefficient sheet. The relevant formulas are:

$$\beta = \tilde{\Lambda}_O I_{\theta, \cos},$$

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1,$$

and in canonical explicit normalization,

$$C_{\parallel} = C_E = C_O = 1, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}.$$

Therefore the sign of β is completely fixed by the sign of the microscopic coupling product and the orientation/overlap factor. There is no remaining structural ambiguity.

CLXIV. 1. EXACT REDUCED ODD COEFFICIENT

The explicit extra odd carrier route gives the exact reduced dictionary

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta, \cos} \tilde{\Lambda}_E, \quad \beta = I_{\theta, \cos} \tilde{\Lambda}_O. \quad (190)$$

Thus the odd coefficient is

$$\beta = \tilde{\Lambda}_O I_{\theta, \cos}, \quad I_{\theta, \cos} := \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \cos \theta(\xi). \quad (191)$$

At operator level, the one-mediator microscopic coefficient point is

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1. \quad (192)$$

Substituting (192) into (191) yields the exact unabsorbed sign formula

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_{\theta, \cos}. \quad (193)$$

Since $M_X^2 > 0$, its sign is

$$\text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O C_O J_1 I_{\theta, \cos}). \quad (194)$$

This is the exact sign rule in the unabsorbed normalization.

CLXV. 2. CANONICAL EXPLICIT NORMALIZATION

In the canonical explicit normalization one chooses

$$C_{\parallel} = C_E = C_O = 1.$$

Then the operator coefficient reduces to

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} J_1. \quad (195)$$

If one keeps J_1 explicit, then

$$\boxed{\beta = -\frac{\beta_X \gamma_O}{M_X^2} J_1 I_{\theta, \cos}}, \quad (196)$$

so

$$\boxed{\text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O J_1 I_{\theta, \cos})}. \quad (197)$$

If one instead absorbs J_1 into the transport-side normalization, the notes give

$$\boxed{\beta = -\frac{\beta_X \gamma_O}{M_X^2} I_{\theta, \cos}}. \quad (198)$$

Hence the fully canonical sign rule is

$$\boxed{\text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O I_{\theta, \cos})}. \quad (199)$$

So the only difference between the two conventions is whether J_1 is regarded as part of the coefficient or as part of the normalization bookkeeping. The physical sign rule is unchanged.

CLXVI. 3. BRANCH DECISION AND PURE-EVEN BRANCH EXCLUSION

The notes also state that the current explicit witness branch is the pure-even one:

$$\boxed{T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0}. \quad (200)$$

Therefore

$$\boxed{\tilde{\Lambda}_O = 0, \quad \beta = 0}. \quad (201)$$

So the pure-even explicit witness branch can never be used to read a nonzero odd coefficient.

To activate the odd channel one must move to a mixed transverse branch with

$$\boxed{\gamma_O \neq 0}. \quad (202)$$

Once $\gamma_O \neq 0$, the odd coefficient is no longer structurally obstructed. It is decided only by the sign of

$$\beta_X \gamma_O C_O J_1 I_{\theta, \cos}$$

(or $\beta_X \gamma_O I_{\theta, \cos}$ in the fully absorbed canonical convention).

CLXVII. 4. ORIENTATION CONVENTION FOR THE DEFECT OVERLAP

The overlap factor is

$$I_{\theta, \cos} = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \cos \theta(\xi).$$

Its sign is not an additional dynamical mystery. It depends on the chosen orientation of the defect coordinate ξ and on the orientation convention for the profile $\theta(\xi)$.

Therefore one may adopt the *canonical overlap orientation convention*

$$\boxed{I_{\theta, \cos} > 0}. \quad (203)$$

Under this convention, the sign rules simplify to

$$\boxed{\text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O C_O J_1)} \quad (\text{unabsorbed normalization}), \quad (204)$$

and

$$\boxed{\text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O J_1)} \quad (\text{canonical with explicit } J_1), \quad (205)$$

or

$$\boxed{\text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O)} \quad (\text{fully absorbed canonical transport notation}). \quad (206)$$

This is a convention choice, not a theorem. It is exactly the correct way to separate physical coupling signs from coordinate-orientation bookkeeping.

CLXVIII. 5. MINIMAL NONZERO AND SIGN-STABLE CRITERION

The notes already give the minimal nonzero criterion:

$$\boxed{\beta_X \neq 0, \quad \gamma_O \neq 0, \quad I_{\theta, \cos} \neq 0}, \quad (207)$$

and, in the unabsorbed operator normalization,

$$\boxed{C_O J_1 \neq 0}. \quad (208)$$

So the sign-stable odd branch is obtained once these nonvanishing conditions hold and the overlap orientation is fixed.

In particular, after choosing the canonical overlap orientation $I_{\theta, \cos} > 0$, the odd coefficient has a fixed sign as soon as

$$\beta_X, \quad \gamma_O, \quad J_1$$

have fixed signs.

Thus the sign problem is no longer structural. It is now only a bookkeeping choice plus the microscopic signs of the couplings.

CLXIX. CONCLUSION

The exact odd-coefficient sign formula is

$$\boxed{\beta = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_{\theta, \cos}}.$$

Hence

$$\boxed{\text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O C_O J_1 I_{\theta, \cos})}.$$

In canonical explicit normalization:

$$\boxed{\beta = -\frac{\beta_X \gamma_O}{M_X^2} J_1 I_{\theta, \cos}, \quad \text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O J_1 I_{\theta, \cos})}.$$

If J_1 is also absorbed into the transport-side normalization, then

$$\boxed{\beta = -\frac{\beta_X \gamma_O}{M_X^2} I_{\theta, \cos}, \quad \text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O I_{\theta, \cos})}.$$

After fixing the canonical defect-orientation convention $I_{\theta,\cos} > 0$, these reduce to the simplest branch sign rules

$$\boxed{\text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O C_O J_1),}$$

or

$$\boxed{\text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O)}$$

in the fully absorbed canonical transport notation.

Therefore the odd-channel route is now completely sign-closed. The current pure-even branch has $\gamma_O = 0$ and hence $\beta = 0$. The next actual branch must satisfy $\gamma_O \neq 0$, after which the sign of β is fixed entirely by the microscopic coupling signs and the chosen overlap orientation convention.

CLXX. FINAL SUMMARY

The explicit odd extra-carrier route gives the exact odd coefficient

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_{\theta,\cos}, \quad I_{\theta,\cos} = \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi).$$

Therefore the exact sign rule is

$$\text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O C_O J_1 I_{\theta,\cos}).$$

In canonical explicit normalization,

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} J_1 I_{\theta,\cos},$$

and if J_1 is absorbed into the transport normalization,

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} I_{\theta,\cos}.$$

Thus the remaining sign choice is only the overlap-orientation convention for $I_{\theta,\cos}$. After fixing $I_{\theta,\cos} > 0$, one gets

$$\text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O C_O J_1),$$

or, in the fully absorbed canonical notation,

$$\text{sgn } \beta = -\text{sgn}(\beta_X \gamma_O).$$

The current pure-even explicit witness branch has

$$\gamma_O = 0,$$

so it necessarily gives

$$\beta = 0.$$

Therefore the next actual branch must be a mixed transverse branch with $\gamma_O \neq 0$, after which the sign of β is completely fixed by microscopic coupling signs and the overlap-orientation convention.

CLXXI. MIXED TRANSVERSE BRANCH UPGRADE: FIRST EXPLICIT BRANCH WITH $\beta \neq 0$

CLXXII. SHORT KOREAN EXPLANATION

The pure-even explicit witness branch

$$\gamma_E = 1, \quad \gamma_O = 0$$

necessarily gives

$$\beta = 0,$$

so it must be abandoned if the goal is to read the odd coefficient.

The correct next branch is the mixed transverse branch

$$T_O \neq 0 \quad \text{equivalently} \quad \gamma_O \neq 0.$$

The simplest canonical mixed sample is

$$T_E = T_O = 1.$$

On this branch, the reduced symbol already becomes an explicit anisotropic Dirac symbol with both even and odd transverse components turned on.

CLXXIII. 1. EXACT MIXED-BRANCH MASTER SYMBOL

The exact mixed-branch reduced symbol is

$$A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta'^3} p_{\parallel} \sigma_3 + \beta_X I_{\theta} T_E (p_1 \sigma_1 + p_2 \sigma_2) + \beta_X I_{\theta} T_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (209)$$

Therefore the reduced coefficient point is

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X L_* I_{\theta'^3}, \quad \alpha = \beta_X I_{\theta} T_E, \quad \beta = \beta_X I_{\theta} T_O. \quad (210)$$

Using the explicit overlap values

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (211)$$

one gets

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \beta_X T_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X T_O \theta_0 \kappa. \quad (212)$$

Thus the odd channel is turned on exactly when

$$T_O \neq 0 \quad \Longleftrightarrow \quad \gamma_O \neq 0.$$

CLXXIV. 2. CANONICAL MINIMAL MIXED SAMPLE

Now choose the simplest canonical mixed branch

$$T_E = T_O = 1. \quad (213)$$

Then (212) becomes

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (214)$$

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa. \quad (215)$$

The transverse resultant amplitude is

$$\boxed{R_{\perp} = \sqrt{T_E^2 + T_O^2} = \sqrt{2}.} \quad (216)$$

The corresponding branch angle is

$$T_E = R_{\perp} \cos \varphi, \quad T_O = R_{\perp} \sin \varphi,$$

so for $T_E = T_O = 1$,

$$\boxed{\varphi = \frac{\pi}{4}, \quad \frac{\beta}{\alpha} = \tan \varphi = 1.} \quad (217)$$

The document also gives the corresponding explicit overlap factor

$$\boxed{\Delta_{00} = \frac{1}{\sqrt{2}}.} \quad (218)$$

Therefore this branch is the first explicit mixed transverse sample on which the odd coefficient is genuinely nonzero.

CLXXV. 3. SIGN RULE ON THE CANONICAL MIXED BRANCH

From (215),

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.$$

Hence, on the canonical mixed sample $T_E = T_O = 1$,

$$\boxed{\text{sgn } \alpha = \text{sgn } \beta = -\text{sgn}(\beta_X \theta_0 \kappa).} \quad (219)$$

So once the defect orientation convention fixes the sign of $\theta_0 \kappa$, the sign of the odd coefficient is determined entirely by the sign of β_X .

In particular, if one adopts the standard positive-defect convention

$$\theta_0 \kappa > 0,$$

then

$$\boxed{\text{sgn } \beta = -\text{sgn}(\beta_X).} \quad (220)$$

CLXXVI. 4. HONEST STATUS AFTER THE MIXED-BRANCH UPGRADE

The mixed transverse branch upgrade does *not* mean that the full microscopic PDE has already been finally derived. What is closed is the reduced branch-level symbol and its coefficient reading.

The final unconditional microscopic PDE is still open, and the remaining frontier is still

$$L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

So the honest statement is:

$$\boxed{\text{the odd coefficient is now explicitly readable on the mixed transverse branch,}} \quad (221)$$

but

$$\boxed{\text{the full microscopic coefficient extraction from the complete TECT operator remains open.}} \quad (222)$$

CLXXVII. FINAL SUMMARY

The pure-even witness branch

$$\gamma_E = 1, \quad \gamma_O = 0$$

necessarily gives

$$\beta = 0.$$

Therefore the correct next branch is the mixed transverse branch

$$\boxed{T_O \neq 0 \quad \Longleftrightarrow \quad \gamma_O \neq 0.}$$

The exact mixed-branch coefficient sheet is

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \beta_X T_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X T_O \theta_0 \kappa.}$$

For the simplest canonical mixed sample

$$\boxed{T_E = T_O = 1,}$$

one gets

$$\boxed{\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad R_{\perp} = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}}, \quad \frac{\beta}{\alpha} = 1.}$$

Thus this is the first explicit branch on which the odd coefficient is genuinely turned on.

Its sign rule is

$$\boxed{\text{sgn } \beta = -\text{sgn}(\beta_X \theta_0 \kappa).}$$

Under the standard convention $\theta_0 \kappa > 0$,

$$\boxed{\text{sgn } \beta = -\text{sgn}(\beta_X).}$$

So the mixed transverse branch upgrade closes the branch-level odd coefficient reading, while the final microscopic PDE coefficient extraction still remains open.

CLXXVIII. MIXED ODD-ACTIVE BRANCH: EXACT MATCHING LAYER TO THE FINITE MICROSCOPIC EXTRACTION CHAIN

CLXXIX. SHORT EXPLANATION

The mixed transverse branch is no longer a structural guess. Its reduced symbol is already fixed exactly, and the next legitimate task is to connect it back to the finite microscopic extraction chain

$$L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta).$$

The cleanest first odd-active sample is the canonical mixed choice

$$T_E = T_O = 1.$$

The correct next layer is therefore:

1. write the exact target coefficient point for this branch,
2. translate it into target conditions on the projected Bloch derivative matrices,
3. attach the branch-consistent same-shell support inequality,
4. state honestly what remains finite insertion rather than open structure.

CLXXX. 1. EXACT MIXED-BRANCH TARGET COEFFICIENT SHEET

The exact mixed-branch reduced symbol is

$$A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta'^3} p_{\parallel} \sigma_3 + \beta_X I_{\theta} T_E (p_1 \sigma_1 + p_2 \sigma_2) + \beta_X I_{\theta} T_O (-p_2 \sigma_1 + p_1 \sigma_2). \quad (223)$$

Using the exact overlap factors

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad (224)$$

the reduced coefficient sheet is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (225)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X T_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X T_O \theta_0 \kappa. \quad (226)$$

The exact nondegeneracy condition is

$$L_* \neq 0, \quad R_{\perp} := \sqrt{T_E^2 + T_O^2} > 0. \quad (227)$$

CLXXXI. 2. CANONICAL MIXED SAMPLE

Choose the first odd-active canonical mixed sample

$$T_E = T_O = 1. \quad (228)$$

Then

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (229)$$

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa. \quad (230)$$

Moreover

$$R_{\perp} = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}}, \quad \frac{\beta}{\alpha} = 1. \quad (231)$$

So this is the first explicit odd-active reduced coefficient point.

CLXXXII. 3. EXACT MATCHING BACK TO THE PROJECTED BLOCH DERIVATIVE MATRICES

Let

$$K_i := \partial_{k_i} L_{\text{Bloch}}(k) \Big|_{k=k_*}, \quad M_i := P_* K_i P_*.$$

Then the microscopic frontier is to compute these M_i explicitly from the complete operator and verify that they match the mixed-branch target.

In the shell-adapted frame $(\hat{n}, e_1, e_2)$, define

$$M_{\parallel} := \hat{n}^i M_i, \quad M_1 := e_1^i M_i, \quad M_2 := e_2^i M_i.$$

The mixed-branch target matrices are

$$\boxed{M_{\parallel}^{\text{target}} = \lambda_{\parallel} \sigma_3,} \quad (232)$$

$$\boxed{M_1^{\text{target}} = \alpha \sigma_1 + \beta \sigma_2, \quad M_2^{\text{target}} = \alpha \sigma_2 - \beta \sigma_1.} \quad (233)$$

Therefore on the canonical mixed sample $T_E = T_O = 1$, one gets

$$\boxed{M_1^{\text{target}} = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa (\sigma_1 + \sigma_2),} \quad (234)$$

$$\boxed{M_2^{\text{target}} = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa (\sigma_2 - \sigma_1).} \quad (235)$$

So the finite microscopic extraction problem is now exactly the finite matrix-matching problem

$$\boxed{P_* K_{\parallel} P_* \stackrel{?}{=} -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3 \sigma_3,} \quad (236)$$

$$\boxed{P_* K_1 P_* \stackrel{?}{=} -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa (\sigma_1 + \sigma_2),} \quad (237)$$

$$\boxed{P_* K_2 P_* \stackrel{?}{=} -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa (\sigma_2 - \sigma_1).} \quad (238)$$

Equivalently, in trace-reading form,

$$\boxed{\frac{1}{2} \text{Tr}(\sigma_3 M_{\parallel}) = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3,} \quad (239)$$

$$\boxed{\frac{1}{4} \text{Tr}(\sigma_1 M_1 + \sigma_2 M_2) = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa,} \quad (240)$$

$$\boxed{\frac{1}{4} \text{Tr}(\sigma_2 M_1 - \sigma_1 M_2) = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.} \quad (241)$$

This is the exact mixed-branch microscopic target.

CLXXXIII. 4. BRANCH-CONSISTENT SAME-SHELL SUPPORT CRITERION

The pure-even same-shell benchmark is already known, and for the canonical mixed branch the branch-consistent lift is also already fixed.

For

$$T_E = T_O = 1,$$

the effective same-shell correction strength becomes

$$a_T = \frac{2\beta_X^2}{M_X^2}. \quad (242)$$

The corresponding overlap package is

$$\mathcal{S}_{ab}^{(\pi/4)} = \frac{1}{2}\mathcal{S}_{ab}^{EE} + \frac{1}{2}\mathcal{S}_{ab}^{OO} + \mathcal{S}_{ab}^{EO}. \quad (243)$$

Hence the correct branch-consistent sufficient same-shell support condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}. \quad (244)$$

So the support-side problem is no longer conceptual either. It is a finite insertion problem for the mixed same-shell constants

$$\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, \quad L_-^{TT}, \quad L_0^{TT}, \quad C_+^{TT}, \quad C_-^{TT}, \quad C_m^{TT}.$$

CLXXXIV. 5. HONEST CURRENT STATUS

The mixed-branch coefficient sheet is exact. The branch-consistent same-shell support inequality is also exact. The remaining frontier is still finite insertion, not conceptual structure.

What is *not* yet closed is the unconditional extraction from the final complete TECT operator. So the honest status remains:

$$\boxed{\text{reduced mixed-branch odd reading is closed,} \quad \text{full microscopic extraction remains open.}} \quad (245)$$

CLXXXV. CONCLUSION

The mixed odd-active branch has now been matched to the exact finite microscopic target.

The canonical mixed sample $T_E = T_O = 1$ fixes

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa,$$

and therefore fixes the target projected derivative matrices

$$M_{\parallel}^{\text{target}} = \lambda_{\parallel} \sigma_3, \quad M_1^{\text{target}} = \alpha(\sigma_1 + \sigma_2), \quad M_2^{\text{target}} = \alpha(\sigma_2 - \sigma_1).$$

So the full microscopic frontier is now a finite matrix-matching and same-shell insertion problem.

CLXXXVI. FINAL SUMMARY

The mixed transverse branch with

$$T_O \neq 0$$

is the correct odd-active upgrade of the pure-even witness branch. For the canonical mixed sample

$$T_E = T_O = 1,$$

the exact reduced coefficient point is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa,$$

with

$$R_{\perp} = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}}.$$

Therefore the exact microscopic extraction target becomes

$$P_* K_{\parallel} P_* \stackrel{?}{=} \lambda_{\parallel} \sigma_3, \quad P_* K_1 P_* \stackrel{?}{=} \alpha(\sigma_1 + \sigma_2), \quad P_* K_2 P_* \stackrel{?}{=} \alpha(\sigma_2 - \sigma_1).$$

The corresponding branch-consistent same-shell sufficient condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p, \rho}^{(1, \pi/4)}.$$

Thus the mixed odd-active branch is no longer structurally open. It is now reduced to:

$$\boxed{\text{finite projected Bloch-derivative matching}}$$

and

$$\boxed{\text{finite same-shell constant insertion.}}$$

What remains open is not the reduced branch logic, but the final unconditional microscopic extraction from the complete TECT operator.

CLXXXVII. EXACT SAME-SHELL SUPPORT SHEET FOR THE CANONICAL MIXED ODD-ACTIVE BRANCH

CLXXXVIII. GOAL

We now fix the first explicit odd-active mixed branch by

$$T_E = T_O = 1, \quad R_{\perp} = \sqrt{2}, \quad \varphi = \frac{\pi}{4}.$$

The reduced coefficient point is already exact:

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.$$

So the nontrivial remaining support-side problem is not structural. It is the same-shell non-enriched benchmark on this mixed branch.

CLXXXIX. 1. EXACT MIXED SAME-SHELL BLOCK

The pure-even same-shell block was

$$H_{\text{non-enr}}^{(1)}(p) = \begin{pmatrix} d_+(p) & m(p) \\ m(p) & d_-(p) \end{pmatrix},$$

with

$$d_{\pm}(p) = E_{z_0}^{\text{dir}}(\pm p) - a_K \mathcal{S}_{\pm\pm}^{EE}(p), \quad m(p) = B_{z_0}(p) - a_K \mathcal{S}_{+-}^{EE}(p).$$

The branch-consistent mixed uplift is therefore

$$\boxed{d_+^{(T)}(p) = E_{z_0}^{\text{dir}}(p) - a_T \mathcal{S}_{++}^{TT}(p)}, \quad (246)$$

$$\boxed{d_-^{(T)}(p) = E_{z_0}^{\text{dir}}(-p) - a_T \mathcal{S}_{--}^{TT}(p),} \quad (247)$$

$$\boxed{m^{(T)}(p) = B_{z_0}(p) - a_T \mathcal{S}_{+-}^{TT}(p),} \quad (248)$$

with

$$\boxed{a_T = \frac{2\beta_X^2}{M_X^2}.} \quad (249)$$

So the mixed same-shell benchmark remains a 2×2 Hermitian problem, but with the EE -family replaced by the mixed TT -family.

CXC. 2. CANONICAL $\pi/4$ OVERLAP PACKAGE

For the canonical mixed branch

$$T_E = T_O = 1, \quad \varphi = \frac{\pi}{4},$$

the overlap family is exactly

$$\boxed{\mathcal{S}_{ab}^{(\pi/4)} = \frac{1}{2} \mathcal{S}_{ab}^{EE} + \frac{1}{2} \mathcal{S}_{ab}^{OO} + \mathcal{S}_{ab}^{EO}.} \quad (250)$$

Hence

$$\boxed{\mathcal{S}_{\text{diag}}^{(\pi/4)} = \max\left\{|\mathcal{S}_{++}^{(\pi/4)}(0)|, |\mathcal{S}_{--}^{(\pi/4)}(0)|, |\mathcal{S}_{+-}^{(\pi/4)}(0)|\right\},} \quad (251)$$

or, if one adopts the symmetric locked-core reduction used in the notes, the same object may be written as the common diagonal benchmark extracted from the mixed family

$$\mathcal{S}_{ab}^{TT}(p; \varphi).$$

Thus for the canonical branch the coefficient entering the support inequality is not a new unknown function class; it is the explicit $\pi/4$ combination of the already known EE , OO , and EO overlap families.

CXCI. 3. EXACT SMALL- p DRIFT BUDGETS

Differentiate the mixed same-shell family at $p = 0$. The direct core drift is unchanged:

$$\boxed{a_{+,i}^{\text{dir}} = \delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right), \quad a_{-,i}^{\text{dir}} = -\delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right), \quad b_i^{\text{dir}} = 0.} \quad (252)$$

Now define the mixed derivative budgets

$$\boxed{\ell_{+,i}^{TT} := \partial_{p_i} \mathcal{S}_{++}^{TT}(p; \varphi)|_{p=0}, \quad \ell_{-,i}^{TT} := \partial_{p_i} \mathcal{S}_{--}^{TT}(p; \varphi)|_{p=0},} \quad (253)$$

$$\boxed{\ell_{0,i}^{TT} := \partial_{p_i} \mathcal{S}_{+-}^{TT}(p; \varphi)|_{p=0}.} \quad (254)$$

Then the full mixed drift coefficients are

$$\boxed{a_{+,i}^{(T)} = \delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right) - a_T \ell_{+,i}^{TT},} \quad (255)$$

$$a_{-,i}^{(T)} = -\delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right) - a_T \ell_{-,i}^{TT}, \quad (256)$$

$$b_i^{(T)} = -a_T \ell_{0,i}^{TT}. \quad (257)$$

Define the norm-level budgets

$$L_+^{TT} = \left(\sum_i |\ell_{+,i}^{TT}|^2 \right)^{1/2}, \quad L_-^{TT} = \left(\sum_i |\ell_{-,i}^{TT}|^2 \right)^{1/2}, \quad L_0^{TT} = \left(\sum_i |\ell_{0,i}^{TT}|^2 \right)^{1/2}. \quad (258)$$

Therefore

$$A_+^{(T)} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_+^{TT}, \quad (259)$$

$$A_-^{(T)} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_-^{TT}, \quad (260)$$

$$B^{(T)} \leq a_T L_0^{TT}. \quad (261)$$

CXCII. 4. EXACT MIXED-BRANCH SMALL- p ERROR BUDGET

Introduce the quadratic remainder constants

$$C_+^{TT}, \quad C_-^{TT}, \quad C_m^{TT}.$$

Then the mixed-branch small- p error satisfies

$$\varepsilon_{\text{small-}p,\rho}^{(1,T)} \leq \left[(\rho A_+^{(T)} + C_+^{TT} \rho^2)^2 + (\rho A_-^{(T)} + C_-^{TT} \rho^2)^2 + 2(\rho B^{(T)} + C_m^{TT} \rho^2)^2 \right]^{1/2}. \quad (262)$$

For the canonical $\pi/4$ branch, this is simply

$$\varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)} = \varepsilon_{\text{small-}p,\rho}^{(1,T)} \Big|_{\varphi=\pi/4}.$$

CXCIII. 5. EXACT BRANCH-CONSISTENT SUPPORT CRITERION

Exactly parallel to the pure-even case, the branch-consistent mixed same-shell sufficient condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2a_T \mathcal{S}_{\text{diag}}^{TT} + \varepsilon_{\text{small-}p,\rho}^{(1,T)}. \quad (263)$$

Substituting

$$a_T = \frac{2\beta_X^2}{M_X^2}$$

and specializing to $\varphi = \pi/4$, one gets

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}. \quad (264)$$

Equivalently, in the scalar-only locked-core form used in the notes,

$$\left| -\mu^2 + \frac{Z q_*^2}{4} - \frac{Y q_*^4}{16} + 12uA^2 \right| - 4uA^2 > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}. \quad (265)$$

This is the sharpest exact support-side inequality currently available for the first odd-active branch.

CXCIV. 6. WHAT IS STILL NOT INSERTED

The notes are explicit that the following quantities are still not inserted numerically:

$$\boxed{\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, \quad L_-^{TT}, \quad L_0^{TT}, \quad C_+^{TT}, \quad C_-^{TT}, \quad C_m^{TT}}. \quad (266)$$

Therefore the support problem is no longer conceptual. It is the finite insertion of these seven mixed same-shell constants into (264).

CXCV. 7. CONNECTION TO THE REMAINING MICROSCOPIC AUDIT

The same-shell support audit is only one of the two final microscopic tasks.

The other remaining task, already isolated in the full-Hessian audit, is the mixed transverse audit of

$$TD, \quad DT, \quad DD.$$

The notes summarize the current conservative status as

$$\ell_0^{(1)} \approx 0, \quad \ell_0^{(3)} \approx \ell_{0;OO}^{(3)}, \quad e_0^{(1)} \approx e_{0;K}^{(1)}, \quad o_0^{(1)} \approx o_{0;K}^{(1)}.$$

Hence, unless the mixed blocks TD, DT, DD cancel the explicit K -sector pair, the reduced mixed-branch coefficient point remains

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3\ell_{0;OO}^{(3)}, \quad \alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa e_{0;K}^{(1)}, \quad \beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa o_{0;K}^{(1)}.$$

So the true remaining microscopic tasks are exactly:

- (i) mixed same-shell constant insertion, (ii) mixed-block transverse audit for TD, DT, DD .

CXCVI. CONCLUSION

For the canonical mixed odd-active branch $T_E = T_O = 1$, the same-shell benchmark is fully reduced to the finite insertion problem

$$\mathcal{S}_{\text{diag}}^{(\pi/4)}, \quad L_+^{TT}, \quad L_-^{TT}, \quad L_0^{TT}, \quad C_+^{TT}, \quad C_-^{TT}, \quad C_m^{TT}.$$

The exact sufficient support criterion is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}.$$

Thus the mixed branch is no longer structurally open. What remains is only finite insertion and the last mixed-block audit.

CXCVII. FINAL SUMMARY

On the canonical mixed odd-active branch

$$T_E = T_O = 1, \quad a_T = \frac{2\beta_X^2}{M_X^2},$$

the mixed same-shell overlap family is

$$\mathcal{S}_{ab}^{(\pi/4)} = \frac{1}{2}\mathcal{S}_{ab}^{EE} + \frac{1}{2}\mathcal{S}_{ab}^{OO} + \mathcal{S}_{ab}^{EO}.$$

The mixed same-shell block is

$$d_+^{(T)}(p) = E_{z_0}^{\text{dir}}(p) - a_T \mathcal{S}_{++}^{TT}(p),$$

$$d_-^{(T)}(p) = E_{z_0}^{\text{dir}}(-p) - a_T \mathcal{S}_{--}^{TT}(p),$$

$$m^{(T)}(p) = B_{z_0}(p) - a_T \mathcal{S}_{+-}^{TT}(p).$$

Its exact small- p support inequality is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}.$$

What is still not inserted are the seven mixed same-shell constants

$$\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, \quad L_-^{TT}, \quad L_0^{TT}, \quad C_+^{TT}, \quad C_-^{TT}, \quad C_m^{TT}.$$

So the mixed same-shell side is reduced completely to finite insertion.
Together with the already isolated full-Hessian mixed-block audit of

$$TD, \quad DT, \quad DD,$$

these are the only genuinely microscopic tasks still left on the mixed odd-active branch.

CXCVIII. NEAR-FINAL EXACT MICROSCOPIC AUDIT SHEET

CXCIX. SHORT KOREAN EXPLANATION

At the present stage, the mixed odd-active branch is no longer conceptually open.
What remains is exactly split into two finite insertion problems:

1. the mixed same-shell support insertion;
2. the mixed-block transverse cancellation audit.

No new theorem structure is missing. Only these explicit finite inputs remain.

CC. 1. EXACT MIXED SAME-SHELL SUPPORT SHEET

For the canonical mixed branch

$$T_E = T_O = 1, \quad R_\perp = \sqrt{2}, \quad \varphi = \frac{\pi}{4},$$

the branch-consistent transverse overlap family is

$$\boxed{\mathcal{S}_{ab}^{(\pi/4)}(p) = \frac{1}{2} \mathcal{S}_{ab}^{EE}(p) + \frac{1}{2} \mathcal{S}_{ab}^{OO}(p) + \mathcal{S}_{ab}^{EO}(p).} \quad (267)$$

The same-shell block is

$$\boxed{d_+^{(T)}(p) = E_{z_0}^{\text{dir}}(p) - a_T \mathcal{S}_{++}^{TT}(p),} \quad (268)$$

$$\boxed{d_-^{(T)}(p) = E_{z_0}^{\text{dir}}(-p) - a_T \mathcal{S}_{--}^{TT}(p),} \quad (269)$$

$$m^{(T)}(p) = B_{z_0}(p) - a_T \mathcal{S}_{+-}^{TT}(p), \quad (270)$$

with

$$a_T = \frac{2\beta_X^2}{M_X^2}. \quad (271)$$

Define the mixed drift budgets

$$\ell_{+,i}^{TT} = \partial_{p_i} \mathcal{S}_{++}^{TT}(0), \quad \ell_{-,i}^{TT} = \partial_{p_i} \mathcal{S}_{--}^{TT}(0), \quad \ell_{0,i}^{TT} = \partial_{p_i} \mathcal{S}_{+-}^{TT}(0),$$

and

$$L_+^{TT} = \left(\sum_i |\ell_{+,i}^{TT}|^2 \right)^{1/2}, \quad L_-^{TT} = \left(\sum_i |\ell_{-,i}^{TT}|^2 \right)^{1/2}, \quad L_0^{TT} = \left(\sum_i |\ell_{0,i}^{TT}|^2 \right)^{1/2}. \quad (272)$$

Then

$$\varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)} \leq \left[(\rho A_+^{(T)} + C_+^{TT} \rho^2)^2 + (\rho A_-^{(T)} + C_-^{TT} \rho^2)^2 + 2(\rho B^{(T)} + C_m^{TT} \rho^2)^2 \right]^{1/2}, \quad (273)$$

with

$$A_+^{(T)} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_+^{TT},$$

$$A_-^{(T)} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_-^{TT},$$

$$B^{(T)} \leq a_T L_0^{TT}.$$

Therefore the exact branch-consistent same-shell sufficient condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}. \quad (274)$$

So the mixed same-shell side is reduced completely to the seven explicit finite constants

$$\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, \quad L_-^{TT}, \quad L_0^{TT}, \quad C_+^{TT}, \quad C_-^{TT}, \quad C_m^{TT}. \quad (275)$$

CCI. 2. EXACT MIXED-BLOCK TRANSVERSE AUDIT SHEET

The background-dependent Hessian transport kernel has already been reduced to the blocks

$$TT, \quad TD, \quad DT, \quad DD,$$

with universal one-derivative form

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i\mathcal{V}_{AB}^\mu(\xi)\partial_{y^\mu} + \cdots, \quad \mathcal{M}_{AB}^\mu := \Pi_Q \mathcal{V}_{AB}^\mu \Pi_Q.$$

The invariant projectors are

$$\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu). \quad (276)$$

At the sharp minimal mixed-block stage,

$$\mathcal{M}_{TD+DT}^\mu := \Pi_Q(\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu)\Pi_Q, \quad (277)$$

and the leading transverse coefficients are

$$e_0^{(1)} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad o_0^{(1)} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}]. \quad (278)$$

In the sharp current reading,

$$\Lambda_E = e_0^{(1)}, \quad \Lambda_O = o_0^{(1)}, \quad (279)$$

or more generally

$$\Lambda_E = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \Lambda_O = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}. \quad (280)$$

The conservative currently supported truncation is

$$\ell_0^{(1)} \approx 0, \quad \ell_0^{(3)} \approx \ell_{0;OO}^{(3)}, \quad e_0^{(1)} \approx e_{0;K}^{(1)}, \quad o_0^{(1)} \approx o_{0;K}^{(1)}. \quad (281)$$

So the exact remaining cancellation test is:

$$\Delta e_{\text{mix}} := e_{0;TD}^{(1)} + e_{0;DT}^{(1)} + e_{0;DD}^{(1)} + \cdots, \quad (282)$$

$$\Delta o_{\text{mix}} := o_{0;TD}^{(1)} + o_{0;DT}^{(1)} + o_{0;DD}^{(1)} + \cdots. \quad (283)$$

Then the actual transverse coefficients are

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}. \quad (284)$$

Thus the witness survives provided the mixed-block correction does not cancel the explicit K -sector pair:

$$\Delta e_{\text{mix}} \neq -e_{0;K}^{(1)}, \quad \Delta o_{\text{mix}} \neq -o_{0;K}^{(1)}. \quad (285)$$

A stronger sufficient condition is

$$|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|. \quad (286)$$

Under this condition the signs of Λ_E, Λ_O are inherited from the explicit K -sector pair.

CCII. 3. REDUCED COEFFICIENT POINT AFTER THE FINAL TWO INSERTIONS

Once the seven same-shell constants are inserted and the mixed-block cancellation audit is completed, the mixed odd-active coefficient point is fully populated as

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3\ell_{0;OO}^{(3)}, \quad (287)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa(e_{0;K}^{(1)} + \Delta e_{\text{mix}}), \quad (288)$$

$$\beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa(o_{0;K}^{(1)} + \Delta o_{\text{mix}}). \quad (289)$$

So the true final microscopic frontier is no longer diffuse. It is exactly:

$$\boxed{\text{(i) insert the seven mixed same-shell constants,} \quad \text{(ii) compute } \Delta e_{\text{mix}}, \Delta o_{\text{mix}}.}$$

Nothing else remains at the reduced microscopic audit level.

CCIII. FINAL SUMMARY

For the canonical mixed odd-active branch

$$T_E = T_O = 1,$$

the exact same-shell support side has been reduced to the finite insertion of the seven constants

$$\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, L_-^{TT}, L_0^{TT}, \quad C_+^{TT}, C_-^{TT}, C_m^{TT},$$

through the sufficient condition

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}.$$

On the coefficient side, the remaining microscopic audit is exactly the mixed-block cancellation problem

$$TD, DT, DD,$$

whose net effect is

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}.$$

Therefore the final populated mixed-branch coefficient point is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3\ell_{0;OO}^{(3)},$$

$$\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa(e_{0;K}^{(1)} + \Delta e_{\text{mix}}),$$

$$\beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa(o_{0;K}^{(1)} + \Delta o_{\text{mix}}).$$

So the only genuinely microscopic tasks left are:

$$\boxed{\text{insert the 7 same-shell constants and compute } (\Delta e_{\text{mix}}, \Delta o_{\text{mix}}).}$$

CCIV. PROJECT: TECT — NEAR-FINAL STATUS FOR THE MIXED ODD-ACTIVE BRANCH

CCV. CURRENT STAGE

The project has now reached the following precise stage:

$$\boxed{\text{the reduced mixed odd-active branch is structurally closed,}} \quad (290)$$

but

$$\boxed{\text{the final microscopic coefficient point is still awaiting two finite audits.}} \quad (291)$$

The correct current branch is no longer the pure-even witness branch. It is the canonical mixed odd-active branch

$$\boxed{T_E = T_O = 1, \quad R_\perp = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}}.} \quad (292)$$

On this branch the reduced coefficient point is already fixed as

$$\boxed{\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3,} \quad (293)$$

$$\boxed{\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.} \quad (294)$$

So the odd coefficient is genuinely turned on at branch level, and the remaining problem is no longer branch choice.

CCVI. WHAT IS ALREADY CLOSED

The following parts are already closed at theorem / exact-reduced-sheet level:

1. corrected residual quartic closure at the exact four-class level;
2. failure of the old (u, v, κ) truncation;
3. exact coarse-orbit quartic matrix M^{avg} ;
4. proof that coarse four-class quartics alone are A/E -symmetric;
5. necessity of the first non-diagonal completion;
6. sign-resolved suborbit refinement and the first transverse-sensitive quartic trigger patterns;
7. explicit odd extra-carrier route;
8. mixed odd-active branch decision $T_O \neq 0$;
9. exact reduced branch-level coefficient sheet for the canonical mixed sample.

Therefore the project is no longer waiting on a new conceptual mechanism.

CCVII. WHAT REMAINS OPEN

Exactly two genuinely microscopic tasks remain:

A. (A) Mixed same-shell support insertion

The canonical mixed same-shell branch has

$$a_T = \frac{2\beta_X^2}{M_X^2}, \quad \mathcal{S}_{ab}^{(\pi/4)} = \frac{1}{2} \mathcal{S}_{ab}^{EE} + \frac{1}{2} \mathcal{S}_{ab}^{OO} + \mathcal{S}_{ab}^{EO}.$$

Its exact branch-consistent sufficient support condition is

$$\boxed{|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}} \quad (295)$$

What is still not numerically inserted are the seven mixed same-shell constants:

$$\boxed{\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, L_-^{TT}, L_0^{TT}, \quad C_+^{TT}, C_-^{TT}, C_m^{TT}.} \quad (296)$$

So the same-shell side is reduced completely to a seven-number finite insertion problem.

B. (B) Mixed-block transverse cancellation audit

The microscopic source of the transverse coefficients is the mixed block family

$$TD, DT, OD, DO, DD,$$

with exact leading reduced extraction

$$e_0^{(1)} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad o_0^{(1)} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}. \quad (297)$$

At the current conservative stage,

$$\ell_0^{(1)} \approx 0, \quad \ell_0^{(3)} \approx \ell_{0;OO}^{(3)}, \quad e_0^{(1)} \approx e_{0;K}^{(1)}, \quad o_0^{(1)} \approx o_{0;K}^{(1)}. \quad (298)$$

Therefore define the mixed-block corrections

$$\Delta e_{\text{mix}} := e_{0;TD}^{(1)} + e_{0;DT}^{(1)} + e_{0;DD}^{(1)} + \cdots, \quad (299)$$

$$\Delta o_{\text{mix}} := o_{0;TD}^{(1)} + o_{0;DT}^{(1)} + o_{0;DD}^{(1)} + \cdots. \quad (300)$$

Then the actual transverse coefficient point is

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}. \quad (301)$$

A strong sufficient survival condition is

$$|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|. \quad (302)$$

If this holds, the mixed blocks do not kill the already explicit K -sector witness pair.

CCVIII. FINAL POPULATED COEFFICIENT POINT ONCE THE TWO AUDITS ARE DONE

Once the seven same-shell constants are inserted and $(\Delta e_{\text{mix}}, \Delta o_{\text{mix}})$ are computed, the final mixed-branch microscopic coefficient point is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)}, \quad (303)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (e_{0;K}^{(1)} + \Delta e_{\text{mix}}), \quad (304)$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (o_{0;K}^{(1)} + \Delta o_{\text{mix}}). \quad (305)$$

Equivalently, in projected-Bloch notation, the exact microscopic target is

$$P_* K_{\parallel} P_* = \lambda_{\parallel} \sigma_3, \quad P_* K_1 P_* = \alpha (\sigma_1 + \sigma_2), \quad P_* K_2 P_* = \alpha (\sigma_2 - \sigma_1). \quad (306)$$

So the project is now one finite support audit and one finite mixed-block cancellation audit away from the fully populated reduced microscopic coefficient point.

CCIX. HONEST FINAL STATUS

The honest status is therefore:

$$\boxed{\text{reduced branch logic is closed,} \quad \text{two finite microscopic audits remain.}} \quad (307)$$

Nothing essential remains open at the level of reduced-branch structure. What remains is only the actual numerical / explicit insertion of the microscopic constants.

CCX. FINAL SUMMARY

The correct current branch is the canonical mixed odd-active sample

$$T_E = T_O = 1, \quad R_\perp = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}},$$

for which

$$\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.$$

The project is now reduced to two finite microscopic tasks:

$$\boxed{\text{(A) insert the seven mixed same-shell constants}}$$

through

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)},$$

and

$$\boxed{\text{(B) compute the mixed-block corrections } (\Delta e_{\text{mix}}, \Delta o_{\text{mix}}).}$$

Once these are done, the final mixed-branch microscopic coefficient point is

$$\begin{aligned} \lambda_\parallel &= -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)}, \\ \alpha &= -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (e_{0;K}^{(1)} + \Delta e_{\text{mix}}), \\ \beta &= -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (o_{0;K}^{(1)} + \Delta o_{\text{mix}}). \end{aligned}$$

Thus the reduced microscopic audit is no longer structurally open. Only two finite insertions remain.

CCXI. EXACT FINITE-REDUCTION OF THE TWO REMAINING MIXED-BRANCH AUDITS

CCXII. CURRENT STAGE

The canonical mixed odd-active branch is already fixed as

$$T_E = T_O = 1, \quad R_\perp = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}},$$

and the reduced coefficient sheet is already

$$\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.$$

Thus the branch-choice problem is closed. What remains is only the final microscopic finite insertion/cancellation stage.

CCXIII. AUDIT A: EXACT REDUCTION OF THE MIXED SAME-SHELL SUPPORT PROBLEM

The exact mixed-branch same-shell overlap package is

$$\mathcal{S}_{ab}^{TT}(p; \varphi) = \cos^2 \varphi \mathcal{S}_{ab}^{EE}(p) + \sin^2 \varphi \mathcal{S}_{ab}^{OO}(p) + 2 \sin \varphi \cos \varphi \mathcal{S}_{ab}^{EO}(p), \quad a, b \in \{+, -\}.$$

For the canonical branch

$$\varphi = \frac{\pi}{4},$$

this becomes

$$\mathcal{S}_{ab}^{(\pi/4)}(p) = \frac{1}{2} \mathcal{S}_{ab}^{EE}(p) + \frac{1}{2} \mathcal{S}_{ab}^{OO}(p) + \mathcal{S}_{ab}^{EO}(p).$$

Therefore the diagonal benchmark constant is exactly

$$\mathcal{S}_{\text{diag}}^{(\pi/4)} = \frac{1}{2} \mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2} \mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO}.$$

Now assume the small- p drift coefficients are defined by the corresponding Taylor data of the same-shell package:

$$\ell_{+,i}^{XY}, \quad \ell_{-,i}^{XY}, \quad \ell_{0,i}^{XY}, \quad C_+^{XY}, \quad C_-^{XY}, \quad C_m^{XY}, \quad XY \in \{EE, OO, EO\}.$$

Then linearity of differentiation gives the exact mixed formulas

$$\ell_{+,i}^{TT} = \frac{1}{2} \ell_{+,i}^{EE} + \frac{1}{2} \ell_{+,i}^{OO} + \ell_{+,i}^{EO},$$

$$\ell_{-,i}^{TT} = \frac{1}{2} \ell_{-,i}^{EE} + \frac{1}{2} \ell_{-,i}^{OO} + \ell_{-,i}^{EO},$$

$$\ell_{0,i}^{TT} = \frac{1}{2} \ell_{0,i}^{EE} + \frac{1}{2} \ell_{0,i}^{OO} + \ell_{0,i}^{EO},$$

and likewise

$$C_+^{TT} = \frac{1}{2} C_+^{EE} + \frac{1}{2} C_+^{OO} + C_+^{EO},$$

$$C_-^{TT} = \frac{1}{2} C_-^{EE} + \frac{1}{2} C_-^{OO} + C_-^{EO},$$

$$C_m^{TT} = \frac{1}{2} C_m^{EE} + \frac{1}{2} C_m^{OO} + C_m^{EO}.$$

Hence the seven missing mixed same-shell constants are reduced exactly to the already-separated EE , OO , and EO data:

$$\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, \quad L_-^{TT}, \quad L_0^{TT}, \quad C_+^{TT}, \quad C_-^{TT}, \quad C_m^{TT}$$

with

$$L_+^{TT} = \left(\sum_i |\ell_{+,i}^{TT}|^2 \right)^{1/2}, \quad L_-^{TT} = \left(\sum_i |\ell_{-,i}^{TT}|^2 \right)^{1/2}, \quad L_0^{TT} = \left(\sum_i |\ell_{0,i}^{TT}|^2 \right)^{1/2}.$$

At the norm level, one always has the rigorous upper bounds

$$L_+^{TT} \leq \frac{1}{2}L_+^{EE} + \frac{1}{2}L_+^{OO} + L_+^{EO},$$

$$L_-^{TT} \leq \frac{1}{2}L_-^{EE} + \frac{1}{2}L_-^{OO} + L_-^{EO},$$

$$L_0^{TT} \leq \frac{1}{2}L_0^{EE} + \frac{1}{2}L_0^{OO} + L_0^{EO}.$$

If all coefficients are sign-coherent in the chosen basis, these may sharpen to equalities, but the inequalities are the honest theorem-safe form.

Therefore the mixed same-shell support condition becomes

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \left(\frac{1}{2}\mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2}\mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO} \right) + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}.$$

Moreover,

$$\varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)} \leq \left[(\rho A_+^{TT} + C_+^{TT} \rho^2)^2 + (\rho A_-^{TT} + C_-^{TT} \rho^2)^2 + 2(\rho B^{TT} + C_m^{TT} \rho^2)^2 \right]^{1/2},$$

with

$$A_+^{TT} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + \frac{2\beta_X^2}{M_X^2} L_+^{TT},$$

$$A_-^{TT} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + \frac{2\beta_X^2}{M_X^2} L_-^{TT},$$

$$B^{TT} \leq \frac{2\beta_X^2}{M_X^2} L_0^{TT}.$$

So Audit A is no longer a conceptual problem. It is exactly the finite insertion of the $EE/OO/EO$ Taylor data into the formulas above.

CCXIV. AUDIT B: EXACT REDUCTION OF THE MIXED-BLOCK TRANSVERSE CANCELLATION PROBLEM

The exact remaining mixed-block corrections are

$$\Delta e_{\text{mix}} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \Delta o_{\text{mix}} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}.$$

Using the invariant projector functionals on the projected one-derivative kernels,

$$\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}(\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}(\Sigma_a \mathcal{M}_\mu),$$

one has the exact blockwise identities

$$e_{0;AB}^{(1)} = \mathfrak{P}_E[\mathcal{M}_{AB,\text{inv}}], \quad o_{0;AB}^{(1)} = \mathfrak{P}_O[\mathcal{M}_{AB,\text{inv}}],$$

hence

$$\Delta e_{\text{mix}} = \sum_{AB} \mathfrak{P}_E[\mathcal{M}_{AB,\text{inv}}], \quad \Delta o_{\text{mix}} = \sum_{AB} \mathfrak{P}_O[\mathcal{M}_{AB,\text{inv}}],$$

with $AB \in \{TD, DT, OD, DO, DD\}$.

The actual transverse coefficient point is therefore

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}.$$

There are then three logically distinct sufficient criteria:

a. Criterion B1: componentwise sign-preserving criterion.

$$|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|.$$

This preserves the nonvanishing of each component separately.

b. Criterion B2: Euclidean noncancellation criterion. Define

$$v_K := (e_{0;K}^{(1)}, o_{0;K}^{(1)}), \quad v_{\text{mix}} := (\Delta e_{\text{mix}}, \Delta o_{\text{mix}}).$$

Then

$$\|v_{\text{mix}}\|_2 < \|v_K\|_2 \implies v_K + v_{\text{mix}} \neq (0, 0).$$

This is weaker than componentwise sign preservation, but sharper as a survival test for $(\Lambda_E, \Lambda_O) \neq (0, 0)$.

c. Criterion B3: exact cancellation test. The only actual failure condition is

$$\Delta e_{\text{mix}} = -e_{0;K}^{(1)}, \quad \Delta o_{\text{mix}} = -o_{0;K}^{(1)}.$$

So once the five mixed blocks are explicitly projected, the transverse audit is completely decided.

CCXV. FINAL COEFFICIENT POINT AFTER THE TWO AUDITS

After Audit A and Audit B, the final microscopic coefficient point is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)},$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (e_{0;K}^{(1)} + \Delta e_{\text{mix}}),$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (o_{0;K}^{(1)} + \Delta o_{\text{mix}}).$$

Thus the project is reduced to the following exact finite checklist:

1. compute the EE , OO , and EO same-shell Taylor packages;
2. assemble

$$\mathcal{S}_{\text{diag}}^{TT}, \ell_{\pm,i}^{TT}, \ell_{0,i}^{TT}, C_+^{TT}, C_-^{TT}, C_m^{TT};$$

3. evaluate the support inequality;
4. project the five mixed blocks

$$TD, DT, OD, DO, DD$$

through $\mathfrak{P}_E, \mathfrak{P}_O$;

5. compare the result against the explicit K -sector vector.

No new conceptual theorem is needed beyond this point. Only finite microscopic insertion remains.

CCXVI. FINAL SUMMARY

Current stage: the mixed odd-active branch is structurally closed,

but the final microscopic coefficient point still depends on exactly two finite audits.

For the canonical mixed branch

$$T_E = T_O = 1, \quad R_\perp = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}},$$

the branch-level reduced coefficients are already fixed.

The first remaining audit is the mixed same-shell support insertion. Its exact overlap package is

$$\mathcal{S}_{ab}^{(\pi/4)} = \frac{1}{2}\mathcal{S}_{ab}^{EE} + \frac{1}{2}\mathcal{S}_{ab}^{OO} + \mathcal{S}_{ab}^{EO},$$

so the missing seven constants are reducible to the $EE/OO/EO$ Taylor data.

The second remaining audit is the mixed-block transverse cancellation problem:

$$\Delta e_{\text{mix}} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \Delta o_{\text{mix}} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}.$$

The actual transverse pair is

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}.$$

Hence the program is no longer waiting on a new mechanism. It is reduced to a finite microscopic insertion-and-comparison problem.

Immediate next step: assemble the TT seven-constant package from $EE/OO/EO$,

then project TD, DT, OD, DO, DD to compute $(\Delta e_{\text{mix}}, \Delta o_{\text{mix}})$.

CCXVII. EXACT CLOSURE OF AUDIT A AND SHARP EXECUTION ORDER FOR AUDIT B

CCXVIII. CURRENT STAGE

The reduced mixed odd-active branch is already fixed at the branch level by

$$T_E = T_O = 1, \quad R_\perp = \sqrt{2}, \quad \varphi = \frac{\pi}{4}, \quad \Delta_{00} = \frac{1}{\sqrt{2}},$$

and the remaining microscopic work is reduced to exactly two finite audits: the mixed same-shell support insertion and the mixed-block transverse cancellation audit.

CCXIX. 1. EXACT MIXED SAME-SHELL PACKAGE

The branch-consistent mixed same-shell overlap family is

$$\mathcal{S}_{ab}^{TT}(p; \varphi) = \cos^2 \varphi \mathcal{S}_{ab}^{EE}(p) + \sin^2 \varphi \mathcal{S}_{ab}^{OO}(p) + 2 \sin \varphi \cos \varphi \mathcal{S}_{ab}^{EO}(p), \quad (308)$$

for $a, b \in \{+, -\}$.

For the canonical mixed branch $\varphi = \pi/4$,

$$\mathcal{S}_{ab}^{(\pi/4)}(p) = \frac{1}{2}\mathcal{S}_{ab}^{EE}(p) + \frac{1}{2}\mathcal{S}_{ab}^{OO}(p) + \mathcal{S}_{ab}^{EO}(p). \quad (309)$$

The same-shell correction strength is

$$a_T = \frac{\beta_X^2}{M_X^2}(T_E^2 + T_O^2) = \frac{2\beta_X^2}{M_X^2}. \quad (310)$$

The mixed same-shell block is therefore

$$d_+^{(T)}(p) = E_{z_0}^{\text{dir}}(p) - a_T \mathcal{S}_{++}^{TT}(p), \quad (311)$$

$$d_-^{(T)}(p) = E_{z_0}^{\text{dir}}(-p) - a_T \mathcal{S}_{--}^{TT}(p), \quad (312)$$

$$m^{(T)}(p) = B_{z_0}(p) - a_T \mathcal{S}_{+-}^{TT}(p). \quad (313)$$

Under the valley-symmetric diagonal subcase,

$$\mathcal{S}_{\text{diag}}^{TT} := \mathcal{S}_{++}^{TT}(0) = \mathcal{S}_{--}^{TT}(0), \quad (314)$$

hence

$$\mathcal{S}_{\text{diag}}^{TT} = \frac{1}{2}\mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2}\mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO}. \quad (315)$$

CCXX. 2. EXACT FIRST-DERIVATIVE VECTORS

Define the mixed first-derivative vectors by

$$\ell_{+,i}^{TT} := \partial_{p_i} \mathcal{S}_{++}^{TT}(p)|_{p=0}, \quad \ell_{-,i}^{TT} := \partial_{p_i} \mathcal{S}_{--}^{TT}(p)|_{p=0}, \quad \ell_{0,i}^{TT} := \partial_{p_i} \mathcal{S}_{+-}^{TT}(p)|_{p=0}. \quad (316)$$

By linearity of differentiation and (309),

$$\ell_{+,i}^{TT} = \frac{1}{2}\ell_{+,i}^{EE} + \frac{1}{2}\ell_{+,i}^{OO} + \ell_{+,i}^{EO}, \quad (317)$$

$$\ell_{-,i}^{TT} = \frac{1}{2}\ell_{-,i}^{EE} + \frac{1}{2}\ell_{-,i}^{OO} + \ell_{-,i}^{EO}, \quad (318)$$

$$\ell_{0,i}^{TT} = \frac{1}{2}\ell_{0,i}^{EE} + \frac{1}{2}\ell_{0,i}^{OO} + \ell_{0,i}^{EO}. \quad (319)$$

Now define the norm package

$$L_+^{TT} := \left(\sum_i |\ell_{+,i}^{TT}|^2 \right)^{1/2}, \quad L_-^{TT} := \left(\sum_i |\ell_{-,i}^{TT}|^2 \right)^{1/2}, \quad L_0^{TT} := \left(\sum_i |\ell_{0,i}^{TT}|^2 \right)^{1/2}. \quad (320)$$

These are not independent constants. They satisfy the rigorous bounds

$$L_+^{TT} \leq \frac{1}{2}L_+^{EE} + \frac{1}{2}L_+^{OO} + L_+^{EO}, \quad (321)$$

$$L_-^{TT} \leq \frac{1}{2}L_-^{EE} + \frac{1}{2}L_-^{OO} + L_-^{EO}, \quad (322)$$

$$L_0^{TT} \leq \frac{1}{2}L_0^{EE} + \frac{1}{2}L_0^{OO} + L_0^{EO}. \quad (323)$$

a. Important honesty point. The vector identities (317)–(319) are exact. The norm formulas (321)–(323) are, in general, only upper bounds.

CCXXI. 3. EXACT SECOND-DERIVATIVE TENSORS AND HONEST C -CONSTANTS

Define the Hessian tensors

$$H_{+,ij}^{TT} := \partial_{p_i} \partial_{p_j} \mathcal{S}_{++}^{TT}(p)|_{p=0}, \quad H_{-,ij}^{TT} := \partial_{p_i} \partial_{p_j} \mathcal{S}_{--}^{TT}(p)|_{p=0}, \quad H_{0,ij}^{TT} := \partial_{p_i} \partial_{p_j} \mathcal{S}_{+-}^{TT}(p)|_{p=0}. \quad (324)$$

Then

$$H_{+,ij}^{TT} = \frac{1}{2} H_{+,ij}^{EE} + \frac{1}{2} H_{+,ij}^{OO} + H_{+,ij}^{EO}, \quad (325)$$

$$H_{-,ij}^{TT} = \frac{1}{2} H_{-,ij}^{EE} + \frac{1}{2} H_{-,ij}^{OO} + H_{-,ij}^{EO}, \quad (326)$$

$$H_{0,ij}^{TT} = \frac{1}{2} H_{0,ij}^{EE} + \frac{1}{2} H_{0,ij}^{OO} + H_{0,ij}^{EO}. \quad (327)$$

Now define valid quadratic Taylor remainder constants by

$$C_+^{TT} := \frac{1}{2} \sup_{|p| \leq \rho} \|\nabla_p^2 d_+^{(T)}(p)\|_{\text{op}}, \quad C_-^{TT} := \frac{1}{2} \sup_{|p| \leq \rho} \|\nabla_p^2 d_-^{(T)}(p)\|_{\text{op}}, \quad (328)$$

$$C_m^{TT} := \frac{1}{2} \sup_{|p| \leq \rho} \|\nabla_p^2 m^{(T)}(p)\|_{\text{op}}. \quad (329)$$

Because these are supremum bounds, the honest branch-consistent estimates are

$$C_+^{TT} \leq \frac{1}{2} C_+^{EE} + \frac{1}{2} C_+^{OO} + C_+^{EO}, \quad (330)$$

$$C_-^{TT} \leq \frac{1}{2} C_-^{EE} + \frac{1}{2} C_-^{OO} + C_-^{EO}, \quad (331)$$

$$C_m^{TT} \leq \frac{1}{2} C_m^{EE} + \frac{1}{2} C_m^{OO} + C_m^{EO}. \quad (332)$$

Again, these are generally sufficient bounds, not exact equalities.

CCXXII. 4. MIXED-BRANCH DRIFT BUDGET

The mixed drift coefficients are

$$a_{+,i}^{TT} = \delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right) - a_T \ell_{+,i}^{TT}, \quad (333)$$

$$a_{-,i}^{TT} = -\delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right) - a_T \ell_{-,i}^{TT}, \quad (334)$$

$$\boxed{b_i^{TT} = -a_T \ell_{0,i}^{TT}} \quad (335)$$

Hence

$$\boxed{A_+^{TT} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_+^{TT},} \quad (336)$$

$$\boxed{A_-^{TT} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_-^{TT},} \quad (337)$$

$$\boxed{B^{TT} \leq a_T L_0^{TT}.} \quad (338)$$

Therefore the small- p budget satisfies

$$\boxed{\varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)} \leq \left[(\rho A_+^{TT} + C_+^{TT} \rho^2)^2 + (\rho A_-^{TT} + C_-^{TT} \rho^2)^2 + 2(\rho B^{TT} + C_m^{TT} \rho^2)^2 \right]^{1/2}.} \quad (339)$$

CCXXIII. 5. FINAL EXACT FORM OF AUDIT A

The sharp mixed same-shell sufficient condition is

$$\boxed{|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{TT} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)},} \quad (340)$$

with

$$\mathcal{S}_{\text{diag}}^{TT} = \frac{1}{2} \mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2} \mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO},$$

and with $L_{\pm}^{TT}, L_0^{TT}, C_{\pm}^{TT}, C_m^{TT}$ given by the exact/bound formulas above.

Thus Audit A is completely reduced to the explicit insertion of the already-separated $EE/OO/EO$ overlap data.

CCXXIV. 6. SHARP EXECUTION ORDER FOR AUDIT B

The exact mixed transverse coefficient routing is

$$\boxed{\Lambda_E = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \Lambda_O = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}} \quad (341)$$

But the structurally principal source is $TD + DT$, while OD, DO, DD are residual corrections. Hence define

$$\boxed{v_{\text{pr}} := \left(e_{0;TD}^{(1)} + e_{0;DT}^{(1)}, o_{0;TD}^{(1)} + o_{0;DT}^{(1)} \right),} \quad (342)$$

$$\boxed{r_{\text{rem}} := \left(e_{0;OD}^{(1)} + e_{0;DO}^{(1)} + e_{0;DD}^{(1)}, o_{0;OD}^{(1)} + o_{0;DO}^{(1)} + o_{0;DD}^{(1)} \right).} \quad (343)$$

Then

$$\boxed{(\Lambda_E, \Lambda_O) = v_{\text{pr}} + r_{\text{rem}}.} \quad (344)$$

Therefore the sharp execution order is:

1. compute $TD + DT$ exactly and obtain v_{pr} ;
2. only then compute or bound $OD + DO + DD$ as r_{rem} ;
3. conclude noncancellation from a dominance test.

Two useful sufficient criteria are:

a. *Componentwise strong criterion.*

$$\boxed{|e_{0;OD}^{(1)} + e_{0;DO}^{(1)} + e_{0;DD}^{(1)}| < |e_{0;TD}^{(1)} + e_{0;DT}^{(1)}|,} \quad (345)$$

$$\boxed{|o_{0;OD}^{(1)} + o_{0;DO}^{(1)} + o_{0;DD}^{(1)}| < |o_{0;TD}^{(1)} + o_{0;DT}^{(1)}|.} \quad (346)$$

b. *Euclidean dominance criterion.*

$$\boxed{\|r_{\text{rem}}\|_2 < \|v_{\text{pr}}\|_2 \implies (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (347)$$

CCXXV. 7. FINAL POPULATED COEFFICIENT POINT

Once Audit A and Audit B are inserted,

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3\ell_{0;OO}^{(3)},} \quad (348)$$

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_E, \quad \beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\Lambda_O,} \quad (349)$$

with

$$(\Lambda_E, \Lambda_O) = v_{\text{pr}} + r_{\text{rem}}.$$

So the remaining frontier is now purely finite:

$$\boxed{\text{assemble the } TT \text{ support package, then evaluate } TD + DT \text{ before all residual blocks.}}$$

CCXXVI. FINAL SUMMARY

The mixed odd-active branch is already structurally fixed. The same-shell support audit is now reduced to the canonical mixed package

$$\mathcal{S}_{ab}^{TT}(p; \varphi) = \cos^2 \varphi \mathcal{S}_{ab}^{EE}(p) + \sin^2 \varphi \mathcal{S}_{ab}^{OO}(p) + 2 \sin \varphi \cos \varphi \mathcal{S}_{ab}^{EO}(p),$$

which at $\varphi = \pi/4$ becomes

$$\mathcal{S}_{ab}^{(\pi/4)} = \frac{1}{2}\mathcal{S}_{ab}^{EE} + \frac{1}{2}\mathcal{S}_{ab}^{OO} + \mathcal{S}_{ab}^{EO}.$$

The diagonal and first-derivative data satisfy exact linear relations:

$$\mathcal{S}_{\text{diag}}^{TT} = \frac{1}{2}\mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2}\mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO},$$

$$\ell_{\bullet}^{TT} = \frac{1}{2}\ell_{\bullet}^{EE} + \frac{1}{2}\ell_{\bullet}^{OO} + \ell_{\bullet}^{EO}.$$

The norm constants $L_{\bullet}^{TT}$ are therefore bounded by triangle inequalities, while the quadratic remainder constants $C_{\bullet}^{TT}$ are honest supremum bounds, not exact linear identities in general.

Hence Audit A is fully reduced to finite insertion of the separated $EE/OO/EO$ overlap data.

For Audit B, the exact coefficient routing is

$$\Lambda_E = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \Lambda_O = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}.$$

But the structurally principal source is $TD + DT$, so the sharp execution order is

$(TD + DT)$ first, $(OD + DO + DD)$ as residual correction second.

Therefore the project is now reduced to:

$$\boxed{\text{(i) assemble } (\mathcal{S}_{\text{diag}}^{TT}, L_+^{TT}, L_-^{TT}, L_0^{TT}, C_+^{TT}, C_-^{TT}, C_m^{TT}),}$$

$$\boxed{\text{(ii) compute } v_{\text{pr}} = (TD + DT) \text{ and only then add the residual correction } r_{\text{rem}}.}$$

No new structural theorem is needed beyond this point. Only finite microscopic insertion remains.

CCXXVII. AUDIT A CLOSURE: EXACT CANONICAL TT SEVEN-CONSTANT PACKAGE

CCXXVIII. CURRENT STAGE

For the canonical mixed odd-active branch,

$$T_E = T_O = 1, \quad \varphi = \frac{\pi}{4}, \quad a_T = \frac{2\beta_X^2}{M_X^2},$$

the same-shell benchmark remains a 2×2 Hermitian problem, but with the pure-even EE overlap family replaced by the mixed TT overlap family.

The key point is that the mixed overlap family is not new data. It is assembled linearly from the already separated EE , OO , and EO packages.

CCXXIX. 1. EXACT CANONICAL MIXED OVERLAP FAMILY

The mixed same-shell overlap family is

$$\boxed{\mathcal{S}_{ab}^{TT}(p; \varphi) = \cos^2 \varphi \mathcal{S}_{ab}^{EE}(p) + \sin^2 \varphi \mathcal{S}_{ab}^{OO}(p) + 2 \sin \varphi \cos \varphi \mathcal{S}_{ab}^{EO}(p).} \quad (350)$$

At the canonical mixed branch $\varphi = \pi/4$,

$$\boxed{\mathcal{S}_{ab}^{TT}(p) = \frac{1}{2} \mathcal{S}_{ab}^{EE}(p) + \frac{1}{2} \mathcal{S}_{ab}^{OO}(p) + \mathcal{S}_{ab}^{EO}(p).} \quad (351)$$

Therefore the diagonal same-shell constant is exactly

$$\boxed{\mathcal{S}_{\text{diag}}^{TT} = \frac{1}{2} \mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2} \mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO}.} \quad (352)$$

This is an exact identity, not a bound.

CCXXX. 2. EXACT FIRST-DERIVATIVE VECTORS

Define

$$\boxed{\ell_{+,i}^{TT} := \partial_{p_i} \mathcal{S}_{++}^{TT}(p)|_{p=0}, \quad \ell_{-,i}^{TT} := \partial_{p_i} \mathcal{S}_{--}^{TT}(p)|_{p=0}, \quad \ell_{0,i}^{TT} := \partial_{p_i} \mathcal{S}_{+-}^{TT}(p)|_{p=0}.} \quad (353)$$

Using (351), differentiation is linear, hence

$$\boxed{\ell_{+,i}^{TT} = \frac{1}{2} \ell_{+,i}^{EE} + \frac{1}{2} \ell_{+,i}^{OO} + \ell_{+,i}^{EO},} \quad (354)$$

$$\ell_{-,i}^{TT} = \frac{1}{2}\ell_{-,i}^{EE} + \frac{1}{2}\ell_{-,i}^{OO} + \ell_{-,i}^{EO}, \quad (355)$$

$$\ell_{0,i}^{TT} = \frac{1}{2}\ell_{0,i}^{EE} + \frac{1}{2}\ell_{0,i}^{OO} + \ell_{0,i}^{EO}. \quad (356)$$

Thus the derivative vectors are also exact linear combinations.

CCXXXI. 3. NORM-LEVEL CONSTANTS $L_+^{TT}, L_-^{TT}, L_0^{TT}$

Define

$$L_+^{TT} = \left(\sum_i |\ell_{+,i}^{TT}|^2 \right)^{1/2}, \quad L_-^{TT} = \left(\sum_i |\ell_{-,i}^{TT}|^2 \right)^{1/2}, \quad L_0^{TT} = \left(\sum_i |\ell_{0,i}^{TT}|^2 \right)^{1/2}. \quad (357)$$

These are generally not exact linear combinations, because norms do not distribute linearly. The honest theorem-level statement is the triangle-inequality bound

$$L_+^{TT} \leq \frac{1}{2}L_+^{EE} + \frac{1}{2}L_+^{OO} + L_+^{EO}, \quad (358)$$

$$L_-^{TT} \leq \frac{1}{2}L_-^{EE} + \frac{1}{2}L_-^{OO} + L_-^{EO}, \quad (359)$$

$$L_0^{TT} \leq \frac{1}{2}L_0^{EE} + \frac{1}{2}L_0^{OO} + L_0^{EO}. \quad (360)$$

If the three contributing vectors are sign/phase aligned in the chosen basis, these bounds can sharpen to equalities, but that is an additional branch-specific property and must not be assumed by default.

CCXXXII. 4. HONEST SECOND-DERIVATIVE PACKAGE AND C -CONSTANTS

The C -constants are quadratic Taylor-remainder bounds, not raw second derivatives. So the correct object is the Hessian/supremum package.

Define the Hessian tensors

$$H_{+,ij}^{TT}(p) := \partial_{p_i} \partial_{p_j} d_+^{(T)}(p), \quad H_{-,ij}^{TT}(p) := \partial_{p_i} \partial_{p_j} d_-^{(T)}(p), \quad H_{m,ij}^{TT}(p) := \partial_{p_i} \partial_{p_j} m^{(T)}(p). \quad (361)$$

Then

$$C_+^{TT} := \frac{1}{2} \sup_{|p| \leq \rho} \|H_+^{TT}(p)\|_{\text{op}}, \quad C_-^{TT} := \frac{1}{2} \sup_{|p| \leq \rho} \|H_-^{TT}(p)\|_{\text{op}}, \quad C_m^{TT} := \frac{1}{2} \sup_{|p| \leq \rho} \|H_m^{TT}(p)\|_{\text{op}}. \quad (362)$$

Because the TT overlap family is linear in $EE/OO/EO$, the Hessian tensors are also linear in those three sectors. However, after taking operator norms and suprema, the correct general statement is again an upper bound:

$$C_+^{TT} \leq \frac{1}{2}C_+^{EE} + \frac{1}{2}C_+^{OO} + C_+^{EO}, \quad (363)$$

$$C_-^{TT} \leq \frac{1}{2}C_-^{EE} + \frac{1}{2}C_-^{OO} + C_-^{EO}, \quad (364)$$

$$C_m^{TT} \leq \frac{1}{2}C_m^{EE} + \frac{1}{2}C_m^{OO} + C_m^{EO}. \quad (365)$$

So the C -package is controlled, but generally not by exact equality.

CCXXXIII. 5. DRIFT COEFFICIENTS AND NORM BUDGETS

The direct locked-core contribution is unchanged:

$$a_{+,i}^{\text{dir}} = \delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right), \quad a_{-,i}^{\text{dir}} = -\delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right), \quad b_i^{\text{dir}} = 0. \quad (366)$$

The mixed-branch full derivative coefficients are therefore

$$a_{+,i}^{(T)} = \delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right) - a_T \ell_{+,i}^{TT}, \quad (367)$$

$$a_{-,i}^{(T)} = -\delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right) - a_T \ell_{-,i}^{TT}, \quad (368)$$

$$b_i^{(T)} = -a_T \ell_{0,i}^{TT}. \quad (369)$$

Hence the norm-level budgets satisfy

$$A_+^{(T)} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_+^{TT}, \quad (370)$$

$$A_-^{(T)} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_-^{TT}, \quad (371)$$

$$B^{(T)} \leq a_T L_0^{TT}. \quad (372)$$

CCXXXIV. 6. EXPLICIT TT SEVEN-CONSTANT TABLE

The canonical TT seven-constant package is therefore

constant	status	canonical formula at $\varphi = \pi/4$
$\mathcal{S}_{\text{diag}}^{TT}$	exact	$\frac{1}{2}\mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2}\mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO}$
L_+^{TT}	bound	$L_+^{TT} \leq \frac{1}{2}L_+^{EE} + \frac{1}{2}L_+^{OO} + L_+^{EO}$
L_-^{TT}	bound	$L_-^{TT} \leq \frac{1}{2}L_-^{EE} + \frac{1}{2}L_-^{OO} + L_-^{EO}$
L_0^{TT}	bound	$L_0^{TT} \leq \frac{1}{2}L_0^{EE} + \frac{1}{2}L_0^{OO} + L_0^{EO}$
C_+^{TT}	bound	$C_+^{TT} \leq \frac{1}{2}C_+^{EE} + \frac{1}{2}C_+^{OO} + C_+^{EO}$
C_-^{TT}	bound	$C_-^{TT} \leq \frac{1}{2}C_-^{EE} + \frac{1}{2}C_-^{OO} + C_-^{EO}$
C_m^{TT}	bound	$C_m^{TT} \leq \frac{1}{2}C_m^{EE} + \frac{1}{2}C_m^{OO} + C_m^{EO}$

(373)

This is the honest fully reduced form of Audit A.

CCXXXV. 7. FINAL MIXED SAME-SHELL SUPPORT INEQUALITY

The mixed-branch same-shell small- p drift budget obeys

$$\varepsilon_{\text{small-}p,\rho}^{(1,T)} \leq \left[(\rho A_+^{(T)} + C_+^{TT} \rho^2)^2 + (\rho A_-^{(T)} + C_-^{TT} \rho^2)^2 + 2(\rho B^{(T)} + C_m^{TT} \rho^2)^2 \right]^{1/2}. \quad (374)$$

Hence the final branch-consistent sufficient support condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > 2a_T \mathcal{S}_{\text{diag}}^{TT} + \varepsilon_{\text{small-}p,\rho}^{(1,T)}. \quad (375)$$

Using

$$a_T = \frac{2\beta_X^2}{M_X^2},$$

this is equivalently

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{TT} + \varepsilon_{\text{small-}p,\rho}^{(1,T)}. \quad (376)$$

CCXXXVI. 8. WHAT REMAINS AFTER AUDIT A

After (373) and (376), Audit A is no longer open structurally.

It is reduced to the explicit finite insertion of the three already separated packages:

$$(EE), \quad (OO), \quad (EO).$$

So the next nonredundant step is to evaluate the principal transverse mixed-block source

$$TD + DT$$

before the residual correction blocks

$$OD, \quad DO, \quad DD.$$

CCXXXVII. FINAL SUMMARY

For the canonical mixed odd-active branch, the same-shell TT benchmark package is now fully reduced.

The mixed overlap family is

$$\mathcal{S}_{ab}^{TT}(p) = \frac{1}{2} \mathcal{S}_{ab}^{EE}(p) + \frac{1}{2} \mathcal{S}_{ab}^{OO}(p) + \mathcal{S}_{ab}^{EO}(p),$$

so the diagonal constant is exactly

$$\mathcal{S}_{\text{diag}}^{TT} = \frac{1}{2} \mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2} \mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO}.$$

The derivative vectors are also exact:

$$\ell_{\bullet}^{TT} = \frac{1}{2} \ell_{\bullet}^{EE} + \frac{1}{2} \ell_{\bullet}^{OO} + \ell_{\bullet}^{EO}.$$

However, the norm constants and Taylor-remainder constants are only controlled by honest upper bounds:

$$L_{\bullet}^{TT} \leq \frac{1}{2} L_{\bullet}^{EE} + \frac{1}{2} L_{\bullet}^{OO} + L_{\bullet}^{EO},$$

$$C_{\bullet}^{TT} \leq \frac{1}{2}C_{\bullet}^{EE} + \frac{1}{2}C_{\bullet}^{OO} + C_{\bullet}^{EO}.$$

Thus the canonical seven-constant package

$$(\mathcal{S}_{\text{diag}}^{TT}, L_+^{TT}, L_-^{TT}, L_0^{TT}, C_+^{TT}, C_-^{TT}, C_m^{TT})$$

is no longer an independent unknown set.

It is a finite assembled package built from the already separated $EE/OO/EO$ data.

Therefore the mixed same-shell sufficient support condition is fixed as

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{TT} + \varepsilon_{\text{small-}p,\rho}^{(1,T)}.$$

So Audit A is structurally closed.

The next nonredundant step is Audit B: first compute the principal transverse source from $TD + DT$, and only afterwards add the residual correction from $OD + DO + DD$.

CCXXXVIII. AUDIT B: EXACT PRINCIPAL TRANSVERSE PAIR FROM THE $TD + DT$ BLOCKS

CCXXXIX. GOAL

At the current stage, the sharp leading transverse audit starts from the mixed block

$$TD + DT,$$

not from the full enlarged correction family. The exact leading-order task is to extract the principal pair

$$\Lambda_E, \quad \Lambda_O,$$

from the projected one-defect first-order Hessian kernel.

CCXL. 1. CANONICAL PROJECTOR EXTRACTION

The background-dependent Hessian is expanded blockwise as

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i \mathcal{V}_{AB}^{\mu}(\xi) \partial_{y^{\mu}} + \cdots,$$

and the tangent-projected first-order kernel is

$$\mathcal{M}_{AB}^{\mu}(\xi) := \Pi_Q \mathcal{V}_{AB}^{\mu}(\xi) \Pi_Q.$$

For the transverse mixed source, define

$$\boxed{\mathcal{M}_{TD+DT}^{\mu} := \Pi_Q (\mathcal{V}_{TD}^{\mu} + \mathcal{V}_{DT}^{\mu}) \Pi_Q.} \quad (377)$$

The invariant projector functionals are

$$\boxed{\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^{\mu}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{\mu}), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^{\mu}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_{\mu}).} \quad (378)$$

Therefore the exact leading transverse coefficients are

$$\boxed{e_0^{(1)} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad o_0^{(1)} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].} \quad (379)$$

At the minimal mixed-block level one identifies

$$\boxed{\Lambda_E = e_0^{(1)}, \quad \Lambda_O = o_0^{(1)}.} \quad (380)$$

CCXLI. 2. EXPLICIT TD AND DT BLOCK FORM

For the defect background

$$P_B = z_B z_B^\dagger,$$

one has

$$\partial_\xi P_B = \theta'(\xi) \left(e_1 z_B^\dagger + z_B e_1^\dagger \right). \quad (381)$$

Hence every mixed shell-defect transport block contains one explicit factor of $\theta'(\xi)$.

The TD block takes the reduced form

$$V_{TD}^\mu = \theta'(\xi) (A_E E^\mu{}_a \Sigma_a + A_O J^\mu{}_a \Sigma_a), \quad (382)$$

and the DT block similarly takes the form

$$V_{DT}^\mu = \theta'(\xi) (B_E E^\mu{}_a \Sigma_a + B_O J^\mu{}_a \Sigma_a). \quad (383)$$

Therefore the combined principal transverse kernel is

$$\mathcal{M}_{TD+DT}^\mu = \theta'(\xi) [(A_E + B_E) E^\mu{}_a \Sigma_a + (A_O + B_O) J^\mu{}_a \Sigma_a]. \quad (384)$$

Define

$$C_E := A_E + B_E, \quad C_O := A_O + B_O. \quad (385)$$

Then

$$\mathcal{M}_{TD+DT}^\mu = \theta'(\xi) \left[C_E E^\mu{}_a \Sigma_a + C_O J^\mu{}_a \Sigma_a \right]. \quad (386)$$

CCXLII. 3. PROJECTION TO THE INVARIANT PAIR

Substituting (386) into (378), one obtains

$$e_0^{(1)} = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B} \left(\Sigma_a \theta'(\xi) [C_E E_{\mu b} \Sigma_b + C_O J_{\mu b} \Sigma_b] \right), \quad (387)$$

$$o_0^{(1)} = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B} \left(\Sigma_a \theta'(\xi) [C_E E_{\mu b} \Sigma_b + C_O J_{\mu b} \Sigma_b] \right). \quad (388)$$

Under the canonical invariant normalization used in the reduced projector formalism, the E - and J -channels decouple and the Pauli trace gives the standard branch-level result

$$e_0^{(1)} = \frac{1}{2} C_E = \frac{1}{2} (A_E + B_E), \quad (389)$$

$$o_0^{(1)} = \frac{1}{2} C_O = \frac{1}{2} (A_O + B_O). \quad (390)$$

Equivalently,

$$\Lambda_E = \frac{1}{2} (A_E + B_E), \quad \Lambda_O = \frac{1}{2} (A_O + B_O), \quad (391)$$

at the minimal $TD + DT$ level.

a. Honest status. Equations (379) and (380) are the exact leading projector formulas. Equations (389)–(391) are the current branch-level reduced realization of that exact audit.

CCXLIII. 4. DICTIONARY TO THE REDUCED PDE COEFFICIENTS

Once Λ_E, Λ_O are extracted, the reduced coefficient dictionary is

$$\boxed{\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O.} \quad (392)$$

Using

$$\boxed{I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,} \quad (393)$$

one gets

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_E, \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \Lambda_O.} \quad (394)$$

Combining with (391),

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_E + B_E), \quad \beta = -\frac{\sqrt{3}\pi}{32} \theta_0 \kappa (A_O + B_O).} \quad (395)$$

CCXLIV. 5. PLACEMENT INSIDE THE FULL MIXED-BLOCK AUDIT

Beyond the minimal truncation, the actual transverse point is

$$\boxed{\Lambda_E = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \Lambda_O = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}.} \quad (396)$$

But the sharp leading-order execution order is:

1. compute $TD + DT$ exactly;
2. define

$$v_{\text{pr}} := (\Lambda_E^{(TD+DT)}, \Lambda_O^{(TD+DT)});$$

3. only afterwards add the residual correction

$$r_{\text{rem}} := \left(\sum_{AB \in \{OD, DO, DD\}} e_{0;AB}^{(1)}, \sum_{AB \in \{OD, DO, DD\}} o_{0;AB}^{(1)} \right).$$

Thus

$$\boxed{(\Lambda_E, \Lambda_O) = v_{\text{pr}} + r_{\text{rem}},} \quad (397)$$

with

$$\boxed{v_{\text{pr}} = \left(\frac{1}{2}(A_E + B_E), \frac{1}{2}(A_O + B_O) \right)} \quad (398)$$

in the current branch-level reduced realization.

CCXLV. 6. STRONG SURVIVAL TEST AFTER THE PRINCIPAL PAIR IS COMPUTED

A strong sufficient survival condition is

$$\left| \sum_{AB \in \{OD, DO, DD\}} e_{0;AB}^{(1)} \right| < \left| \frac{1}{2}(A_E + B_E) \right|, \quad (399)$$

$$\left| \sum_{AB \in \{OD, DO, DD\}} o_{0;AB}^{(1)} \right| < \left| \frac{1}{2}(A_O + B_O) \right|. \quad (400)$$

A weaker Euclidean sufficient condition is

$$\|r_{\text{rem}}\|_2 < \|v_{\text{pr}}\|_2. \quad (401)$$

CCXLVI. CONCLUSION

The $TD + DT$ principal transverse audit is now completely reduced to the finite pair

$$(A_E + B_E, A_O + B_O).$$

At the current branch level,

$$\Lambda_E = \frac{1}{2}(A_E + B_E), \quad \Lambda_O = \frac{1}{2}(A_O + B_O),$$

hence

$$\alpha = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_E + B_E), \quad \beta = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_O + B_O).$$

So the next nonredundant microscopic task is no longer structural: it is the explicit evaluation of the residual correction vector from

$$OD, \quad DO, \quad DD.$$

CCXLVII. FINAL SUMMARY

The principal transverse audit is now reduced exactly to the projected mixed block

$$\mathcal{M}_{TD+DT}^\mu = \Pi_Q(\mathcal{V}_{TD}^\mu + \mathcal{V}_{DT}^\mu)\Pi_Q.$$

Its leading transverse coefficient point is extracted by

$$\Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].$$

At the current branch-level reduced realization,

$$V_{TD}^\mu = \theta'(\xi)(A_E E^\mu{}_a \Sigma_a + A_O J^\mu{}_a \Sigma_a),$$

$$V_{DT}^\mu = \theta'(\xi)(B_E E^\mu{}_a \Sigma_a + B_O J^\mu{}_a \Sigma_a),$$

so the principal pair is

$$\Lambda_E = \frac{1}{2}(A_E + B_E), \quad \Lambda_O = \frac{1}{2}(A_O + B_O).$$

Therefore

$$\alpha = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_E + B_E), \quad \beta = -\frac{\sqrt{3}\pi}{32}\theta_0\kappa(A_O + B_O).$$

Beyond the minimal truncation, the full transverse point is

$$\Lambda_E = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \Lambda_O = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}.$$

But the sharp execution order is: first compute $TD + DT$, then add the residual correction from OD, DO, DD .

Thus the next nonredundant task is the explicit computation of

$$r_{\text{rem}} = \left(\sum_{AB \in \{OD, DO, DD\}} e_{0;AB}^{(1)}, \sum_{AB \in \{OD, DO, DD\}} o_{0;AB}^{(1)} \right),$$

and the survival test

$$(\Lambda_E, \Lambda_O) = v_{\text{pr}} + r_{\text{rem}} \neq (0, 0).$$

CCXLVIII. RESIDUAL CORRECTION VECTOR FROM $OD + DO + DD$: EXACT PROJECTOR FORMULATION

CCXLIX. GOAL

The principal transverse source has already been isolated as

$$TD + DT,$$

with leading coefficient point

$$\Lambda_E^{\text{pr}} = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O^{\text{pr}} = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].$$

What remains is the residual correction carried by

$$OD, \quad DO, \quad DD.$$

The correct task is therefore not to invent a new mechanism, but to write the exact correction vector

$$r_{\text{rem}}$$

in the same projector language and derive sharp sufficient non-cancellation criteria.

CCL. 1. EXACT BLOCKWISE RESIDUAL COEFFICIENTS

For every candidate transverse block

$$AB \in \{OD, DO, DD\},$$

define the tangent-projected first-order kernel

$$\boxed{\mathcal{M}_{AB}^\mu(\xi) := \Pi_Q \mathcal{V}_{AB}^\mu(\xi) \Pi_Q.} \quad (402)$$

The invariant projector functionals are

$$\boxed{\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}_{Q_B}(\Sigma_a \mathcal{M}_\mu).} \quad (403)$$

Hence the exact residual blockwise coefficients are

$$\boxed{e_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \mathfrak{P}_E[\mathcal{M}_{AB}^{\text{inv}}], \quad o_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \mathfrak{P}_O[\mathcal{M}_{AB}^{\text{inv}}],} \quad (404)$$

for $AB \in \{OD, DO, DD\}$.

This is the exact same executable definition already used for the full mixed-block audit.

CCLI. 2. EXACT RESIDUAL CORRECTION VECTOR

Define the residual correction pair by

$$\Delta e_{\text{rem}} := e_{0;OD}^{(1)} + e_{0;DO}^{(1)} + e_{0;DD}^{(1)}, \quad (405)$$

$$\Delta o_{\text{rem}} := o_{0;OD}^{(1)} + o_{0;DO}^{(1)} + o_{0;DD}^{(1)}. \quad (406)$$

Equivalently, in vector form,

$$r_{\text{rem}} := (\Delta e_{\text{rem}}, \Delta o_{\text{rem}}) = \sum_{AB \in \{OD, DO, DD\}} (e_{0;AB}^{(1)}, o_{0;AB}^{(1)}). \quad (407)$$

This is the exact residual correction vector.

CCLII. 3. PRINCIPAL-PLUS-RESIDUAL SPLIT

The full transverse coefficient point is

$$\Lambda_E = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \Lambda_O = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}. \quad (408)$$

Define the principal pair

$$r_{\text{pr}} := (\Lambda_E^{\text{pr}}, \Lambda_O^{\text{pr}}) = (e_{0;TD}^{(1)} + e_{0;DT}^{(1)}, o_{0;TD}^{(1)} + o_{0;DT}^{(1)}). \quad (409)$$

Then the full transverse point decomposes exactly as

$$(\Lambda_E, \Lambda_O) = r_{\text{pr}} + r_{\text{rem}}. \quad (410)$$

At the current conservative stage one already uses

$$e_0^{(1)} \approx e_{0;K}^{(1)}, \quad o_0^{(1)} \approx o_{0;K}^{(1)},$$

so the same split may be written as

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}, \quad (411)$$

with the understanding that the residual part is precisely the $OD+DO+DD$ completion of the mixed-block correction family.

CCLIII. 4. STRONG SUFFICIENT SURVIVAL CRITERIA

A strong componentwise sufficient condition is

$$|\Delta e_{\text{rem}}| < |\Lambda_E^{\text{pr}}|, \quad |\Delta o_{\text{rem}}| < |\Lambda_O^{\text{pr}}|. \quad (412)$$

Under this condition, the residual blocks cannot kill the principal transverse witness pair componentwise.

At the conservative K -sector stage this reduces to

$$|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|. \quad (413)$$

This is exactly the already-stated strong sufficient survival condition.

CCLIV. 5. EUCLIDEAN SUFFICIENT SURVIVAL CRITERION

The componentwise condition is strong but not necessary. A weaker rotationally invariant sufficient criterion follows from the triangle inequality.

If

$$\|r_{\text{rem}}\|_2 < \|r_{\text{pr}}\|_2, \quad (414)$$

then

$$r_{\text{pr}} + r_{\text{rem}} \neq (0, 0), \quad (415)$$

hence

$$(\Lambda_E, \Lambda_O) \neq (0, 0). \quad (416)$$

Indeed,

$$\|r_{\text{pr}} + r_{\text{rem}}\|_2 \geq \left| \|r_{\text{pr}}\|_2 - \|r_{\text{rem}}\|_2 \right| > 0.$$

Thus the Euclidean criterion is a sharp and basis-independent sufficient test for survival of the transverse pair.

CCLV. 6. NORM BOUNDS IN TERMS OF BLOCKWISE BUDGETS

Define the blockwise magnitudes

$$\rho_{AB} := \sqrt{(e_{0;AB}^{(1)})^2 + (o_{0;AB}^{(1)})^2}, \quad AB \in \{OD, DO, DD\}. \quad (417)$$

Then by the triangle inequality,

$$\|r_{\text{rem}}\|_2 \leq \rho_{OD} + \rho_{DO} + \rho_{DD}. \quad (418)$$

Therefore a practical sufficient condition for transverse survival is

$$\rho_{OD} + \rho_{DO} + \rho_{DD} < \|r_{\text{pr}}\|_2. \quad (419)$$

This reduces the correction audit to three finite blockwise magnitude estimates.

CCLVI. 7. DICTIONARY TO THE REDUCED PDE COEFFICIENTS

Once the full transverse point is known,

$$\alpha = I_\theta \Lambda_E, \quad \beta = I_\theta \Lambda_O. \quad (420)$$

Using

$$I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa,$$

one gets

$$\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \Lambda_E, \quad \beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa \Lambda_O. \quad (421)$$

Combining with (410),

$$\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa (\Lambda_E^{\text{pr}} + \Delta e_{\text{rem}}), \quad (422)$$

$$\beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_O^{\text{pr}} + \Delta o_{\text{rem}}\right). \quad (423)$$

In the conservative K -sector notation this is

$$\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(e_{0;K}^{(1)} + \Delta e_{\text{mix}}\right), \quad \beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(o_{0;K}^{(1)} + \Delta o_{\text{mix}}\right). \quad (424)$$

CCLVII. 8. HONEST CONCLUSION

What is now fully closed is the exact residual correction formalism:

1. blockwise exact definitions of $e_{0;AB}^{(1)}, o_{0;AB}^{(1)}$ for $AB = OD, DO, DD$;
2. the exact residual vector r_{rem} ;
3. the exact principal-plus-residual split

$$(\Lambda_E, \Lambda_O) = r_{\text{pr}} + r_{\text{rem}};$$

4. strong and Euclidean sufficient survival conditions.

What is not yet inserted numerically is the explicit microscopic evaluation of

$$\mathcal{M}_{OD}^{\text{inv}}, \quad \mathcal{M}_{DO}^{\text{inv}}, \quad \mathcal{M}_{DD}^{\text{inv}}.$$

So the project is now reduced to three literal blockwise projector computations.

CCLVIII. FINAL SUMMARY

The residual transverse correction is now fully reduced to the three blockwise projector computations

$$OD, \quad DO, \quad DD.$$

For each residual block $AB \in \{OD, DO, DD\}$, define

$$e_{0;AB}^{(1)} = \lim \mathfrak{P}_E[\mathcal{M}_{AB}^{\text{inv}}], \quad o_{0;AB}^{(1)} = \lim \mathfrak{P}_O[\mathcal{M}_{AB}^{\text{inv}}].$$

Then the exact residual correction vector is

$$r_{\text{rem}} = (\Delta e_{\text{rem}}, \Delta o_{\text{rem}}) = \sum_{AB \in \{OD, DO, DD\}} (e_{0;AB}^{(1)}, o_{0;AB}^{(1)}).$$

The full transverse coefficient point splits exactly as

$$(\Lambda_E, \Lambda_O) = r_{\text{pr}} + r_{\text{rem}},$$

where r_{pr} is the already isolated principal $TD + DT$ source.

Hence the strong componentwise sufficient survival condition is

$$|\Delta e_{\text{rem}}| < |\Lambda_E^{\text{pr}}|, \quad |\Delta o_{\text{rem}}| < |\Lambda_O^{\text{pr}}|.$$

A weaker Euclidean sufficient condition is

$$\|r_{\text{rem}}\|_2 < \|r_{\text{pr}}\|_2.$$

Finally, the reduced PDE coefficients are

$$\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_E^{\text{pr}} + \Delta e_{\text{rem}}\right), \quad \beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left(\Lambda_O^{\text{pr}} + \Delta o_{\text{rem}}\right).$$

Thus the project is now reduced to three literal residual block computations:

$$\mathcal{M}_{OD}^{\text{inv}}, \quad \mathcal{M}_{DO}^{\text{inv}}, \quad \mathcal{M}_{DD}^{\text{inv}}.$$

No further structural theorem is needed.

CCLIX. RESIDUAL-BLOCK CLOSURE BY PARITY AUDIT: OD, DO, DD

CCLX. SHORT EXPLANATION

At the current stage, the honest status is:

$$\boxed{TD + DT \text{ are the proven leading transverse source blocks,}} \quad (425)$$

while

$$\boxed{OD, DO, DD \text{ are candidate residual corrections beyond the minimal truncation.}} \quad (426)$$

Therefore the correct next step is not to assert that OD, DO, DD vanish identically, but to derive a sharp *selection-rule audit* telling us exactly when they vanish and when they must be computed.

CCLXI. 1. EXACT RESIDUAL-BLOCK DEFINITIONS

For each residual block

$$AB \in \{OD, DO, DD\},$$

define the tangent-projected first-order kernel

$$\boxed{\mathcal{M}_{AB}^\mu(\xi) := \Pi_Q \mathcal{V}_{AB}^\mu(\xi) \Pi_Q.} \quad (427)$$

The executable projector formulas are

$$\boxed{e_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \mathfrak{P}_E[\mathcal{M}_{AB, \text{inv}}], \quad o_{0;AB}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \mathfrak{P}_O[\mathcal{M}_{AB, \text{inv}}],} \quad (428)$$

with

$$\boxed{\mathfrak{P}_E[\mathcal{M}] = \frac{1}{4} E^\mu{}_a \text{Tr}(\Sigma_a \mathcal{M}_\mu), \quad \mathfrak{P}_O[\mathcal{M}] = \frac{1}{4} J^\mu{}_a \text{Tr}(\Sigma_a \mathcal{M}_\mu).} \quad (429)$$

Thus the exact residual correction vector is

$$\boxed{r_{\text{rem}} = (\Delta e_{\text{rem}}, \Delta o_{\text{rem}}) = \sum_{AB \in \{OD, DO, DD\}} (e_{0;AB}^{(1)}, o_{0;AB}^{(1)}).} \quad (430)$$

CCLXII. 2. REFLECTION-PARITY SELECTION RULE

For a reflection-symmetric defect background, the low modes satisfy

$$u_a^+(-\xi) = +u_a^+(\xi), \quad u_a^-(-\xi) = -u_a^-(\xi),$$

and the projected mixed symbol is

$$(M_{AB}^\mu)_{ab} = \int d\xi (u_a^+)^\dagger(\xi) \mathcal{M}_{AB}^\mu(\xi) u_b^-(\xi).$$

Now the integrand parity is determined entirely by the parity of $\mathcal{M}_{AB}^\mu$:

- if

$$\mathcal{M}_{AB}^\mu(-\xi) = +\mathcal{M}_{AB}^\mu(\xi),$$

then

$$(u_a^+)^\dagger \mathcal{M}_{AB}^\mu u_b^-$$

is odd and integrates to zero;

- if

$$\mathcal{M}_{AB}^\mu(-\xi) = -\mathcal{M}_{AB}^\mu(\xi),$$

then

$$(u_a^+)^\dagger \mathcal{M}_{AB}^\mu u_b^-$$

is even and can survive.

Therefore:

Lemma CCLXII.1 (Residual parity-vanishing criterion). *For any block $AB \in \{OD, DO, DD\}$, if the projected first-order kernel is reflection-even,*

$$\boxed{\mathcal{M}_{AB,\text{inv}}^\mu(-\xi) = +\mathcal{M}_{AB,\text{inv}}^\mu(\xi),} \quad (431)$$

then

$$\boxed{e_{0;AB}^{(1)} = 0, \quad o_{0;AB}^{(1)} = 0.} \quad (432)$$

Proof. Because u^+ is even and u^- is odd, an even kernel produces an odd integrand, hence every mixed matrix element vanishes. The projector contractions $\mathfrak{P}_E, \mathfrak{P}_O$ are linear in those matrix elements, so both coefficients vanish. $\square$

Corollary CCLXII.2 (Residual survival criterion). *If instead*

$$\boxed{\mathcal{M}_{AB,\text{inv}}^\mu(-\xi) = -\mathcal{M}_{AB,\text{inv}}^\mu(\xi),} \quad (433)$$

then the block is parity-allowed to contribute:

$$\boxed{e_{0;AB}^{(1)}, o_{0;AB}^{(1)} \text{ need not vanish and must be computed explicitly.}} \quad (434)$$

CCLXIII. 3. WHY $TD + DT$ ARE LEADING WHILE OD, DO, DD ARE RESIDUAL

The reason the sharp current audit starts from $TD + DT$ is structural:

1. the transverse source is generated by blocks mixing shell-gradient/current data with defect/projector/tangent data;
2. the explicit current Class II source splits into a transverse part and a primitive longitudinal part;
3. the leading transverse kernel is therefore assigned to $TD + DT$;
4. OD, DO, DD are allowed as corrections, but not yet established as leading.

So the correct hierarchy is

$$\boxed{TD + DT \text{ first,} \quad OD + DO + DD \text{ second.}} \quad (435)$$

CCLXIV. 4. BLOCKWISE AUDIT LOGIC FOR OD AND DO

The OD -type sector has already been reduced in earlier work to an explicit projected self-energy derivative test. In particular, under $O \leftrightarrow D$ exchange symmetry one obtains a 2×2 reduced subsector

$$M_Z^{OD} = \begin{pmatrix} (M_Z^{OD})_{OO} & (M_Z^{OD})_{OD} \\ (M_Z^{OD})_{DO} & (M_Z^{OD})_{DD} \end{pmatrix},$$

with the exchange-symmetric reduction

$$(M_Z^{OD})_{OO} = (M_Z^{OD})_{DD}, \quad (M_Z^{OD})_{OD} = (M_Z^{OD})_{DO}.$$

Then the physical traceless mode is controlled by

$$\boxed{\lambda_{OD}^{\text{phys}} = (M_Z^{OD})_{OO} - (M_Z^{OD})_{OD}.} \quad (436)$$

This is not yet the same thing as the final transverse pair $(e_{0;OD}^{(1)}, o_{0;OD}^{(1)})$, but it tells us the honest current status:

$$\boxed{OD/DO \text{ are not theorem-level zero.}} \quad (437)$$

What is closed is only that their effect has been reduced to an explicit projected contraction/sign test.

CCLXV. 5. BLOCKWISE AUDIT LOGIC FOR DD

The present block assignment states that

$$DD$$

can

$$\boxed{\text{renormalize or directly support the final } Q_B\text{-endomorphism-valued doublet structure after Schur reduction.}} \quad (438)$$

Therefore the honest statement is again:

$$\boxed{DD \text{ is not proved to vanish identically.}} \quad (439)$$

However, by the parity criterion above, DD vanishes automatically if its projected first-order kernel is reflection-even:

$$\mathcal{M}_{DD,\text{inv}}^\mu(-\xi) = +\mathcal{M}_{DD,\text{inv}}^\mu(\xi) \implies e_{0;DD}^{(1)} = o_{0;DD}^{(1)} = 0.$$

So the DD -audit is reduced to one binary question:

$$\boxed{\text{is } \mathcal{M}_{DD,\text{inv}}^\mu \text{ even or odd under } \xi \mapsto -\xi?}$$

CCLXVI. 6. EXACT RESIDUAL CLASSIFICATION TABLE

The sharp blockwise status is therefore

block	current theorem-level status	next exact test
OD	candidate correction, not proved zero	$\mathcal{M}_{OD,\text{inv}}^\mu(-\xi) \stackrel{?}{=} \pm \mathcal{M}_{OD,\text{inv}}^\mu(\xi)$
DO	candidate correction, not proved zero	$\mathcal{M}_{DO,\text{inv}}^\mu(-\xi) \stackrel{?}{=} \pm \mathcal{M}_{DO,\text{inv}}^\mu(\xi)$
DD	candidate correction, not proved zero	$\mathcal{M}_{DD,\text{inv}}^\mu(-\xi) \stackrel{?}{=} \pm \mathcal{M}_{DD,\text{inv}}^\mu(\xi)$

If the answer is $+$ for a given block, its contribution vanishes exactly. If the answer is $-$, its contribution is parity-allowed and must be computed.

CCLXVII. 7. FINAL RESIDUAL CLOSURE STATEMENT

Hence the final residual correction vector splits as follows.

a. Case I: full even residual sector. If all three residual kernels are reflection-even,

$$\boxed{\mathcal{M}_{OD,\text{inv}}^\mu, \mathcal{M}_{DO,\text{inv}}^\mu, \mathcal{M}_{DD,\text{inv}}^\mu \text{ are all even,}} \quad (440)$$

then

$$\boxed{r_{\text{rem}} = 0.} \quad (441)$$

b. Case II: partial even/odd mixture. If only some are even, then only those vanish. The remaining odd blocks form the true residual correction:

$$r_{\text{rem}} = \sum_{\substack{AB \in \{OD, DO, DD\} \\ \mathcal{M}_{AB} \text{ odd}}} (e_{0;AB}^{(1)}, o_{0;AB}^{(1)}).$$

c. Case III: no parity cancellation. If all three are odd, then no residual block is killed by parity and the full

$$OD + DO + DD$$

projector computation is required.

CCLXVIII. 8. DICTIONARY TO THE FINAL REDUCED COEFFICIENTS

The final transverse coefficient point is

$$\boxed{(\Lambda_E, \Lambda_O) = r_{\text{pr}} + r_{\text{rem}},} \quad (442)$$

hence

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa(\Lambda_E^{\text{pr}} + \Delta e_{\text{rem}}),} \quad (443)$$

$$\boxed{\beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa(\Lambda_O^{\text{pr}} + \Delta o_{\text{rem}}).} \quad (444)$$

So the final microscopic closure question for the transverse sector has now been reduced to a parity audit followed, only if needed, by literal blockwise projection.

CCLXIX. FINAL SUMMARY

The residual mixed blocks OD, DO, DD are not theorem-level zero at the current stage. They are candidate corrections beyond the proven leading $TD + DT$ transverse source.

However, there is now an exact parity-based vanishing criterion.

For any residual block $AB \in \{OD, DO, DD\}$, if

$$\mathcal{M}_{AB,\text{inv}}^\mu(-\xi) = +\mathcal{M}_{AB,\text{inv}}^\mu(\xi),$$

then

$$e_{0;AB}^{(1)} = o_{0;AB}^{(1)} = 0.$$

If instead

$$\mathcal{M}_{AB,\text{inv}}^\mu(-\xi) = -\mathcal{M}_{AB,\text{inv}}^\mu(\xi),$$

then the block is parity-allowed to contribute and must be computed explicitly.

Thus the residual transverse closure is reduced to a three-block parity audit:

$$OD, \quad DO, \quad DD.$$

If all three projected kernels are even, then

$$r_{\text{rem}} = 0.$$

Otherwise, only the parity-odd subset survives and contributes to

$$(\Lambda_E, \Lambda_O) = r_{\text{pr}} + r_{\text{rem}}.$$

Therefore the final transverse microscopic closure is no longer a structural problem. It is a literal blockwise parity classification followed, if needed, by explicit projector evaluation.

CCLXX. RESIDUAL-BLOCK AUDIT I: HONEST STATUS OF THE OD BLOCK

CCLXXI. GOAL

The immediate question is whether the residual correction block

$$OD$$

is automatically killed by the already-closed parity/oddness selection rule, or whether it remains a genuine explicit computation.

CCLXXII. 1. WHY THE PREVIOUSLY CLOSED ODDNESS THEOREM DOES *NOT* AUTOMATICALLY KILL OD

The already-closed parity fact for the Class-II working branch states that first-order transport contributions extracted from the linearized

$$J^{(1)}, \quad K^{(1)}$$

sector are automatically odd in ξ , because the defect-background gradient enters through

$$\partial_\xi P_B, \quad \partial_\xi z_B, \quad \theta'(\xi).$$

Therefore the corresponding first-order transport source belongs to the odd transport sector.

However, the OD -mixed block is not introduced in the current notes as a direct $J^{(1)}/K^{(1)}$ transport source. Instead, it is generated from the mixed quartic interaction

$$\lambda_{OD} \rho_O \rho_D$$

on the background

$$\bar{\Phi}_{\text{min}} = (0, \bar{n}_B(\xi), \bar{\chi}_B(\xi)).$$

Expanding this interaction gives the background-generated cubic vertices

$$\boxed{\Gamma_{\nu\nu\zeta}^{(3)}[\bar{\Phi}_{\text{min}}] \sim \lambda_{OD} \bar{\chi}_B(\xi), \quad \Gamma_{\nu\zeta\zeta}^{(3)}[\bar{\Phi}_{\text{min}}] \sim \lambda_{OD} \bar{n}_B(\xi).} \quad (445)$$

These vertices then feed the projected bubble self-energies

$$\Sigma_O^{(OD)}(p; \bar{\Phi}_{\text{min}}), \quad \Sigma_D^{(OD)}(p; \bar{\Phi}_{\text{min}}), \quad (446)$$

rather than the already-isolated direct odd transport source.

Therefore the existing oddness theorem for the direct transport sector does not, by itself, imply

$$OD = 0.$$

CCLXXIII. 2. EXACT CURRENT MEANING OF THE OD CONTRIBUTION

The notes reduce the OD -mixed piece to the explicit kinetic-flow matrix

$$(M_Z^{OD})_{IJ} = - \left. \frac{\partial}{\partial \delta_J} \frac{\partial}{\partial p^2} \Sigma_I^{(OD)}(p; \bar{\Phi}_{\min}) \right|_{p=0, \delta=0}, \quad I, J \in \{O, D\}. \quad (447)$$

Hence the OD -block is already known to be a genuine projected self-energy derivative test. Its sign is not yet theorem-level fixed by the currently available truncation.

So the honest statement is

$$\boxed{OD \text{ is not proved to vanish identically.}} \quad (448)$$

CCLXXIV. 3. EXCHANGE-SYMMETRIC REDUCTION

If the projected background and regulator preserve

$$O \leftrightarrow D$$

exchange symmetry, then

$$\boxed{(M_Z^{OD})_{OO} = (M_Z^{OD})_{DD}, \quad (M_Z^{OD})_{OD} = (M_Z^{OD})_{DO}.} \quad (449)$$

In that case the physical traceless mode is controlled by the single eigenvalue

$$\boxed{\lambda_{OD}^{\text{phys}} = (M_Z^{OD})_{OO} - (M_Z^{OD})_{OD}.} \quad (450)$$

Thus the entire OD -mixed bottleneck reduces to one explicit inequality:

$$\boxed{(M_Z^{OD})_{OO} - (M_Z^{OD})_{OD} < 0} \quad (451)$$

if one wants the OD sector to be isotropizing rather than destabilizing.

CCLXXV. 4. CONSEQUENCE FOR THE RESIDUAL TRANSVERSE AUDIT

Because OD is not killed automatically by the already-closed odd transport theorem, the correction pair

$$(e_{0;OD}^{(1)}, o_{0;OD}^{(1)})$$

must be regarded as parity-allowed unless one performs an additional block-specific parity computation for the actual projected kernel.

Hence the current honest classification is:

$$\boxed{OD \text{ is a candidate residual correction that must be explicitly computed.}}$$

In particular, one must *not* replace it by zero merely by citing the direct $\theta'(\xi)$ -oddness of the J/K -generated first-order transport source.

CCLXXVI. 5. WHAT IS ACTUALLY CLOSED NOW

The strongest closed statement is therefore:

1. the already-proved direct odd transport source controls the leading $TD + DT$ audit;
2. the OD -mixed block is structurally different, because it is generated through background-dependent cubic vertices and projected bubble self-energies;
3. its sign and magnitude are reduced to the explicit projected contraction (447);
4. under $O \leftrightarrow D$ symmetry, the full OD -question reduces to the single inequality (451).

Thus OD is not conceptually open any more, but it is still computationally open.

CCLXXVII. FINAL SUMMARY

The OD -block is not automatically killed by the already-closed direct odd transport theorem.

That theorem applies to the direct first-order transport source generated from the background-gradient structure

$$\partial_\xi P_B, \quad \partial_\xi z_B, \quad \theta'(\xi),$$

inside the J/K -based transport sector.

By contrast, the OD -mixed block arises from the mixed quartic interaction

$$\lambda_{OD}\rho_O\rho_D,$$

which generates the background-dependent cubic vertices

$$\Gamma_{\nu\nu\zeta}^{(3)} \sim \lambda_{OD}\bar{\chi}_B(\xi), \quad \Gamma_{\nu\zeta\zeta}^{(3)} \sim \lambda_{OD}\bar{n}_B(\xi),$$

and therefore feeds a projected bubble self-energy derivative test.

Hence the honest exact current statement is

$$\boxed{OD \text{ is not theorem-level zero.}}$$

Its contribution is reduced to the explicit projected contraction

$$(M_Z^{OD})_{IJ} = - \frac{\partial}{\partial \delta_J} \frac{\partial}{\partial p^2} \Sigma_I^{(OD)}(p; \bar{\Phi}_{\min}) \Big|_{p=0, \delta=0}, \quad I, J \in \{O, D\}.$$

If $O \leftrightarrow D$ exchange symmetry holds, the problem reduces further to the single inequality

$$\lambda_{OD}^{\text{phys}} = (M_Z^{OD})_{OO} - (M_Z^{OD})_{OD} < 0.$$

Therefore the OD -block is a genuine residual correction that must be explicitly computed, not discarded by automatic parity cancellation.

CCLXXVIII. RESIDUAL-BLOCK AUDIT II: HONEST STATUS OF THE DO BLOCK

CCLXXIX. GOAL

After the OD -audit, the next question is whether the residual block

$$DO$$

is automatically killed, or whether it is simply the exchange partner of OD and therefore remains an explicit projected correction.

CCLXXX. 1. DO IS ALREADY PART OF THE EXACT MIXED-BLOCK AUDIT

The exact transverse candidate family is

$$\boxed{TD, DT, OD, DO, DD.} \tag{452}$$

The practical audit order is explicitly fixed as

$$TD, DT, OD, DO, DD.$$

Therefore DO is not a block that has been removed by a theorem-level symmetry argument. It is already built into the exact mixed-block audit architecture.

Hence the honest starting point is

$$\boxed{DO \text{ is a candidate residual correction block.}} \tag{453}$$

CCLXXXI. 2. WHY THE DIRECT ODD-TRANSPORT THEOREM DOES NOT AUTOMATICALLY KILL DO

The already-closed direct oddness theorem applies to the explicit first-order transport source generated through the defect-gradient structures

$$\partial_\xi P_B, \quad \partial_\xi z_B, \quad \theta'(\xi),$$

in the direct J/K -generated transport sector.

But DO is not introduced in the current audit as a direct isolated $\theta'(\xi)$ -transport source. It belongs to the same mixed residual family as OD , namely the correction blocks beyond the proven leading source $TD + DT$.

Therefore the previously closed direct oddness theorem does not, by itself, imply

$$DO = 0.$$

CCLXXXII. 3. EXCHANGE SYMMETRY WITH OD

In the symmetry-reduced mixed OD analysis, the projected flow matrix satisfies

$$\boxed{(M_Z^{OD})_{OO} = (M_Z^{OD})_{DD}, \quad (M_Z^{OD})_{OD} = (M_Z^{OD})_{DO}}, \quad (454)$$

under $O \leftrightarrow D$ exchange symmetry.

This has a direct consequence:

$$\boxed{\text{if the projected background and regulator preserve } O \leftrightarrow D, \text{ then } DO \text{ is not independent of } OD.} \quad (455)$$

In that case, the DO -block is the exchange partner of the OD -block, and both are controlled by the same reduced projected test.

Thus the honest theorem-level statement is

$$\boxed{DO \text{ is not proved to vanish identically,}} \quad (456)$$

but rather

$$\boxed{DO \text{ is exchange-equivalent to } OD \text{ whenever } O \leftrightarrow D \text{ symmetry holds.}} \quad (457)$$

CCLXXXIII. 4. CONSEQUENCE FOR THE RESIDUAL TRANSVERSE PAIR

Therefore the correction pair

$$(e_{0;DO}^{(1)}, o_{0;DO}^{(1)})$$

must be treated as parity-allowed unless one performs either:

1. a separate blockwise parity computation for $\mathcal{M}_{DO,inv}^\mu$, or
2. an exchange-symmetry reduction identifying it with the corresponding OD contribution.

Hence the strongest honest current classification is:

$$\boxed{DO \text{ is a genuine residual correction block that must be explicitly accounted for.}} \quad (458)$$

CCLXXXIV. 5. MOST ECONOMICAL REDUCTION UNDER EXCHANGE SYMMETRY

If the $O \leftrightarrow D$ exchange symmetry is exact in the projected mixed sector, then one may reduce the residual pair by

$$\boxed{e_{0;DO}^{(1)} = e_{0;OD}^{(1)}, \quad o_{0;DO}^{(1)} = o_{0;OD}^{(1)},} \quad (459)$$

at the symmetry-reduced level.

Then the residual vector becomes

$$\boxed{r_{\text{rem}} = 2(e_{0;OD}^{(1)}, o_{0;OD}^{(1)}) + (e_{0;DD}^{(1)}, o_{0;DD}^{(1)})}. \quad (460)$$

This is the sharpest reduction available without overclaiming blockwise vanishing.

CCLXXXV. CONCLUSION

The correct conclusion is not that DO vanishes. The correct conclusion is:

1. DO remains inside the exact residual mixed-block audit;
2. the direct odd-transport theorem does not eliminate it automatically;
3. under $O \leftrightarrow D$ symmetry, it is exchange-equivalent to OD ;
4. otherwise it must be computed separately.

So DO is computationally open but structurally reduced.

CCLXXXVI. FINAL SUMMARY

The DO -block is not theorem-level zero.

It is already part of the exact mixed-block transverse audit:

$$TD, DT, OD, DO, DD.$$

Therefore it cannot be discarded automatically.

Moreover, the already-closed direct odd transport theorem does not eliminate DO , because that theorem applies to the direct J/K -generated first-order transport source, whereas DO belongs to the residual mixed-block correction family.

If the projected mixed sector preserves $O \leftrightarrow D$ exchange symmetry, then

$$(M_Z^{OD})_{OD} = (M_Z^{OD})_{DO},$$

so DO is exchange-equivalent to OD . In that symmetry-reduced case one may write

$$e_{0;DO}^{(1)} = e_{0;OD}^{(1)}, \quad o_{0;DO}^{(1)} = o_{0;OD}^{(1)},$$

and therefore

$$r_{\text{rem}} = 2(e_{0;OD}^{(1)}, o_{0;OD}^{(1)}) + (e_{0;DD}^{(1)}, o_{0;DD}^{(1)}).$$

Thus the honest current status is:

$$\boxed{DO \text{ is a genuine residual correction block,}}$$

but

$$\boxed{\text{under}} O \leftrightarrow D \text{ symmetry, it is not independent of } OD.$$

So the next irreducible task is the DD -audit.

CCLXXXVII. RESIDUAL-BLOCK AUDIT III: HONEST FINAL STATUS OF THE DD BLOCK

CCLXXXVIII. GOAL

The final residual question is whether the block

$$DD$$

is automatically absent, or whether it remains a genuine transverse correction even after all currently closed reductions.

CCLXXXIX. 1. DD SURVIVES EVERY CURRENT REDUCTION LAYER

Across the currently available notes, there is a mild drift in how aggressively the residual mixed-block audit is compressed.

In the broad mixed-block audit, the candidate transverse family is

$$\boxed{TD, DT, OD, DO, DD.} \quad (461)$$

But in the conservative near-final full-Hessian snapshot, the remaining unresolved blocks are reduced to

$$\boxed{TT, TD, DT, DD.} \quad (462)$$

This is an important consistency fact: even when the audit is compressed conservatively, the DD -block still survives. Therefore the honest theorem-level statement is

$$\boxed{DD \text{ is not a dispensable block in the present microscopic program.}} \quad (463)$$

CCXC. 2. WHY DD IS STRUCTURALLY DIFFERENT FROM OD AND DO

The DD -block acts entirely inside the defect-tangent doublet sector. Hence, after tangent projection, it directly defines a Q_B -endomorphism-valued first-order kernel.

Write

$$\boxed{\mathcal{M}_{DD,\text{inv}}^\mu = a_{\parallel;DD}(\theta) n^\mu \sigma_3 + a_{E;DD}(\theta) E^\mu{}_a \sigma_a + a_{O;DD}(\theta) J^\mu{}_a \sigma_a.} \quad (464)$$

Therefore the executable projector formulas are

$$\boxed{e_{0;DD}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} E^\mu{}_a \text{Tr}(\sigma_a \mathcal{M}_\mu^{DD}),} \quad (465)$$

$$\boxed{o_{0;DD}^{(1)} = \lim_{\theta, \theta', \dots \rightarrow 0} \frac{1}{4} J^\mu{}_a \text{Tr}(\sigma_a \mathcal{M}_\mu^{DD}).} \quad (466)$$

So DD is not merely an exchange partner of some other block. It carries its own direct transverse projection channels.

CCXCI. 3. HONEST PHYSICAL ROLE OF THE DD -BLOCK

The current notes already isolate the two structurally allowed roles of DD :

1. direct support of the transverse doublet;
2. renormalization of the already-induced TD/DT transport.

This means that the correct current statement is not

$$DD = 0,$$

but rather

$$\boxed{DD \text{ is a genuine transverse candidate,}} \quad (467)$$

with the additional structural expectation that

$$\boxed{DD \text{ is more likely to renormalize the mixed-shell carrier than to generate it from nothing.}} \quad (468)$$

The second statement is an honest structural expectation from the present notes, not yet a theorem of vanishing or dominance.

CCXCII. 4. WHY THERE IS NO AUTOMATIC PARITY-KILLING THEOREM FOR DD AT PRESENT

A parity-killing theorem would require a closed proof that

$$\mathcal{M}_{DD,\text{inv}}^\mu(-\xi) = +\mathcal{M}_{DD,\text{inv}}^\mu(\xi)$$

on the actual projected one-defect background.

But the current background-dependent Hessian library does not provide such a theorem. Instead, the DD -sector contains

$$\mathcal{H}_{DD} = -Z_{D,k}\partial^2 + \mathcal{M}_{DD}^{\text{pot}}(\xi) + Y_k\mathcal{K}_{DD}^{\nabla\text{ani}}(\xi, \partial) + \mathcal{T}_{DD}(\xi, \partial_y),$$

so the actual first-order kernel inherits contributions from:

1. the defect-background-dependent potential tensor,
2. derivative-anisotropy terms,
3. the explicit first-order transport operator $\mathcal{T}_{DD}$.

Hence the parity of $\mathcal{M}_{DD,\text{inv}}^\mu$ is not theorem-level fixed by the current notes. Therefore the strongest honest statement is

$$\boxed{DD \text{ is not proved to vanish by parity.}} \quad (469)$$

CCXCIII. 5. EXACT PLACE OF DD INSIDE THE FINAL TRANSVERSE PAIR

The exact full transverse coefficient point is

$$\Lambda_E = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad \Lambda_O = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}. \quad (470)$$

In the sharper conservative reduction, this becomes

$$\boxed{e_0^{(1)} = e_{0;K}^{(1)} + e_{0;TD}^{(1)} + e_{0;DT}^{(1)} + e_{0;DD}^{(1)},} \quad (471)$$

$$\boxed{o_0^{(1)} = o_{0;K}^{(1)} + o_{0;TD}^{(1)} + o_{0;DT}^{(1)} + o_{0;DD}^{(1)}. \quad (472)$$

So among all residual blocks, DD is the one that persists even in the conservative compression of the audit.

CCXCIV. 6. STRONGEST HONEST DD -CLASSIFICATION

Combining all currently closed facts, the sharp classification is:

 DD is not theorem-level zero,

 DD has direct projector-visible transverse channels (E, O) ,

 DD survives even the conservative reduced audit,

and therefore

 DD must be explicitly computed unless a new block-specific parity theorem is proved.

CCXCV. 7. FINAL CLOSURE STATEMENT FOR THE RESIDUAL SECTOR

The residual sector is now sharply separated into two layers.

a. Exchange-type residuals. The blocks OD and DO are residual mixed blocks, and under $O \leftrightarrow D$ exchange symmetry they are not independent.

b. Intrinsic defect-tangent residual. The block DD is different: it acts intrinsically inside the defect-tangent doublet sector and couples directly to the transverse projectors.

Thus the final residual correction vector can be written schematically as

$$r_{\text{rem}} = r_{OD/DO} + (e_{0;DD}^{(1)}, o_{0;DD}^{(1)}), \quad (473)$$

where $r_{OD/DO}$ may simplify under exchange symmetry, but the DD -piece remains genuinely intrinsic.

CCXCVI. CONCLUSION

The correct final conclusion is not that DD vanishes. The correct final conclusion is:

1. DD remains present in both the broad and conservative residual audits;
2. DD has direct transverse projector channels through $a_{E;DD}$ and $a_{O;DD}$;
3. no theorem-level parity cancellation is currently available for DD ;
4. therefore DD is the final unavoidable explicit residual projector computation.

CCXCVII. FINAL SUMMARY

The DD -block is the final intrinsically surviving residual block.

It remains present in both the broad mixed-block audit

$$TD, DT, OD, DO, DD,$$

and in the conservative reduced audit

$$TT, TD, DT, DD.$$

Moreover, the projected DD -kernel already has the explicit transverse decomposition

$$\mathcal{M}_{DD,\text{inv}}^\mu = a_{\parallel;DD} n^\mu \sigma_3 + a_{E;DD} E^\mu{}_a \sigma_a + a_{O;DD} J^\mu{}_a \sigma_a,$$

so the corresponding coefficients

$$e_{0;DD}^{(1)}, \quad o_{0;DD}^{(1)}$$

are direct projector-visible transverse candidates.

Therefore the honest theorem-level conclusion is

$$\boxed{DD \text{ is not automatically zero,}}$$

and

$$\boxed{DD \text{ must be explicitly computed unless a new block-specific parity theorem is established.}}$$

Structurally, the sharp current expectation is that DD is more likely to renormalize the already-induced TD/DT transverse carrier than to generate it from nothing, but this is not yet a theorem of vanishing or dominance.

So the residual sector is now fully classified: OD and DO are exchange-type residual corrections, while DD is the final intrinsic defect-tangent residual block.

CCXCVIII. FINAL SUMMARY

The DD -block is the final intrinsically surviving residual block.

It remains present in both the broad mixed-block audit

$$TD, DT, OD, DO, DD,$$

and in the conservative reduced audit

$$TT, TD, DT, DD.$$

Moreover, the projected DD -kernel already has the explicit transverse decomposition

$$\mathcal{M}_{DD,\text{inv}}^\mu = a_{\parallel;DD} n^\mu \sigma_3 + a_{E;DD} E^\mu{}_a \sigma_a + a_{O;DD} J^\mu{}_a \sigma_a,$$

so the corresponding coefficients

$$e_{0;DD}^{(1)}, \quad o_{0;DD}^{(1)}$$

are direct projector-visible transverse candidates.

Therefore the honest theorem-level conclusion is

$$\boxed{DD \text{ is not automatically zero,}}$$

and

$$\boxed{DD \text{ must be explicitly computed unless a new block-specific parity theorem is established.}}$$

Structurally, the sharp current expectation is that DD is more likely to renormalize the already-induced TD/DT transverse carrier than to generate it from nothing, but this is not yet a theorem of vanishing or dominance.

So the residual sector is now fully classified: OD and DO are exchange-type residual corrections, while DD is the final intrinsic defect-tangent residual block.

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.$$

$$\left[i\partial_t - \lambda_{\parallel} \sigma_3 (-i\partial_{\parallel}) - \alpha \left(\sigma_1 [-i(\partial_1 - \partial_2)] + \sigma_2 [-i(\partial_1 + \partial_2)] \right) \right] \chi = 0,$$

$$\left[i\partial_t + \frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3 \sigma_3 i\partial_{\parallel} + \frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa \left(\sigma_1 i(\partial_1 - \partial_2) + \sigma_2 i(\partial_1 + \partial_2) \right) \right] \chi = 0.$$

$$v_{\perp}^{\text{can}} = \sqrt{2} \alpha = -\frac{\sqrt{6}\pi}{16} \beta_X \theta_0 \kappa.$$

$$\left[i\partial_t - \lambda_{\parallel} \sigma_3 (-i\partial_{\parallel}) - v_{\perp}^{\text{can}} (\sigma_1 (-i\partial_{q_1}) + \sigma_2 (-i\partial_{q_2})) \right] \chi = 0,$$

$$\left[i\partial_t + \frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3 \sigma_3 i\partial_{\parallel} + \frac{\sqrt{6}\pi}{16} \beta_X \theta_0 \kappa (\sigma_1 i\partial_{q_1} + \sigma_2 i\partial_{q_2}) \right] \chi = 0.$$

$$E(p)^2 = \lambda_{\parallel}^2 p_{\parallel}^2 + 2\alpha^2 (p_1^2 + p_2^2).$$

$$E(p)^2 = \left(\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3 \right)^2 p_{\parallel}^2 + \frac{3\pi^2}{128} \beta_X^2 \theta_0^2 \kappa^2 (p_1^2 + p_2^2).$$

$$\rho_{\text{can}} := \frac{v_{\parallel}^{\text{can}}}{v_{\perp}^{\text{can}}} = \frac{3}{32\sqrt{2}} \frac{\alpha_X L_*}{\beta_X} (\theta_0 \kappa)^2.$$

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}.$$

$$\left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}.$$

$$\mathcal{D}_{\text{can}}(p) = \rho_1 \otimes A_1(p) = \lambda_{\parallel} p_{\parallel} \rho_1 \otimes \sigma_3 + (\alpha p_1 - \beta p_2) \rho_1 \otimes \sigma_1 + (\alpha p_2 + \beta p_1) \rho_1 \otimes \sigma_2.$$

$$\mathcal{D}_{\text{can}}(p) = \lambda_{\parallel} p_{\parallel} \rho_1 \otimes \sigma_3 + \alpha(p_1 - p_2) \rho_1 \otimes \sigma_1 + \alpha(p_1 + p_2) \rho_1 \otimes \sigma_2.$$

$$\mathcal{D}_{\text{can}}(p) = \lambda_{\parallel} p_{\parallel} \rho_1 \otimes \sigma_3 + v_{\perp}^{\text{can}} q_1 \rho_1 \otimes \sigma_1 + v_{\perp}^{\text{can}} q_2 \rho_1 \otimes \sigma_2.$$

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)},$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (e_{0;K}^{(1)} + \Delta e_{\text{mix}}),$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (o_{0;K}^{(1)} + \Delta o_{\text{mix}}).$$

CCXCIX. COMPLETED CANONICAL MIXED-BRANCH PDE

For the canonical mixed odd-active branch

$$T_E = T_O = 1, \quad R_\perp = \sqrt{2}, \quad \varphi = \frac{\pi}{4}, \quad \Delta_{00} = \frac{1}{\sqrt{2}},$$

the exact mixed-branch master symbol is

$$A_1(p) = \frac{1}{2} \alpha_X L_* I_{\theta'^3} p_\parallel \sigma_3 + \beta_X I_\theta T_E (p_1 \sigma_1 + p_2 \sigma_2) + \beta_X I_\theta T_O (-p_2 \sigma_1 + p_1 \sigma_2),$$

with

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.$$

Therefore the coefficient sheet is

$$\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.$$

Hence

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + \alpha \left[(p_1 - p_2) \sigma_1 + (p_1 + p_2) \sigma_2 \right].$$

Equivalently, introducing rotated transverse momenta

$$q_1 := \frac{p_1 - p_2}{\sqrt{2}}, \quad q_2 := \frac{p_1 + p_2}{\sqrt{2}},$$

one obtains

$$A_1(p) = \lambda_\parallel p_\parallel \sigma_3 + v_\perp^{\text{can}} (q_1 \sigma_1 + q_2 \sigma_2),$$

with

$$v_\perp^{\text{can}} = \sqrt{2} \alpha = -\frac{\sqrt{6}\pi}{16} \beta_X \theta_0 \kappa.$$

Therefore the completed reduced PDE on the current working branch is

$$\left[i\partial_t - \lambda_\parallel \sigma_3 (-i\partial_\parallel) - v_\perp^{\text{can}} (\sigma_1 (-i\partial_{q_1}) + \sigma_2 (-i\partial_{q_2})) \right] \chi = 0.$$

The corresponding dispersion relation is

$$E(p)^2 = \lambda_\parallel^2 p_\parallel^2 + (v_\perp^{\text{can}})^2 (q_1^2 + q_2^2) = \lambda_\parallel^2 p_\parallel^2 + \frac{3\pi^2}{128} \beta_X^2 \theta_0^2 \kappa^2 (p_1^2 + p_2^2).$$

The anisotropy ratio is

$$\rho_{\text{can}} = \frac{\lambda_\parallel}{v_\perp^{\text{can}}} = \frac{3}{32\sqrt{2}} \frac{\alpha_X L_*}{\beta_X} (\theta_0 \kappa)^2.$$

The canonical mixed-branch same-shell acceptance condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}.$$

Finally, the most complete correction-dressed microscopic target remains

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)},$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (e_{0;K}^{(1)} + \Delta e_{\text{mix}}), \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (o_{0;K}^{(1)} + \Delta o_{\text{mix}}).$$

CCC. FINITE-SCALAR-UPGRADED WORKING PDE

CCCI. SHORT EXPLANATION

The reduced Dirac-type PDE is already closed at branch level. To go one step further, the correct move is to replace the abstract coefficient triplet

$$(\lambda_{\parallel}, \alpha, \beta)$$

by the finite scalar packages that are already isolated in the notes.

At the current honest stage, this gives the strongest currently justified microscopic PDE: a branch-consistent reduced Dirac PDE whose coefficients are explicitly expressed in terms of the finite scalar audits

$$\mathcal{N}_{JJ}, \quad m_{\parallel}, \quad u_{\parallel}, \quad A_{G_*}, \quad \mathfrak{T}_E, \quad \mathfrak{T}_O.$$

CCCII. 1. EXACT OVERLAP DICTIONARY

The exact local overlap dictionary is

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta^3} \lambda_c, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (474)$$

Using the explicit defect overlaps,

$$I_{\theta^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad (475)$$

one gets

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad (476)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa. \quad (477)$$

CCCI. 2. LONGITUDINAL FINITE SCALAR REDUCTION

At the current working branch level, the longitudinal source is reduced to the JJ -type finite scalar package:

$$\lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*}. \quad (478)$$

Therefore the longitudinal coefficient becomes

$$\lambda_{\parallel}^{(JJ)} \sim -\frac{3\sqrt{3}\pi}{512} \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*} \theta_0^3 \kappa^3. \quad (479)$$

Here the symbol $\sim$ means: this is the current finite-scalar working-branch reduction, not yet the unconditional final exact coefficient of complete TECT.

a. Executable subformulas. The shell-normal asymmetry is

$$u_{\parallel} = \frac{1}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)) = \frac{1}{2} \left[(\hat{n}^j \mathcal{U}_j)_{11} - (\hat{n}^j \mathcal{U}_j)_{22} \right]. \quad (480)$$

The shell asymmetry is

$$\Delta_{00} = \overline{A_1^{(+)} A_1^{(-)}} - \overline{A_2^{(+)} A_2^{(-)}}. \quad (481)$$

The longitudinal geometric scalar is

$$L_* = m_{\parallel} u_{\parallel}. \quad (482)$$

Equivalently, at the branch-level notation used earlier,

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (483)$$

where L_* is the finite longitudinal matrix element extracted from the shell-normal projected J -sector.

CCCIV. 3. TRANSVERSE FINITE SCALAR REDUCTION

The transverse source is reduced to the finite scalar contractions

$$\Lambda_E = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E, \quad \Lambda_O = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O. \quad (484)$$

Hence

$$\alpha \sim -\frac{\sqrt{3}\pi}{16} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E \theta_0 \kappa, \quad (485)$$

$$\beta \sim -\frac{\sqrt{3}\pi}{16} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O \theta_0 \kappa. \quad (486)$$

Again, $\mathfrak{T}_E, \mathfrak{T}_O$ are the remaining branch-specific finite scalar contractions of the principal mixed $JK + KJ$ source and its completion.

CCCV. 4. FINITE-SCALAR-UPGRADED REDUCED PDE

The reduced 2×2 symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (487)$$

Substituting (479), (485), (486), the strongest currently justified microscopic PDE becomes

$$\left[i\partial_t - v_{\parallel} \sigma_3 (-i\partial_{\parallel}) - v_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - v_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0, \quad (488)$$

with

$$v_{\parallel} \sim -\frac{3\sqrt{3}\pi}{512} \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*} \theta_0^3 \kappa^3, \quad (489)$$

$$v_E \sim -\frac{\sqrt{3}\pi}{16} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E \theta_0 \kappa, \quad (490)$$

$$v_O \sim -\frac{\sqrt{3}\pi}{16} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O \theta_0 \kappa. \quad (491)$$

This is the finite-scalar-upgraded working PDE.

CCCVI. 5. DISPERSION RELATION

Squaring the reduced symbol gives

$$E(p)^2 = v_{\parallel}^2 p_{\parallel}^2 + (v_E^2 + v_O^2)(p_1^2 + p_2^2). \quad (492)$$

So the transverse speed is

$$v_{\perp} = \sqrt{v_E^2 + v_O^2} \sim \frac{\sqrt{3}\pi}{16} \frac{|\alpha_X \beta_X|}{M_X^2} \sqrt{\mathfrak{T}_E^2 + \mathfrak{T}_O^2} |\theta_0 \kappa|. \quad (493)$$

The exact anisotropy ratio is therefore

$$\rho_{\text{fs}} := \frac{v_{\parallel}}{v_{\perp}} \sim \frac{3}{32} \frac{\alpha_X}{\beta_X} \frac{\mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*}}{\sqrt{\mathfrak{T}_E^2 + \mathfrak{T}_O^2}} (\theta_0 \kappa)^2. \quad (494)$$

CCCVII. 6. EXACT FINITE NONDEGENERACY CRITERION

The reduced Dirac phase is nondegenerate if and only if

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (495)$$

At the finite-scalar working level this becomes

$$\mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*} \neq 0, \quad (\mathfrak{T}_E, \mathfrak{T}_O) \neq (0, 0). \quad (496)$$

So the PDE is fully populated up to a finite scalar audit, not an unknown functional derivation.

CCCVIII. 7. SUPPORT-SIDE CONDITION

The same-shell support side remains

$$\boxed{|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}}. \quad (497)$$

Therefore the honest final acceptance statement for the current working PDE is:

$$\boxed{\text{finite-scalar nondegeneracy} + \text{same-shell support} + \text{remote-gap support}.} \quad (498)$$

CCCIX. FINAL SUMMARY

The branch-level reduced PDE can be pushed one step further to a finite-scalar-upgraded working PDE. The exact overlap dictionary is

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \lambda_c, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O,$$

with

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

At the current finite-scalar working level,

$$\lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*},$$

and

$$\Lambda_E = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E, \quad \Lambda_O = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O.$$

Therefore the strongest currently justified microscopic PDE is

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0,$$

with

$$v_{\parallel} \sim -\frac{3\sqrt{3}\pi}{512} \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*} \theta_0^3 \kappa^3,$$

$$v_E \sim -\frac{\sqrt{3}\pi}{16} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E \theta_0 \kappa, \quad v_O \sim -\frac{\sqrt{3}\pi}{16} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O \theta_0 \kappa.$$

Its dispersion is

$$E(p)^2 = v_{\parallel}^2 p_{\parallel}^2 + (v_E^2 + v_O^2)(p_1^2 + p_2^2).$$

The exact finite nondegeneracy condition is

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0),$$

which at the present finite-scalar working level becomes

$$\mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*} \neq 0, \quad (\mathfrak{T}_E, \mathfrak{T}_O) \neq (0, 0).$$

So the PDE is no longer blocked by missing structure. It is blocked only by the finite scalar audit.

CCCX. ACTUAL MICROSCOPIC CARRIER PDE AFTER THE FINAL FINITE AUDIT

CCCXI. SHORT EXPLANATION

For the intended Class II / canonical working branch, the remaining obstruction is no longer structural. The carrier-level certificate is already closed in form, and what remains is the finite microscopic audit of the branch-dependent scalar packages.

Therefore the strongest honest current endpoint is the following:

$$\boxed{\text{if the finite audit passes, then the actual microscopic carrier PDE on the intended branch is closed.}} \quad (499)$$

This is not yet the full unconditional closure of all complete TECT branches, but it is the correct actual branch-level PDE completion target.

CCCXII. 1. EXACT COMPLETION PROTOCOL

The actual PDE completion protocol is

$$\boxed{L(k) \longrightarrow K_i := \partial_{k_i} L|_{G_*} \longrightarrow P_* K_i P_* \longrightarrow (\lambda_{\parallel}, \alpha, \beta).} \quad (500)$$

Equivalently, in blockwise Hessian language,

$$\boxed{\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.} \quad (501)$$

The overlap-completed reduced dictionary is

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.} \quad (502)$$

Using the explicit defect overlaps,

$$\boxed{I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3.} \quad (503)$$

CCCXIII. 2. LONGITUDINAL FINITE-SCALAR INSERTION

The longitudinal finite scalar package has already been reduced to

$$\boxed{\lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A_{G_*}.} \quad (504)$$

Moreover, on the explicit equal-amplitude BCC working branch,

$$\boxed{A_{G_*} = A.} \quad (505)$$

In the canonical locked basis, the cleanest active branch is

$$\boxed{T_{\star} = T^1, \quad u_{\parallel}(T^1) = \frac{1}{2} \Re(\kappa_J).} \quad (506)$$

Therefore the branch-level longitudinal coefficient becomes

$$\boxed{\lambda_{\parallel} \sim -\frac{3\sqrt{3}\pi}{512} \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A \theta_0^3 \kappa^3.} \quad (507)$$

Using $u_{\parallel}(T^1) = \frac{1}{2} \Re(\kappa_J)$, this may be written more explicitly as

$$\boxed{\lambda_{\parallel} \sim -\frac{3\sqrt{3}\pi}{1024} \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} \Re(\kappa_J) |G_*| A \theta_0^3 \kappa^3.} \quad (508)$$

CCCXIV. 3. TRANSVERSE FINITE-SCALAR INSERTION

The transverse finite scalar package is

$$\Lambda_E = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E, \quad \Lambda_O = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O. \quad (509)$$

Hence

$$\alpha \sim -\frac{\sqrt{3}\pi}{16} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E \theta_0 \kappa, \quad (510)$$

$$\beta \sim -\frac{\sqrt{3}\pi}{16} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O \theta_0 \kappa. \quad (511)$$

CCCXV. 4. THE ACTUAL CARRIER PDE AFTER AUDIT COMPLETION

Substituting (507), (510), (511) into the reduced symbol

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2),$$

the actual carrier PDE on the intended branch is

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0, \quad (512)$$

with

$$v_{\parallel} \sim -\frac{3\sqrt{3}\pi}{512} \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A \theta_0^3 \kappa^3, \quad (513)$$

$$v_E \sim -\frac{\sqrt{3}\pi}{16} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E \theta_0 \kappa, \quad (514)$$

$$v_O \sim -\frac{\sqrt{3}\pi}{16} \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O \theta_0 \kappa. \quad (515)$$

CCCXVI. 5. DISPERSION AND NONDEGENERACY

The corresponding dispersion is

$$E(p)^2 = v_{\parallel}^2 p_{\parallel}^2 + (v_E^2 + v_O^2)(p_1^2 + p_2^2). \quad (516)$$

Thus the exact finite nondegeneracy criterion is

$$\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0). \quad (517)$$

At the finite-scalar working level, this becomes

$$\mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A \neq 0, \quad (\mathfrak{T}_E, \mathfrak{T}_O) \neq (0, 0). \quad (518)$$

CCCXVII. 6. ACTUAL PASS CERTIFICATE

The actual microscopic carrier closes once the following finite audit passes.

a. Gate 1: support / gap

$$\mathfrak{G}_1 := \Delta_{\text{bench},\rho}^{\text{fin}} - \eta_{R,\rho} > 0. \quad (519)$$

b. Gate 2: longitudinal margin

$$\mathfrak{G}_2 := |\Delta_{00} \alpha_X L_* A| - |R_N| > 0, \quad L_* = m_{\parallel} u_{\parallel}. \quad (520)$$

c. Gate 3: transverse support

$$\mathfrak{G}_{3,\text{tr}} := |c_E| + |c_O| > 0, \quad \mathfrak{G}_{3,\text{cal}} := |C_E| + |C_O| > 0. \quad (521)$$

d. Gate 4: constant-sector protection

$$[Q_*, H_{\Psi\Psi}^{\text{const}}] = 0, \quad [Q_*, \delta H_{\text{Class II}}^{\text{const}}] = 0, \quad [Q_*, \Sigma(0,0)] = 0. \quad (522)$$

Therefore

$$\text{Gate 1} \wedge \text{Gate 2} \wedge \text{Gate 3} \wedge \text{Gate 4} \implies \text{actual branch-level protected anisotropic massless Dirac carrier.} \quad (523)$$

CCCXVIII. 7. ONE-DEFECT ACCEPTANCE FORM

At the one-defect level, the decisive inequalities reduce to

$$\mu_{\infty} > 0, \quad \xi_E g_E > \xi_c g_c, \quad (524)$$

with

$$g_c = \frac{\lambda_{\parallel}}{C_c}, \quad g_E = \frac{\alpha}{C_E}, \quad g_O = \frac{\beta}{C_O}.$$

The exact longitudinal burden ratio is

$$R_{\text{exact}} = \frac{\xi_c}{\xi_E} \frac{C_E}{C_c} \frac{3}{32} (\theta_0 \kappa)^2 \frac{\lambda_c}{\Lambda_E}. \quad (525)$$

So the one-defect problem is also reduced to the same finite scalar insertion layer.

CCCXIX. CONCLUSION

The remaining frontier is no longer a missing PDE architecture. It is the explicit evaluation of the finite scalar set

$$\mu_{\infty}, \quad \mathfrak{T}_E, \quad \mathfrak{T}_O, \quad \mathcal{N}_{JJ}, \quad m_{\parallel}, \quad u_{\parallel}, \quad \kappa_*.$$

Once these are inserted and the four gates pass, the intended branch-level actual microscopic carrier PDE is closed.

CCCXX. FINAL SUMMARY

Yes: for the intended Class II / canonical working branch, the actual microscopic carrier PDE is now reduced to a finite scalar audit.

The exact completion protocol is

$$L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta),$$

with dictionary

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.$$

Using the current finite scalar reductions,

$$\lambda_c^{(JJ)} \sim \frac{\alpha_X^2}{M_X^2} \mathcal{N}_{JJ} m_{\parallel} u_{\parallel} |G_*| A,$$

$$\Lambda_E = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_E, \quad \Lambda_O = \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{T}_O,$$

the actual branch-level carrier PDE becomes

$$\left[i\partial_t - v_{\parallel} \sigma_3(-i\partial_{\parallel}) - v_E(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - v_O(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

It closes once the finite audit passes:

$$\mathfrak{G}_1 > 0, \quad \mathfrak{G}_2 > 0, \quad \mathfrak{G}_{3,\text{tr}} > 0, \quad \mathfrak{G}_{3,\text{cal}} > 0,$$

together with the constant-sector commutator protection test.

So the remaining frontier is no longer structural. It is the explicit evaluation of the finite scalar set

$$\mu_{\infty}, \quad \mathfrak{T}_E, \quad \mathfrak{T}_O, \quad \mathcal{N}_{JJ}, \quad m_{\parallel}, \quad u_{\parallel}, \quad \kappa_*.$$

CCCXXI. EXPLICIT CANONICAL FINITE AUDIT AND THE FULLY INSERTED ONE-DEFECT PASS SHEET

CCCXXII. CURRENT EXPLICIT BRANCH

The currently most explicit one-defect audit branch is

$$\boxed{T_{\star} = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0.}$$

On this branch, the Bloch/Class-II transport coefficients are already fixed as

$$\boxed{g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3,} \tag{526}$$

$$\boxed{g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0.} \tag{527}$$

Therefore the reduced PDE on this explicit branch is the pure- E working branch

$$\boxed{\left[i\partial_t - \lambda_{\parallel} \sigma_3(-i\partial_{\parallel}) - \alpha(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) \right] \chi = 0,} \tag{528}$$

with $\beta = 0$.

CCCXXIII. EXACT MATCHING DICTIONARY

The radial core coefficients are

$$\boxed{a_0 = \frac{\xi_E g_E + \xi_c g_c}{z_{0,0}}, \quad b_0 = \frac{\xi_E g_E}{z_{0,0}}, \quad c_0 = \frac{\xi_O g_O}{z_{0,0}}.} \quad (529)$$

Since $g_O = 0$ on this branch,

$$\boxed{c_0 = 0.} \quad (530)$$

Substituting (526)–(527) into (529), one gets

$$\boxed{b_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa,} \quad (531)$$

and

$$\boxed{a_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa \left[1 + \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 \right].} \quad (532)$$

Define the exact longitudinal burden ratio

$$\boxed{R_c := \frac{\xi_c g_c}{\xi_E g_E} = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2.} \quad (533)$$

Then

$$\boxed{a_0 = b_0 (1 + R_c).} \quad (534)$$

CCCXXIV. BULK-MASS SIGN AUDIT

The exact constitutive bulk mass is

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}.} \quad (535)$$

In the asymptotic vacuum branch $\chi_\infty = 0$,

$$\boxed{\mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}.} \quad (536)$$

Hence, assuming $z_0(\sigma_\infty) > 0$, the first audit reduces to

$$\boxed{m(\sigma_\infty) > 0.} \quad (537)$$

CCCXXV. LOWEST-CHANNEL REDUCTION

The practically relevant lowest-channel structure is already fixed:

$$\boxed{\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1.} \quad (538)$$

The general one-mode sufficient criterion

$$\mu_\infty > 0, \quad a_0 > 0, \quad b_0 > 0, \quad \frac{|\kappa_*| b_0}{a_0} > \frac{1}{2}$$

then reduces to

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.} \quad (539)$$

Using (533), this is

$$\boxed{-1 < \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2 < 1,} \quad (540)$$

together with

$$\boxed{\frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.} \quad (541)$$

Equivalently, the decisive transport bridge is

$$\boxed{\xi_E g_E > \xi_c g_c.} \quad (542)$$

CCCXXVI. UNIT-BENCHMARK NUMERICAL REDUCTION

For the unit benchmark

$$\boxed{\alpha_X = \beta_X = M_X = \xi_E = z_{0,0} = 1, \quad \chi_\infty = 0,} \quad (543)$$

the coefficients become

$$\boxed{b_0 \approx 0.34008738079 (\theta_0 \kappa),} \quad (544)$$

$$\boxed{R_c = 0.09375 \xi_c (\theta_0 \kappa)^2,} \quad (545)$$

$$\boxed{a_0 \approx 0.34008738079 (\theta_0 \kappa) [1 + 0.09375 \xi_c (\theta_0 \kappa)^2],} \quad (546)$$

and the R_c -window gives

$$\boxed{|\xi_c| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2} = \frac{10.666 \dots}{(\theta_0 \kappa)^2}.} \quad (547)$$

Therefore:

$$\boxed{\theta_0 \kappa = 0.1 \implies |\xi_c| < 1066.67,} \quad (548)$$

$$\boxed{\theta_0 \kappa = 0.3 \implies |\xi_c| < 118.52,} \quad (549)$$

$$\boxed{\theta_0 \kappa = 1 \implies |\xi_c| < 10.67.} \quad (550)$$

CCCXXVII. ACTUAL FINITE UNKNOWNNS THAT REMAIN

After all these substitutions, the remaining unknown branch data are only

$$\boxed{\xi_E, \quad \xi_c, \quad \mu_\infty, \quad \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.} \quad (551)$$

Equivalently, assuming the canonical sign choice

$$\frac{\beta_X}{M_X^2} \theta_0 \kappa > 0,$$

the branch passes if and only if the following four finite checks hold:

$$\boxed{m(\sigma_\infty) > 0}, \tag{552}$$

$$\boxed{\frac{\beta_X}{M_X^2} \theta_0 \kappa > 0}, \tag{553}$$

$$\boxed{\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}}, \tag{554}$$

$$\boxed{\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}}. \tag{555}$$

Thus the explicit branch-level one-defect acceptance problem is completely reduced to four finite scalar insertions.

CCCXXVIII. FINAL SUMMARY

For the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the transport coefficients are already fixed:

$$g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0.$$

Therefore the radial coefficients are

$$b_0 = \frac{\xi_E}{z_{0,0}} \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad c_0 = 0,$$

$$a_0 = b_0(1 + R_c), \quad R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2.$$

The bulk mass is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)},$$

which reduces to

$$\mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}$$

in the vacuum asymptotic branch $\chi_\infty = 0$.

For the practically relevant lowest channel

$$\kappa_\star = -1, \quad \kappa_{\text{comp}} \geq 1,$$

the one-defect acceptance criterion reduces exactly to

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

Hence the explicit branch-level problem is reduced to four finite scalar checks:

$$m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

So the one-defect side is no longer structurally open. Only finite constitutive insertion remains.

CCCXXIX. THE μ_∞ -FIRST AUDIT IN FULLY EXECUTABLE FORM

CCCXXX. SHORT EXPLANATION

For the explicit canonical branch, the earliest kill condition is the bulk positivity gate

$$\mu_\infty > 0.$$

At the current stage this is no longer a conceptual issue. It is a literal finite constitutive sign test.

CCCXXXI. 1. EXACT ASYMPTOTIC CONSTITUTIVE SCALAR

The asymptotic bulk mass parameter is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (556)$$

Define the asymptotic bulk numerator

$$\Delta_\mu := m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty). \quad (557)$$

Then

$$\mu_\infty = \frac{\Delta_\mu}{z_0(\sigma_\infty)}. \quad (558)$$

CCCXXXII. 2. EXACT SIGN REDUCTION

Assume the standard normalization

$$z_0(\sigma_\infty) > 0. \quad (559)$$

Then

$$\text{sgn } \mu_\infty = \text{sgn } \Delta_\mu. \quad (560)$$

Therefore exactly one of the following holds:

$$\Delta_\mu > 0 \iff \mu_\infty > 0, \quad (561)$$

$$\Delta_\mu = 0 \iff \mu_\infty = 0, \quad (562)$$

$$\Delta_\mu < 0 \iff \mu_\infty < 0. \quad (563)$$

So the bulk gate is completely executable:

$$\text{compute } \Delta_\mu \text{ and check its sign.} \quad (564)$$

CCCXXXIII. 3. SIMPLEST VACUUM BRANCH

If the asymptotic branch is vacuum-like,

$$\boxed{\chi_\infty = 0,} \quad (565)$$

then

$$\boxed{\Delta_\mu = m(\sigma_\infty), \quad \mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}.} \quad (566)$$

Under $z_0(\sigma_\infty) > 0$,

$$\boxed{\mu_\infty > 0 \iff m(\sigma_\infty) > 0.} \quad (567)$$

Thus on the simplest vacuum branch the entire bulk audit collapses to a single scalar sign.

CCCXXXIV. 4. NON-VACUUM ASYMPTOTIC BRANCH

If $\chi_\infty \neq 0$, then the exact condition is

$$\boxed{m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty).} \quad (568)$$

This yields the branchwise subcases:

a. Case A: $u_X(\sigma_\infty) \geq 0$, $\chi_\infty \geq 0$. Then the Class-II asymptotic correction lowers the gap, so one needs

$$\boxed{m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty) \geq 0.} \quad (569)$$

b. Case B: $u_X(\sigma_\infty) \leq 0$, $\chi_\infty \geq 0$. Then the Class-II correction helps the gap, so the exact condition remains

$$\boxed{m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty),} \quad (570)$$

which is weaker than demanding $m(\sigma_\infty) > 0$.

c. Case C: sign-indefinite asymptotic sector. Then no further universal simplification is possible, but the audit is still finite:

$$\boxed{\text{sgn } \mu_\infty = \text{sgn}(m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)).} \quad (571)$$

CCCXXXV. 5. EARLIEST REJECT THEOREM

Under $z_0(\sigma_\infty) > 0$, if

$$\boxed{\Delta_\mu \leq 0,} \quad (572)$$

then

$$\boxed{\mu_\infty \leq 0,} \quad (573)$$

so the branch is rejected before any insertion of

$$g_E, \quad g_C, \quad \kappa_*, \quad \text{competitor exclusion.}$$

Thus the correct finite audit order really is

$$\boxed{\mu_\infty \longrightarrow g_E \longrightarrow g_C \longrightarrow \kappa_* \longrightarrow \text{competitor exclusion.}} \quad (574)$$

CCCXXXVI. 6. COMBINATION WITH THE EXPLICIT CANONICAL BRANCH

For the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the already extracted transport coefficients satisfy

$$g_O = 0, \quad g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad (575)$$

and

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2. \quad (576)$$

Hence the explicit branch acceptance sheet is

$$\Delta_\mu > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}. \quad (577)$$

This is the fully inserted practical finite audit form.

CCCXXXVII. 7. NUMERICAL BENCHMARK FORM

For the unit benchmark

$$\alpha_X = \beta_X = M_X = \xi_E = z_{0,0} = 1,$$

the R_c -window becomes

$$R_c = 0.09375 \xi_c (\theta_0 \kappa)^2, \quad |\xi_c| < \frac{10.666\dots}{(\theta_0 \kappa)^2}. \quad (578)$$

So, in the small-defect regime, the longitudinal burden window is broad, and the earliest genuinely nontrivial check is the bulk sign

$$\Delta_\mu > 0.$$

CCCXXXVIII. FINAL SUMMARY

The μ_∞ -first audit is now completely executable.

Define

$$\Delta_\mu := m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty).$$

Then

$$\mu_\infty = \frac{\Delta_\mu}{z_0(\sigma_\infty)}.$$

Under the standard normalization $z_0(\sigma_\infty) > 0$,

$$\mu_\infty > 0 \iff \Delta_\mu > 0.$$

Thus exactly one of the following holds:

$$\Delta_\mu > 0 \iff \mu_\infty > 0, \quad \Delta_\mu = 0 \iff \mu_\infty = 0, \quad \Delta_\mu < 0 \iff \mu_\infty < 0.$$

On the simplest vacuum branch $\chi_\infty = 0$, this reduces further to

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0.$$

For the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the full practical finite audit sheet is

$$\Delta_\mu > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.$$

So the bulk side is no longer conceptually open. It is reduced to an explicit constitutive scalar sign test.

CCCXXXIX. CLOSURE OF THE μ_∞ GATE ON THE SIMPLEST VACUUM ASYMPTOTIC BRANCH

CCCXL. CURRENT STAGE

At the explicit canonical one-defect stage, the earliest kill condition is

$$\mu_\infty > 0.$$

The exact constitutive formula is

$$\mu_\infty = \frac{m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (579)$$

Under the standard normalization

$$z_0(\sigma_\infty) > 0, \quad (580)$$

this is equivalent to

$$\mu_\infty > 0 \iff m(\sigma_\infty) - 3\chi_\infty u_X(\sigma_\infty) > 0. \quad (581)$$

CCCXLI. 1. SIMPLEST VACUUM ASYMPTOTIC BRANCH

Now fix the asymptotic branch by

$$X_{\perp, \infty} = X_{\parallel, \infty} = 0. \quad (582)$$

Then

$$\chi_\infty = \frac{1}{3}(2X_{\perp, \infty} + X_{\parallel, \infty}) = 0. \quad (583)$$

Substituting (583) into (579), one obtains

$$\mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}. \quad (584)$$

Therefore, under (580),

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0. \quad (585)$$

CCCXLII. 2. HONEST CONSEQUENCE

This is the strongest executable closure of the bulk gate currently available from the notes.

On a generic non-vacuum asymptotic Class-II branch, the sign condition is the balance inequality

$$m(\sigma_\infty) > 3\chi_\infty u_X(\sigma_\infty).$$

But on the simplest vacuum branch, the asymptotic Class-II correction disappears identically, so the bulk gate is no longer a balance problem. It is a one-sign problem.

Hence the constitutive bulk gate is closed once the chosen branch satisfies

$$\boxed{m(\sigma_\infty) > 0.} \quad (586)$$

CCCXLIII. 3. EXPLICIT CANONICAL ONE-DEFECT ACCEPTANCE SHEET AFTER BULK CLOSURE

For the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the practical lowest-channel acceptance problem had already been reduced to

$$\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0\kappa)^2}, \quad \frac{\beta_X}{M_X^2} \theta_0\kappa > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

Using (585), this becomes

$$\boxed{\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0\kappa)^2}, \quad \frac{\beta_X}{M_X^2} \theta_0\kappa > 0, \quad m(\sigma_\infty) > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.} \quad (587)$$

So the explicit one-defect acceptance theorem on the simplest vacuum branch is reduced to four finite scalar checks.

CCCXLIV. 4. THE RESULTING EXPLICIT WORKING PDE

On the same explicit canonical branch, the transport coefficients are already fixed as

$$\boxed{g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3,} \quad (588)$$

$$\boxed{g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0.} \quad (589)$$

Therefore the explicit reduced carrier PDE on this asymptotic branch is

$$\boxed{\left[i\partial_t - \lambda_\parallel \sigma_3(-i\partial_\parallel) - \alpha(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) \right] \chi = 0,} \quad (590)$$

with

$$\boxed{\beta = 0.} \quad (591)$$

So the branch is now bulk-closed up to the single constitutive sign $m(\sigma_\infty)$.

CCCXLV. 5. HONEST STATUS AFTER THIS STEP

What is now closed:

1. the μ_∞ -gate is reduced to a single scalar sign on the simplest vacuum branch;
2. the explicit one-defect acceptance sheet is reduced to the four checks in (587);
3. the bulk-side frontier is no longer conceptual.

What is still not inserted numerically:

1. the actual sign of $m(\sigma_\infty)$;
2. the sign of $\beta_X \theta_0 \kappa$;
3. the burden ratio ξ_c/ξ_E ;
4. the remote-gap margin $\Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}$.

CCCXLVI. FINAL SUMMARY

The strongest executable bulk closure is obtained by fixing the asymptotic constitutive branch to the simplest vacuum branch

$$X_{\perp,\infty} = X_{\parallel,\infty} = 0.$$

Then

$$\chi_\infty = 0, \quad \mu_\infty = \frac{m(\sigma_\infty)}{z_0(\sigma_\infty)}.$$

Under the standard normalization $z_0(\sigma_\infty) > 0$, the bulk gate becomes

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0.$$

Therefore, for the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

the practical one-defect acceptance problem reduces to the four finite scalar checks

$$\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad m(\sigma_\infty) > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

So the bulk side is closed on this branch. What remains is only finite insertion.

CCCXLVII. CLOSURE OF THE LONGITUDINAL BURDEN GATE ON THE EXPLICIT CANONICAL BRANCH

CCCXLVIII. SHORT EXPLANATION

After the bulk gate $\mu_\infty > 0$, the next genuine finite audit is the longitudinal burden ratio. On the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0,$$

this gate is no longer structural. It is a literal finite inequality on the ratio ξ_c/ξ_E .

CCCXLIX. 1. EXACT RATIO FORMULA

The dimensionless burden ratio is

$$R_c := \frac{\xi_c g_c}{\xi_E g_E}. \quad (592)$$

On the explicit canonical branch, the extracted coefficients are

$$g_c = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad g_E = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad g_O = 0.$$

Therefore

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2. \quad (593)$$

CCCL. 2. EXACT LOWEST-CHANNEL ACCEPTANCE WINDOW

For the practically most relevant lowest channel

$$\kappa_* = -1, \quad \kappa_{\text{comp}} \geq 1,$$

the one-defect criterion reduces to

$$-1 < R_c < 1. \quad (594)$$

Using (593), this is exactly equivalent to

$$\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}. \quad (595)$$

So the burden gate is now a single explicit finite ratio bound.

CCCLI. 3. EQUIVALENT TRANSPORT-DOMINANCE FORM

The sharp reduced inequality can also be written directly in transport variables as

$$\xi_E g_E > \xi_c g_c. \quad (596)$$

Equivalently, using the canonical matrix normalization,

$$\frac{\xi_E}{C_E} \alpha > \frac{\xi_c}{C_{\parallel}} \lambda_{\parallel}. \quad (597)$$

Thus the physical meaning of the gate is simple:

$$\boxed{\text{the matched transverse splitting must dominate the matched longitudinal burden.}} \quad (598)$$

CCCLII. 4. AUTOMATIC-PASS REGION IN THE SMALL-DEFECT REGIME

From (595), the gate is automatically satisfied whenever

$$\boxed{(\theta_0 \kappa)^2 < \frac{32}{3|\alpha_X|} \left| \frac{\xi_E}{\xi_c} \right|}. \quad (599)$$

Equivalently,

$$\boxed{|\theta_0 \kappa| < \sqrt{\frac{32}{3|\alpha_X|} \left| \frac{\xi_E}{\xi_c} \right|}}. \quad (600)$$

Hence the burden window is naturally broad in the small-defect regime. Unless $|\xi_c/\xi_E|$ is abnormally large, the burden gate is not fine-tuned.

CCCLIII. 5. EXPLICIT BENCHMARK VALUES

Under the unit benchmark

$$\alpha_X = \beta_X = M_X = \xi_E = z_{0,0} = 1,$$

the ratio formula becomes

$$\boxed{R_c = 0.09375 \xi_c (\theta_0 \kappa)^2}. \quad (601)$$

So the exact window is

$$\boxed{|\xi_c| < \frac{32}{3(\theta_0 \kappa)^2}}. \quad (602)$$

Representative values are:

$$\boxed{\theta_0 \kappa = 0.1 \implies |\xi_c/\xi_E| < 1066.666 \dots} \quad (603)$$

$$\boxed{\theta_0 \kappa = 0.3 \implies |\xi_c/\xi_E| < 118.518 \dots} \quad (604)$$

$$\boxed{\theta_0 \kappa = 1 \implies |\xi_c/\xi_E| < 10.666 \dots} \quad (605)$$

CCCLIV. 6. COMBINATION WITH THE ALREADY-CLOSED BULK GATE

On the simplest vacuum asymptotic branch, the bulk gate has already reduced to

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0.$$

Therefore the explicit canonical one-defect acceptance sheet now becomes

$$\boxed{m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}}. \quad (606)$$

So after the μ_∞ -closure, the burden gate is also fully executable.

CCCLV. 7. HONEST STATUS AFTER THIS STEP

What is now closed:

1. the bulk gate is reduced to the sign of $m(\sigma_\infty)$ on the simplest vacuum branch;
2. the longitudinal burden gate is reduced to the explicit ratio window (595);
3. the physical content of the inequality is the dominance condition (597);
4. in the small-defect regime, the burden gate is naturally broad.

What is still not inserted numerically:

1. the actual branch value of $m(\sigma_\infty)$;
2. the actual matched ratio ξ_c/ξ_E ;
3. the remote-gap margin $\Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}$.

CCCLVI. FINAL SUMMARY

The longitudinal burden gate is now fully executable on the explicit canonical branch. Its exact form is

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2,$$

and in the practically relevant lowest-channel case

$$\kappa_* = -1,$$

the exact acceptance window is

$$\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}.$$

Equivalently, the physical transport meaning is

$$\xi_E g_E > \xi_c g_c,$$

or in normalized coefficient form

$$\frac{\xi_E}{C_E} \alpha > \frac{\xi_c}{C_{\parallel}} \lambda_{\parallel}.$$

In the small-defect regime, the window is naturally broad:

$$|\theta_0 \kappa| < \sqrt{\frac{32}{3|\alpha_X|} \left| \frac{\xi_E}{\xi_c} \right|}.$$

Combining this with the already-closed bulk gate on the simplest vacuum branch, the explicit canonical one-defect acceptance sheet is

$$m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}.$$

So the remaining frontier is reduced to finite insertion of branch data.

CCCLVII. EXPLICIT REMOTE-GAP PASS CERTIFICATE ON THE CANONICAL EXPLICIT BRANCH

CCCLVIII. SHORT EXPLANATION

At this stage, Gate 1 is no longer a structural theorem problem. On the explicit canonical branch, it is a single finite scalar margin test.

The exact support condition remains

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}} > \eta_{R,\rho}}, \quad (607)$$

or, on the explicitly inserted branch notation,

$$\boxed{\Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}}. \quad (608)$$

CCCLIX. 1. BENCHMARK SIDE

The exact benchmark side is

$$\boxed{\Delta_{\text{bench},\rho}^{\text{fin}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)} \right\}}. \quad (609)$$

On the explicit canonical branch $T_\star = T^1$, $\eta = z_0$, $\Lambda_O^{(0)} = 0$, the sharpest same-shell usable insertion is

$$\boxed{\Xi_{\text{non}}^{(1)}(\rho) := \left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 - 2a_K \mathcal{S}_{\text{diag}}^{EE} - \varepsilon_{\text{small-}p,\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)}}, \quad (610)$$

with

$$\boxed{a_K = \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2}. \quad (611)$$

Thus, for the explicit branch,

$$\boxed{\Delta_{\text{bench},\rho}^{\text{full}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Xi_{\text{non}}^{(1)}(\rho) \right\}}. \quad (612)$$

CCCLX. 2. RESIDUAL SIDE

The exact residual decomposition is

$$\boxed{\eta_{R,\rho} \leq \eta_{R,\rho}^{\text{tr}} + \eta_{\text{tail},\rho} + \eta_{\text{diag},\rho}}. \quad (613)$$

For the explicit branch, the practically inserted upper bound is

$$\boxed{\eta_{\text{rem},\rho} \leq 2 \left[\kappa \rho (|C_E| + |C_O|) + \frac{1}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2) \right] + 2C_{\text{high,non}}\rho + \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} + \eta_{\text{diag},\rho}}. \quad (614)$$

CCCLXI. 3. SIGNED MARGIN

Define the actual remote-gap margin

$$\mathfrak{G}_{\text{gap}}(\rho) := \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}. \quad (615)$$

Substituting (612) and (614), one gets

$$\mathfrak{G}_{\text{gap}}(\rho) = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Xi_{\text{non}}^{(1)}(\rho) \right\} - \eta_{\text{rem},\rho}. \quad (616)$$

Therefore Gate 1 passes if and only if

$$\mathfrak{G}_{\text{gap}}(\rho) > 0. \quad (617)$$

CCCLXII. 4. SPECTRAL CONSEQUENCE

If $\mathfrak{G}_{\text{gap}}(\rho) > 0$, then the remote block has a uniform positive spectral gap:

$$\text{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \emptyset \quad (|p| \leq \rho), \quad (618)$$

with

$$\Delta_\rho \geq \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho} = \mathfrak{G}_{\text{gap}}(\rho) > 0. \quad (619)$$

So the low shell slot is spectrally isolated once this one scalar margin is positive.

CCCLXIII. 5. PRACTICAL TWO-SUBPROBLEM REDUCTION

The same-shell non-enriched benchmark is already reduced to a finite Hermitian eigenvalue problem:

$$\Delta_{\text{non-enr},\rho}^{(N)} = \inf_{|p| \leq \rho} \min_j |\lambda_j^{(N)}(p)|, \quad (620)$$

and the usable entry is

$$\Delta_{\text{non-enr},\rho}^{(N)} - \varepsilon_{\text{non-enr},\rho}^{(N)}. \quad (621)$$

Hence the real unresolved Gate-1 tasks are exactly:

$$(i) \text{ one finite same-shell non-enriched eigenvalue computation,} \quad (622)$$

$$(ii) \text{ one remote-mixing norm bound } \eta_{\text{rem},\rho}. \quad (623)$$

Everything else in $\mathfrak{G}_{\text{gap}}(\rho)$ is already explicit.

CCCLXIV. 6. FINAL EXPLICIT ONE-DEFECT ACCEPTANCE SHEET

Combining the already closed bulk gate and burden gate on the simplest vacuum asymptotic branch, the explicit canonical one-defect pass sheet becomes

$$m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \mathfrak{G}_{\text{gap}}(\rho) > 0. \quad (624)$$

This is the fully executable practical pass certificate for the explicit canonical branch.

CCCLXV. 7. HONEST STATUS

What is now closed:

1. the remote-gap gate is reduced to the single scalar margin $\mathfrak{G}_{\text{gap}}(\rho)$;
2. the benchmark side is explicit;
3. the residual side is explicit as an upper bound;
4. the one-defect pass sheet is reduced to four scalar checks.

What is still not inserted numerically:

1. $\Xi_{\text{non}}^{(1)}(\rho)$, equivalently the same-shell non-enriched finite eigenvalue slot;
2. the residual constants $C_{\text{high,non}}, K_a, \tilde{K}_a, r_a, \sigma_a, \eta_{\text{diag},\rho}$.

CCCLXVI. FINAL SUMMARY

The remote-gap gate is now reduced to a single scalar margin

$$\mathfrak{G}_{\text{gap}}(\rho) = \Delta_{\text{bench},\rho}^{\text{full}} - \eta_{\text{rem},\rho}.$$

Gate 1 passes if and only if

$$\mathfrak{G}_{\text{gap}}(\rho) > 0.$$

On the explicit canonical branch, the benchmark side is

$$\Delta_{\text{bench},\rho}^{\text{full}} = \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Xi_{\text{non}}^{(1)}(\rho) \right\},$$

with

$$\Xi_{\text{non}}^{(1)}(\rho) = \left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 - 2a_K \mathcal{S}_{\text{diag}}^{EE} - \varepsilon_{\text{small-}p,\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)}.$$

The residual side is bounded by

$$\eta_{\text{rem},\rho} \leq 2 \left[\kappa \rho (|C_E| + |C_O|) + \frac{1}{M_X^2} (\alpha_X^2 + |\alpha_X \beta_X| + \beta_X^2) \right] + 2C_{\text{high,non}}\rho + \sum_{a \in \mathcal{A}_{\text{act}}} K_a \rho^{r_a} + \sum_{a \in \mathcal{A}_{\text{act}}^{(H)}} \frac{\tilde{K}_a}{M_a^{\sigma_a}} \rho^{r_a} + \eta_{\text{diag},\rho}.$$

Therefore the explicit canonical one-defect acceptance sheet is now

$$m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \mathfrak{G}_{\text{gap}}(\rho) > 0.$$

So the only genuinely unresolved Gate-1 tasks are: one same-shell non-enriched finite eigenvalue computation and one remote-mixing residual bound.

CCCLXVII. A SHARP SUFFICIENT POSITIVITY CUT FOR $\Xi_{\text{non}}^{(1)}(\rho)$

CCCLXVIII. SHORT EXPLANATION

The remaining Gate-1 obstruction on the explicit canonical branch is the same-shell non-enriched benchmark slot

$$\Xi_{\text{non}}^{(1)}(\rho).$$

At this stage, the correct move is not another structural reduction. It is to isolate a clean static margin and compare it against the total ρ -dependent drift budget.

CCCLXIX. 1. EXACT STARTING POINT

The explicit canonical-branch benchmark slot is

$$\Xi_{\text{non}}^{(1)}(\rho) = \left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 - 2a_K \mathcal{S}_{\text{diag}}^{EE} - \varepsilon_{\text{small-}p,\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)}. \quad (625)$$

Define the static core margin

$$\Delta_{\text{core}}^{(1)} := \left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 - 2a_K \mathcal{S}_{\text{diag}}^{EE}. \quad (626)$$

Then

$$\Xi_{\text{non}}^{(1)}(\rho) = \Delta_{\text{core}}^{(1)} - \varepsilon_{\text{small-}p,\rho}^{(1)} - \varepsilon_{\text{non-enr},\rho}^{(1)}. \quad (627)$$

So the exact positivity condition is

$$\Delta_{\text{core}}^{(1)} > \varepsilon_{\text{small-}p,\rho}^{(1)} + \varepsilon_{\text{non-enr},\rho}^{(1)}. \quad (628)$$

CCCLXX. 2. FIRST SHARP SUFFICIENT CONDITION

Therefore a first clean sufficient condition is simply

$$\Delta_{\text{core}}^{(1)} > 0 \quad \text{and} \quad \varepsilon_{\text{small-}p,\rho}^{(1)} + \varepsilon_{\text{non-enr},\rho}^{(1)} < \Delta_{\text{core}}^{(1)}. \quad (629)$$

Equivalently,

$$\Xi_{\text{non}}^{(1)}(\rho) > 0 \quad \Longleftarrow \quad \varepsilon_{\text{small-}p,\rho}^{(1)} + \varepsilon_{\text{non-enr},\rho}^{(1)} < \Delta_{\text{core}}^{(1)}. \quad (630)$$

CCCLXXI. 3. EXPLICIT DRIFT ESTIMATE

For the minimal same-shell non-enriched block, the small- p drift satisfies

$$\varepsilon_{\text{small-}p,\rho}^{(1)} \leq \left[(\rho A_+ + C_+ \rho^2)^2 + (\rho A_- + C_- \rho^2)^2 + 2(\rho B + C_m \rho^2)^2 \right]^{1/2}. \quad (631)$$

At leading order this is

$$\varepsilon_{\text{small-}p,\rho}^{(1)} = \rho L_{\text{non}}^{(1)} + O(\rho^2), \quad L_{\text{non}}^{(1)} := \sqrt{A_+^2 + A_-^2 + 2B^2}. \quad (632)$$

Introduce any quadratic constant $C_{\text{non}}^{(1)}$ such that

$$\varepsilon_{\text{small-}p,\rho}^{(1)} \leq \rho L_{\text{non}}^{(1)} + C_{\text{non}}^{(1)} \rho^2. \quad (633)$$

Now absorb the second remainder into a similar bound

$$\varepsilon_{\text{non-enr},\rho}^{(1)} \leq \rho \tilde{L}_{\text{non}}^{(1)} + \tilde{C}_{\text{non}}^{(1)} \rho^2. \quad (634)$$

Then

$$\varepsilon_{\text{small-}p,\rho}^{(1)} + \varepsilon_{\text{non-enr},\rho}^{(1)} \leq \rho L_{\text{tot}}^{(1)} + C_{\text{tot}}^{(1)} \rho^2, \quad (635)$$

where

$$L_{\text{tot}}^{(1)} := L_{\text{non}}^{(1)} + \tilde{L}_{\text{non}}^{(1)}, \quad C_{\text{tot}}^{(1)} := C_{\text{non}}^{(1)} + \tilde{C}_{\text{non}}^{(1)}. \quad (636)$$

CCCLXXII. 4. FINAL EXPLICIT SUFFICIENT CUT

Substituting (635) into (628), one gets the clean sufficient criterion

$$\boxed{\rho L_{\text{tot}}^{(1)} + C_{\text{tot}}^{(1)} \rho^2 < \Delta_{\text{core}}^{(1)}} \quad (637)$$

This is the sharpest practical positivity cut in scalar form.

CCCLXXIII. 5. EXPLICIT ρ -WINDOW

If $C_{\text{tot}}^{(1)} > 0$, then (637) is equivalent to

$$\boxed{0 < \rho < \rho_{\text{max}}^{(1)}}, \quad (638)$$

with

$$\boxed{\rho_{\text{max}}^{(1)} = \frac{-L_{\text{tot}}^{(1)} + \sqrt{(L_{\text{tot}}^{(1)})^2 + 4C_{\text{tot}}^{(1)}\Delta_{\text{core}}^{(1)}}}{2C_{\text{tot}}^{(1)}}}. \quad (639)$$

If instead $C_{\text{tot}}^{(1)} = 0$, then the sufficient window reduces to

$$\boxed{0 < \rho < \frac{\Delta_{\text{core}}^{(1)}}{L_{\text{tot}}^{(1)}}}. \quad (640)$$

So the same-shell non-enriched positivity gate is completely reduced to:

$$\boxed{\Delta_{\text{core}}^{(1)} > 0 \quad \text{and} \quad \rho < \rho_{\text{max}}^{(1)}}. \quad (641)$$

CCCLXXIV. 6. MOST TRANSPARENT STATIC CONDITION

The static positivity of the slot requires

$$\boxed{\left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| > 4uA^2 + 2a_K \mathcal{S}_{\text{diag}}^{EE}}. \quad (642)$$

Using

$$\boxed{a_K = \frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2}, \quad (643)$$

this becomes

$$\boxed{\left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| > 4uA^2 + 2\frac{\beta_X^2}{M_X^2} (\Lambda_E^{(0)})^2 \mathcal{S}_{\text{diag}}^{EE}}. \quad (644)$$

This is the clean static benchmark-side dominance condition.

CCCLXXV. 7. FINAL GATE-1 PASS CERTIFICATE

Hence the explicit canonical branch Gate-1 certificate can be written as

$$\boxed{\mathfrak{G}_{\text{gap}}(\rho) > 0 \iff \min \left\{ 4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2 \left| (\lambda + \mu) + \frac{g^2}{M_\sigma^2 + c_\sigma^2 q_*^2} \right|, \Delta_{\text{core}}^{(1)} - \rho L_{\text{tot}}^{(1)} - C_{\text{tot}}^{(1)} \rho^2 \right\} > \eta_{\text{rem}, \rho}}. \quad (645)$$

This is the fully executable scalar pass condition.

CCCLXXVI. 8. COMBINATION WITH THE ALREADY CLOSED BRANCH GATES

Therefore the explicit canonical one-defect acceptance sheet is now

$$\boxed{m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \mathfrak{G}_{\text{gap}}(\rho) > 0,} \quad (646)$$

and a sufficient concrete version is

$$\boxed{m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \Delta_{\text{core}}^{(1)} > 0, \quad \rho < \rho_{\text{max}}^{(1)}, \quad \min\{4K - |r_*|, \sqrt{M_\sigma^2 + c_\sigma^2(q_* - \rho)^2}, q_*^2\} > 0,} \quad (647)$$

CCCLXXVII. FINAL SUMMARY

The same-shell non-enriched benchmark slot is now reduced to a clean scalar positivity cut. Define

$$\Delta_{\text{core}}^{(1)} := \left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 - 2a_K \mathcal{S}_{\text{diag}}^{EE}.$$

Then

$$\Xi_{\text{non}}^{(1)}(\rho) = \Delta_{\text{core}}^{(1)} - \varepsilon_{\text{small-}p, \rho}^{(1)} - \varepsilon_{\text{non-enr}, \rho}^{(1)}.$$

A sharp practical sufficient condition is

$$\rho L_{\text{tot}}^{(1)} + C_{\text{tot}}^{(1)} \rho^2 < \Delta_{\text{core}}^{(1)},$$

which implies

$$\Xi_{\text{non}}^{(1)}(\rho) > 0.$$

Hence the Gate-1 margin

$$\mathfrak{G}_{\text{gap}}(\rho) = \Delta_{\text{bench}, \rho}^{\text{full}} - \eta_{\text{rem}, \rho}$$

is fully reduced to a scalar comparison problem.

Therefore the explicit canonical one-defect acceptance sheet is now

$$m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \mathfrak{G}_{\text{gap}}(\rho) > 0.$$

So the remaining frontier is no longer structural. It is only finite scalar insertion and comparison.

CCCLXXVIII. PRACTICAL FINAL ACCEPTANCE THEOREM ON THE EXPLICIT CANONICAL BRANCH

CCCLXXIX. GOAL

We now bundle the previously separated finite scalar checks into a single practical acceptance theorem for the explicit canonical working branch.

The branch is fixed as

$$\boxed{T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0.} \quad (648)$$

The exact reduced one-defect lowest-channel acceptance conditions are already known to be

$$\boxed{-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench}, \rho}^{\text{full}} > \eta_{\text{rem}, \rho}.} \quad (649)$$

We now rewrite these conditions entirely in explicit scalar form.

CCCLXXX. 1. EXPLICIT BRANCH DICTIONARIES

The burden ratio is

$$R_c = \frac{\xi_c}{\xi_E} \frac{3}{32} \alpha_X (\theta_0 \kappa)^2. \quad (650)$$

On the simplest vacuum asymptotic branch

$$X_{\perp, \infty} = X_{\parallel, \infty} = 0, \quad z_0(\sigma_\infty) > 0,$$

the bulk gate reduces to

$$\mu_\infty > 0 \iff m(\sigma_\infty) > 0. \quad (651)$$

The transverse sign condition becomes

$$\alpha > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0. \quad (652)$$

The remote-gap gate is encoded by the scalar margin

$$\mathfrak{G}_{\text{gap}}(\rho) := \Delta_{\text{bench}, \rho}^{\text{full}} - \eta_{\text{rem}, \rho}, \quad (653)$$

and passes iff

$$\mathfrak{G}_{\text{gap}}(\rho) > 0. \quad (654)$$

CCCLXXXI. 2. FINAL EXPLICIT SCALAR SHEET

Using (650), the window

$$-1 < R_c < 1$$

is equivalent to

$$\left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}. \quad (655)$$

Therefore the invariant sheet (649) is equivalent to the explicit scalar sheet

$$m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0 \kappa)^2}, \quad \mathfrak{G}_{\text{gap}}(\rho) > 0. \quad (656)$$

CCCLXXXII. 3. WITNESS NONDEGENERACY

On the same explicit branch, the carrier coefficients are

$$\lambda_{\parallel} = \frac{3\sqrt{3}\pi}{512} \Gamma_{JJ} m_{\parallel} u_{\parallel}(T^1) \theta_0^3 \kappa^3, \quad (657)$$

$$\alpha = \frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad \beta = 0. \quad (658)$$

Hence a practical sufficient witness nondegeneracy condition is

$$\Gamma_{JJ} \neq 0, \quad \beta_X \neq 0, \quad m_{\parallel} \neq 0, \quad u_{\parallel}(T^1) \neq 0, \quad \theta_0 \kappa \neq 0. \quad (659)$$

Under (659),

$$\lambda_{\parallel} \neq 0, \quad \alpha \neq 0, \quad \beta = 0,$$

so the explicit canonical branch is already a nondegenerate carrier witness branch.

CCCLXXXIII. 4. PRACTICAL FINAL THEOREM

Theorem CCCLXXXIII.1 (Practical explicit-branch one-defect pass theorem). *Assume:*

1. the explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0;$$

2. the lowest projected channel satisfies

$$|\kappa_\star| = 1;$$

3. the asymptotic branch is the simplest vacuum branch,

$$X_{\perp,\infty} = X_{\parallel,\infty} = 0, \quad z_0(\sigma_\infty) > 0;$$

4. the explicit scalar sheet (656) holds;

5. the witness nondegeneracy condition (659) holds.

Then the branch passes the practical one-defect one-mode audit and defines an explicit nondegenerate carrier witness branch.

Proof. By (651), the bulk gate is equivalent to $m(\sigma_\infty) > 0$. By (652), the sign condition $\alpha > 0$ is equivalent to $\frac{\beta_X}{M_X^2}\theta_0\kappa > 0$. By (655), the lowest-channel window $-1 < R_c < 1$ is equivalent to the explicit ratio bound. By (654), the remote-gap support is equivalent to $\mathfrak{G}_{\text{gap}}(\rho) > 0$. Hence (656) is exactly equivalent to the invariant practical acceptance sheet (649).

Finally, (659) implies

$$\lambda_\parallel \neq 0, \quad \alpha \neq 0, \quad \beta = 0,$$

so the accepted branch is a nondegenerate explicit carrier witness branch. $\square$

CCCLXXXIV. 5. HONEST STATUS

This theorem closes the *explicit canonical branch* at the practical one-defect witness level.

It does *not* yet imply that the canonical mixed odd-active branch has been fully populated, because that branch still awaits the two remaining finite microscopic audits:

$$\mathcal{S}_{\text{diag}}^{TT}, L_{\pm}^{TT}, L_0^{TT}, C_{\pm}^{TT}, C_m^{TT}, \quad \Delta e_{\text{mix}}, \Delta o_{\text{mix}}.$$

So the explicit canonical branch now has a practical final pass theorem, whereas the mixed odd-active branch still has a finite coefficient-population problem.

CCCLXXXV. FINAL SUMMARY

The explicit canonical branch

$$T_\star = T^1, \quad \gamma_E = 1, \quad \gamma_O = 0$$

now admits a practical final acceptance theorem.

Its invariant one-defect lowest-channel acceptance sheet

$$-1 < R_c < 1, \quad \alpha > 0, \quad \mu_\infty > 0, \quad \Delta_{\text{bench},\rho}^{\text{full}} > \eta_{\text{rem},\rho}$$

is exactly equivalent to the explicit scalar sheet

$$m(\sigma_\infty) > 0, \quad \frac{\beta_X}{M_X^2}\theta_0\kappa > 0, \quad \left| \frac{\xi_c}{\xi_E} \right| < \frac{32}{3|\alpha_X|(\theta_0\kappa)^2}, \quad \mathfrak{G}_{\text{gap}}(\rho) > 0.$$

If, in addition, the witness nondegeneracy conditions

$$\Gamma_{JJ} \neq 0, \quad \beta_X \neq 0, \quad m_{\parallel} \neq 0, \quad u_{\parallel}(T^1) \neq 0, \quad \theta_0 \kappa \neq 0$$

hold, then

$$\lambda_{\parallel} \neq 0, \quad \alpha \neq 0, \quad \beta = 0,$$

so the branch is a nondegenerate explicit carrier witness branch.

This closes the explicit canonical branch at the practical one-defect witness level.

However, this still does not fully populate the canonical mixed odd-active branch, which remains two finite microscopic audits away from its final reduced coefficient point.

CCCLXXXVI. CONDITIONAL FINAL ACCEPTANCE THEOREM FOR THE CANONICAL MIXED ODD-ACTIVE BRANCH

CCCLXXXVII. GOAL

We now state the sharpest honest final theorem currently available for the canonical mixed odd-active branch. The branch data are fixed as

$$T_E = T_O = 1, \quad R_{\perp} = \sqrt{2}, \quad \varphi = \frac{\pi}{4}, \quad \Delta_{00} = \frac{1}{\sqrt{2}}. \quad (660)$$

The exact reduced mixed-branch coefficient sheet is already fixed:

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (661)$$

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa. \quad (662)$$

What remains are exactly two finite microscopic audits.

CCCLXXXVIII. 1. AUDIT A: MIXED SAME-SHELL SUPPORT

For the canonical mixed branch one has

$$a_T = \frac{2\beta_X^2}{M_X^2}, \quad \mathcal{S}_{ab}^{(\pi/4)} = \frac{1}{2} \mathcal{S}_{ab}^{EE} + \frac{1}{2} \mathcal{S}_{ab}^{OO} + \mathcal{S}_{ab}^{EO}. \quad (663)$$

The exact branch-consistent sufficient same-shell support condition is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}. \quad (664)$$

Equivalently, in scalar locked-core form,

$$\left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}. \quad (665)$$

The only not-yet-inserted same-shell constants are

$$\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, L_-^{TT}, L_0^{TT}, \quad C_+^{TT}, C_-^{TT}, C_m^{TT}. \quad (666)$$

CCCLXXXIX. 2. AUDIT B: MIXED-BLOCK TRANSVERSE CANCELLATION

The exact leading transverse coefficients are

$$e_0^{(1)} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad o_0^{(1)} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}. \quad (667)$$

At the current conservative stage,

$$\ell_0^{(1)} \approx 0, \quad \ell_0^{(3)} \approx \ell_{0;OO}^{(3)}, \quad e_0^{(1)} \approx e_{0;K}^{(1)}, \quad o_0^{(1)} \approx o_{0;K}^{(1)}. \quad (668)$$

Define the mixed-block correction pair

$$\Delta e_{\text{mix}} := e_{0;TD}^{(1)} + e_{0;DT}^{(1)} + e_{0;DD}^{(1)} + \cdots, \quad (669)$$

$$\Delta o_{\text{mix}} := o_{0;TD}^{(1)} + o_{0;DT}^{(1)} + o_{0;DD}^{(1)} + \cdots. \quad (670)$$

Then the actual transverse coefficient point is

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}. \quad (671)$$

A strong sufficient survival condition is

$$|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|. \quad (672)$$

CCCXC. 3. FINAL POPULATED COEFFICIENT POINT AFTER THE TWO AUDITS

Once the seven same-shell constants are inserted and

$$(\Delta e_{\text{mix}}, \Delta o_{\text{mix}})$$

are computed, the final mixed-branch microscopic coefficient point is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)}, \quad (673)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (e_{0;K}^{(1)} + \Delta e_{\text{mix}}), \quad (674)$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (o_{0;K}^{(1)} + \Delta o_{\text{mix}}). \quad (675)$$

Equivalently,

$$P_* K_{\parallel} P_* = \lambda_{\parallel} \sigma_3, \quad P_* K_1 P_* = \alpha (\sigma_1 + \sigma_2), \quad P_* K_2 P_* = \alpha (\sigma_2 - \sigma_1), \quad (676)$$

in the canonical symmetric mixed case $\alpha = \beta$.

CCCXCI. 4. CONDITIONAL FINAL THEOREM

Theorem CCCXCI.1 (Conditional final acceptance theorem for the canonical mixed odd-active branch). *Assume:*

1. the canonical mixed branch data (660);
2. the same-shell support inequality (664);
3. the strong mixed-block survival condition (672);
4. the longitudinal primitive seed survives, i.e.

$$\ell_{0;OO}^{(3)} \neq 0;$$

5. the constant-sector protection and remote-gap support already closed at working-branch level remain valid.

Then the canonical mixed odd-active branch defines a nondegenerate odd-active reduced microscopic carrier, with final coefficient point (673)–(675).

Proof. By (664), the same-shell slot remains spectrally admissible on the canonical mixed branch. By (672), the mixed-block corrections cannot cancel the already explicit K -sector transverse witness pair. Hence $(\Lambda_E, \Lambda_O) \neq (0, 0)$. If in addition $\ell_{0;OO}^{(3)} \neq 0$, then $\lambda_{\parallel} \neq 0$. Therefore the final reduced microscopic coefficient point is nondegenerate and given by (673)–(675). $\square$

CCCXCII. 5. HONEST STATUS

This theorem is the sharpest currently justified endpoint for the canonical mixed odd-active branch. It says:

$$\boxed{\text{no new structural theorem is missing.}} \quad (677)$$

What remains is only:

$$\boxed{\text{insert the seven same-shell constants and compute } (\Delta e_{\text{mix}}, \Delta o_{\text{mix}}).} \quad (678)$$

So the branch is two finite microscopic audits away from a fully populated reduced microscopic coefficient point.

CCCXCIII. FINAL SUMMARY

The canonical mixed odd-active branch is now structurally closed. Its branch data are fixed by

$$T_E = T_O = 1, \quad R_{\perp} = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}},$$

and the only remaining microscopic tasks are exactly two finite audits:

- (A) mixed same-shell support insertion, (B) mixed-block transverse cancellation audit.

If the branch-consistent same-shell support inequality

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}$$

holds, if the mixed-block survival condition

$$|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|$$

holds, and if the primitive longitudinal seed survives

$$\ell_{0;OO}^{(3)} \neq 0,$$

then the branch defines a nondegenerate odd-active reduced microscopic carrier.

Its final coefficient point is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)},$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (e_{0;K}^{(1)} + \Delta e_{\text{mix}}), \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (o_{0;K}^{(1)} + \Delta o_{\text{mix}}).$$

Therefore no new structural theorem is missing. The branch is two finite microscopic audits away from a fully populated reduced microscopic coefficient point.

CCCXCIV. AUDIT A IN FULLY EXECUTABLE FORM: THE CANONICAL TT SEVEN-CONSTANT TABLE

CCCXCV. CURRENT STAGE

For the canonical mixed odd-active branch

$$T_E = T_O = 1, \quad R_{\perp} = \sqrt{2}, \quad \varphi = \frac{\pi}{4},$$

the same-shell support side is already reduced to the branch-consistent inequality

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}. \quad (679)$$

The only uninserted same-shell data are

$$\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, L_-^{TT}, L_0^{TT}, \quad C_+^{TT}, C_-^{TT}, C_m^{TT}. \quad (680)$$

CCCXCVI. 1. EXACT CANONICAL TT -OVERLAP PACKAGE

The branch-consistent mixed overlap family is

$$\mathcal{S}_{ab}^{TT}(p; \varphi) = \cos^2 \varphi \mathcal{S}_{ab}^{EE}(p) + \sin^2 \varphi \mathcal{S}_{ab}^{OO}(p) + 2 \sin \varphi \cos \varphi \mathcal{S}_{ab}^{EO}(p), \quad (681)$$

and at $\varphi = \pi/4$,

$$\mathcal{S}_{ab}^{(\pi/4)}(p) = \frac{1}{2} \mathcal{S}_{ab}^{EE}(p) + \frac{1}{2} \mathcal{S}_{ab}^{OO}(p) + \mathcal{S}_{ab}^{EO}(p). \quad (682)$$

Therefore the diagonal benchmark quantity is exactly

$$\mathcal{S}_{\text{diag}}^{TT} = \mathcal{S}_{++}^{TT}(0) = \mathcal{S}_{--}^{TT}(0) = \frac{1}{2} \mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2} \mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO}, \quad (683)$$

whenever the valley-symmetric diagonal subcase holds.

CCCXCVII. 2. EXACT FIRST-DERIVATIVE VECTORS

Define

$$\ell_{+,i}^{TT} := \partial_{p_i} \mathcal{S}_{++}^{TT}(p)|_{p=0}, \quad \ell_{-,i}^{TT} := \partial_{p_i} \mathcal{S}_{--}^{TT}(p)|_{p=0}, \quad \ell_{0,i}^{TT} := \partial_{p_i} \mathcal{S}_{+-}^{TT}(p)|_{p=0}. \quad (684)$$

By linearity of differentiation and (682),

$$\ell_{+,i}^{TT} = \frac{1}{2} \ell_{+,i}^{EE} + \frac{1}{2} \ell_{+,i}^{OO} + \ell_{+,i}^{EO}, \quad (685)$$

$$\ell_{-,i}^{TT} = \frac{1}{2} \ell_{-,i}^{EE} + \frac{1}{2} \ell_{-,i}^{OO} + \ell_{-,i}^{EO}, \quad (686)$$

$$\ell_{0,i}^{TT} = \frac{1}{2} \ell_{0,i}^{EE} + \frac{1}{2} \ell_{0,i}^{OO} + \ell_{0,i}^{EO}. \quad (687)$$

Thus the three vector-valued derivative packages are exact linear combinations.

CCCXCVIII. 3. NORM-LEVEL BUDGETS

Define

$$L_+^{TT} = \left(\sum_i |\ell_{+,i}^{TT}|^2 \right)^{1/2}, \quad L_-^{TT} = \left(\sum_i |\ell_{-,i}^{TT}|^2 \right)^{1/2}, \quad L_0^{TT} = \left(\sum_i |\ell_{0,i}^{TT}|^2 \right)^{1/2}. \quad (688)$$

Since norms are not linear, the honest branch-level statement is the triangle-inequality bound

$$L_+^{TT} \leq \frac{1}{2} L_+^{EE} + \frac{1}{2} L_+^{OO} + L_+^{EO}, \quad (689)$$

$$L_-^{TT} \leq \frac{1}{2} L_-^{EE} + \frac{1}{2} L_-^{OO} + L_-^{EO}, \quad (690)$$

$$L_0^{TT} \leq \frac{1}{2} L_0^{EE} + \frac{1}{2} L_0^{OO} + L_0^{EO}. \quad (691)$$

These three bounds are the correct scalar inputs for the drift budget.

CCCXCIX. 4. SECOND-DERIVATIVE REMAINDER CONSTANTS

The mixed-branch drift coefficients are

$$a_{+,i}^{(T)} = \delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right) - a_T \ell_{+,i}^{TT}, \quad (692)$$

$$a_{-,i}^{(T)} = -\delta_{i\parallel} q_* \left(Z - \frac{Y q_*^2}{2} \right) - a_T \ell_{-,i}^{TT}, \quad (693)$$

$$b_i^{(T)} = -a_T \ell_{0,i}^{TT}, \quad a_T = \frac{2\beta_X^2}{M_X^2}. \quad (694)$$

Hence

$$A_+^{(T)} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_+^{TT}, \quad (695)$$

$$A_-^{(T)} \leq \left| q_* \left(Z - \frac{Y q_*^2}{2} \right) \right| + a_T L_-^{TT}, \quad (696)$$

$$B^{(T)} \leq a_T L_0^{TT}. \quad (697)$$

Now define the quadratic Taylor-remainder constants by

$$C_+^{TT} := \frac{1}{2} \sup_{|p| \leq \rho} \left\| \nabla_p^2 d_+^{(T)}(p) \right\|_{\text{op}}, \quad C_-^{TT} := \frac{1}{2} \sup_{|p| \leq \rho} \left\| \nabla_p^2 d_-^{(T)}(p) \right\|_{\text{op}}, \quad (698)$$

$$C_m^{TT} := \frac{1}{2} \sup_{|p| \leq \rho} \left\| \nabla_p^2 m^{(T)}(p) \right\|_{\text{op}}. \quad (699)$$

Because $\mathcal{S}^{TT}$ is linear in $\mathcal{S}^{EE}, \mathcal{S}^{OO}, \mathcal{S}^{EO}$, their Hessians add linearly, but the operator norm and supremum produce only a bound:

$$C_+^{TT} \leq \frac{1}{2} C_+^{EE} + \frac{1}{2} C_+^{OO} + C_+^{EO}, \quad (700)$$

$$C_-^{TT} \leq \frac{1}{2} C_-^{EE} + \frac{1}{2} C_-^{OO} + C_-^{EO}, \quad (701)$$

$$C_m^{TT} \leq \frac{1}{2} C_m^{EE} + \frac{1}{2} C_m^{OO} + C_m^{EO}. \quad (702)$$

CD. 5. THE FULLY EXECUTABLE SEVEN-CONSTANT TABLE

The canonical TT seven-constant package is therefore:

constant	status	explicit canonical formula
$\mathcal{S}_{\text{diag}}^{TT}$	exact	$\frac{1}{2} \mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2} \mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO}$
L_+^{TT}	bound	$L_+^{TT} \leq \frac{1}{2} L_+^{EE} + \frac{1}{2} L_+^{OO} + L_+^{EO}$
L_-^{TT}	bound	$L_-^{TT} \leq \frac{1}{2} L_-^{EE} + \frac{1}{2} L_-^{OO} + L_-^{EO}$
L_0^{TT}	bound	$L_0^{TT} \leq \frac{1}{2} L_0^{EE} + \frac{1}{2} L_0^{OO} + L_0^{EO}$
C_+^{TT}	bound	$C_+^{TT} \leq \frac{1}{2} C_+^{EE} + \frac{1}{2} C_+^{OO} + C_+^{EO}$
C_-^{TT}	bound	$C_-^{TT} \leq \frac{1}{2} C_-^{EE} + \frac{1}{2} C_-^{OO} + C_-^{EO}$
C_m^{TT}	bound	$C_m^{TT} \leq \frac{1}{2} C_m^{EE} + \frac{1}{2} C_m^{OO} + C_m^{EO}$

(703)

This is the fully executable same-shell insertion table.

CDI. 6. FULLY INSERTED DRIFT BUDGET

The small- p drift satisfies

$$\varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)} \leq \left[(\rho A_+^{(T)} + C_+^{TT} \rho^2)^2 + (\rho A_-^{(T)} + C_-^{TT} \rho^2)^2 + 2(\rho B^{(T)} + C_m^{TT} \rho^2)^2 \right]^{1/2}, \quad (704)$$

with $A_+^{(T)}, A_-^{(T)}, B^{(T)}$ bounded by (695)–(697).

Therefore the branch-consistent same-shell support inequality becomes

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \left(\frac{1}{2} \mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2} \mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO} \right) + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}. \quad (705)$$

Equivalently, in locked-core scalar form,

$$\left| -\mu^2 + \frac{Zq_*^2}{4} - \frac{Yq_*^4}{16} + 12uA^2 \right| - 4uA^2 > \frac{4\beta_X^2}{M_X^2} \left(\frac{1}{2} \mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2} \mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO} \right) + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}. \quad (706)$$

CDII. 7. CONDITIONAL CLOSURE STATEMENT FOR AUDIT A

Hence Audit A closes once the $EE/OO/EO$ scalar packages are inserted into (703). No new theorem is needed. The only remaining task is the literal numerical or symbolic insertion of the already-defined $EE/OO/EO$ benchmark/Taylor data.

CDIII. FINAL SUMMARY

Audit A is now fully reduced to an explicit insertion table.

For the canonical mixed branch,

$$\mathcal{S}_{ab}^{TT}(p) = \frac{1}{2} \mathcal{S}_{ab}^{EE}(p) + \frac{1}{2} \mathcal{S}_{ab}^{OO}(p) + \mathcal{S}_{ab}^{EO}(p).$$

Hence the diagonal benchmark quantity is exactly

$$\mathcal{S}_{\text{diag}}^{TT} = \frac{1}{2} \mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2} \mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO}.$$

Likewise, the derivative vectors satisfy exact linear identities, while the norm-level quantities

$$L_+^{TT}, L_-^{TT}, L_0^{TT}, C_+^{TT}, C_-^{TT}, C_m^{TT}$$

are controlled by honest upper bounds.

Therefore the fully inserted same-shell support inequality is

$$|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \left(\frac{1}{2} \mathcal{S}_{\text{diag}}^{EE} + \frac{1}{2} \mathcal{S}_{\text{diag}}^{OO} + \mathcal{S}_{\text{diag}}^{EO} \right) + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}.$$

So Audit A is no longer structurally open. What remains is only the literal insertion of the already-defined $EE/OO/EO$ scalar packages.

CDIV. AUDIT B BEYOND THE FORMAL SPLIT: A KK-RESIDUAL DOMINANCE REDUCTION

CDV. SHORT EXPLANATION

At the current stage, the exact transverse extraction is already closed as a blockwise sum:

$$e_0^{(1)} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} e_{0;AB}^{(1)}, \quad o_0^{(1)} = \sum_{AB \in \{TD, DT, OD, DO, DD\}} o_{0;AB}^{(1)}.$$

The remaining issue is therefore not how to define the correction pair

$$(\Delta e_{\text{mix}}, \Delta o_{\text{mix}}),$$

but how to dominate it sharply enough so that the already explicit K -sector witness survives.

The strongest further reduction supported by the current notes is a source-level dominance criterion: the principal transverse source is $JK + KJ$, while the KK -sector is residual.

CDVI. 1. EXACT PRINCIPAL/RESIDUAL SPLIT

Define

$$r_{\text{pr}} := (e_{0;TD}^{(1)} + e_{0;DT}^{(1)}, o_{0;TD}^{(1)} + o_{0;DT}^{(1)}), \quad (707)$$

and

$$r_{\text{rem}} := (\Delta e_{\text{mix}}, \Delta o_{\text{mix}}) = \sum_{AB \in \{OD, DO, DD\}} (e_{0;AB}^{(1)}, o_{0;AB}^{(1)}). \quad (708)$$

Then the full transverse point is

$$(\Lambda_E, \Lambda_O) = r_{\text{pr}} + r_{\text{rem}}. \quad (709)$$

At the current conservative stage,

$$\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}. \quad (710)$$

CDVII. 2. SOURCE-LEVEL CLASSIFICATION

The exact source-by-source classification is already fixed:

$$JJ \rightarrow \lambda_c, \quad JK + KJ \rightarrow (\Lambda_E, \Lambda_O), \quad KK \rightarrow \text{transverse residual corrections}. \quad (711)$$

Thus the principal transverse carrier is the mixed $JK + KJ$ source, while the KK -sector should be treated as a correction sector rather than a principal carrier source.

CDVIII. 3. EXACT QUADRATIC FORMS IN RESPONSE-VECTOR LANGUAGE

The exact finite quadratic forms are

$$W_T = \alpha_X^2 \mathcal{N}_T^{JJ} + 2\alpha_X \beta_X \mathcal{N}_T^{JK} + \beta_X^2 \mathcal{N}_T^{KK}, \quad (712)$$

$$W_s = \alpha_X^2 \mathcal{N}_s^{JJ} + 2\alpha_X \beta_X \mathcal{N}_s^{JK} + \beta_X^2 \mathcal{N}_s^{KK}. \quad (713)$$

Accordingly define the principal transverse weights

$$W_T^{\text{pr}} := 2\alpha_X \beta_X \mathcal{N}_T^{JK}, \quad W_s^{\text{pr}} := 2\alpha_X \beta_X \mathcal{N}_s^{JK}, \quad (714)$$

and the residual KK -weights

$$W_T^{\text{KK}} := \beta_X^2 \mathcal{N}_T^{KK}, \quad W_s^{\text{KK}} := \beta_X^2 \mathcal{N}_s^{KK}. \quad (715)$$

CDIX. 4. SHELL-SUPPORT SUM RULE

For each mixed shell-defect Hessian block B , if the shell-support property holds for the transverse probe, then the exact mixed-sector sum rule is

$$\boxed{g_{B,+}^2 + g_{B,-}^2 = g_{s_B}^2.} \quad (716)$$

Hence shell-defect mixing cannot enhance the total mixed-sector transverse weight beyond the corresponding shell-slot bare weight.

This gives the sharpest currently available principle for the residual audit: the KK -sector may renormalize the transverse pair, but it cannot produce an arbitrarily larger mixed-sector weight than its own bare shell slot.

CDX. 5. PRACTICAL SUFFICIENT DOMINANCE CRITERION

The strong coefficient-level survival condition is

$$\boxed{|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|.} \quad (717)$$

A source-level sufficient reduction of this is to demand that the residual KK -weights be dominated by the principal JK -weights:

$$\boxed{|W_T^{\text{KK}}| < |W_T^{\text{pr}}|, \quad |W_s^{\text{KK}}| < |W_s^{\text{pr}}|.} \quad (718)$$

Equivalently,

$$\boxed{\beta_X^2 |\mathcal{N}_T^{\text{KK}}| < 2|\alpha_X \beta_X| |\mathcal{N}_T^{\text{JK}}|, \quad \beta_X^2 |\mathcal{N}_s^{\text{KK}}| < 2|\alpha_X \beta_X| |\mathcal{N}_s^{\text{JK}}|.} \quad (719)$$

A rotationally invariant combined sufficient criterion is

$$\boxed{\sqrt{(W_T^{\text{KK}})^2 + (W_s^{\text{KK}})^2} < \sqrt{(W_T^{\text{pr}})^2 + (W_s^{\text{pr}})^2}.} \quad (720)$$

a. Honest status. The inequalities (718)–(720) are not exact identities between

$$(\Delta e_{\text{mix}}, \Delta o_{\text{mix}}) \quad \text{and} \quad (W_T^{\text{KK}}, W_s^{\text{KK}}),$$

but they are the strongest source-level sufficient dominance criteria currently justified by the exact response-weight decomposition and shell-support sum rule.

CDXI. 6. EXCHANGE-SYMMETRIC RESIDUAL SIMPLIFICATION

If the projected residual sector preserves

$$O \leftrightarrow D$$

exchange symmetry, then

$$\boxed{e_{0;DO}^{(1)} = e_{0;OD}^{(1)}, \quad o_{0;DO}^{(1)} = o_{0;OD}^{(1)}.} \quad (721)$$

Hence

$$\boxed{r_{\text{rem}} = 2(e_{0;OD}^{(1)}, o_{0;OD}^{(1)}) + (e_{0;DD}^{(1)}, o_{0;DD}^{(1)}).} \quad (722)$$

So once exchange symmetry is imposed, Audit B reduces to:

1. one OD -pair,
2. one DD -pair,
3. the KK -dominance comparison against the principal JK -pair.

CDXII. 7. CONDITIONAL FINAL REDUCTION OF AUDIT B

Therefore the sharpest current practical reduction of Audit B is the following:

$$\boxed{\text{Audit B closes if either the coefficient-level strong condition (717) is checked directly,}} \quad (723)$$

or, more conservatively,

$$\boxed{\text{the source-level } KK\text{-dominance bounds (719) or (720) are satisfied.}} \quad (724)$$

In that case the already explicit K -sector witness pair survives, and hence

$$(\Lambda_E, \Lambda_O) \neq (0, 0).$$

CDXIII. 8. FINAL MIXED-BRANCH COEFFICIENT POINT UNDER THE TWO AUDITS

Once Audit A is inserted and Audit B passes, the final mixed-branch microscopic coefficient point is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)}, \quad (725)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (e_{0;K}^{(1)} + \Delta e_{\text{mix}}), \quad (726)$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (o_{0;K}^{(1)} + \Delta o_{\text{mix}}). \quad (727)$$

So no new structural theorem remains. What remains is only literal finite insertion and comparison.

CDXIV. FINAL SUMMARY

The sharpest further reduction of Audit B supported by the current notes is a source-level dominance criterion. The exact source classification is

$$JJ \rightarrow \lambda_c, \quad JK + KJ \rightarrow (\Lambda_E, \Lambda_O), \quad KK \rightarrow \text{transverse residual corrections.}$$

Therefore the principal transverse weights are

$$W_T^{\text{pr}} = 2\alpha_X \beta_X \mathcal{N}_T^{JK}, \quad W_s^{\text{pr}} = 2\alpha_X \beta_X \mathcal{N}_s^{JK},$$

while the residual KK -weights are

$$W_T^{\text{KK}} = \beta_X^2 \mathcal{N}_T^{KK}, \quad W_s^{\text{KK}} = \beta_X^2 \mathcal{N}_s^{KK}.$$

A practical sufficient condition for residual non-cancellation is

$$\beta_X^2 |\mathcal{N}_T^{KK}| < 2|\alpha_X \beta_X| |\mathcal{N}_T^{JK}|, \quad \beta_X^2 |\mathcal{N}_s^{KK}| < 2|\alpha_X \beta_X| |\mathcal{N}_s^{JK}|.$$

Equivalently, in Euclidean form,

$$\sqrt{(W_T^{\text{KK}})^2 + (W_s^{\text{KK}})^2} < \sqrt{(W_T^{\text{pr}})^2 + (W_s^{\text{pr}})^2}.$$

If, in addition, the residual sector preserves $O \leftrightarrow D$ exchange symmetry, then

$$r_{\text{rem}} = 2(e_{0;OD}^{(1)}, o_{0;OD}^{(1)}) + (e_{0;DD}^{(1)}, o_{0;DD}^{(1)}),$$

so Audit B reduces to one OD -pair, one DD -pair, and one KK -dominance comparison.

Thus the branch is now reduced to: one same-shell finite insertion audit, and one source-level residual dominance audit.

CDXV. AUDIT B AS AN EXPLICIT SOURCE-SPLIT COEFFICIENT FORMULA

CDXVI. SHORT EXPLANATION

The strongest further reduction of Audit B is not another abstract survival statement. It is the explicit source-split formula for the transverse coefficients themselves.

The exact Schur-corrected effective operator already has the source decomposition

$$\Delta H_{\text{eff}} = -\frac{\alpha_X^2}{M_X^2}(\delta J \delta J^\dagger) - \frac{\alpha_X \beta_X}{M_X^2}(\delta J \delta K^\dagger + \delta K \delta J^\dagger) - \frac{\beta_X^2}{M_X^2}(\delta K \delta K^\dagger).$$

Therefore every reduced coefficient splits into JJ , $JK + KJ$, and KK pieces.

CDXVII. 1. EXACT SOURCE DECOMPOSITION OF THE REDUCED COEFFICIENTS

The reduced invariants satisfy

$$\Delta \lambda_{\parallel} = -\frac{\alpha_X^2}{M_X^2} \Lambda_c^{JJ} - \frac{\alpha_X \beta_X}{M_X^2} \Lambda_c^{JK} - \frac{\beta_X^2}{M_X^2} \Lambda_c^{KK}, \quad (728)$$

$$\Delta \alpha = -\frac{\alpha_X^2}{M_X^2} \Lambda_E^{JJ} - \frac{\alpha_X \beta_X}{M_X^2} \Lambda_E^{JK} - \frac{\beta_X^2}{M_X^2} \Lambda_E^{KK}, \quad (729)$$

$$\Delta \beta = -\frac{\alpha_X^2}{M_X^2} \Lambda_O^{JJ} - \frac{\alpha_X \beta_X}{M_X^2} \Lambda_O^{JK} - \frac{\beta_X^2}{M_X^2} \Lambda_O^{KK}. \quad (730)$$

Now insert the already-closed routing logic:

$$\Lambda_E^{JJ} = \Lambda_O^{JJ} = 0, \quad \Lambda_c^{JK} = 0, \quad \Lambda_c^{KK} = 0 \quad \text{at principal shell order.} \quad (731)$$

Therefore the principal reduced source structure collapses to

$$\lambda_c \sim \lambda_c^{JJ}, \quad (732)$$

$$\Lambda_E \sim \Lambda_E^{JK} + \Lambda_E^{KK}, \quad \Lambda_O \sim \Lambda_O^{JK} + \Lambda_O^{KK}. \quad (733)$$

CDXVIII. 2. EXPLICIT MIXED-BRANCH COEFFICIENT FORMULAS

After overlap insertion, the canonical mixed-branch coefficient dictionary is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa.$$

Using (732) and (733), one gets the sharper source-split formulas

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \frac{\alpha_X^2}{M_X^2} \Lambda_c^{JJ} \theta_0^3 \kappa^3, \quad (734)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} \Lambda_E^{JK} + \frac{\beta_X^2}{M_X^2} \Lambda_E^{KK} \right], \quad (735)$$

$$\beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left[\frac{\alpha_X\beta_X}{M_X^2}\Lambda_O^{JK} + \frac{\beta_X^2}{M_X^2}\Lambda_O^{KK}\right]. \quad (736)$$

These are more explicit than the earlier

$$\alpha = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa(e_{0;K}^{(1)} + \Delta e_{\text{mix}}), \quad \beta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa(o_{0;K}^{(1)} + \Delta o_{\text{mix}}),$$

because they identify the principal source and the residual source separately.

CDXIX. 3. IDENTIFICATION OF THE PRINCIPAL AND RESIDUAL TRANSVERSE PIECES

Comparing (735)–(736) with the conservative mixed-branch form, one may identify

$$e_{0;K}^{(1)} \sim \frac{\alpha_X\beta_X}{M_X^2}\Lambda_E^{JK}, \quad o_{0;K}^{(1)} \sim \frac{\alpha_X\beta_X}{M_X^2}\Lambda_O^{JK}, \quad (737)$$

and

$$\Delta e_{\text{mix}} \sim \frac{\beta_X^2}{M_X^2}\Lambda_E^{KK}, \quad \Delta o_{\text{mix}} \sim \frac{\beta_X^2}{M_X^2}\Lambda_O^{KK}, \quad (738)$$

at the source-classification level.

a. Honest status. These are not exact equalities block-by-block in the already-expanded

$$TD, DT, OD, DO, DD$$

basis. They are the sharpest source-level identifications currently justified by the exact Schur split and the already-closed routing logic.

CDXX. 4. PRACTICAL DOMINANCE CRITERIA

The strong coefficient-level survival condition

$$|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|$$

is therefore implied by the explicit source-level inequalities

$$\left|\frac{\beta_X^2}{M_X^2}\Lambda_E^{KK}\right| < \left|\frac{\alpha_X\beta_X}{M_X^2}\Lambda_E^{JK}\right|, \quad (739)$$

$$\left|\frac{\beta_X^2}{M_X^2}\Lambda_O^{KK}\right| < \left|\frac{\alpha_X\beta_X}{M_X^2}\Lambda_O^{JK}\right|. \quad (740)$$

Equivalently,

$$|\beta_X||\Lambda_E^{KK}| < |\alpha_X||\Lambda_E^{JK}|, \quad |\beta_X||\Lambda_O^{KK}| < |\alpha_X||\Lambda_O^{JK}|. \quad (741)$$

A rotationally invariant sufficient version is

$$|\beta_X|\sqrt{(\Lambda_E^{KK})^2 + (\Lambda_O^{KK})^2} < |\alpha_X|\sqrt{(\Lambda_E^{JK})^2 + (\Lambda_O^{JK})^2}. \quad (742)$$

CDXXI. 5. WHAT THIS MEANS FOR THE MIXED ODD-ACTIVE BRANCH

Thus the canonical mixed odd-active branch closes once the following hold:

1. the same-shell support inequality from Audit A;
2. the longitudinal primitive seed

$$\Lambda_c^{JJ} \neq 0;$$

3. the mixed-source transverse pair survives

$$(\Lambda_E^{JK}, \Lambda_O^{JK}) \neq (0, 0);$$

4. the KK -residual is dominated by the JK -source in the sense of (741) or (742).

Under these conditions the final coefficient point is

$$\begin{aligned} \lambda_{\parallel} &= -\frac{3\sqrt{3}\pi}{512} \frac{\alpha_X^2}{M_X^2} \Lambda_c^{JJ} \theta_0^3 \kappa^3, \\ \alpha &= -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} \Lambda_E^{JK} + \frac{\beta_X^2}{M_X^2} \Lambda_E^{KK} \right], \\ \beta &= -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} \Lambda_O^{JK} + \frac{\beta_X^2}{M_X^2} \Lambda_O^{KK} \right]. \end{aligned}$$

So Audit B is no longer an amorphous mixed-block remainder. It is a concrete comparison between a principal JK -pair and a residual KK -pair.

CDXXII. FINAL SUMMARY

Audit B can now be reduced to an explicit source-split coefficient formula.

Using the exact Schur decomposition and the already-closed routing logic, the longitudinal coefficient is sourced only by the JJ -sector:

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \frac{\alpha_X^2}{M_X^2} \Lambda_c^{JJ} \theta_0^3 \kappa^3.$$

The transverse coefficients split into a principal JK -pair and a residual KK -pair:

$$\begin{aligned} \alpha &= -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} \Lambda_E^{JK} + \frac{\beta_X^2}{M_X^2} \Lambda_E^{KK} \right], \\ \beta &= -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} \Lambda_O^{JK} + \frac{\beta_X^2}{M_X^2} \Lambda_O^{KK} \right]. \end{aligned}$$

Therefore Audit B is no longer an undifferentiated mixed-block correction problem. It is a concrete competition between the principal JK -source and the residual KK -source.

A practical sufficient dominance criterion is

$$|\beta_X| |\Lambda_E^{KK}| < |\alpha_X| |\Lambda_E^{JK}|, \quad |\beta_X| |\Lambda_O^{KK}| < |\alpha_X| |\Lambda_O^{JK}|,$$

or equivalently

$$|\beta_X| \sqrt{(\Lambda_E^{KK})^2 + (\Lambda_O^{KK})^2} < |\alpha_X| \sqrt{(\Lambda_E^{JK})^2 + (\Lambda_O^{JK})^2}.$$

Hence the canonical mixed odd-active branch closes once: same-shell support holds, the longitudinal JJ -seed survives, the principal JK -transverse pair is nonzero, and the residual KK -pair is dominated by the JK -pair.

CDXXIII. EXPLICIT JK -DRIVEN PRINCIPAL TRANSVERSE COEFFICIENTS ON THE FROZEN CORRECTED BRANCH

CDXXIV. SHORT EXPLANATION

The source routing was already known:

$$JJ \rightarrow \lambda_c, \quad JK + KJ \rightarrow (\Lambda_E, \Lambda_O), \quad KK \rightarrow \text{transverse residual corrections.}$$

The genuinely new forward step is that, on the current refined locked + Class II working branch, the JK -driven principal transverse pair is already reduced to an explicit finite contraction.

CDXXV. 1. BACKGROUND CURRENTS

On the locked one-defect background

$$\Psi_B(\xi, x_{\parallel}) = \phi_0(x_{\parallel}) z_B(\xi), \quad P_B(\xi) = z_B(\xi) z_B^{\dagger}(\xi),$$

the microscopic currents are

$$J_i^a[\Psi_B] = 2(\partial_i \phi_0) \phi_0 m_a(\theta) + \delta_{i\xi} \phi_0^2 \mathcal{A}_{\xi}^a(\theta, \theta'), \quad (743)$$

with

$$m_a(\theta) = z_B^{\dagger} T_a z_B, \quad \mathcal{A}_{\xi}^a = (\partial_{\xi} z_B^{\dagger}) T_a z_B + z_B^{\dagger} T_a (\partial_{\xi} z_B), \quad (744)$$

and

$$K_i^a[P_B, \psi] = \delta_{i\xi} \theta'(\xi) k_a(\theta) \mathcal{O}[\psi], \quad k_a(\theta) := \text{Tr} \left[T_a (e_1 z_B^{\dagger} + z_B e_1^{\dagger}) \right]. \quad (745)$$

Thus the background J -sector carries the shell gradient, while the K -sector carries the defect/projector derivative.

CDXXVI. 2. PRINCIPAL JK -PAIR

At the current refined locked + Class II branch, the mixed $JK + KJ$ source yields

$$c_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad c_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (746)$$

Hence the principal transverse pair may be identified as

$$\Lambda_E^{JK} = (j \cdot k) \Lambda_E^{(0)}, \quad \Lambda_O^{JK} = (j \cdot k) \Lambda_O^{(0)}. \quad (747)$$

This is the sharpest explicit source-level reduction currently available for the principal transverse source.

CDXXVII. 3. MIXED-BRANCH COEFFICIENT FORMULAS WITH EXPLICIT JK -PAIR

The exact reduced coefficient dictionary is

$$\alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O, \quad I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

Separating principal JK and residual KK pieces, one gets

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)} + \frac{\beta_X^2}{M_X^2} \Lambda_E^{KK} \right], \quad (748)$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)} + \frac{\beta_X^2}{M_X^2} \Lambda_O^{KK} \right]. \quad (749)$$

So the principal/transverse competition is now completely explicit:

principal JK vs. residual KK .

CDXXVIII. 4. SHARP PRACTICAL DOMINANCE CRITERION

The KK -residual cannot kill the principal transverse pair if

$$|\beta_X| |\Lambda_E^{KK}| < |\alpha_X| |j \cdot k| |\Lambda_E^{(0)}|, \quad (750)$$

$$|\beta_X| |\Lambda_O^{KK}| < |\alpha_X| |j \cdot k| |\Lambda_O^{(0)}|. \quad (751)$$

A rotationally invariant sufficient condition is

$$|\beta_X| \sqrt{(\Lambda_E^{KK})^2 + (\Lambda_O^{KK})^2} < |\alpha_X| |j \cdot k| \sqrt{(\Lambda_E^{(0)})^2 + (\Lambda_O^{(0)})^2}. \quad (752)$$

This is the sharpest currently justified scalar dominance cut for Audit B.

CDXXIX. 5. HONEST ODD-CHANNEL STATUS

There is one important caveat. The simplest local real Hermitian JK bilinear can vanish identically by Pauli anticommutation, so a nonzero odd transverse witness is not automatic from the naive local shortcut alone. Moreover, at the minimal rigid-internal first-shell truncation,

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0,$$

so orbit-amplitude corrections alone do not activate the odd channel.

Therefore:

$$\Lambda_O^{(0)} \neq 0 \text{ requires genuinely derivative-resolved and/or phase-resolved microscopic structure.} \quad (753)$$

So the honest remaining transverse frontier is not the whole JK -pair, but specifically the explicit activation of the odd slot.

CDXXX. 6. FINAL MIXED-BRANCH SOURCE-SPLIT COEFFICIENTS

Combining the already-closed longitudinal source with the new principal JK -pair, the mixed odd-active branch coefficient point is now reduced to

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \frac{\alpha_X^2}{M_X^2} \Lambda_c^{JJ} \theta_0^3 \kappa^3, \quad (754)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)} + \frac{\beta_X^2}{M_X^2} \Lambda_E^{KK} \right], \quad (755)$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)} + \frac{\beta_X^2}{M_X^2} \Lambda_O^{KK} \right]. \quad (756)$$

This is the sharpest explicitly source-split form currently available.

CDXXXI. CONCLUSION

The principal transverse source is no longer abstract. It is controlled by the finite background contraction

$$(j \cdot k)$$

and the shell-doublet invariants

$$\Lambda_E^{(0)}, \quad \Lambda_O^{(0)}.$$

So the remaining literal microscopic tasks are now exactly:

1. evaluate $j \cdot k$ on the frozen corrected branch;
2. determine whether $\Lambda_O^{(0)}$ is activated by the genuinely derivative/phase-resolved microscopic slot;
3. estimate $(\Lambda_E^{KK}, \Lambda_O^{KK})$ and check the dominance inequality.

No larger structural ambiguity remains.

CDXXXII. FINAL SUMMARY

The principal transverse JK -source is now reduced to an explicit finite contraction on the frozen corrected branch:

$$\Lambda_E^{JK} = (j \cdot k) \Lambda_E^{(0)}, \quad \Lambda_O^{JK} = (j \cdot k) \Lambda_O^{(0)}.$$

Therefore the mixed odd-active branch coefficients are

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \frac{\alpha_X^2}{M_X^2} \Lambda_c^{JJ} \theta_0^3 \kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)} + \frac{\beta_X^2}{M_X^2} \Lambda_E^{KK} \right],$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)} + \frac{\beta_X^2}{M_X^2} \Lambda_O^{KK} \right].$$

A practical sufficient residual-dominance condition is

$$|\beta_X| \sqrt{(\Lambda_E^{KK})^2 + (\Lambda_O^{KK})^2} < |\alpha_X| |j \cdot k| \sqrt{(\Lambda_E^{(0)})^2 + (\Lambda_O^{(0)})^2}.$$

So the remaining literal microscopic tasks are: evaluate $j \cdot k$, determine whether $\Lambda_O^{(0)}$ is actually activated, and estimate the residual KK -pair.

CDXXXIII. MINIMAL MICROSCOPIC ACTIVATION ROUTE FOR THE ODD TRANSVERSE SLOT

CDXXXIV. CURRENT STAGE

The remaining sharp microscopic question is not whether an odd transverse slot is logically allowed. That has already been established.

The actual question is:

what is the *minimal* microscopic slot that activates $\Lambda_O^{(0)}$ on the intended branch?

(757)

The honest answer splits into two different notions:

1. the minimal odd source *inside the current frozen corrected Class II working branch*;
2. the minimal odd source *after enlarging the complete action by an explicit odd carrier slot*.

CDXXXV. 1. WHY THE NAIVE LOCAL JK ROUTE IS NOT THE ANSWER

At the shell-adapted Pauli level,

$$J \sim j \Sigma_3, \quad K \sim k_E \Sigma_1 + k_O \Sigma_2.$$

Then the naive local real Hermitian JK bilinear gives

$$\boxed{\Sigma_3(k_E \Sigma_1 + k_O \Sigma_2) + (k_E \Sigma_1 + k_O \Sigma_2) \Sigma_3 = 0} \quad (758)$$

by Pauli anticommutation.

Therefore

$$\boxed{\text{the naive local real Hermitian } JK \text{ bilinear does } \textit{not} \text{ activate } \Lambda_O^{(0)}}. \quad (759)$$

So a nonzero odd transverse coefficient cannot be claimed from a local algebraic shortcut. A genuinely nonzero odd channel requires derivative-resolved and/or phase-resolved microscopic structure.

CDXXXVI. 2. FIRST ACTIVE ODD SOURCE ON THE FROZEN CORRECTED BRANCH

The exact odd transverse source on the current working branch is

$$\boxed{X_2(\delta) = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)]}. \quad (760)$$

Its corrected odd transverse Hessian response is

$$\boxed{H_{rs}^{(2)} := \partial_{\delta_r} \partial_{\delta_s} X_2(\delta)|_{\delta=0} = \partial_{\delta_r} \partial_{\delta_s} \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)]_{\delta=0}}. \quad (761)$$

On the refined locked + Class II frozen working branch, the leading mixed-channel realization is

$$\boxed{H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[\partial_{rs}(\delta J \delta K^\dagger + \delta K \delta J^\dagger)_{\delta=0}] + \text{subleading completion terms.}} \quad (762)$$

Moreover, on the sign-resolved first-shell lift of the frozen corrected seed, the exact transverse rows satisfy

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.$$

Hence there is no linear odd transverse activation at the frozen corrected seed.

Therefore the first genuine odd source on the current branch is not linear but quadratic:

$$\boxed{\Lambda_O^{(0)} \text{ first appears through the quadratic Hessian response } H_{rs}^{(2)}}. \quad (763)$$

At the level of the corrected branch displacement δ^{corr} , the natural leading-order estimate is

$$\boxed{\Lambda_O^{(0)} \sim \frac{1}{2} \delta_r^{\text{corr}} H_{rs}^{(2)} \delta_s^{\text{corr}}}, \quad (764)$$

up to higher-order completion terms. This is an inference from the Hessian-response definition and the absence of linear activation.

CDXXXVII. 3. THE MINIMAL ODD SLOT AFTER COMPLETE-ACTION ENLARGEMENT

The complete mixing sector can include an explicit odd carrier slot

$$\boxed{\mathcal{L}_{\text{mix}}^{\text{comp}} \supset \gamma_O \mathcal{I}_O + \text{h.c.}} \quad (765)$$

in addition to the J - and K -couplings.

On the embedded defect branch,

$$q_O(\theta) = \text{Tr}(S_O M_O) = \cos \theta,$$

so the explicit odd carrier density is locally nonzero:

$$\boxed{\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi.} \quad (766)$$

The induced tangent transport block is

$$\boxed{V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right].} \quad (767)$$

Hence the local odd density is

$$\boxed{b_O^{(EO)} = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi).} \quad (768)$$

Therefore, in the enlarged complete-action sense,

$$\boxed{\text{the minimal } \textit{linear} \text{ odd microscopic slot is the explicit odd carrier } \mathcal{I}_O.} \quad (769)$$

CDXXXVIII. 4. HONEST TWO-ROUTE CONCLUSION

We can now distinguish the two notions sharply:

a. (A) *Minimal odd source inside the current frozen corrected branch.*

$$\boxed{\text{quadratic mixed } JK + KJ \text{ Hessian response } H_{rs}^{(2)}.} \quad (770)$$

b. (B) *Minimal odd source after complete-action enlargement.*

$$\boxed{\text{explicit odd carrier slot } \mathcal{I}_O \text{ generating } V_{EO}^\mu \propto \theta'(\xi) \cos \theta(\xi) J^\mu{}_a \Sigma_a.} \quad (771)$$

So the current branch and the completed action do not have the same minimal odd activation mechanism.

CDXXXIX. 5. CONSEQUENCE FOR THE MIXED ODD-ACTIVE COEFFICIENT SHEET

Accordingly, the odd coefficient in the mixed odd-active branch can be written in two logically distinct ways.

a. *Current frozen corrected branch.*

$$\boxed{\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)} + \frac{\beta_X^2}{M_X^2} \Lambda_O^{KK} \right],} \quad (772)$$

with

$$\Lambda_O^{(0)} \sim \frac{1}{2} \delta_r^{\text{corr}} H_{rs}^{(2)} \delta_s^{\text{corr}}$$

as the first genuinely active odd source.

b. *Enlarged complete-action route.*

$$\boxed{\beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa \left[\frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)} + \frac{\beta_X^2}{M_X^2} \Lambda_O^{KK} + \tilde{\Lambda}_O^{(EO)} \right],} \quad (773)$$

where $\tilde{\Lambda}_O^{(EO)}$ is the coefficient induced by the explicit odd carrier slot.

CDXL. CONCLUSION

The minimal microscopic activation route for the odd transverse slot is now sharply identified:

$$\boxed{\text{inside the current branch: quadratic Hessian response;}} \quad (774)$$

$$\boxed{\text{inside the enlarged complete action: explicit odd carrier } \mathcal{I}_O.} \quad (775)$$

So the real remaining microscopic frontier is no longer “does an odd slot exist?” It is:

$$\boxed{\text{which of these two routes is actually realized in the physically intended complete TECT branch?}} \quad (776)$$

CDXLI. FINAL SUMMARY

The minimal microscopic activation mechanism for the odd transverse slot is now sharply separated into two routes.

Inside the current frozen corrected Class II working branch, the naive local real Hermitian JK bilinear vanishes. There is no linear odd activation at the frozen corrected seed. Hence the first genuinely active odd source is the quadratic Hessian response

$$H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_{rs} (\delta J \delta K^\dagger + \delta K \delta J^\dagger)_{\delta=0} \right] + \dots,$$

so that

$$\Lambda_O^{(0)} \sim \frac{1}{2} \delta_r^{\text{corr}} H_{rs}^{(2)} \delta_s^{\text{corr}}.$$

By contrast, after enlarging the complete action, the explicit odd carrier slot

$$\gamma_O \mathcal{I}_O$$

provides a linear odd route through

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right],$$

with local odd density

$$b_O^{(EO)} = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi).$$

Therefore the true remaining microscopic frontier is not the existence of an odd slot, but which of these two activation routes is physically realized in the intended complete TECT branch.

CDXLII. ACTUAL CONTRACTION OF THE ODD HESSIAN RESPONSE ON THE FROZEN CORRECTED BRANCH

CDXLIII. SHORT EXPLANATION

The corrected branch is frozen at

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (\delta_A, \delta_E, \delta_O) = (0.0096, -0.0288, 0.0096).$$

The odd transverse coefficient starts at quadratic order:

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta^{\text{final}}) + O(\delta^3).$$

So the correct next step is to contract the quadratic form explicitly.

CDXLIV. 1. EXACT QUADRATIC CONTRACTION ON δ^{final}

By definition,

$$Q_2(\delta) = \frac{1}{2} \sum_{r,s \in \{A,E,O\}} H_{rs}^{(2)} \delta_r \delta_s. \quad (777)$$

For

$$\delta_A = \delta_O = d, \quad \delta_E = -3d, \quad d = 0.0096,$$

one has

$$d^2 = 9.216 \times 10^{-5}.$$

Therefore

$$Q_2(\delta^{\text{final}}) = \frac{d^2}{2} \left[H_{AA}^{(2)} + 9H_{EE}^{(2)} + H_{OO}^{(2)} - 6H_{AE}^{(2)} + 2H_{AO}^{(2)} - 6H_{EO}^{(2)} \right] \quad (778)$$

$$= 4.608 \times 10^{-5} \left[H_{AA}^{(2)} + 9H_{EE}^{(2)} + H_{OO}^{(2)} - 6H_{AE}^{(2)} + 2H_{AO}^{(2)} - 6H_{EO}^{(2)} \right]. \quad (779)$$

Define the contracted odd Hessian combination

$$\mathfrak{H}_O^{\text{corr}} := H_{AA}^{(2)} + 9H_{EE}^{(2)} + H_{OO}^{(2)} - 6H_{AE}^{(2)} + 2H_{AO}^{(2)} - 6H_{EO}^{(2)}. \quad (780)$$

Then

$$Q_2(\delta^{\text{final}}) = 4.608 \times 10^{-5} \mathfrak{H}_O^{\text{corr}}. \quad (781)$$

Hence the corrected odd coefficient is

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 (4.608 \times 10^{-5}) \mathfrak{H}_O^{\text{corr}} + O(\delta^3). \quad (782)$$

So the odd slot is activated on the frozen corrected branch if and only if

$$\mathfrak{H}_O^{\text{corr}} \neq 0. \quad (783)$$

CDXLV. 2. EXACT SIX-ROW MICROSCOPIC FORMULA

The common exact working-branch definition of the six Hessian rows is

$$H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{rs} K_0^\dagger + J_r K_s^\dagger + J_s K_r^\dagger + J_0 K_{rs}^\dagger + K_{rs} J_0^\dagger + K_r J_s^\dagger + K_s J_r^\dagger + K_0 J_{rs}^\dagger \right] + \Delta_{rs}^{\text{comp}}. \quad (784)$$

Substituting (784) into (780), one gets

$$\mathfrak{H}_O^{\text{corr}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{X}_O + \mathfrak{D}_O, \quad (785)$$

where

$$\mathfrak{X}_O := \mathfrak{P}_O \left[\mathcal{C}_{AA} + 9\mathcal{C}_{EE} + \mathcal{C}_{OO} - 6\mathcal{C}_{AE} + 2\mathcal{C}_{AO} - 6\mathcal{C}_{EO} \right], \quad (786)$$

with

$$\mathcal{C}_{rs} := J_{rs} K_0^\dagger + J_r K_s^\dagger + J_s K_r^\dagger + J_0 K_{rs}^\dagger + K_{rs} J_0^\dagger + K_r J_s^\dagger + K_s J_r^\dagger + K_0 J_{rs}^\dagger, \quad (787)$$

and

$$\mathfrak{D}_O := \Delta_{AA}^{\text{comp}} + 9\Delta_{EE}^{\text{comp}} + \Delta_{OO}^{\text{comp}} - 6\Delta_{AE}^{\text{comp}} + 2\Delta_{AO}^{\text{comp}} - 6\Delta_{EO}^{\text{comp}}. \quad (788)$$

Therefore the corrected odd coefficient becomes

$$\beta^{\text{corr}} = \frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 (4.608 \times 10^{-5}) \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{X}_O - \frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 (4.608 \times 10^{-5}) \mathfrak{D}_O + O(\delta^3). \quad (789)$$

This is the fully contracted source-split odd-slot formula on the frozen corrected branch.

CDXLVI. 3. FAST PRACTICAL APPROXIMATION

Under the fast practical approximation

$$J_{rs} \approx 0, \quad K_{rs} \approx 0, \quad \Delta_{rs}^{\text{comp}} \approx 0,$$

all six rows reduce to

$$H_{rs}^{(2),\text{fast}} \approx -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger]. \quad (790)$$

Then

$$\mathfrak{H}_O^{\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{X}_O^{\text{fast}}, \quad (791)$$

with

$$\mathfrak{X}_O^{\text{fast}} := \mathfrak{P}_O [\mathcal{B}_{AA} + 9\mathcal{B}_{EE} + \mathcal{B}_{OO} - 6\mathcal{B}_{AE} + 2\mathcal{B}_{AO} - 6\mathcal{B}_{EO}], \quad (792)$$

where

$$\mathcal{B}_{rs} := J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger. \quad (793)$$

Hence

$$\beta_{\text{fast}}^{\text{corr}} \approx \frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 (4.608 \times 10^{-5}) \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{X}_O^{\text{fast}}. \quad (794)$$

So even in the fast approximation, the odd-slot question has been reduced to one scalar:

$$\mathfrak{X}_O^{\text{fast}}.$$

CDXLVII. 4. A CLEAN SUFFICIENT NONZERO CRITERION

A sharp sufficient condition for odd-slot activation is

$$\left| \frac{\alpha_X \beta_X}{M_X^2} \mathfrak{X}_O \right| > |\mathfrak{D}_O|. \quad (795)$$

Indeed, under (785), this implies

$$\mathfrak{H}_O^{\text{corr}} \neq 0$$

and therefore $\beta^{\text{corr}} \neq 0$ up to the overall prefactor $\theta_0 \kappa A_0 \neq 0$.

In the fast approximation this reduces simply to

$$\mathfrak{X}_O^{\text{fast}} \neq 0 \implies \beta_{\text{fast}}^{\text{corr}} \neq 0 \quad (796)$$

provided

$$\alpha_X \beta_X \theta_0 \kappa A_0 \neq 0.$$

CDXLVIII. 5. A USEFUL DOMINANCE CUT DIRECTLY ON THE SIX ROWS

Because the EE -entry carries the largest weight 9, one convenient sufficient criterion is

$$\boxed{|H_{EE}^{(2)}| > \frac{1}{9}(|H_{AA}^{(2)}| + |H_{OO}^{(2)}|) + \frac{2}{3}(|H_{AE}^{(2)}| + |H_{EO}^{(2)}|) + \frac{2}{9}|H_{AO}^{(2)}|}. \quad (797)$$

If, moreover, the signs are such that the EE -term contributes constructively, then

$$\mathfrak{H}_O^{\text{corr}} \neq 0.$$

This is not necessary, but it is a practical row-level sufficient test.

CDXLIX. 6. HONEST STATUS RELATIVE TO EARLIER TRUNCATIONS

At the minimal rigid-internal first-shell truncation one had

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0,$$

so orbit-amplitude corrections alone do not activate the odd witness.

The present formula is stronger: it shows exactly where the first nonzero odd response can appear on the corrected branch, namely in the contracted six-row Hessian combination $\mathfrak{H}_O^{\text{corr}}$.

So the odd-slot frontier is no longer vague. It is a literal six-row microscopic overlap table, or one scalar contraction thereof.

CDL. FINAL SUMMARY

On the frozen corrected branch

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$$

the odd transverse response has now been fully contracted.

The quadratic odd response is

$$Q_2(\delta^{\text{final}}) = 4.608 \times 10^{-5} \left[H_{AA}^{(2)} + 9H_{EE}^{(2)} + H_{OO}^{(2)} - 6H_{AE}^{(2)} + 2H_{AO}^{(2)} - 6H_{EO}^{(2)} \right].$$

Hence

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa A_0(4.608 \times 10^{-5})\mathfrak{H}_O^{\text{corr}} + O(\delta^3),$$

with

$$\mathfrak{H}_O^{\text{corr}} = H_{AA}^{(2)} + 9H_{EE}^{(2)} + H_{OO}^{(2)} - 6H_{AE}^{(2)} + 2H_{AO}^{(2)} - 6H_{EO}^{(2)}.$$

Using the exact six-row source formula,

$$H_{rs}^{(2)} = -\frac{\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O[\dots] + \Delta_{rs}^{\text{comp}},$$

this becomes

$$\mathfrak{H}_O^{\text{corr}} = -\frac{\alpha_X\beta_X}{M_X^2}\mathfrak{X}_O + \mathfrak{D}_O.$$

Therefore a sharp sufficient odd-slot activation criterion is

$$\left| \frac{\alpha_X\beta_X}{M_X^2}\mathfrak{X}_O \right| > |\mathfrak{D}_O|.$$

So the odd transverse frontier is no longer vague: it is a literal six-row microscopic overlap table, or equivalently one contracted scalar $\mathfrak{H}_O^{\text{corr}}$.

CDLI. ACTUAL FIRST NONZERO ODD ROUTE: FIXING THE E -SECTOR AS THE DECISIVE MICROSCOPIC TEST

CDLII. SHORT EXPLANATION

At this point the odd-slot frontier is no longer a general six-row ambiguity. The hierarchy is already fixed. In the minimal rigid-internal first-shell truncation,

$$K_A = K_E = K_O = 0,$$

hence all fast six-row coefficients vanish identically and therefore

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.$$

So the only way forward is to leave the rigid truncation. Once this is done, the corrected odd shift is dominated first by $H_{EE}^{(2)}$, then by $H_{AE}^{(2)}$ and $H_{EO}^{(2)}$.

CDLIII. 1. FAST SIX-ROW SYSTEM

Under the fast working approximation,

$$H_{rs}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger]. \quad (798)$$

Therefore, explicitly,

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger], \quad (799)$$

$$H_{AE}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_A K_E^\dagger + J_E K_A^\dagger + K_A J_E^\dagger + K_E J_A^\dagger], \quad (800)$$

$$H_{EO}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_E K_O^\dagger + J_O K_E^\dagger + K_E J_O^\dagger + K_O J_E^\dagger]. \quad (801)$$

Likewise,

$$H_{AA}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_A K_A^\dagger + K_A J_A^\dagger], \quad (802)$$

$$H_{OO}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_O K_O^\dagger + K_O J_O^\dagger], \quad (803)$$

$$H_{AO}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_A K_O^\dagger + J_O K_A^\dagger + K_A J_O^\dagger + K_O J_A^\dagger]. \quad (804)$$

CDLIV. 2. WHY THE RIGID TRUNCATION MUST BE ABANDONED

The completed orbit-current pairs are explicit:

$$\boxed{J_{r,i}^a = q_0^a \partial_i N_r + (\partial_i \bar{\rho})(\eta_r c^a + \bar{\eta}_r \bar{c}^a) + \bar{\rho}(c^a \partial_i \eta_r + \bar{c}^a \partial_i \bar{\eta}_r),} \quad (805)$$

$$\boxed{K_{r,i}^a = (c^a \partial_i \eta_r + \bar{c}^a \partial_i \bar{\eta}_r) \bar{\chi}} \quad (806)$$

in the non-rigid completion picture. :contentReference[oaicite:3]index=3

But in the minimal rigid-internal first-shell truncation one has

$$Q_r^a = 0 \quad \forall r \in \{A, E, O\},$$

hence

$$K_A = K_E = K_O = 0.$$

Therefore every fast coefficient vanishes:

$$\boxed{H_{AA}^{(2),\text{fast}} = H_{EE}^{(2),\text{fast}} = H_{OO}^{(2),\text{fast}} = H_{AE}^{(2),\text{fast}} = H_{AO}^{(2),\text{fast}} = H_{EO}^{(2),\text{fast}} = 0,} \quad (807)$$

and consequently

$$\boxed{Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.} \quad (808)$$

So the first genuine microscopic conclusion is:

$$\boxed{\text{odd-slot activation is impossible unless one leaves the rigid first-shell truncation.}} \quad (809)$$

CDLV. 3. WEIGHT HIERARCHY FIXES THE PRIORITY ORDER

On the frozen corrected branch,

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096),$$

the corrected odd shift is

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)}.$$

Hence the exact weight hierarchy is

$$\boxed{|EE| : |AE| : |EO| : |AO| : |AA| : |OO| = 9 : 6 : 6 : 2 : 1 : 1.} \quad (810)$$

Therefore the correct microscopic priority order is

$$\boxed{H_{EE}^{(2)} \text{ first, } H_{AE}^{(2)}, H_{EO}^{(2)} \text{ second, } H_{AO}^{(2)} \text{ third.}} \quad (811)$$

So the first decisive test for whether $\Lambda_O \neq 0$ beyond the rigid truncation is not the whole six-row system but the single coefficient $H_{EE}^{(2)}$.

CDLVI. 4. EXACT E -SECTOR SOURCE SPLIT

The exact non-rigid E -sector current pair is

$$\boxed{J_{E,i}^a = \partial_i M_E^a}, \quad (812)$$

$$\boxed{K_{E,i}^a = \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_E}. \quad (813)$$

Therefore

$$\boxed{H_{EE}^{(2),\text{fast}} = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)}}, \quad (814)$$

where

$$\boxed{H_{EE,\text{proj}}^{(2)} \text{ comes from } \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi}}, \quad (815)$$

and

$$\boxed{H_{EE,\text{qgrad}}^{(2)} \text{ comes from } (\partial_i \bar{q}^a) \chi_E}. \quad (816)$$

Hence the first nonzero odd transverse source beyond the rigid truncation can come from either

$$\boxed{\partial_i F_E^a \neq 0 \quad (\text{non-rigid internal/projector texture}),} \quad (817)$$

or

$$\boxed{\partial_i \bar{q}^a \neq 0 \quad (\text{background charge-gradient source}).} \quad (818)$$

CDLVII. 5. SHARP SUFFICIENT CRITERION FOR THE FIRST NONZERO ODD SLOT

A practical sufficient condition for first odd activation is:

$$\boxed{H_{EE}^{(2)} \neq 0 \implies \beta^{\text{corr}} \neq 0 \text{ unless the subleading rows cancel it with the tuned opposite sign.}} \quad (819)$$

A stronger row-dominance version is

$$\boxed{|H_{EE}^{(2)}| > \max \left\{ \frac{2.7648}{4.1472} |H_{AE}^{(2)}|, \frac{2.7648}{4.1472} |H_{EO}^{(2)}|, \frac{0.9216}{4.1472} |H_{AO}^{(2)}|, \frac{0.4608}{4.1472} |H_{AA}^{(2)}|, \frac{0.4608}{4.1472} |H_{OO}^{(2)}| \right\}}, \quad (820)$$

which implies

$$\boxed{\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 (4.1472 \times 10^{-4}) H_{EE}^{(2)}}. \quad (821)$$

CDLVIII. CONCLUSION

So the actual next microscopic calculation is uniquely fixed:

$$\boxed{\text{leave the rigid truncation, compute } (J_E, K_E), \text{ and test } H_{EE}^{(2)} \neq 0.} \quad (822)$$

If $H_{EE}^{(2)} \neq 0$, then the odd slot is activated generically before any full six-row computation is required. The six-row system is still exact, but the E -sector is the first decisive microscopic test.

CDLIX. FINAL SUMMARY

The odd-slot frontier is now sharply localized.
In the minimal rigid-internal first-shell truncation,

$$K_A = K_E = K_O = 0,$$

so

$$H_{AA}^{(2),\text{fast}} = H_{EE}^{(2),\text{fast}} = H_{OO}^{(2),\text{fast}} = H_{AE}^{(2),\text{fast}} = H_{AO}^{(2),\text{fast}} = H_{EO}^{(2),\text{fast}} = 0,$$

and therefore

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.$$

Hence odd-slot activation is impossible unless the rigid truncation is abandoned.
Beyond that truncation, the corrected odd shift carries the exact weight hierarchy

$$|EE| : |AE| : |EO| : |AO| : |AA| : |OO| = 9 : 6 : 6 : 2 : 1 : 1,$$

so the first decisive microscopic test is

$$H_{EE}^{(2)}.$$

The exact non-rigid E -sector current pair is

$$J_{E,i}^a = \partial_i M_E^a, \quad K_{E,i}^a = \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_E.$$

Therefore the first nonzero odd transverse source can come from either

$$\partial_i F_E^a \neq 0 \quad (\text{projector/internal texture}),$$

or

$$\partial_i \bar{q}^a \neq 0 \quad (\text{background charge-gradient source}).$$

So the next unique microscopic task is: leave the rigid truncation, compute the E -sector pair (J_E, K_E) , and test whether $H_{EE}^{(2)} \neq 0$.

CDLX. FIRST ACTUAL DECISION BETWEEN THE TWO E -SECTOR ODD SOURCES

CDLXI. SHORT EXPLANATION

The rigid first-shell truncation is already dead for the odd channel:

$$K_A = K_E = K_O = 0 \implies H_{AA}^{(2),\text{fast}} = H_{EE}^{(2),\text{fast}} = H_{OO}^{(2),\text{fast}} = H_{AE}^{(2),\text{fast}} = H_{AO}^{(2),\text{fast}} = H_{EO}^{(2),\text{fast}} = 0,$$

hence

$$Q_2(\delta_{\text{corr}}) = 0, \quad \beta^{\text{corr}} = 0.$$

So the actual odd-slot question begins only beyond the rigid truncation. At that point, the first decisive coefficient is $H_{EE}^{(2)}$, and it splits into

$$H_{EE}^{(2),\text{fast}} = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)}.$$

CDLXII. 1. EXACT E -SECTOR SPLIT

The exact fast E -sector coefficient is

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O[J_E K_E^\dagger + K_E J_E^\dagger]. \quad (823)$$

The completed E -sector orbit-current pair is

$$J_{E,i}^a = \partial_i M_E^a, \quad (824)$$

$$K_{E,i}^a = \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_E. \quad (825)$$

Therefore

$$H_{EE}^{(2),\text{fast}} = H_{EE,\text{proj}}^{(2)} + H_{EE,\text{qgrad}}^{(2)}, \quad (826)$$

with

$$H_{EE,\text{proj}}^{(2)} \propto \mathfrak{P}_O \left[(\partial_i M_E^a) ((\partial_i Q_E^a) \bar{\chi})^\dagger + \text{h.c.} \right], \quad (827)$$

and

$$H_{EE,\text{qgrad}}^{(2)} \propto \mathfrak{P}_O \left[(\partial_\xi M_E^a) (\theta'(\xi) \partial_\theta m_a(\theta) \chi_E)^\dagger + \text{h.c.} \right]. \quad (828)$$

CDLXIII. 2. WHICH SOURCE IS FIRST GENERIC?

The projector/internal-texture piece (827) requires

$$\partial_i Q_E^a \neq 0,$$

that is, genuinely non-rigid internal/projector texture.

By contrast, the q-gradient piece (828) is already carried by the defect background itself, through the background charge gradient and the defect profile:

$$\partial_\xi M_E^a, \quad \theta'(\xi), \quad \partial_\theta m_a(\theta).$$

Therefore the first generic candidate is:

$$H_{EE,\text{qgrad}}^{(2)}. \quad (829)$$

This is precisely why, once the rigid truncation is abandoned, the corrected odd coefficient is led by the q-gradient E -sector source.

CDLXIV. 3. WEIGHTED INSERTION INTO THE CORRECTED ODD COEFFICIENT

For the frozen corrected seed

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096),$$

the corrected odd shift is

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)}.$$

Since the EE -slot carries the largest weight, once no large cancellation occurs,

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 (4.1472 \times 10^{-4}) H_{EE,\text{qgrad}}^{(2)}. \quad (830)$$

So the first actual odd activation test is reduced to the sign and nonvanishing of $H_{EE,\text{qgrad}}^{(2)}$.

CDLXV. 4. FULLY EXPLICIT SHELL-OVERLAP FORM

The q-gradient route has already been reduced to a literal shell-overlap test:

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{4} \frac{\alpha_X \beta_X}{M_X^2} \theta_0 \kappa A_0 (4.1472 \times 10^{-4}) \mathfrak{P}_O \left[\sum_Q Q_\xi \theta'(\xi) k_a(\theta) \Im \left(\widehat{M}_E^a(Q) \overline{\widehat{\chi}_E(-Q)} \right) \right]. \quad (831)$$

Therefore the literal frozen-branch odd decision is

$$\mathfrak{P}_O \left[\sum_Q Q_\xi \theta'(\xi) k_a(\theta) \Im \left(\widehat{M}_E^a(Q) \overline{\widehat{\chi}_E(-Q)} \right) \right] \stackrel{?}{\neq} 0. \quad (832)$$

CDLXVI. 5. HONEST PRACTICAL CONCLUSION

So there are now two different honest forward routes:

a. Route A: stay inside the current JK-route. Then the first generic odd source is $H_{EE, \text{qgrad}}^{(2)}$, and the final decision is the finite shell-overlap test (832). This still requires the actual frozen branch harmonics

$$\bar{A}_G, \quad B_G^{(E)}.$$

b. Route B: move to the explicit odd extra-carrier route. Then the odd slot is explicitly present already at lowest order through

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi, \quad (833)$$

with induced tangent transport block

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right]. \quad (834)$$

So the odd transport density is directly nonzero whenever

$$\tilde{\Lambda}_O \neq 0, \quad \theta'(\xi) \neq 0, \quad \cos \theta(\xi) \neq 0.$$

CDLXVII. CONCLUSION

Inside the current frozen corrected branch, the first generic odd source is the E -sector q-gradient term:

$$H_{EE, \text{qgrad}}^{(2)}$$

But its literal sign test still requires the unavailable frozen harmonics.

Therefore the honest status is:

$$\text{current } JK\text{-route: reduced to one finite shell-overlap test;} \quad (835)$$

$$\text{explicit extra-carrier route: already forward-computable and explicitly odd at lowest order.} \quad (836)$$

CDLXVIII. FINAL SUMMARY

Inside the current frozen corrected branch, the first generic odd source is not the projector/internal-texture term but the E -sector q-gradient term:

$$\boxed{H_{EE,\text{qgrad}}^{(2)}}$$

Accordingly, the corrected odd coefficient is led by

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16}\theta_0\kappa A_0 (4.1472 \times 10^{-4}) H_{EE,\text{qgrad}}^{(2)}.$$

Moreover, the JK -route has already been reduced to a single literal shell-overlap test:

$$\mathfrak{P}_O \left[\sum_Q Q_\xi \theta'(\xi) k_a(\theta) \Im \left(\widehat{M}_E^a(Q) \overline{\widehat{\chi}_E(-Q)} \right) \right] \stackrel{?}{\neq} 0.$$

So the current JK -route is no longer structurally vague. It is one finite shell-overlap decision. However, this test still requires the actual frozen branch harmonics

$$\bar{A}_G, \quad B_G^{(E)},$$

which are not instantiated in the present context.

Therefore the honest forward-computable route at the current stage is the explicit odd extra-carrier route, where

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi$$

induces

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right],$$

and hence gives a directly nonzero odd transport density whenever

$$\tilde{\Lambda}_O \neq 0, \quad \theta'(\xi) \neq 0, \quad \cos \theta(\xi) \neq 0.$$

CDLXIX. ADOPTING THE EXPLICIT ODD EXTRA-CARRIER ROUTE AS THE ACTIVE FORWARD CLOSURE

CDLXX. CURRENT STAGE

The JK -route has already been reduced to a single shell-overlap decision, but that literal sign test still requires unavailable frozen harmonics. Therefore the correct forward move is not to stall there, but to switch to the already explicit odd extra-carrier route.

The complete minimal extra mixing sector is

$$\boxed{\mathcal{L}_{\text{mix}}^{\text{extra}} = \gamma_L \mathcal{I}_L + \gamma_E \mathcal{I}_E + \gamma_O \mathcal{I}_O + \text{h.c.},} \quad (837)$$

with the odd transverse carrier slot

$$\boxed{\mathcal{I}_O = X_i^{(O)} \mathcal{O}^{ia}[P, \Psi, \sigma].} \quad (838)$$

This slot is not optional at the level of complete first-order principal-symbol closure: the complete action needs one independent generator for each of the three carrier directions L, E, O .

CDLXXI. 1. EXPLICIT ODD CARRIER ON THE EMBEDDED DEFECT BRANCH

On the embedded defect branch one has

$$q_O(\theta) = \text{Tr}(S_O M_O) = \cos \theta. \quad (839)$$

Therefore the explicit odd extra carrier is

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi. \quad (840)$$

Hence the odd carrier is locally nonzero whenever

$$\cos \theta(\xi) \neq 0. \quad (841)$$

CDLXXII. 2. INDUCED TANGENT TRANSPORT BLOCK

The induced tangent transport block from the explicit extra E/O slots is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right]. \quad (842)$$

Therefore the local tangent transport densities are

$$a_3^{(EO)} = 0, \quad b_E^{(EO)} = \tilde{\Lambda}_E \theta'(\xi) \cos \theta(\xi), \quad b_O^{(EO)} = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi). \quad (843)$$

So the odd transport density is explicitly nonzero at lowest order whenever

$$\tilde{\Lambda}_O \neq 0, \quad \theta'(\xi) \neq 0, \quad \cos \theta(\xi) \neq 0. \quad (844)$$

CDLXXIII. 3. EXACT REDUCED COEFFICIENT SHEET

Define the overlap integrals

$$I_3 := \int d\xi f_+ f_- \theta'(\xi)^3, \quad I_c := \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi). \quad (845)$$

Then the exact reduced coefficient dictionary is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O. \quad (846)$$

Equivalently, with $I_{\theta, \cos} = I_c$,

$$\alpha = I_{\theta, \cos} \tilde{\Lambda}_E, \quad \beta = I_{\theta, \cos} \tilde{\Lambda}_O. \quad (847)$$

CDLXXIV. 4. MICROSCOPIC ONE-MEDIATOR COEFFICIENT POINT

At operator level, the one-mediator audit gives

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1. \quad (848)$$

If the shell basis is canonically normalized so that

$$C_{\parallel} = C_E = C_O = 1,$$

then

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} J_1, \quad (849)$$

and, in the transport-side canonical explicit normalization where J_1 is absorbed,

$$\tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}. \quad (850)$$

The same formulas hold for the even slot:

$$\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2} \quad (851)$$

in canonical explicit normalization.

CDLXXV. 5. EXPLICIT ODD COEFFICIENT

Substituting (848) into (846), one gets

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_c. \quad (852)$$

In the canonical explicit normalization this reduces to

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} I_c \quad (853)$$

if J_1 is absorbed, or

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} J_1 I_c \quad (854)$$

if J_1 is kept explicit.

Likewise,

$$\alpha = -\frac{\beta_X \gamma_E}{M_X^2} I_c \quad (855)$$

in the same canonical explicit normalization.

CDLXXVI. 6. RESULTING REDUCED PDE

Therefore the solved lowest-order Branch-B reduced PDE becomes

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) - I_c \tilde{\Lambda}_E (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - I_c \tilde{\Lambda}_O (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0. \quad (856)$$

In canonical explicit normalization this is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) + \frac{\beta_X \gamma_E}{M_X^2} I_c (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) + \frac{\beta_X \gamma_O}{M_X^2} I_c (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0. \quad (857)$$

CDLXXVII. 7. MINIMAL NONZERO CRITERION

From (852), the odd transverse channel is nonzero exactly when

$$\boxed{\beta_X \neq 0, \quad \gamma_O \neq 0, \quad I_c \neq 0,} \quad (858)$$

and, if the operator normalization is not absorbed,

$$\boxed{C_O J_1 \neq 0.} \quad (859)$$

So the odd channel is no longer waiting on unavailable frozen harmonics. It is reduced to one coefficient sign/product condition and one geometric overlap condition.

CDLXXVIII. 8. HONEST RELATION TO THE PREVIOUS MIXED ODD-ACTIVE BRANCH

This move changes the active forward route.

The earlier canonical mixed odd-active branch was structurally closed but still two finite microscopic audits away from a fully populated coefficient point. By contrast, the explicit odd extra-carrier route already supplies the odd channel at lowest order.

So the honest current forward statement is:

stop stalling on the unavailable frozen-harmonic JK overlap and use the explicit odd extra-carrier route as the active closure route
--

(860)

CDLXXIX. FINAL SUMMARY

The correct active forward route is now the explicit odd extra-carrier route.

The JK -route has already been reduced to a single shell-overlap decision, but that literal sign test still requires unavailable frozen harmonics. By contrast, the explicit odd route is already concrete.

Its odd carrier is

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi,$$

and the induced tangent transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E^\mu{}_a \Sigma_a + \tilde{\Lambda}_O J^\mu{}_a \Sigma_a \right].$$

Therefore the reduced coefficients are

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O,$$

with

$$I_3 = \int d\xi f_+ f_- \theta'(\xi)^3, \quad I_c = \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi).$$

At one-mediator level,

$$\tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1,$$

so in canonical explicit normalization

$$\beta = -\frac{\beta_X \gamma_O}{M_X^2} I_c.$$

Hence the odd channel is nonzero exactly when

$$\beta_X \neq 0, \quad \gamma_O \neq 0, \quad I_c \neq 0,$$

and, in the unabsorbed normalization, also

$$C_O J_1 \neq 0.$$

So the odd channel is no longer waiting on unavailable frozen harmonics. The active forward-computable closure route is already explicit.

CDLXXX. COMPLETE LOWEST-ORDER PDE ON THE EXPLICIT ODD EXTRA-CARRIER ROUTE

CDLXXXI. ACTIVE BRANCH CHOICE

We now adopt the explicit odd extra-carrier route as the active forward closure route and fix

$$\boxed{\gamma_O \neq 0.} \tag{861}$$

If one also wants a nonzero even transverse slot on the same route, one keeps

$$\boxed{\gamma_E \neq 0.} \tag{862}$$

This does not yet fix the numerical values of γ_E, γ_O , but it completely fixes the lowest-order coefficient architecture.

CDLXXXII. 1. EXACT OVERLAP DICTIONARY

The reduced coefficient dictionary on the explicit odd extra-carrier route is

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O.} \tag{863}$$

The geometric overlap integrals are

$$\boxed{I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3, \quad I_c = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \cos \theta(\xi).} \tag{864}$$

CDLXXXIII. 2. ONE-MEDIATOR COEFFICIENT POINT

At one-mediator level, the explicit odd/even carrier coefficients are

$$\boxed{\tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1, \quad \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.} \tag{865}$$

In canonical explicit normalization,

$$C_{\parallel} = C_E = C_O = 1,$$

and if J_1 is absorbed into the coefficient normalization, this reduces to

$$\boxed{\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}.} \tag{866}$$

CDLXXXIV. 3. FINAL LOWEST-ORDER COEFFICIENT SHEET

Substituting (865) into (863), one gets the general explicit coefficient sheet

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad (867)$$

$$\alpha = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1 I_c, \quad \beta = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_c. \quad (868)$$

In canonical explicit normalization with J_1 absorbed, this becomes

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = -\frac{\beta_X \gamma_E}{M_X^2} I_c, \quad \beta = -\frac{\beta_X \gamma_O}{M_X^2} I_c. \quad (869)$$

CDLXXXV. 4. COMPLETE LOWEST-ORDER REDUCED PDE

The reduced principal symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

Therefore the complete lowest-order reduced PDE is

$$\left[i\partial_t - \lambda_{\parallel} \sigma_3(-i\partial_{\parallel}) - \alpha(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - \beta(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0. \quad (870)$$

Using (868), this is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{\parallel}) + \frac{\beta_X \gamma_E}{M_X^2} C_E J_1 I_c (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + \frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_c (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0. \quad (871)$$

In canonical explicit normalization with J_1 absorbed:

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{\parallel}) + \frac{\beta_X \gamma_E}{M_X^2} I_c (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + \frac{\beta_X \gamma_O}{M_X^2} I_c (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0. \quad (872)$$

CDLXXXVI. 5. DISPERSION RELATION

The resulting dispersion is

$$E(p)^2 = \lambda_{\parallel}^2 p_{\parallel}^2 + (\alpha^2 + \beta^2)(p_1^2 + p_2^2). \quad (873)$$

Using (868),

$$E(p)^2 = \frac{\lambda_c^2 I_3^2}{4} p_{\parallel}^2 + \frac{\beta_X^2 I_c^2}{M_X^4} \left[(\gamma_E C_E J_1)^2 + (\gamma_O C_O J_1)^2 \right] (p_1^2 + p_2^2). \quad (874)$$

In canonical explicit normalization with J_1 absorbed,

$$E(p)^2 = \frac{\lambda_c^2 I_3^2}{4} p_{\parallel}^2 + \frac{\beta_X^2 I_c^2}{M_X^4} (\gamma_E^2 + \gamma_O^2) (p_1^2 + p_2^2). \quad (875)$$

CDLXXXVII. 6. EXACT NONZERO CONDITIONS

The longitudinal channel is nonzero iff

$$\boxed{\lambda_c \neq 0, \quad I_3 \neq 0.} \quad (876)$$

The even transverse channel is nonzero iff

$$\boxed{\beta_X \neq 0, \quad \gamma_E \neq 0, \quad I_c \neq 0,} \quad (877)$$

and, in unabsorbed normalization, also $C_E J_1 \neq 0$.

The odd transverse channel is nonzero iff

$$\boxed{\beta_X \neq 0, \quad \gamma_O \neq 0, \quad I_c \neq 0,} \quad (878)$$

and, in unabsorbed normalization, also $C_O J_1 \neq 0$.

CDLXXXVIII. 7. PRACTICAL FINAL THEOREM

Theorem CDLXXXVIII.1 (Complete lowest-order odd-route carrier theorem). *Assume:*

1. *the explicit odd extra-carrier route is adopted;*
2. $\gamma_O \neq 0$;
3. $\lambda_c \neq 0, I_3 \neq 0, I_c \neq 0$;
4. $\beta_X \neq 0$;
5. *if the operator normalization is unabsorbed, $C_E J_1 \neq 0$ and $C_O J_1 \neq 0$.*

Then the reduced lowest-order carrier PDE is exactly (871), and the odd transverse slot is nonzero:

$$\beta \neq 0.$$

If in addition $\gamma_E \neq 0$, then $\alpha \neq 0$ as well, and the branch carries the full longitudinal-even-odd coefficient triplet at lowest order.

Proof. This follows directly from the explicit carrier dictionary (863), the one-mediator coefficient point (865), and the nonzero conditions (876)–(878). $\square$

CDLXXXIX. FINAL SUMMARY

Adopting the explicit odd extra-carrier route and choosing $\gamma_O \neq 0$ fixes the complete lowest-order reduced coefficient sheet.

The exact coefficient dictionary is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O,$$

with

$$I_3 = \int d\xi f_+ f_- \theta'(\xi)^3, \quad I_c = \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi).$$

At one-mediator level,

$$\tilde{\Lambda}_E = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1, \quad \tilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1.$$

Hence the complete lowest-order reduced PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) - \alpha (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) - \beta (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0,$$

with

$$\alpha = -\frac{\beta_X \gamma_E}{M_X^2} C_E J_1 I_c, \quad \beta = -\frac{\beta_X \gamma_O}{M_X^2} C_O J_1 I_c.$$

In canonical explicit normalization this simplifies to

$$\alpha = -\frac{\beta_X \gamma_E}{M_X^2} I_c, \quad \beta = -\frac{\beta_X \gamma_O}{M_X^2} I_c.$$

Therefore the odd transverse slot is nonzero exactly when

$$\beta_X \neq 0, \quad \gamma_O \neq 0, \quad I_c \neq 0,$$

and, if the normalization is not absorbed, also

$$C_O J_1 \neq 0.$$

So the odd channel is now closed at complete lowest order on the explicit odd extra-carrier route.

CDXC. REFERENCE PDE ON THE EXPLICIT MIXED TRANSVERSE ROUTE AND ITS MATCHING TO THE CANONICAL MIXED ODD-ACTIVE BRANCH

CDXCI. SHORT EXPLANATION

The explicit pure-even witness branch

$$\gamma_E = 1, \quad \gamma_O = 0$$

is no longer useful for reading the odd coefficient, because it forces

$$\beta = 0.$$

So the next active branch must satisfy

$$\boxed{\gamma_O \neq 0.} \tag{879}$$

We therefore adopt the explicit odd extra-carrier route as the reference low-order PDE route.

CDXCII. 1. EXACT REDUCED COEFFICIENT SHEET ON THE EXPLICIT ODD ROUTE

The reduced coefficient dictionary is

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \tilde{\Lambda}_E I_c, \quad \beta = \tilde{\Lambda}_O I_c,} \tag{880}$$

with

$$\boxed{I_3 = \int d\xi f_+ f_- \theta'(\xi)^3, \quad I_c = \int d\xi f_+ f_- \theta'(\xi) \cos \theta(\xi).} \tag{881}$$

At one-mediator level,

$$\boxed{\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2} C_E, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2} C_O.} \tag{882}$$

In canonical explicit normalization

$$C_{\parallel} = C_E = C_O = 1,$$

this becomes

$$\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}. \quad (883)$$

Therefore the explicit mixed transverse coefficient sheet is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = -\frac{\beta_X \gamma_E}{M_X^2} I_c, \quad \beta = -\frac{\beta_X \gamma_O}{M_X^2} I_c. \quad (884)$$

CDXCIII. 2. REFERENCE PDE

The reference reduced symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

Hence the reference PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) + \frac{\beta_X \gamma_E}{M_X^2} I_c (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) + \frac{\beta_X \gamma_O}{M_X^2} I_c (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0. \quad (885)$$

The corresponding dispersion is

$$E(p)^2 = \frac{\lambda_c^2 I_3^2}{4} p_{\parallel}^2 + \frac{\beta_X^2 I_c^2}{M_X^4} (\gamma_E^2 + \gamma_O^2) (p_1^2 + p_2^2). \quad (886)$$

CDXCIV. 3. MINIMAL NONZERO CONDITIONS

The longitudinal slot is nonzero iff

$$\lambda_c \neq 0, \quad I_3 \neq 0. \quad (887)$$

The even transverse slot is nonzero iff

$$\beta_X \neq 0, \quad \gamma_E \neq 0, \quad I_c \neq 0. \quad (888)$$

The odd transverse slot is nonzero iff

$$\beta_X \neq 0, \quad \gamma_O \neq 0, \quad I_c \neq 0. \quad (889)$$

So the odd route is already closed at lowest order once γ_O is chosen nonzero.

CDXCV. 4. MATCHING TO THE CANONICAL MIXED ODD-ACTIVE BRANCH

The exact branch-level mixed odd-active coefficient sheet is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \beta_X T_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X T_O \theta_0 \kappa. \quad (890)$$

Therefore the explicit odd route matches the mixed odd-active sheet provided

$$\frac{\lambda_c}{2} I_3 = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (891)$$

$$\boxed{\frac{\gamma_E}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} T_E \theta_0 \kappa, \quad \frac{\gamma_O}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} T_O \theta_0 \kappa.} \quad (892)$$

In particular, for the canonical mixed odd-active target

$$T_E = T_O = 1,$$

the most natural matched explicit route is

$$\boxed{\gamma_E = \gamma_O =: \gamma_\perp \neq 0,} \quad (893)$$

together with

$$\boxed{\frac{\gamma_\perp}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} \theta_0 \kappa.} \quad (894)$$

Then

$$\boxed{\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa,} \quad (895)$$

which is exactly the canonical mixed odd-active transverse sheet.

CDXCVI. 5. LEADING ANISOTROPY RATIO

In the one-mediator truncation the leading anisotropy ratio is

$$\boxed{\rho_{\text{LO}} = -\frac{3}{32} \frac{\alpha_X}{\sqrt{\gamma_E^2 + \gamma_O^2}} (\theta_0 \kappa)^2} \quad (896)$$

in canonical explicit normalization.

For the matched canonical mixed choice $\gamma_E = \gamma_O = \gamma_\perp$,

$$\boxed{\rho_{\text{LO}} = -\frac{3}{32\sqrt{2}} \frac{\alpha_X}{|\gamma_\perp|} (\theta_0 \kappa)^2.} \quad (897)$$

So the explicit odd route not only reproduces the odd channel, but also gives a direct low-order anisotropy formula.

CDXCVII. 6. PRACTICAL FINAL THEOREM

Theorem CDXCVII.1 (Reference-PDE theorem on the explicit mixed transverse route). *Assume canonical explicit normalization, and choose*

$$\gamma_O \neq 0.$$

Then the explicit odd extra-carrier route defines the exact lowest-order reduced PDE (885).

If, in addition,

$$\lambda_c \neq 0, \quad I_3 \neq 0, \quad I_c \neq 0, \quad \beta_X \neq 0,$$

then

$$\lambda_\parallel \neq 0, \quad \beta \neq 0.$$

If also $\gamma_E \neq 0$, then $\alpha \neq 0$ as well.

Finally, if the matching conditions (891)–(892) hold, then the explicit odd route reproduces the canonical mixed odd-active coefficient sheet.

Proof. The first part follows directly from (884). The second part follows from the nonzero conditions (887)–(889). The final statement follows by direct substitution of the matching relations into (884). $\square$

CDXCVIII. FINAL SUMMARY

We now fix the explicit odd extra-carrier route as the reference low-order PDE route and choose

$$\gamma_O \neq 0.$$

In canonical explicit normalization, the exact reduced coefficient sheet is

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = -\frac{\beta_X \gamma_E}{M_X^2} I_c, \quad \beta = -\frac{\beta_X \gamma_O}{M_X^2} I_c.$$

Therefore the reference PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{\parallel}) + \frac{\beta_X \gamma_E}{M_X^2} I_c (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + \frac{\beta_X \gamma_O}{M_X^2} I_c (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

This route has

$$\beta \neq 0 \iff \beta_X \neq 0, \gamma_O \neq 0, I_c \neq 0.$$

Moreover, it matches the canonical mixed odd-active branch provided

$$\frac{\gamma_E}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} T_E \theta_0 \kappa, \quad \frac{\gamma_O}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} T_O \theta_0 \kappa.$$

For the canonical mixed target $T_E = T_O = 1$, the natural matched choice is

$$\gamma_E = \gamma_O =: \gamma_{\perp} \neq 0,$$

which reproduces

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.$$

So the explicit odd route is now fixed as the reference low-order PDE route, and it can be matched directly onto the canonical mixed odd-active coefficient sheet.

CDXCIX. FIXING THE CURRENT REFERENCE PDE BY THE SYMMETRIC MIXED TRANSVERSE CHOICE

D. REFERENCE BRANCH CHOICE

We fix the active branch parameters by

$$\boxed{\gamma_E = \gamma_O =: \gamma_{\perp} \neq 0.} \tag{898}$$

This is the minimal symmetric mixed transverse choice that keeps both transverse channels alive and matches naturally onto the canonical mixed odd-active target.

DI. 1. EXACT COEFFICIENT SHEET ON THE EXPLICIT ODD ROUTE

The explicit odd extra-carrier route has the exact reduced dictionary

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O.} \tag{899}$$

At one-mediator level, in canonical explicit normalization,

$$\boxed{\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}.} \tag{900}$$

Therefore under (898),

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c, \quad \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c. \quad (901)$$

Thus

$$\alpha = \beta. \quad (902)$$

DII. 2. REFERENCE PDE IN THE ORIGINAL TRANSVERSE COORDINATES

The reduced symbol is

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2).$$

Using (901), the reference PDE becomes

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) + \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c (\sigma_1 (-i\partial_1) + \sigma_2 (-i\partial_2)) + \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c (-\sigma_1 (-i\partial_2) + \sigma_2 (-i\partial_1)) \right] \chi = 0. \quad (903)$$

DIII. 3. ROTATED TRANSVERSE FORM

Because $\alpha = \beta$, introduce the rotated transverse coordinates

$$q_1 := \frac{p_1 - p_2}{\sqrt{2}}, \quad q_2 := \frac{p_1 + p_2}{\sqrt{2}}. \quad (904)$$

Then the symbol becomes

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + v_{\perp}^{\text{ref}} (q_1 \sigma_1 + q_2 \sigma_2), \quad (905)$$

with

$$v_{\perp}^{\text{ref}} = \sqrt{2} \alpha = -\sqrt{2} \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c. \quad (906)$$

Hence the reference PDE takes the isotropic transverse form

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) - v_{\perp}^{\text{ref}} (\sigma_1 (-i\partial_{q_1}) + \sigma_2 (-i\partial_{q_2})) \right] \chi = 0. \quad (907)$$

DIV. 4. EXACT MATCHING TO THE CANONICAL MIXED ODD-ACTIVE BRANCH

The branch-level canonical mixed odd-active sheet is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa. \quad (908)$$

Therefore the explicit odd route matches the canonical mixed odd-active target iff

$$\frac{\lambda_c}{2} I_3 = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (909)$$

and

$$\boxed{\frac{\gamma_{\perp}}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} \theta_0 \kappa.} \quad (910)$$

Under (910),

$$\boxed{\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa,} \quad (911)$$

so the transverse reference sheet coincides exactly with the canonical mixed odd-active branch.

DV. 5. DISPERSION RELATION

The corresponding dispersion relation is

$$\boxed{E(p)^2 = \frac{\lambda_c^2 I_3^2}{4} p_{\parallel}^2 + 2 \left(\frac{\beta_X \gamma_{\perp}}{M_X^2} \right)^2 I_c^2 (p_1^2 + p_2^2).} \quad (912)$$

Equivalently,

$$\boxed{E(p)^2 = \frac{\lambda_c^2 I_3^2}{4} p_{\parallel}^2 + (v_{\perp}^{\text{ref}})^2 (q_1^2 + q_2^2).} \quad (913)$$

DVI. 6. REFERENCE-BRANCH FIXATION THEOREM

Theorem DVI.1 (Current reference-PDE fixation theorem). *Assume canonical explicit normalization and choose*

$$\gamma_E = \gamma_O =: \gamma_{\perp} \neq 0.$$

Then the explicit odd extra-carrier route defines the reference lowest-order reduced PDE (907) with coefficient sheet (901).

If, in addition,

$$\lambda_c \neq 0, \quad I_3 \neq 0, \quad I_c \neq 0, \quad \beta_X \neq 0,$$

then all three reduced channels are nonzero:

$$\lambda_{\parallel} \neq 0, \quad \alpha \neq 0, \quad \beta \neq 0.$$

Finally, if the matching relations (909) and (910) hold, then the reference PDE reproduces the canonical mixed odd-active coefficient sheet exactly.

Proof. Directly from (901), (907), and the matching relations. □

DVII. FINAL SUMMARY

We now fix the current reference branch by

$$\gamma_E = \gamma_O =: \gamma_{\perp} \neq 0.$$

On the explicit odd extra-carrier route, this gives the exact lowest-order coefficient sheet

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3, \quad \alpha = \beta = -\frac{\beta_X \gamma_{\perp}}{M_X^2} I_c.$$

Hence the current reference PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3 (-i\partial_{\parallel}) - v_{\perp}^{\text{ref}} (\sigma_1 (-i\partial_{q_1}) + \sigma_2 (-i\partial_{q_2})) \right] \chi = 0,$$

with

$$v_{\perp}^{\text{ref}} = -\sqrt{2} \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c.$$

This reference route matches the canonical mixed odd-active branch exactly if

$$\frac{\lambda_c}{2} I_3 = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad \frac{\gamma_{\perp}}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} \theta_0 \kappa.$$

Therefore the explicit odd extra-carrier route is now fixed as the current reference low-order PDE route, and it reproduces the canonical mixed odd-active transverse sheet after the matching is imposed.

DVIII. PROJECT: TECT — MASTER CONTENTS FOR CONTINUATION (REFERENCE PDE FIXED AND MIXED-BRANCH MATCHING STAGE)

DIX. CURRENT STAGE

The current research stage is the following:

$$\boxed{\text{the explicit odd extra-carrier route is now fixed as the current reference low-order PDE route,}} \quad (914)$$

while

$$\boxed{\text{the canonical mixed odd-active branch remains the structural target branch,}} \quad (915)$$

and the remaining task is to match the reference route onto that target branch and continue the finite coefficient extraction.

The sharp current separation is:

$$\boxed{\text{Reference route: explicit odd extra-carrier route (forward-computable now),}} \quad (916)$$

$$\boxed{\text{Target route: canonical mixed odd-active branch (structurally closed, two finite audits remain).}} \quad (917)$$

DX. WHAT IS ALREADY FIXED

The canonical mixed odd-active target branch is

$$\boxed{T_E = T_O = 1, \quad R_{\perp} = \sqrt{2}, \quad \Delta_{00} = \frac{1}{\sqrt{2}}.} \quad (918)$$

Its exact reduced branch-level coefficient sheet is

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3,} \quad (919)$$

$$\boxed{\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa.} \quad (920)$$

Thus the target mixed branch is no longer waiting on branch-choice logic. What remains there is only microscopic coefficient population.

DXI. WHY THE ACTIVE FORWARD ROUTE CHANGED

The earlier JK -route for the odd slot was reduced to a finite shell-overlap decision, but its literal sign test still requires unavailable frozen harmonics. Therefore the correct forward move was to stop stalling there and switch to the explicit odd extra-carrier route.

The decisive explicit odd slot is

$$\mathcal{I}_O = X_i^{(O)} \cos \theta(\xi) J_j D_j \psi. \quad (921)$$

Its induced tangent transport block is

$$V_{EO}^\mu = \theta'(\xi) \cos \theta(\xi) \left[\tilde{\Lambda}_E E_a^\mu \Sigma_a + \tilde{\Lambda}_O J_a^\mu \Sigma_a \right]. \quad (922)$$

Hence the local odd density is

$$b_O(\xi) = \tilde{\Lambda}_O \theta'(\xi) \cos \theta(\xi), \quad (923)$$

so the odd channel is explicitly realizable at lowest order once

$$\tilde{\Lambda}_O \neq 0, \quad I_c \neq 0.$$

DXII. CURRENT REFERENCE BRANCH CHOICE

From now on, the current reference branch is fixed by

$$\gamma_E = \gamma_O =: \gamma_\perp \neq 0. \quad (924)$$

This is the minimal symmetric mixed-transverse choice that keeps both transverse channels alive and matches naturally to the target condition

$$T_E = T_O = 1.$$

DXIII. REFERENCE LOW-ORDER COEFFICIENT DICTIONARY

The explicit odd-route reduced coefficient dictionary is

$$\lambda_\parallel = \frac{\lambda_c}{2} I_3, \quad \alpha = I_c \tilde{\Lambda}_E, \quad \beta = I_c \tilde{\Lambda}_O. \quad (925)$$

The overlap integrals are

$$I_3 = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi)^3, \quad I_c = \int d\xi f_+(\xi) f_-(\xi) \theta'(\xi) \cos \theta(\xi). \quad (926)$$

At one-mediator level, in canonical explicit normalization,

$$\tilde{\Lambda}_E^{(1)} = -\frac{\beta_X \gamma_E}{M_X^2}, \quad \tilde{\Lambda}_O^{(1)} = -\frac{\beta_X \gamma_O}{M_X^2}. \quad (927)$$

Under the fixed reference choice $\gamma_E = \gamma_O = \gamma_\perp$,

$$\lambda_\parallel = \frac{\lambda_c}{2} I_3, \quad \alpha = \beta = -\frac{\beta_X \gamma_\perp}{M_X^2} I_c. \quad (928)$$

DXIV. CURRENT REFERENCE PDE

The current reference reduced PDE is

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{\parallel}) + \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0. \quad (929)$$

Introducing rotated transverse coordinates

$$q_1 := \frac{p_1 - p_2}{\sqrt{2}}, \quad q_2 := \frac{p_1 + p_2}{\sqrt{2}}, \quad (930)$$

the same reference PDE becomes

$$\left[i\partial_t - \frac{\lambda_c}{2} I_3 \sigma_3(-i\partial_{\parallel}) - v_{\perp}^{\text{ref}} (\sigma_1(-i\partial_{q_1}) + \sigma_2(-i\partial_{q_2})) \right] \chi = 0, \quad (931)$$

with

$$v_{\perp}^{\text{ref}} = -\sqrt{2} \frac{\beta_X \gamma_{\perp}}{M_X^2} I_c. \quad (932)$$

The corresponding dispersion is

$$E(p)^2 = \frac{\lambda_c^2 I_3^2}{4} p_{\parallel}^2 + 2 \left(\frac{\beta_X \gamma_{\perp}}{M_X^2} \right)^2 I_c^2 (p_1^2 + p_2^2). \quad (933)$$

DXV. REFERENCE-ROUTE NONZERO CONDITIONS

The reference route has

$$\lambda_{\parallel} \neq 0 \iff \lambda_c \neq 0, \quad I_3 \neq 0, \quad (934)$$

and

$$\alpha \neq 0, \quad \beta \neq 0 \iff \beta_X \neq 0, \quad \gamma_{\perp} \neq 0, \quad I_c \neq 0. \quad (935)$$

So the current reference branch is nondegenerate as soon as

$$\lambda_c \neq 0, \quad I_3 \neq 0, \quad I_c \neq 0, \quad \beta_X \neq 0, \quad \gamma_{\perp} \neq 0.$$

DXVI. MATCHING TO THE CANONICAL MIXED ODD-ACTIVE TARGET

The reference route matches the canonical mixed odd-active target iff

$$\frac{\lambda_c}{2} I_3 = -\frac{3\sqrt{3}\pi}{512} \alpha_X L_* \theta_0^3 \kappa^3, \quad (936)$$

and

$$\frac{\gamma_{\perp}}{M_X^2} I_c = \frac{\sqrt{3}\pi}{16} \theta_0 \kappa. \quad (937)$$

Under (937),

$$\alpha = \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \theta_0 \kappa, \quad (938)$$

so the transverse reference sheet reproduces the canonical mixed odd-active target exactly.

DXVII. WHAT REMAINS OPEN ON THE TARGET MIXED BRANCH

The canonical mixed odd-active branch is structurally closed, but its fully populated reduced microscopic coefficient point still awaits exactly two finite microscopic audits.

A. (A) Mixed same-shell support insertion

The branch-consistent sufficient support condition is

$$\boxed{|E_{z_0}^{\text{dir}}(0)| - |B_{z_0}(0)| > \frac{4\beta_X^2}{M_X^2} \mathcal{S}_{\text{diag}}^{(\pi/4)} + \varepsilon_{\text{small-}p,\rho}^{(1,\pi/4)}}. \quad (939)$$

The only uninserted same-shell quantities are

$$\boxed{\mathcal{S}_{\text{diag}}^{TT}, \quad L_+^{TT}, L_-^{TT}, L_0^{TT}, \quad C_+^{TT}, C_-^{TT}, C_m^{TT}}. \quad (940)$$

B. (B) Mixed-block transverse cancellation audit

At the current conservative stage,

$$\boxed{\Lambda_E = e_{0;K}^{(1)} + \Delta e_{\text{mix}}, \quad \Lambda_O = o_{0;K}^{(1)} + \Delta o_{\text{mix}}}. \quad (941)$$

A strong sufficient survival condition is

$$\boxed{|\Delta e_{\text{mix}}| < |e_{0;K}^{(1)}|, \quad |\Delta o_{\text{mix}}| < |o_{0;K}^{(1)}|}. \quad (942)$$

Once the two audits are done, the final mixed-branch coefficient point becomes

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3 \ell_{0;OO}^{(3)}}. \quad (943)$$

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (e_{0;K}^{(1)} + \Delta e_{\text{mix}}), \quad \beta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa (o_{0;K}^{(1)} + \Delta o_{\text{mix}})}. \quad (944)$$

DXVIII. PRACTICAL ACCEPTANCE CONTEXT

At practical working-branch theorem level, carrier acceptance is already conditionally closed, but full unconditional microscopic closure still depends on the remaining finite audits and explicit overlap evaluations. In particular, the actual overlap evaluations $C_{\parallel}, C_E, C_O$, the corrected residual-symmetry audit, and the compatible parameter region where all gates hold simultaneously still matter.

practical theorem level: closed conditionally, full unconditional microscopic closure: not yet.

DXIX. IMMEDIATE NEXT TASKS FOR THE NEXT CONVERSATION

The exact next tasks are:

$$\boxed{\text{Task 1: treat (931) as the current reference PDE and keep all notation in that frame.}} \quad (945)$$

Task 2: continue finite extraction of I_c , I_3 , λ_c , $\gamma_\perp$ and their sign choices. (946)

Task 3: use the matching conditions (936)–(937) as the bridge to the canonical mixed odd-active target. (947)

Task 4: if returning to the target branch directly, continue Audit A and Audit B only. (948)

DXX. SHORTEST HONEST STATUS SENTENCE

Current reference route is fixed, target mixed branch is structurally closed, and what remains is only finite coefficient insertion and matching.